CHAPTER 2

Respiratory System and Tuberculosis

INTRODUCTION, BASIC APPROACH, SYMPTOMATOLOGY, AND INVESTIGATIONS

Q. Write short note on the normal arterial blood gas (ABG) levels.

Table 2.1 depicts normal arterial blood gas levels.

TABLE 2.1: Normal arterial blood gas (ABG) levels.

Arterial oxygen tension (PaO_2)	90–104 mm Hg
Arterial carbon dioxide ($PaCO_2$)	35–45 mm Hg
Arterial oxygen saturation (SaO_2)	95–99%
Arterial blood pH (pH)	7.35–7.45 units
Arterial bicarbonate (HCO_3)	22–30 mEq/L
Base excess (BE)	0 ± 2 mmol/L

HYPERCAPNIA

Q. CT2.4 Describe and discuss the physiology and pathophysiology of hypercapnia.

Definition

- **Hypercapnia** is defined as a raised **arterial carbon dioxide** ($PaCO_2$) **of 45 mm Hg** (6 kPa) at rest.
- **Hypercapnic encephalopathy:** When **$PaCO_2$ exceeds 90 mm Hg** (12 kPa), severe hypercapnia causes confusion, progressive drowsiness, and CO_2 narcosis.

Causes of Hypercapnia (Table 2.2)

Usually results from **alveolar hypoventilation**.

TABLE 2.2: Causes of hypercapnia.

Central depression of respiratory drive: ➢ Brainstem lesions ➢ Central sleep apnea, obesity hypoventilation ➢ *Sedatives*: Morphine	**Reduced alveolar ventilation due to pathology within the lungs:** ➢ Chronic bronchitis ➢ Emphysema
Neuromuscular: ➢ Peripheral neuropathy, GB syndrome, poliomyelitis ➢ Myasthenia gravis ➢ Myopathies	**Reduced chest wall** (including diaphragmatic) **movements:** ➢ Trauma ➢ Kyphoscoliosis and pectus excavatum ➢ Ankylosing spondylitis

(GB: Guillain–Barré syndrome)

Clinical Features

- **Headache:** Severe generalized or bilateral frontal or occipital headache (especially **severe on waking up**)
- **Depresses the level of consciousness:**
  - In mild cases, it produces intermittent drowsiness, indifference or inattention, reduction of psychomotor activity, and forgetfulness.
  - In severe cases, it causes mental dullness, drowsiness, confusion, seizures, stupor, and coma. If left untreated, acute hypercapnic respiratory failure can cause death.

Signs

See **Box 2.1**.

Box 2.1: Signs of hypercapnia.

- Asterixis
- Altered sensorium
- Flushed warm extremities
- Bounding peripheral pulses
- Fasciculations
- Papilledema

Investigations

- **Arterial blood gas (ABG) study:** For confirmation of hypercapnia.
- **Other relevant investigations** depending on the underlying cause.

Precaution

In chronic hypercapnia [e.g., chronic obstructive pulmonary disease (COPD)], oxygen should be administered in a controlled manner (about 2 L/minute) and morphine and other sedatives should be avoided.

Treatment
- Mechanical ventilation with intermittent positive-pressure respirator. Noninvasive ventilation (NIV) may be sufficient in patients with mild symptoms. BiLevel positive airway pressure (BiPAP): 8–12 cmH_2O (inspiratory pressure) and 3–5 cmH_2O (expiratory pressure)
- Maintenance of acid–base and electrolyte balance
- Treatment of the underlying cause.

HYPOXEMIA

Q. CT2.4 **Describe and discuss the physiology and pathophysiology of hypoxia.**

Definition

Hypoxemia is defined as **arterial oxygen tension (PaO_2) of less than 80 mm Hg** (10.6 kPa) in a young healthy adult. As the age advances, the normal arterial oxygen tension (PaO_2) falls gradually.

Causes of Hypoxemia (Table 2.3)

Hypoxemia is rare due to reduced diffusion.

TABLE 2.3: Causes of hypoxemia.

Ventilation-perfusion mismatch in the lung: ➢ Chronic bronchitis ➢ Emphysema ➢ Acute asthma ➢ Pulmonary embolism ➢ Interstitial lung diseases	**Hypoventilation:** ➢ **Central depression of respiratory drive:** Brainstem lesions and central sleep apnea ➢ **Peripheral neuromuscular disease:** Myasthenia gravis peripheral, neuropathy, myasthenia gravis, and myopathies ➢ Malfunctioning of mechanical ventilators
Right-to-left shunts: ➢ Congenital cyanotic heart diseases ➢ Pulmonary collapse/atelectasis ➢ Consolidation, e.g., pneumonia	**Decreased inspired oxygen concentration:** ➢ High altitudes ➢ Malfunctioning of mechanical ventilators

Clinical Features

- **Acute hypoxemia:** It closely resembles acute alcoholism and is characterized by impaired judgment and motor incoordination. On examination, patient has tachypnea, cyanosis, cold peripheries, thready pulse, and hypotension.
- **Chronic hypoxemia:** On long standing, it presents with fatigue, drowsiness, inattentiveness, apathy, delayed reaction time, and reduced work capacity.
- **Consequences:** Centers in the brainstem gets affected and death may occur due to respiratory failure.

Treatment
Treat the underlying cause or mechanism.

Investigations

- **Arterial blood gas (ABG) study:** For confirmation of hypoxemia.
- **Calculate the alveolar-arterial O_2 (PAO_2-PaO_2) gradient:** To differentiate various causes of hypoxemia.

CLUBBING

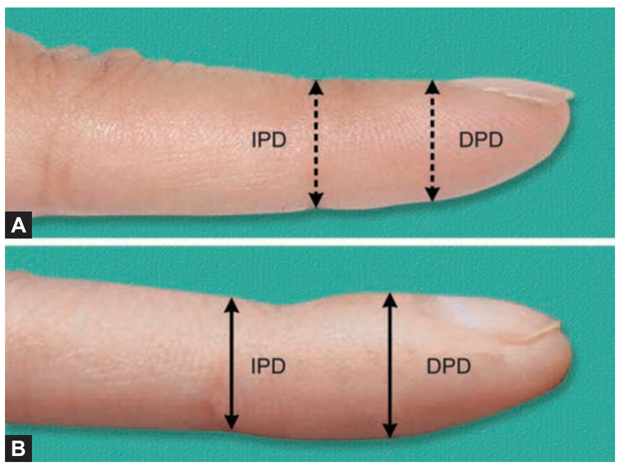
Q. Define clubbing. Enumerate the causes and give the mechanism of clubbing. Enumerate cardiac causes.

Definition

Clubbing (also called Hippocrates fingers) is defined as a selective bulbous enlargement of the distal segment of a digit (fingers and toes) due to an increase in connective tissue especially on the dorsal aspect resulting in loss of the onychonychial angle **(Figs. 2.1A and B).**

Mechanisms

Some of hypotheses are:
- **Role of platelets:** Megakaryocytes from the bone marrow normally break up to the platelets in the pulmonary capillaries. Whenever there are shunts or abnormal circulation (e.g., neoplasm), these megakaryocytes bypass the pulmonary capillaries and reach the systemic circulation. These megakaryocytes preferably lodge in the tips of the digits and locally release platelet-derived growth factor (PDGF) and vascular endothelial growth factor (VEGF). These growth factors along with other mediators increase endothelial permeability and activate and cause proliferation of connective tissue cells (e.g., fibroblasts).



FIGS. 2.1A AND B: (A) Normal finger; (B) Clubbing offinger.
(IPD: Interphalangeal distance; DPD: Distal phalangeal distance)

- **Humoral:** Unknown humoral substances dilate vessels in the fingertip (e.g., acromegaly).
- **Persistent hypoxia:** It causes opening of deep arteriovenous fistula in fingers (e.g., tetralogy of Fallot).
- **Reduced ferritin** in systemic circulation: It causes dilation of arteriovenous anastomoses.
- **Vagal theory:** Persistent vagal stimulation causes vasodilation and clubbing (e.g., lung carcinoma).
- **Toxic:** Subacute bacterial endocarditis (SABE).
- **Metabolic:** Thyrotoxicosis.

Grades of Clubbing (Table 2.4)

The process of clubbing usually takes years but in few conditions (e.g., lung abscess and empyema), it may develop quite fast.

Causes (Table 2.5)

Clubbing may be hereditary, idiopathic, or acquired and associated with a variety of disorders. Various methods of eliciting clubbing are listed in **Box 2.2**.

Phalangeal depth ratio: It is defined by the ratio of digit's depth measured at the junction between skin and nail (nail bed) and at the distal interphalangeal joint. Normally, the depth at distal interphalangeal joint is more than the depth at nail bed. In clubbing fingers, there is reversal of this ratio. A phalangeal depth ratio of over 1 indicates clubbing. It can be measured by a caliper or a digital photograph.

Pseudoclubbing: It is **increase in longitudinal curvature of the nail with loss of nail and nail plate material.** It is characterized clinically by asymmetrical involvement of fingers and radiographically by resorption of the terminal tufts (acroosteolysis). There is no overgrowth of connective tissue as observed in clubbing.

Causes of pseudoclubbing: (1) Subungual tumor or cyst and (2) subperiosteal bone resorption (e.g., scleroderma, thyroid acropachy, vinyl chloride poisoning, acromegaly, hyperparathyroidism, leprosy, chronic renal failure, acrometastasis, etc.).

TABLE 2.4: Grades of clubbing.

Grade	*Features*
1	Increased fluctuation due to softening of the nail bed (increased ballotability)
2	Loss of the normal (<165°) onychodermal angle (Lovibond angle), between the nail bed and the nail fold (cuticle)
3	Thickening of the whole distal (end part of the) finger (resembling a parrot beak or drumstick)
4	Hypertrophic osteoarthropathy (pain and tenderness at distal end of long bones due to subperiosteal new bone formation
5 (Lung India)	Glossy or shiny changes in the nail with longitudinal striations

TABLE 2.5: Causes of clubbing.

Respiratory:
- *Neoplasms:* Bronchogenic carcinoma (especially squamous cell carcinoma), metastasis to lung, mesothelioma
- *Suppurative lung disease:* Bronchiectasis, lung abscess, cystic fibrosis, empyema
- Asbestosis (with mesothelioma)
- Sarcoidosis
- Cystic fibrosis
- Pulmonary AV fistula
- Interstitial lung diseases

Cardiovascular:
- Infective endocarditis
- Cyanotic congenial heart diseases, atrial myxoma, and Eisenmenger's syndrome

Gastrointestinal:
- Inflammatory bowel disease (ulcerative colitis and Crohn's disease)
- Primary biliary cirrhosis
- Hepatoma

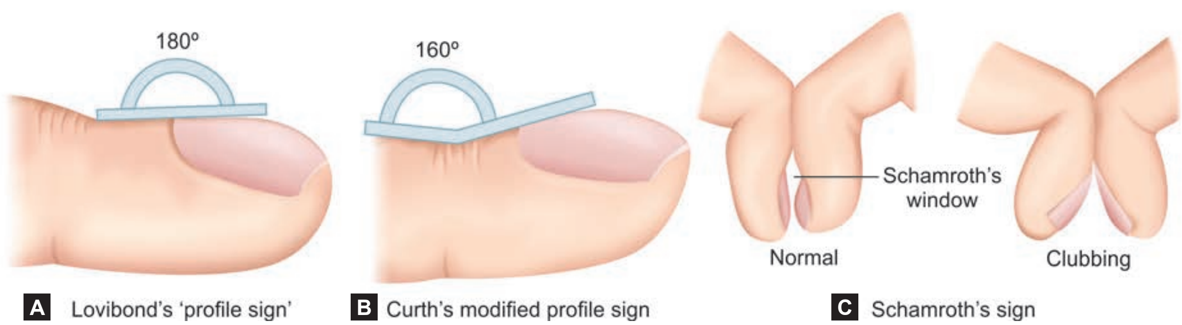
Miscellaneous:
- Hereditary, idiopathic (pachydermoperiostosis or Touraine–Solente–Gole syndrome)
- Unidigital clubbing occurs in repeated trauma

Unilateral clubbing:
- Hemiplegia (long standing)
- Vascular disease
  - *Aneurysm:* Subclavian artery, brachiocephalic trunk
  - Pre-subclavian coarctation of aorta (left-sided clubbing)
  - Pancoast tumor
  - Unilateral erythromelalgia
  - AV fistula used for hemodialysis
  - Infected arterial graft

Box 2.2: Various methods of eliciting the signs of clubbing.

- Lovibond's profile sign (**Fig. 2.2A**)
- Curth's modified profile sign (**Fig. 2.2B**)
- Fluctuation test
- Schamroth's window test (**Fig. 2.2C**)
- Phalangeal depth ratio

Primary hypertrophic osteoarthropathy (PHO) is a rare hereditary disorder with digital clubbing, subperiosteal new bone formation, and arthropathy. It is associated with mutations in the 15-hydroxyprostaglandin dehydrogenase (15-PGDH).



FIGS. 2.2A TO C: Various signs of clubbing.

ABNORMALITIES IN NAILS DUE TO SYSTEMIC DISEASES (TABLE 2.6)

Q. Write short note/essay on abnormalities in nails due to systemic diseases.

TABLE 2.6: Abnormalities in nails due to systemic diseases.

Abnormality and description	*Associated systemic disease*
Shape or growth change	
➢ **Clubbing**	➢ Discussed above (**Table 2.5**)
➢ **Koilonychia:** Spoon-shaped nails (transverse and longitudinal concavity)	➢ Iron deficiency anemia, rarely in hemochromatosis, Raynaud's disease, SLE, hypothyroidism, or hyperthyroidism
➢ **Pitting:** Punctate depressions in nails	➢ Psoriasis, Reiter's syndrome, pemphigus, lichen planus, alopecia areata, and rheumatoid arthritis
➢ **Onycholysis:** Distal nail plate separated from nail bed and white discoloration of the affected part of the nail	➢ Psoriasis, local infection, hyperthyroidism, sarcoidosis, trauma, amyloidosis, connective tissue disorders, and pellagra
➢ **Beau's lines:** Transverse linear depressions over nails, move distally with the growth of nail	➢ Any severe systemic illness that disrupts nail growth, Raynaud's disease, pemphigus, and trauma
➢ **Onychomadesis:** Proximal separation of nail plate from nail bed	➢ Trauma, drug sensitivity, poor nutrition, pemphigus vulgaris, and Kawasaki disease
➢ **Yellow nails:** Nail has a "heaped-up" or thickened appearance, yellow in color, with obliteration of lunula	➢ Lymphedema, pleural effusion, immunodeficiency, bronchiectasis, sinusitis, rheumatoid arthritis, nephrotic syndrome, thyroiditis, tuberculosis, and Raynaud's disease
Color change	
➢ **Terry's (white) nails:** Most of the nail plate turns white with obliteration of lunula, uniformly affects all nails	➢ Liver failure, cirrhosis, diabetes mellitus, CHF, hyperthyroidism, and malnutrition
➢ **Half-and-half nails (Lindsay's nails):** Proximal portion of nail bed is white because of nail-bed edema (half-brown and half-white appearance)	➢ Renal failure, HIV infection, and Crohn's disease
➢ **Azure lunula (blue nails)**	➢ Hepatolenticular degeneration (Wilson's disease), silver poisoning, and quinacrine therapy
➢ **Mees' lines:** Transverse white bands affecting multiple nails, move distally with nail growth	➢ Arsenic poisoning, Hodgkin's lymphoma, CHF, leprosy, malaria, chemotherapy
➢ **Muehrcke's lines:** Pairs of transverse white lines that disappear on applying pressure, do not move with growth of nail	➢ Specific for hypoalbuminemia of any cause
➢ **Dark longitudinal streaks**	➢ Melanoma, benign nevus, chemical staining, and normal variant in darkly pigmented people
➢ **Splinter hemorrhage:** Longitudinal, thin, reddish lines occurring beneath the nail plate	➢ Subacute bacterial endocarditis, SLE, rheumatoid arthritis, peptic ulcer disease, malignancies, oral contraceptive use, pregnancy, psoriasis, and trauma
➢ **Telangiectasia:** Irregular, twisted, and dilated vessels at the distal portion of cuticle covering the nail bed	➢ Rheumatoid arthritis, SLE, dermatomyositis, and scleroderma
➢ **Longitudinal striations**	➢ Alopecia areata, vitiligo, atopic dermatitis, and psoriasis

CYANOSIS (FIG. 2.3)

Definition: Bluish discoloration of the skin and mucous membrane. It results from an increased concentration of reduced hemoglobin/**deoxyhemoglobin (>5 g/dL)** or abnormal hemoglobin derivatives (e.g., methemoglobin, sulfhemoglobin, etc.) in the capillary blood perfusing the area. Cyanosis is normally detected when the oxygen saturation (SaO_2) is <85%.

Types of Cyanosis (Table 2.7)

TABLE 2.7: Types of cyanosis and difference between them.

Features	*Central cyanosis*	*Peripheral cyanosis*
Mechanism	Inadequate oxygenation of systemic arterial blood (hypoxic hypoxia)	Sluggish peripheral circulation (stagnant hypoxia)
Sites to look	Tongue and oral mucosa	Acral: Tip of the nose, ear lobules, outer aspect of lips, finger tips, nail bed, and extremities
Associations	Clubbing and polycythemia	
Extremities	Warm	Cold
Warming extremities	No change	Cyanosis disappears
Oxygen inhalation	Slight improvement	No change
ABG PaO_2	Low: <85%	Normal: 85–100%
Pulse volume	May be high	Usually low
Dyspnea	Usually present	Usually absent
Exercise	Worsens	May improve

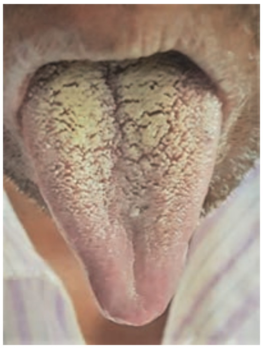


FIG. 2.3: Cyanosis characterized by bluish discoloration of mucous membrane of tongue.

Causes of Cyanosis

Central Cyanosis

- **Cardiac:**
  - Congenital cyanotic heart diseases (*Remember 5 'T's and 2 'E's*): Transposition of great arteries, tetralogy of Fallot, truncus arteriosus, tricuspid valve abnormalities, total anomalous pulmonary venous return (TAPVR), Eisenmenger's syndrome (cyanosis tardive), and Ebstein's anomaly.
  - Acute pulmonary edema (due to left-sided heart failure)
- **Pulmonary:** Acute severe asthma, chronic obstructive pulmonary disease (COPD), cor pulmonale, respiratory failure, respiratory depression, lobar and bronchopneumonia, tension pneumothorax, acute laryngeal edema, and acute pulmonary embolism
- **High altitude** (due to low partial pressure of oxygen)
- **Polycythemia**
- **Enterogenous or pigment cyanosis:** Methemoglobinemia and sulfhemoglobinemia

Peripheral Cyanosis

- **Low-cardiac output:** Congestive heart failure
- **Local vasoconstriction:** Cold, frostbite, Raynaud's phenomenon, and shock
- **Arterial obstruction:** Peripheral vascular diseases (atherosclerosis and Buerger's disease)
- **Venous obstruction:** Superior vena caval syndrome
- **Hyperviscosity syndrome:** Multiple myeloma, polycythemia, and macroglobulinemia
- **Others:** Cryoglobulinemia and mitral stenosis.

Mixed Cyanosis

- All causes of central cyanosis may also cause peripheral cyanosis
- Cardiogenic shock with pulmonary edema
- Congestive cardiac failure due to left-sided heart failure
- Polycythemia (rarely).

Differential cyanosis: Cyanosis only in lower limbs seen in PDA with reversal of shunt.

Cyanosis only in upper limbs (reverse differential cyanosis): Coarctation of aorta (ductal type) with transposition of great arteries.

Cyanosis in left upper limb and both lower limbs: PDA with reversal of shunt and preductal coarctation of aorta.

Intermittent cyanosis: Ebstein's anomaly.

Cyclical cyanosis: Bilateral choanal atresia.

Orthocyanosis: Development of cyanosis only in upright position due to hypoxia occurring in erect posture. It is seen in pulmonary arteriovenous malformation.

Cyanosis absent despite of sufficient reduced hemoglobin: In severe anemia, carbon monoxide poisoning.

Hyperoxia test (cardiac versus pulmonary cyanosis): After giving 100% oxygen for 10 minutes, a repeat ABG is done and if PaO_2 is <150 mm Hg then the cause is cardiac and if the PaO_2 improves to >200 mm Hg, the cause is respiratory.

Pseudocyanosis: Caused by metals (gold, silver, mercury, and arsenic) and drugs (minocycline, phenothiazines, chloroquine, and amiodarone).

RESPIRATORY (PULMONARY) FUNCTION TESTS (TABLE 2.8)

Q. CT2.11 Describe, discuss and interpret pulmonary function tests.

Q. Write short note on FEV_1 (forced expiratory volume in 1 second).

Abbreviations used in pulmonary function tests are presented in **Table 2.9**.

TABLE 2.8: Pulmonary function test (PFT).

PFT tracings have: ➢ **Four lung volumes:** Tidal volume, inspiratory reserve volume, expiratory reserve volume, and residual volume ➢ **Five capacities:** Inspiratory capacity, expiratory capacity, vital capacity, functional residual capacity, and total lung capacity	➢ Flow–volume curves ➢ Blood gases and pulse oximetry ➢ Transfer factor (diffusion) ➢ Exercise tests ➢ Exhaled nitric oxide

TABLE 2.9: Abbreviations used in pulmonary function tests.

Abbreviations	*Explanation*
FVC (forced vital capacity)	Volume of air expired with a maximal effort after deep inspiration
FEV_1 (forced expiratory volume in 1 second)	Volume of air expired in the first second after deep inspiration
VC (vital capacity)	Maximum amount of air that can be expelled from the lungs after the deepest possible breath TLC minus RV or maximum volume of air exhaled from maximal inspiratory level (60–70 mL/kg) (3,100–4,800 mL). VC decreases with age
PEF (peak expiratory flow)	Volume of forcibly expired air during the first 10 seconds after deep inspiration
TLC (total lung capacity)	Sum of all volume compartments or volume of air in lungs after maximum inspiration (4–6 L)
FRC (functional residual capacity)	Sum of RV and ERV or the volume of air in the lungs at end-expiratory tidal position (30–35 mL/kg) (2,300–3,300 mL) measured with multiple-breath closed-circuit helium dilution, multiple-breath open-circuit nitrogen washout, or body plethysmography. It cannot be measured by spirometry
RV (residual volume)	Volume of air remaining in lungs after maximum exhalation (20–25 mL/kg) (1,700–2,100 mL), indirectly measured (FRC-ERV), it cannot be measured by spirometry. RV increases with age
IRV (inspiratory reserve volume)	Maximum volume of air inhaled from the end-inspiratory tidal position (1,900–3,300 mL)
ERV (expiratory reserve volume)	Maximum volume of air that can be exhaled from resting end-expiratory tidal position (700–1,000 mL)
TV (tidal volume)	Volume of air inhaled or exhaled with each breath during quiet breathing (6–8 mL/kg)
IC (inspiratory capacity)	Sum of IRV and TV or the maximum volume of air that can be inhaled from the end-expiratory tidal position (2,400–3,800 mL)
EC (expiratory capacity)	TV + ERV
DLCO (diffusing capacity of lung for carbon monoxide)	Also known as transfer factor, DLCO measures the transfer of inhaled gases into the pulmonary capillaries

Measurement of Airway Obstruction (Fig. 2.4)

Ventilatory Capacity

Ventilatory capacity includes **FEV1** (volume of air expired in the first second after deep inspiration) and **FVC** (volume of air expired with a maximal effort after deep inspiration) and **PEF** (peak expiratory flow rate).

Normally, FEV_1/FVC ratio is around 75%.

Patterns of Abnormalities

- **Obstructive ventilatory defect:** It is characterized by narrowing of airways during expiration (e.g., bronchial asthma, chronic bronchitis, and emphysema). These disorders show **markedly reduced FEV_1, reduced or normal VC, and reduced FEV_1/VC.** In airflow limitation, the FEV_1 is reduced as a percentage of FVC. With increasing airflow limitation, FEV falls proportionately more than FVC, so the FEV_1/FVC ratio is reduced.
  - **Reversibility of airflow limitation:** When FEV_1 is disproportionately reduced, resulting in FEV_1/FVC ratio of less than 70%, spirometry should be repeated following inhaled short-acting β_2-adrenoceptor agonists (e.g., salbutamol). A large improvement in FEV_1 (over 400 mL) is seen in bronchial asthma and to some extent in chronic bronchitis.
- **Restrictive ventilatory defect:** In restrictive lung disease, **FEV_1 and FVC are reduced proportionately and the FEV_1/ FVC ratio may be normal** or may even increase because of enhanced elastic recoil. This pattern is seen in interstitial inflammation and/or fibrosis that lead to progressive loss of lung volume.

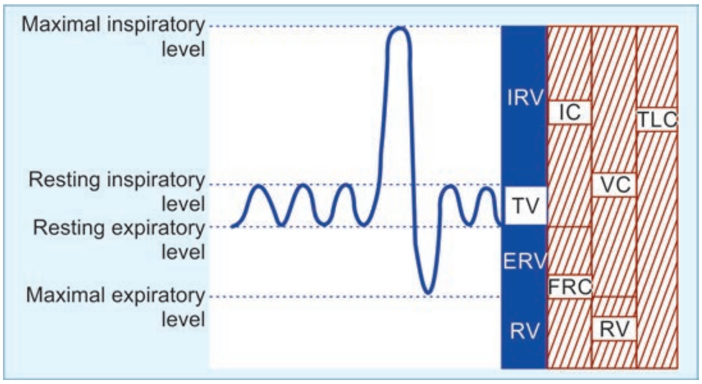


FIG. 2.4: Diagrammatic representation of various lung volume measurements. (TLC: Total lung capacity; VC: Vital capacity; RV: Residual volume; IC: Inspiratory capacity; FRC: Functional residual capacity; IRV: Inspiratory reserve volume; TV: Tidal volume; ERV: Expiratory reserve volume).

Peak Expiratory Flow Rate (PEFR)

Peak flow meters are cheap and simpler than spirometer. Patient is asked to take a full inspiration to total lung capacity and then blowout forcefully into the peak flow meter. PEFR measures the volume of forcibly expired air during the first 10 seconds after deep inspiration. Reduced values are found in airflow obstruction and are not useful in assessing restrictive ventilatory defect. It is mainly useful for diagnosis, to monitor exacerbations and response, and for treatment in asthma.

Lung Volumes

Lung volumes include total lung capacity (TLC) and residual volume (RV). Both can be measured by spirometry.

Total lung capacity (TLC) is the total amount of air in the lungs after taking the deepest breath possible.

Vital capacity (VC) is the maximum amount of air that can be expelled from the lungs after the deepest possible breath and is recorded separately. Patient is asked to make a full but unhurried ("relaxed") exhalation into the spirometer.

Residual volume (RV) is the volume of air in the lungs at the end of full expiration. It is calculated by subtracting the vital capacity (VC) from the total lung capacity (TLC).

Interpretation

- In obstructive lung disease, both TLC and RV are increased.
- In restrictive lung diseases (due to parenchymal lung disease), both TLC and RV are reduced.
- In extraparenchymal diseases with restriction during both inspiration and expiration (ankylosing spondylitis and kyphoscoliosis), RV is increased while TLC is reduced.

Flow–Volume Loops (Table 2.10 and Fig. 2.5)

Flow–volume loops measure flow rates against expired volume and show the site of airflow limitation (obstruction) within the lung. At the start of expiration from TLC, maximum resistance is from the large airways. This affects the flow rate for the first 25% of the curve. As air is exhaled, lung volume reduces, and the flow rate depends on the resistance offered by smaller airways.

TABLE 2.10: Flow–volume loops in various disorders.

Disease states	*FVC*	*FEV_1*	*FEV_1/FVC*
Obstructive	Normal	Reduced	Reduced
Stiff lungs	Reduced	Reduced	Normal
Respiratory muscle weakness	Reduced	Reduced	Normal

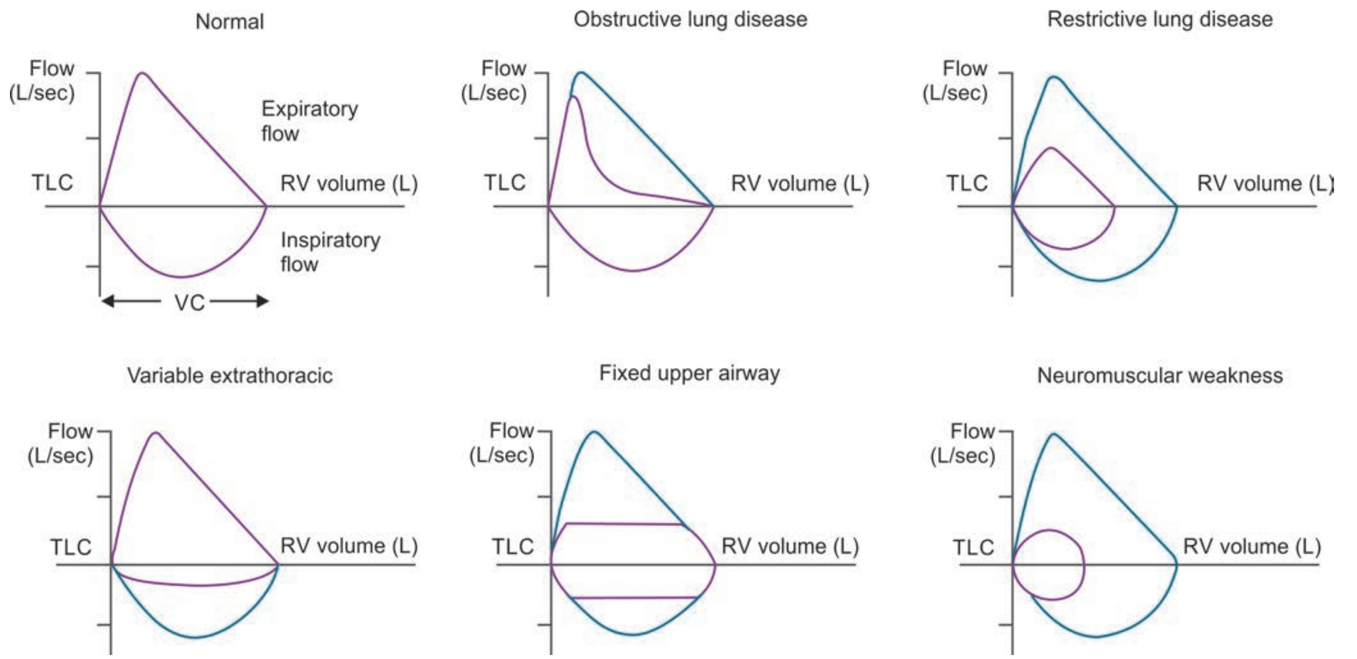


FIG. 2.5: Flow–volume loop for common respiratory diseases.

- **FEV_1** is the volume exhaled during the first second of the FVC maneuver. **It is decreased in both obstructive and restrictive lung disorders**.
- FEF 25–75% is the mean expiratory flow during the middle half of the FVC maneuver; it reflects flow through the small (<2 mm in diameter) airways.
- Interpretation of percent predicted: Normal (>79%), mild obstruction (60–79%), moderate obstruction (40–59%), severe obstruction (<40%).
- **FEV_1/FVC** is the ratio of FEV_1 to FVC × 100 (expressed as a percent); it is an important value because a reduction of this ratio from expected values is specific for obstructive rather than restrictive diseases. Normal value (FEV_1/FVC) is 75–85%, <70% of predicted value in mild obstruction, <60% of predicted value in moderate obstruction, and <50% of predicted value in severe obstruction.

Spirometry interpretation: Obstructive versus restrictive defect (**Table 2.11**).

Arterial blood gases (ABG) and oximetry (*See* Table 2.1)
These tests include measurement of: (i) hydrogen ion (H^+) concentration, (ii) PaO_2 and $PaCO_2$, and (iii) oxygen saturation and bicarbonate concentration (derived from the above values) in an arterial blood. These are performed by automatic analyzers.

Uses: (1) To assess the degree and type of respiratory failure and (2) for measuring acid–based status.

TABLE 2.11: Spirometry interpretation—obstructive versus restrictive defect.

Obstructive disorders	*Restrictive disorders*
Limitation of expiratory airflow as airways cannot empty as rapidly compared to normal (e.g., narrowed airways from bronchospasm, inflammation, etc.)	Characterized by reduced lung volumes/decreased lung compliance
➢ FVC normal or ↓	➢ FVC ↓ (significantly decreased)
➢ FEV_1 ↓ (significantly decreased)	➢ FEV_1 ↓
➢ FEF 25–75% ↓	➢ FEF 25–75% normal to ↓
➢ FEV_1/FVC ↓ (<0.7)	➢ FEV_1/FVC normal to ↑ (>0.7)
➢ TLC normal or ↑	➢ TLC ↓
Examples: Asthma, emphysema, and cystic fibrosis	Examples: Interstitial fibrosis, scoliosis, obesity, lung resection, neuromuscular diseases, and cystic fibrosis

Tests for Gas Exchange Function

a. **Alveolar–arterial O_2 tension gradient:**
   - Sensitive indicator of detecting regional V/Q inequality
   - It is the difference between the amount of the oxygen in the alveoli [i.e., the alveolar oxygen tension (PaO_2)] and the amount of oxygen dissolved in the plasma (PaO_2).

b. **Dyspnea differentiation index (DDI):**
   - To differentiate dyspnea due to respiratory/cardiac diseases

$$DDI = \frac{PEFR \times PaCO_2}{1000}$$

   - DDI—lower in respiratory pathology

c. **Diffusing capacity of lung (DL):** Defined as the rate at which gas enters into blood divided by its driving pressure.
   - The **diffusing capacity for carbon monoxide (DLCO)** is also known as the transfer factor. It measures the ability of the lungs to transfer gas from inhaled air in the alveoli to the red blood cells in pulmonary capillaries.
   - Diseases associated with reduced and increased gas transfer are listed in **Table 2.12**.

TABLE 2.12: Diseases associated with reduced and increased diffusing capacity for carbon monoxide (DLCO).

Increased DLCO [blood flow is in excess of ventilation (a low VQ—SHUNT— "wasted blood")]	*Reduced DLCO [ventilation is in excess of blood flow (a high VQ—DEAD SPACE—"wasted air")]*
Exercise (due to recruitment of capillaries)	Postexercise
Supine position (due to increased pulmonary capillary blood volume)	Standing
Müller maneuver (inspiration against closed mouth and nose after forced expiration)	Valsalva maneuver
Pulmonary hemorrhage	Lung resection
Polycythemia	Pulmonary emphysema (affecting capillary or alveolar bed)
Left-to-right shunt (e.g., atrial septal defect)	Pulmonary vascular disease, including pulmonary arterial hypertension and chronic venous thromboembolism
Obesity	Interstitial lung diseases
Asthma	Anemia
Chronic bronchitis without major emphysema	Evening
Morning	Drugs (e.g., amiodarone, bleomycin, methotrexate)
Pregnancy	Pulmonary lymphangitic carcinomatosis

Tests for Cardiopulmonary Interactions

Reflects gas exchange, ventilation, tissue O_2, and CO_2
- **Qualitative:** History, examination, ABG, and stair climbing test
- **Quantitative:** 6-minute walk test

Stair climbing test: If able to climb three flights of stairs without stopping/dyspnea at his/her own pace-reduced morbidity and mortality. If not able to climb two flights, high-risk.

6-minute walk test: Gold standard. Cardiopulmonary reserve is measured by estimating maximum O_2 uptake (VO_2 maximum) during exercise. Modified, if patient cannot walk—bicycle/arm exercises. If patient is able to walk for >2,000 feet during 6-minute period, VO_2 maximum >15 mL/kg/min, if 1,080 feet in 1 min, VO_2 of 12 mL/kg/min. Simultaneously, oximetry is done and if SpO_2 falls >4%, high-risk.

Exhaled Nitric Oxide

- Nitric oxide is produced by the bronchial epithelium. Its production increases in asthma and other diseases associated with inflammation of airway.
- Measurement of exhaled NO may be helpful in asthma that is difficult to control. Measuring **fractional exhaled nitric oxide** (FeNO) helps to identify patients who are likely to benefit from treatment with corticosteroids. FeNO also predicts

the likelihood of steroid responsiveness more consistently than spirometry, bronchodilator response, peak flow variation, or airway hyper-responsiveness to methacholine. A FeNO greater than 50 ppb in adults or greater than 35 ppb in children suggests eosinophilic airway inflammation.

- **Airway functioning:** Spirometry, impulse oscillometry, body plethysmography, PEFR
- **Airway sensitivity:** Methacholine challenge
- **Airway inflammation:** FeNO
- **Diffusion adequacy:** DLCO, 6min walk test, A-a gradient
    - Spirometry
    - Lung volumes
    - DLCO

BRONCHIAL ASTHMA

Q. Define and classify bronchial asthma.

Q. PE28.19 Describe the etiopathogenesis, clinical features, diagnosis, management and prevention of asthma in children.

Q. PE31.5 Discuss the etiopathogenesis, clinical types, presentations, management and prevention of childhood asthma.

Definition

Asthma is a **chronic inflammatory disorder** of the airways (bronchial tree) in which **breathing is periodically rendered difficult** by **widespread narrowing of the bronchi** (reversible bronchoconstriction).

It is **clinically characterized by recurrent episodes** (paroxysms) **of wheezing, breathlessness** (dyspnea)**, tightness of the chest, and cough**.

Global Initiative for Asthma (GINA)—2022 Definition of Asthma

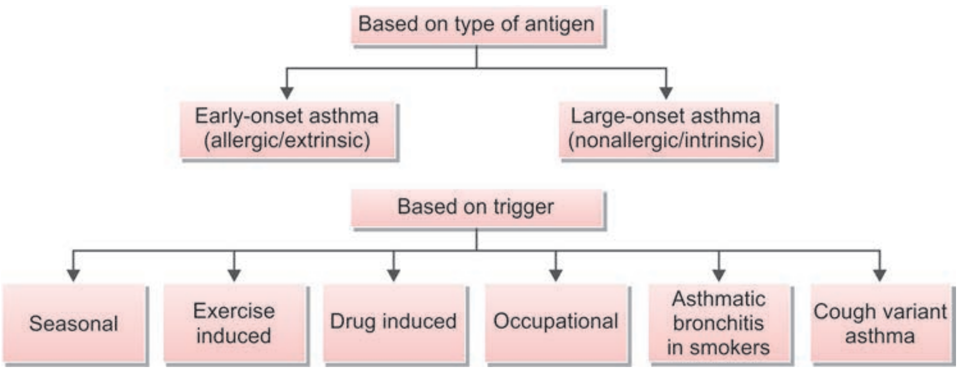
"Asthma is a heterogeneous disease, usually characterized by chronic airway inflammation. It is defined by the history of respiratory symptoms such as wheeze, shortness of breath, chest tightness, and cough that vary over time and in intensity, together with variable expiratory airflow limitation." *These episodes are usually associated with widespread but variable airflow obstruction that is often reversible either spontaneously or with treatment.*

Characteristics of Asthma

1. **Airflow narrowing:** It is due to combination of muscle edema and viscid bronchial secretion. It is generally reversible spontaneously or with treatment.
2. **Airway hyper-reactivity (AHR)/bronchial hyper-responsiveness (BHR):** It is characterized by increased tendency for airway (tracheobronchial tree) narrowing in response to triggers (stimuli) that have little or no effect in normal individuals.
3. **Bronchial inflammation:** Inflammation of the bronchial walls by T lymphocytes, mast cells, eosinophils with associated plasma exudation, edema, smooth muscle hypertrophy, mucus plugging, and epithelial damage.

Classification (Flowchart 2.1)

FLOWCHART 2.1: Classification of asthma.



Q. How will you differentiate early-onset (atopic) asthma from late-onset (non-atopic) asthma?

Differentiating features of early-onset asthma and late-onset asthma are presented in **Table 2.13**.

TABLE 2.13: Differentiating features of early-onset asthma and late-onset asthma.

Features	***Early-onset (atopic) asthma***	***Late-onset (nonatopic) asthma***
Onset	Early age (usually begins in childhood)	Late age
Individuals	Atopic individuals	Nonatopic individuals
Role of external allergens	Have strong role	No role
Family history	Positive history of asthma or allergic diseases (e.g., eczema, urticaria, or hay fever)	Less common or absent
Triggering events	Environmental allergens (e.g., dusts, pollens, animal dander, and foods)	Respiratory infections due to viruses (e.g., rhinovirus and parainfluenza virus) Inhaled air pollutants (e.g., smoke and fumes)
Serum level of IgE	Increased	Normal
Skin hypersensitivity test to common inhalant allergens	Positive	Negative
Response to provocation tests	Positive	Negative

Q. CT2.7 **Describe and discuss allergic and non-allergic precipitants of obstructive airway disease.**

Risk Factors and Triggers (Table 2.14)

Risk Factors (Fig. 2.6)

Endogenous factors

- **Genetic predisposition:** Major etiological factor in atopic asthma is genetic predisposition to **type I hypersensitivity (atopy) reaction** and exposure to environmental trigger. One of the **susceptibility loci is on the chromosome 5 (5q)** → several genes involved in regulation of IgE synthesis and mast cell and eosinophil growth and differentiation.
- **Atopy:** Atopic individuals tend to **have higher serum IgE levels**, and a positive family history of allergy is found in 50% of atopic individuals. Patients with asthma commonly suffer from other atopic diseases (e.g., allergic rhinitis and atopic dermatitis/eczema).

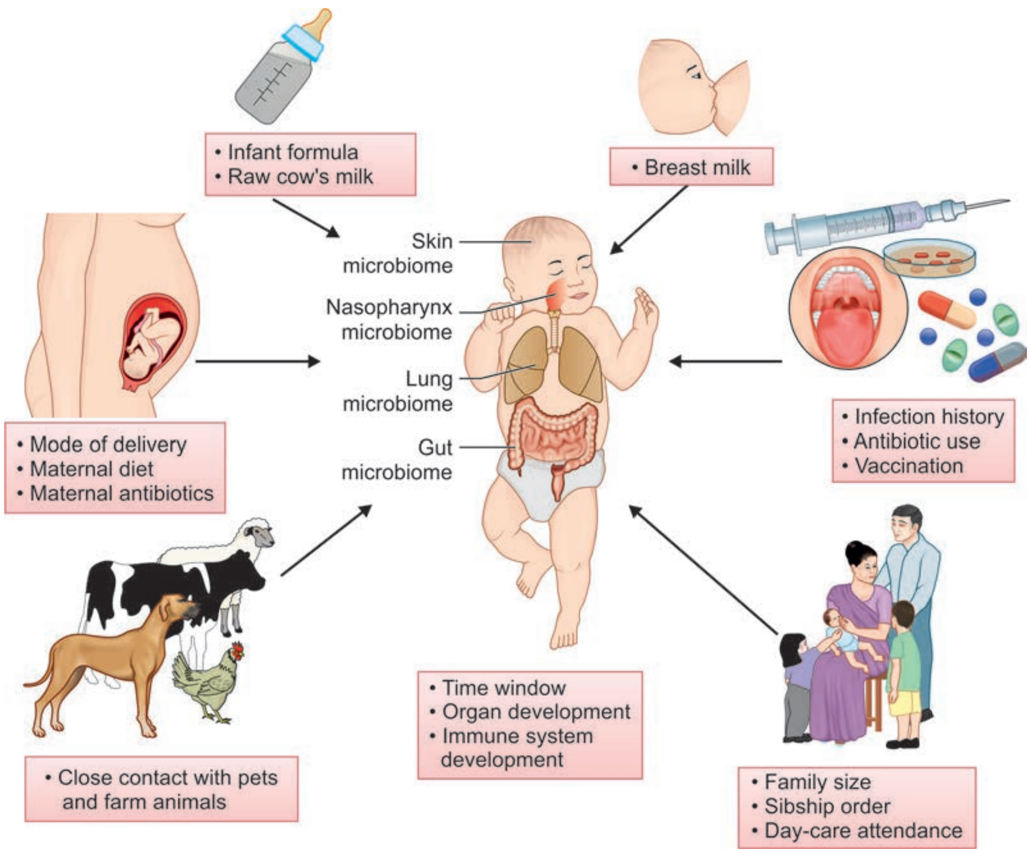


FIG. 2.6: Risk factors for asthma.

- **Airway hyper-responsiveness:** It is an abnormality in which there is an **excessive tendency for airways to contract** (bronchoconstrictor) too easily in response to multiple inhaled triggers that usually do not have any effect on normal individuals.
- **Gender and age:** More common in boys than girls and, after puberty, women slightly more commonly than men. Most cases **begin before the age of 25 years**.

Environmental factors

Hygiene hypothesis proposes that individuals with **lack of infections in early childhood** are more prone to asthma than children brought up on farms who are exposed to a high level of endotoxin. Intestinal parasite infection may also be associated with a decreased risk of asthma. Conversely, early childhood in a "dirtier" environment (exposure to inhaled and ingested products of microorganisms) may allow the immune system to avoid developing allergic responses.

TABLE 2.14: Risk factors and triggers involved in asthma.

Risk factors	
Endogenous factors ➢ Genetic predisposition ➢ Atopy ➢ Airway hyper-responsiveness ➢ Gender and age ➢ Ethnicity ➢ Obesity ➢ Early viral infections	**Environmental factors** ➢ Allergens (indoor/outdoor) ➢ Occupational sensitizers ➢ Respiratory infections
Triggers	
➢ Inhaled allergens ➢ Upper respiratory tract viral infections ➢ Air pollution (e.g., sulfur dioxide, irritant gases) ➢ Passive smoking (tobacco smoke)	➢ Drugs (β-blockers and aspirin) ➢ *Physical factors:* Exercise, cold air, and hyperventilation ➢ Emotional stress ➢ Irritants (household sprays paint fumes)

Pathogenesis (Pathophysiology) of Asthma

Airway Inflammation

- **Inflammation** is chronic and involves many cell types and inflammatory mediators **(Fig. 2.7)**.
- **Strong T_H2 response: Genetic predisposition** with susceptibility genes makes individuals prone to develop **strong T_H2** (type of T lymphocytes) **reactions against environmental antigens** (allergens).
- **T_H2 cells secrete cytokines,** which promote allergic inflammation and **stimulate B-cells to produce IgE**.

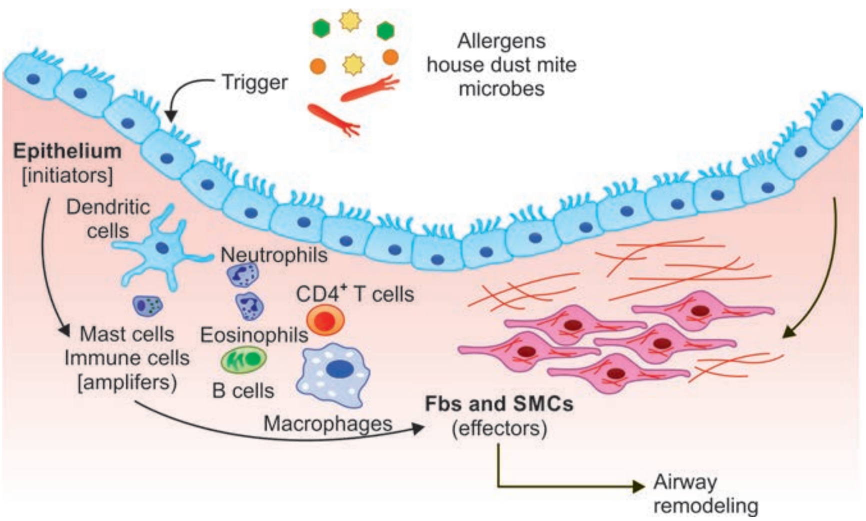


FIG. 2.7: Cells and mediators of asthma.

Cells involved in the inflammatory response

Important cells involved in asthma are—**mast cells, eosinophils, dendritic cells (macrophages), and lymphocytes.**

1. **Mast cells:**
   - **Early reaction** is characterized by **bronchoconstriction**, **increased mucus production**, and **vasodilation with increased vascular permeability** (causes edema).
   - **Late phase reaction:** It is characterized by **inflammation and airway remodeling.**
2. **Eosinophils:**
   - **Mediators released from eosinophils:** LTC_4 and basic proteins such as major basic protein (MBP), eosinophil cationic protein (ECP), and eosinophils peroxidase (EPX). They are toxic to epithelial cells.
   - Corticosteroid rapidly decreases the number and reduces the activation of eosinophils.

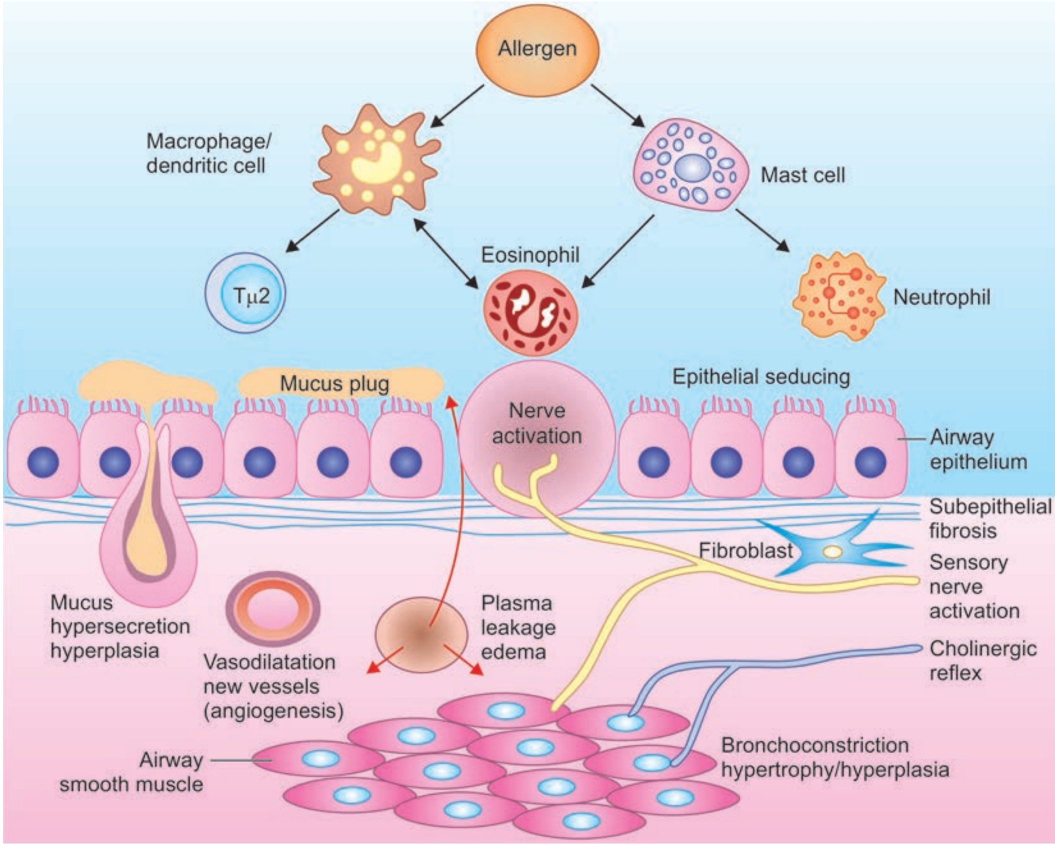


FIG. 2.8: Pathogenesis of asthma.

- Sputum eosinophilia is of diagnostic help and is a biomarker of response to therapy.

3. Dendritic cells and lymphocytes: They release prostaglandin, thromboxane, LTC_4, LTB_4, and platelet-activating factor (PAF).

Airway Remodeling

Airway remodeling is the **group of structural** and **functional changes in the bronchial wall due to repeated bouts of inflammation** observed in chronic asthma.

- An **increase in size and number** (hypertrophy/hyperplasia) of the **submucosal glands**
- **Hypertrophy and/or hyperplasia of the bronchial wall smooth muscle**
- **Increased vascularity**
- **Deposition of subepithelial collagen** accompanied by fibrosis and thickening of the basement membrane.

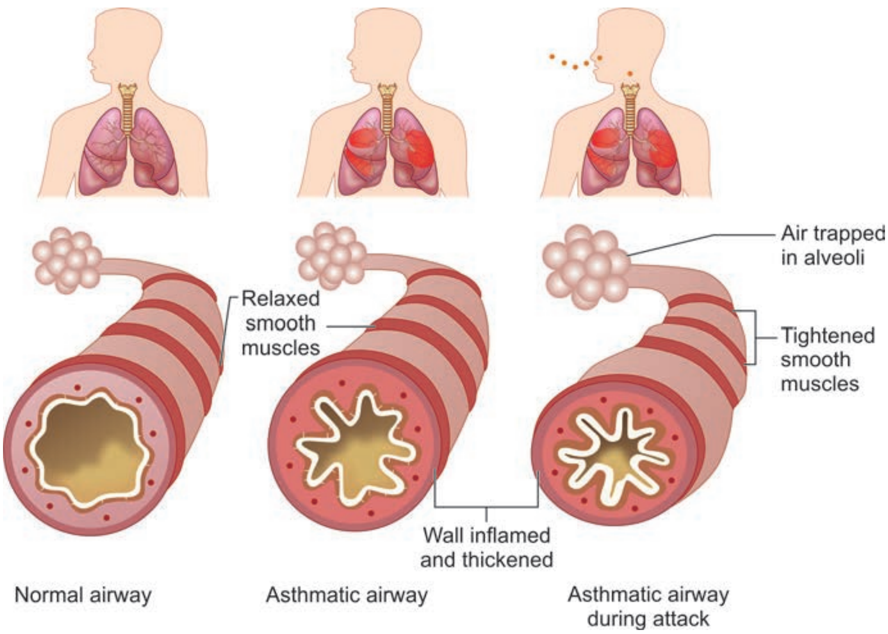


FIG. 2.9: Pathology of asthma.

Pathogenesis of asthma is summarized in **Figure 2.8** and **Flowchart 2.2.** Pathology of asthma is summarized in **Figure 2.9.**

Clinical Features

Q. Write a short note on the clinical presentation of severe acute asthma (status asthmaticus).

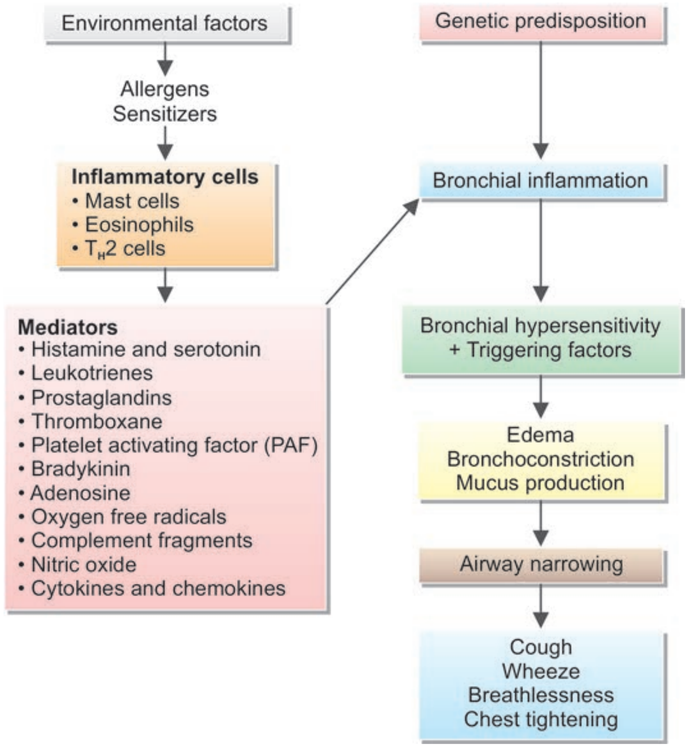
- Clinical features are divided into three headings—namely (1) episodic, (2) severe acute (status asthmaticus), and (3) chronic asthma.
- Usually, atopic individuals develop episodic asthma and nonatopic individuals develop chronic asthma.

1. ***Episodic asthma***
   - Occurs as episodes with asymptomatic period between asthmatic attacks
   - Frequency and duration of attacks vary
   - Presents with relatively sudden onset of paroxysms of **wheezing and dyspnea**
   - May develop spontaneous or triggered by allergens, exercise, or viral infections
   - It may be mild to severe and may last for hours, days, or even weeks
2. ***Severe acute asthma (status asthmaticus)***

Q. Write a short note on the clinical presentation of severe acute asthma (status asthmaticus).

   - It is the **most severe form of asthma** in which the severe acute paroxysm persists for days and even weeks.
   - Presents with severe dyspnea and unproductive cough
   - During this attack, patients prefer an upright position fixing the shoulder girdle to assist the accessory muscles of respiration.
   - Physical signs include sweating, central cyanosis, tachycardia, and pulsus paradoxus. The **bronchoconstriction** and asthmatic symptoms **do not respond** despite the initial administration of standard **acute asthma therapy**.
   - It may cause **severe airflow obstruction leading to severe cyanosis and even death**.
3. ***Chronic asthma***
   - Chronic persistent symptoms include chest tightness, wheeze, and breathlessness on exertion.
   - It is characterized by episodes of spontaneous cough and wheeze, worst during the night.
   - Chronic productive cough with mucoid sputum, punctuated by recurrent attacks of purulent expectoration from frank infection. They are prone to repeated attacks of "severe acute asthma". Features sometimes resemble those of chronic bronchitis.

FLOWCHART 2.2: Pathogenesis of asthma.



Physical Signs

- **During an attack:**
  - *Inspection:* Increased respiratory rate with use of accessory muscles for respiration
  - *Percussion:* Hyper-resonant percussion note over the lungs
  - *Auscultation*
    - Breath sounds are vesicular with prolonged expiration.
    - High-pitched polyphonic expiratory and inspiratory rhonchi
    - Very severe attacks may result in a silent chest, which is an ominous sign.
- **In-between the attacks:** Chest may not reveal any abnormal physical signs.
- **Chronic asthmatics:** Usually reveal few scattered rhonchi.

Table 2.15 shows classification of asthma severity and initiating treatment in persons ≥12 years of age.

Acute exacerbation of bronchial asthma

- Loss of control of any class or variant of asthma
- It is again classified as:
  - *Mild*: The patient is dyspneic, but can complete a sentence in one breath.

TABLE 2.15: Classification of asthma severity and initiating treatment in persons ≥12 years of age.

		Classification of asthma severity (>12 years of age)			
			Persistent		
Components of severity		***Intermittent***	***Mild***	***Moderate***	***Severe***
Impairment Normal FEV_1/FVC: 8–19 years: 85% 20–39 years: 80% 40–59 years: 75% 60–80 years: 70%	Symptoms	≤2 days/week	>2 days/week but not daily	Daily	Throughout the day
	Nighttime awakenings	≤2×/month	3–4×/month	>1/week but not nightly	Often 7×/week
	Short-acting $beta_2$-agonist use for symptom control	≤2 days/week	>2 days/week but not daily, and not more than 1× on any day	Daily	Several times per day
	Interference with normal activity	None	Minor limitation	Some limitation	Extremely limited
	Lung function	➢ Normal FEV_1 between exacerbations ➢ FEV_1 >80% predicted ➢ FEV_1/FVC normal	➢ FEV_1 >80% predicted ➢ FEV_1/FVC normal	➢ FEV_1 >60% but <80% predicted ➢ FEV_1/FVC reduced 5%	➢ FEV1 <60% predicted ➢ FEV_1/FVC reduced >5%
Risk	Exacerbations requiring oral systemic corticosteroids	0–1/year	← ≥2/years → ← Consider severity and interval since last exacerbation → Frequency and severity may fluctuate over time for patients in any severity category Relative annual risk of exacerbations may be related to FEV_1		
Recommended step for initiating treatment		Step 1	Step 2	Step 3	Step 4 or 5
				and consider short course of oral systemic corticosteroids	
		In 2–6 weeks, evaluate level of asthma control that is achieved and adjust therapy accordingly			

- *Moderate*: The patient is more dyspneic and cannot complete a sentence in one breath.
- *Severe (severe acute asthma)*: The patient is severely dyspneic, talks in words and may be restless, even unconscious.

Various Causes of Wheeze

See **Box 2.3**.

Brittle Asthma

- Chaotic variations in lung function despite taking appropriate therapy.
- Type 1: Persistent pattern of variability and may require oral corticosteroids, or continuous infusion of β2 agonists.
- Type 2: Has a normal or near-normal lung function but precipitous, unpredictable falls in lung function that may result in death.
- Does not respond well to corticosteroids and does not reverse well with inhaled bronchodilators.

Box 2.3: Various causes (differential diagnosis) of wheeze.

- Bronchial asthma
- Cardiac asthma
- Chronic obstructive lung disease (COPD)
- Recurrent pulmonary emboli
- Systemic vasculitis, including Churg–Strauss syndrome and polyarteritis nodosa
- Carcinoid tumors
- *Others:* Eosinophilic pneumonia, GERD, foreign body aspiration, mediastinal mass, tracheomalacia, and vocal cord dysfunction

Investigations

The diagnosis is mainly clinical and based on a characteristic history. There is no single satisfactory diagnostic test for asthma.

1. ***Lung function tests***

   Pulmonary function tests useful in asthma are FEV_1, VC, and PEFR.

   - **Spirometry:** It is useful, especially in assessing reversibility. Simple spirometry is useful in confirming the airflow limitation with a reduced FEV_1, FEV_1/FVC ratio, and PEF. Asthma can be diagnosed if there **is greater than 12% and >200 mL improvement in FEV_1 from baseline or PEFR following the inhalation of a bronchodilator**.
   - **Peak expiratory flow rate (PEFR):** It is useful in demonstrating the variable airflow limitation. PEFR measurements to be done on waking, prior to taking a bronchodilator and before bed after a bronchodilator. The diurnal variation in PEFR of more than 20% (the lowest values typically being recorded in the morning) is considered diagnostic. It also provides good measure of disease severity.
   - **DLCO:** Increased in asthma

   Exercise tests: It is used in the diagnosis of asthma in children. The child is asked to run for 6 minutes on a treadmill (heart rate should be above 160 beats per minute). A negative test does not rule out asthma.

Airway responsiveness (AHR): AHR is sensitive but nonspecific. **Histamine or methacholine bronchial provocation test:**

- To detect the presence of airway hyper-responsiveness (a feature of asthma)
- Useful in patients whom cough is the only/main symptom and is useful in the differential diagnosis of chronic cough.
- Contraindicated in individuals who have poor lung function (FEV <1.5 L) or a history of "brittle" asthma.

Indirect challenge tests are release of endogenous mediators that cause the contraction of airway smooth muscle. These include exercise eucapnic voluntary hyperpnea (EVH), ultrasonically nebulizer hypertonic saline, and dry powder mannitol.

2. ***Imaging***
   - **Chest X-ray:**
     - Usually normal between attacks without any diagnostic features
     - During an acute episode or in chronic severe disease, there may be hyperinflated lungs (overinflation).
     - It may be helpful in excluding complications such as pneumothorax, lobar collapse (if mucus occludes large bronchus), or in detecting the pulmonary infiltrates associated with allergic bronchopulmonary aspergillosis.
   - **High-resolution computed tomography (CT):** It may show areas of bronchiectasis (complication) and thickening of the bronchial walls, but these changes are not diagnostic of asthma.
3. ***Measurement of allergic status***
   - **Skin prick tests (SPT):** SPT is performed by intradermal injections of common allergens (house dust mite, cat fur, and grass pollen) and checking the development of a wheal and flare reaction. They are positive in allergic asthma and negative in intrinsic asthma.
   - **Elevated serum IgE levels:** Measurement of total and allergen-specific IgE in serum may be seen.
4. ***Blood and sputum tests***
   - Patients with asthma may show increased numbers of eosinophils (eosinophilia) in peripheral blood ($>0.4 \times 10^9$/L), but sputum eosinophils are a more specific diagnostic finding. Sputum examination may reveal Curschmann's spirals, Creola bodies, and Charcot–Leyden crystals.
   - **Serum periostin** is also a marker of T_H2-associated airway inflammation and a better predictor of airway eosinophilia than blood eosinophil counts or FENO.
5. ***Fractional exhaled nitric oxide (FENO):*** Noninvasive test of airway inflammation. Role of FeNO in asthma
   - **Diagnostic:** >50 ppb (GINA 2020) suggest eosinophilic inflammation likely
   - **Predictive:** >50 ppb predicts corticosteroid responsiveness, improved compliance, response to biologicals [anti-IgE (>20 ppb), IL4/13(>25), anti-TSLptezepelumab].
   - **Prognostication:** Higher values associated with poor control, lower lung function and increased risk of asthma-related exacerbations.
6. ***Trial of corticosteroids***
   Patients with severe airflow limitation should be given a formal trial of corticosteroids. Prednisolone 30 mg orally/day is given for 2 weeks with lung function measured before and immediately after the course. A substantial improvement in FEV (>15%) confirms the presence of a reversible airflow obstruction and indicates that the administration of inhaled steroids will be beneficial to the patient.
7. ***Arterial blood gas analysis***
   - Hypoxia and hypocarbia during acute attack
   - Hypercarbia during sever acute asthma

Differential Diagnosis of Asthma (Table 2.16)

Q. Give the differential diagnosis of a 45-year-old male presenting with acute breathlessness.

Differences between bronchial asthma and cardiac asthma are present in **Table 2.17**.

TABLE 2.16: Differential diagnosis of asthma.

Disease	*Features*
Chronic obstructive pulmonary disease (COPD)	➢ History of smoking ➢ Less reversible airflow obstruction ➢ Reduced DLCO (diffusing capacity of lung for carbon monoxide)
Bronchiectasis	➢ May be secondary to many disorders ➢ Copious purulent sputum ➢ Computed tomography usually diagnostic
Reactive airways viral syndrome	Transient, usually resolves after several weeks
Rhinosinusitis	➢ Nasal congestion and postnasal drip ➢ Common comorbid illness accompanying asthma
Gastroesophageal reflux disease	Common comorbid illness accompanying asthma
Congestive heart failure	➢ Exertional dyspnea ➢ Moist basilar rales, S_3 gallop, orthopnea, and pink frothy/blood-tinged sputum ➢ Echocardiography is helpful
Laryngeal dysfunction	➢ Stridor ➢ May coexist with asthma
Upper airway obstruction	➢ May exhibit stridor ➢ Flow–volume loop is helpful ➢ Endoscopy diagnostic
Recurrent episodes of bronchospasm	Carcinoid tumors and recurrent pulmonary emboli, Churg–Strauss syndrome

TABLE 2.17: Differences between bronchial asthma and cardiac asthma.

Features	*Bronchial asthma*	*Cardiac asthma*
Pathology	Bronchospasm	Pulmonary edema
Age	Young	Elderly (above 50–60 years)
Sex/gender	Both genders	Mostly male
Past history	Eczema, urticaria (allergy) susceptibility to cold, allergy to pollen, groundnuts, and eggs	No history of allergy, history of left ventricular failure, and right ventricular failure
Family history	Other family members may have similar disease	Hypertension may run in families
Personal history	Highly sensitive individual	Nil
Onset	Acute, usually in early hours of morning or late hours of night	Acute, usually at midnight (very specific) 2–3 hours after sleep
Symptoms		
Dyspnea	Expiratory dyspnea	Both expiratory and inspiratory dyspnea
Expectoration	Scanty and mucoid sputum	Profuse and frothy sputum
Palpitation	Absent	Present
On examination		
Respiratory system	Expiratory wheeze present	Basal crepitations present
Cyanosis	Present during acute severe asthma	Cyanosis present
Pulse rate	May be high	Very high (may be pulsus alternans)
Blood pressure	Normal or slightly more systolic	BP usually high
Heart sounds	Heart sounds are distant	S3, gallop rhythm may be present

Q. Write short essay/note on:

- **The drugs used in the treatment of chronic bronchial asthma.**
- **Management of acute severe asthma.**
- **List the drugs used for prophylaxis against asthma.**
- **Status asthmaticus.**
- **Long-term complications of asthma.**

Management

Management is discussed under following headings namely:

Avoid Identified Aggravating Factors/Allergens

- This is important in the management of occupational asthma and atopic asthma.
- Avoid causative allergens such as pets, moulds, and certain foodstuffs particularly in childhood.
- If it is due to single allergen, it is easy to reduce or avoid the exposure. However, when multiple allergens are responsible, avoidance is difficult.

Control of Risk Factors Causing Exacerbation

The rapid identification and removal of extrinsic causes of asthma and risk factors that exacerbate asthma should be done.

- Active and passive smoking should be avoided.
- Control, if there is associated rhinitis and gastroesophageal reflux disease (GERD)
- Control obesity
- Individuals intolerant to aspirin should avoid NSAIDs.
- Avoid inadequate use of inhaled corticosteroids.
- Avoid overuse of inhaled short-acting b-agonists (e.g., more than one canister of 200 doses/month).
- Follow proper inhalation techniques.

Desensitization or Immunotherapy

Desensitization is performed by repeated subcutaneous injections of gradually increasing doses of the extracts of allergen(s). However, its benefit is doubtful. GINA, 2017 recommends adding **SLIT (sublingual immunotherapy)** in adult HDM (house dust mite)-sensitive patients with suboptimally controlled asthma despite low-high dose ICS provided FEV-1 is 70% predicted. **Subcutaneous immunotherapy (SCIT)** is also practiced.

Drug Therapy (Table 2.18)

Drug therapy is used to control or suppress clinical manifestations.

The drugs useful in asthma can be divided into bronchodilators (rapidly relieve of symptoms through relaxation of airway smooth muscle), and controllers (inhibit the underlying inflammatory process).

Bronchodilators

1. ***β_2-adrenoreceptor agonists***

- **β-adrenoreceptor:** There are two types of β-adrenoreceptor namely, β_1 and β_2-adrenoreceptors. β_1-adrenoreceptors are expressed in the heart and β_2-adrenoreceptors are widely expressed in the airways (in bronchial smooth muscles).
- **β_2-adrenoreceptors agonists** (β_2-agonists) can be divided into **short-acting β_2-agonists** (SABAs) (e.g., salbutamol, levosalbutamol, and terbutaline) and **long-acting β_2-agonists** (LABAs) (e.g., bambuterol, salmeterol, and formoterol).
  - **Catecholamines:** Catecholamines used are adrenaline, isoprenaline, and isoetharine. **Adrenaline:** Most commonly used agent in this group. However, it is not a β_2-selective and produces significant undesirable cardiovascular side effects. The usual dose is 0.3–0.5 mL of a 1:1000 solution administered subcutaneously. It may be repeated thrice at an interval of 20 minutes. They are useful in children.
  - **Salbutamol,** levosalbutamol**, terbutaline, and fenoterol:** These drugs are highly selective for β_2-adrenoreceptors and act predominantly on the respiratory tract.
    - **Powerful and rapidly but short-acting** bronchodilators that relax bronchial smooth muscles.
    - **Routes of administration:** They are active by inhalation, oral, intravenous, subcutaneous route of administration, but the **preferred route is inhalation**. Inhalation is extremely effective, since, it rapidly decreases airflow obstruction. Intravenous administration has no advantages over inhalation. Other routes of administration are preferably avoided and reserved for selected indications.
    - **Dose:**
      - *Salbutamol:* 2–4 mg thrice a day orally or two puffs of 100 µg each as required.
      - *Terbutaline:* 2.5–5 mg thrice a day or two puffs of 100 µg each as required.
      - *Levosalbutamol:* Two puffs of 50 µg each as required.
    - **Side effects:** Main untoward effects are tremor and palpitation. Prolonged use of β_2-adrenoreceptor agonists are preferably avoided because they worsen bronchial hyper-responsiveness. Tachycardia, which is less with levosalbutamol compared to salbutamol.
  - **Bambuterol:** It is a long-acting β_2-adrenoreceptor agonist, which is converted into terbutaline in the body.
    - **Dose:** 10–20 mg once a day, orally
    - **Side effects:** More than with inhaled β-agonists and includes tachycardia, palpitations, and tremors
  - **Salmeterol and formoterol:** They are highly selective, potent, and long-acting β_2-adrenoreceptor agonist. They are given once or twice a day by inhalation (either as aerosol or dry powder).
    - *Uses:* Routinely used in place of short-acting β_2-stimulants when the patient requires regular β_2-stimulant therapy. Not to be used as monotherapy but to be used as add-on therapy along with inhaled corticosteroids (ICS) when the response to ICs is suboptimal.
    - Salmeterol has a slow onset of action whereas formoterol has a rapid action. Hence, formoterol is suitable for immediate control of symptoms as well.
    - **Dose:**
      - *Salmeterol:* Two puffs of 25 µg each two to three times a day.
      - *Formoterol:* Two puffs of 6 µg each one to three times a day.

TABLE 2.18: Drugs useful in asthma.

Relievers	***Controllers***
➢ Short-acting inhaled beta2-agonist bronchodilators (SABA) ➢ Low-dose ICS-formoterol ➢ Short-acting anticholinergics	➢ Inhaled corticosteroids (ICS) ➢ ICS and long-acting beta-2 agonist (LABA), bronchodilator combinations (ICS-LABA) ➢ Leukotriene modifiers ➢ Chromones
Add-on controller medications	
➢ **Long-acting anticholinergic** (at Step 4 or 5 with a history of exacerbations despite ICS ± LABA) – Tiotropium ➢ **Anti-IgE** (with severe allergic asthma uncontrolled on high dose ICS-LABA) – Omalizumab ➢ **Anti-IL5 and anti-IL5R** (severe eosinophilic asthma uncontrolled on high-dose ICS-LABA) – Mepolizumab and reslizumab – Benralizumab ➢ **Anti-IL4R** (severe eosinophilic asthma uncontrolled on high-dose ICS-LABA, or requiring maintenance OCS) – Dupilumab ➢ **Systemic corticosteroids:** – Prednisolone – Hydrocortisone – Methylprednisolone	

Q. Write a short note on action of methylxanthines.

2. ***Methylxanthines***

They are of little value as monotherapy but they are beneficial as add-on therapy in patients not controlled with inhaled corticosteroids (ICS). Methylxanthines as an add-on therapy are less effective than long-acting inhaled β_2-agonists.

a. **Theophylline:**
- Theophylline is a medium-potency bronchodilator.
- **Actions:** (i) It improves the movement of airway mucus, (ii) improves diaphragm contractility, and (iii) reduces the release of mediators.
- **Route of administration:** Intravenous, oral, or as suppository. Therapeutic plasma concentrations of theophylline range from 10 to 20 μg/mL. However, the dose required to achieve this concentration varies from patient to patient.
- **Type of preparation:**
  - Acute attacks are treated with short-acting theophylline preparations.
  - For maintenance therapy, long-acting theophylline preparations are used. They are given once or twice a day. Single daily dose in the evening controls nocturnal asthma.
- **Dose:** Usual dose is 100–200 mg (of plain preparation) three times/day, and 300 mg twice/day or 450–600 mg once/day for sustained-release preparation.
- **Side effects:** Nervousness, nausea, vomiting, anorexia, and headache. When plasma levels exceed 30 μg/mL, seizures and cardiac arrhythmias can occur.
- **Precautions:** Theophylline (and aminophylline) clearance is decreased in elderly, liver disease, congestive heart failure, and with concurrent use of erythromycin, allopurinol, and cimetidine. Its clearance is increased with concurrent use of phenobarbitone and phenytoin, and in smokers.

Q. Write short essay/note on adverse effects of and precautions in using aminophylline.

b. **Aminophylline:**
- Aminophylline is a bronchodilator that is effective when given orally, intravenously, and as a suppository. The preferred route of administration is intravenous and may have some role in the management of status asthmaticus (severe acute asthma).
- **Mechanism of action:** Bronchodilator effect is by inhibition of phosphodiesterases in airway smooth-muscle cells, which increase cyclic AMP.
- **Dose:** Loading dose of 5 mg/kg is given slowly intravenously over 20 minutes. This is followed by a maintenance dose of 0.5 mg/kg/h delivered as a continuous intravenous infusion. Patients are already on theophylline; loading dose is preferably withheld or in extreme cases, it is given in a reduced amount at 0.5 mg/kg.
- Rapid infusion of the bolus can lead to sudden death due to cardiac arrhythmias.

c. **Doxophylline (doxofylline):**
- **Mechanism of action:** Bronchodilator effect is by inhibition of phosphodiesterases in airway smooth-muscle cells, which increase cyclic AMP. It has a better safety profile than theophylline. It also inhibits PAF-induced bronchoconstriction and subsequent production of thromboxane A_2.
- **Dose:** 400 mg twice a day.

3. ***Anticholinergics:***
- **Antimuscarinic bronchodilators:** They are less effective than β_2-agonists in asthma therapy, but may be used as an additional bronchodilator in patients with asthma that is not controlled by inhaled corticosteroids (ICS) and LABA combinations. They may be useful during asthma exacerbations, but are less useful in stable asthma, for example, ipratropium and tiotropium.

Controller Therapies

Q. Write short essay/note on inhaled corticosteroid therapy.

1. **Inhaled corticosteroids (ICS):**
- Inhaled corticosteroids are the most effective controllers for asthma.
- **Mechanism of action:** Corticosteroids are not bronchodilators, but they are the most effective anti-inflammatory agents used in asthma, which reduce number of inflammatory cells as well as their activation in the airways. They decrease bronchial hyper-responsiveness and relieve or prevent airflow obstruction. They also reverse β_2-receptor downregulation produced by long-term use of β_2-agonists.
- **Uses:** These are beneficial in treating asthma of any severity and age. They are now given as first line of therapy for persistent asthma.
- **Dose:** These are usually given twice daily. Higher doses may be necessary in severe cases.
  - **Beclomethasone dipropionate (200 μg), budesonide (200 μg), or fluticasone (125 μg) is given twice daily as aerosols or dry powder form.**
  - **Ciclesonide is given in a dose of 80–160** μg once a day. **Others include flunisolide and mometasone.**

- **Advantages:**
  - **Rapid improvement** of the **symptoms** and **lung function** (within several days).
    - They are effective in preventing asthma symptoms, exercise-induced asthma (EIA), and nocturnal exacerbations and they also prevent severe exacerbations.
    - Early treatment with ICS can prevent irreversible changes in airway function that develops in chronic asthma.
  - **Reduces airway responsiveness** (AHR)
  - Reduces the number of courses of oral corticosteroid therapy (OCS)
- **Side effects:**
  - **Local:** Hoarseness (dysphonia/husky voice) and oropharyngeal candidiasis. These side effects can be minimized by the use of a spacing device along with the metered-dose inhaler and gargling with water after use.
  - **Systemic:** Relatively free from systemic side effects at conventional doses. Long-term use may result in osteoporosis, skin thinning, and adrenal suppression.

2. **Systemic corticosteroids:**
   a. **Oral corticosteroids and steroid-sparing agents:**
   - **Oral corticosteroids (OCS):** Oral corticosteroids are necessary in patients controlled by inhaled corticosteroids (ICS).
   - **Dose:** It should be kept as low as possible to minimize side effects. Prednisolone is started as a single morning dose of 40–60 mg orally/day. Thereafter, the dose is reduced by half every 6 hours. Methylprednisolone is given in a dose of 40–125 mg every 6 hours.
   - **Steroid-sparing agents:** Some patients may require continuing treatment with oral corticosteroids. Various immunomodulatory treatments can be used in these patients with severe asthma who have serious side effects with this therapy. Treatment of these patients with low doses of methotrexate (15 mg weekly) can reduce the dose of oral steroids needed to control the disease. Ciclosporin also improves lung function in few steroid-dependent asthmatics.

   b. **Parenteral corticosteroids:**
   - Corticosteroids are used intravenously (hydrocortisone or methylprednisolone) for the treatment of acute severe asthma.
   - **Dose:**
     - Hydrocortisone: Loading dose of 4 mg/kg intravenously followed by 2–3 mg/kg every 6 hours
     - Methylprednisolone: 40–125 mg every 6 hours.
     - Indications for corticosteroids in bronchial asthma
   - Acute asthma which does not respond to or even worsen despite bronchodilator therapy
   - Severe acute asthma (status asthmaticus).

Corticosteroid-resistant asthma
- Complete resistance to corticosteroids:
  - <1/1,000 patients
  - Failure to respond to a high dose of oral prednisone/prednisolone (40 mg OD over 2 weeks), ideally with a 2-week run-in with matched placebo
- Corticosteroid-dependent asthma:
  - More common
  - Reduced responsiveness to steroids
  - Control of asthma requires oral corticosteroids

  Mechanisms:
  - Excess of transcription factor AP-1
  - Increase in the alternately spliced form of GR-β
  - Abnormal pattern of histone acetylation in response to corticosteroids
  - Reduction in histone deacetylase activity, as in COPD.

3. **Cromones (anti-inflammatory drugs):**
   - Cromones are not bronchodilators. They inhibit the degranulation of mast cells, thereby preventing release of mediators from these cells.
   - Cromones are not beneficial in the long-term control of asthma due to their short duration of action.
   - Only used as a prophylactic treatment of asthma. For example, **sodium cromoglycate, cromolyn, nedocromil, and ketotifen**

- **Sodium cromoglycate:** It is useful in children with atopic asthma and in few cases of nonatopic asthma. Sodium cromoglycate is administered as an inhalation. Therapy is started between the attacks or in periods of relative remission. If there is no response within 4–6 weeks, the drug can be discontinued.
- **Nedocromil sodium:** It is given as an inhalation at a dose of 4 mg, two to four times daily.
- **Ketotifen** is not a chromone. It is an antihistaminic that inhibits release of mediators. It is useful in the prophylactic treatment of asthma at a dose of 1–2 mg twice daily, orally. The main side effects are drowsiness and weight gain.

4. **Anticholinergics:**

Q. Write short note/essay on tiotropium/ipratropium bromide uses and side effects.

- Anticholinergics such as atropine sulfate and atropine methyl nitrate were previously used, but they are presently not used because of their systemic side effects.
- Currently used anticholinergics are **ipratropium bromide and tiotropium.** These are nonadsorbable quaternary ammonium compounds with minimal side effects. These are administered as aerosol or in dry powder form. Ipratropium is also given as nebulization solution.
- **Uses:** They are useful in two situations:
  1. Patients with coexisting heart disease, in whom methylxanthines and β_2-adrenoreceptor agonists cause significant tachycardia.
  2. In refractory cases, bronchodilator action of β_2-adrenoreceptor agonists is enhanced by the addition of ipratropium bromide or tiotropium.
- **Dose:**
  - *Ipratropium:* Two puffs of 20 mg each, four times/day
  - *Tiotropium:* Two puffs of 9 µg each, once a day
  - *Ipratropium:* 250–500 µg nebulization; may be repeated, if necessary
- **Side effects:** Dryness of mouth and bitter taste

5. **Leukotriene modifiers:**
- These include leukotriene receptor antagonists—**LTRAs** (montelukast, zafirlukast, and pranlukast) and 5-lipoxygenase inhibitors (Zileuton).
- **Uses:** Used as add-on therapy.
  - In patients who do not respond to the conventional agents
  - In patients who require high doses of inhaled steroids (ICS). They can be used as a second choice to inhaled corticosteroids in mild persistent asthma.
- **Dose:**
  - *Zafirlukast:* 20 mg BID
  - *Montelukast:* 10 mg once a day in the evening
- **Side effects:** Uncommon and include headache, abdominal pain, skin rashes, angioedema, pulmonary eosinophilia, and arthralgia. Zileuton may cause liver damage.

Anti-IgE—Monoclonal Antibodies

- Biologicals are indicated for severe after ensuring optimization of technique, other causes have been ruled out
- **Omalizumab,** recombinant humanized monoclonal antibody against IgE, that neutralizes/chelates free circulating IgE without binding to cell-bound IgE. It prevents the binding of circulating IgE to receptors on mast cells and basophils, and decreases release of mediators. Thus, it inhibits IgE-mediated reactions. Monoclonal antibodies (**mepolizumab and reslizumab**) against interleukin-5 (IL-5), a potent chemoattractant for eosinophils, are indicated for the treatment of severe eosinophilic asthma poorly controlled with conventional therapy.
  - Useful in patients with allergic asthma
  - *Disadvantage:* Treatment is very expensive. Patients should be given a 3–4-month trial of therapy to show objective benefit.
  - *Administration:* Given as a subcutaneous injection once every 2–4 weeks, depending on total serum IgE level and body weight
  - *Side effects:* No significant side effects, very rarely can produce anaphylaxis
- **Anti-TNF therapy** (infliximab or etanercept)—may be beneficial in severe corticosteroid refractory asthma.
- **Methotrexate and cyclosporine** have shown glucocorticoid-sparing effects.
- **Macrolide** antibiotics have both antimicrobial and anti-inflammatory actions raising the possibility of benefit in severe asthma.

Anti-IL-5 therapy:
Interleukin (IL)-5 is a proeosinophilic cytokine that is potent and reduces exacerbations in patients with severe asthma who have blood eosinophil counts of 150/μL or greater.
- Mepolizumab and reslizumab are anti-IL-5 monoclonal antibodies
- Benralizumab is an anti-IL-5 receptor alpha antibody

Dosage:
- Mepolizumab is administered subcutaneously, 100 mg every 4 weeks
- Reslizumab is administered 3 mg/kg by intravenous infusion
- Benralizumab is given subcutaneously, 30 mg every 4 weeks first three doses, and then 30 mg every 8 weeks

Anti-IL-4 receptor alpha subunit antibody:
Dupilumab is a fully human monoclonal antibody that binds to the alpha subunit of the IL-4 receptor. It is effective in patients with frequent exacerbations, high type 2 markers (eosinophil, FeNO >25 ppb) or requirement for oral steroids.
Dose: An initial dose of 400 mg (two 200 mg subcutaneous injections), followed by 200 mg given every other week.

Anti-TSLP therapy:
TSLP protein works on allergic, eosinophilic and neutrophilic inflammatory pathways. Tezepelumab is effective in patients with severe asthma and history of severe exacerbations. Can be used in all phenotypes.
Dose: 210 mg every 4 weeks.

Miscellaneous

Proton pump inhibitor (PPI) may be used in patients with symptomatic gastroesophageal reflux disease and suboptimally controlled asthma.

Bronchial Thermoplasty

- This is invasive procedure for severe asthma. In this therapy, controlled thermal energy is delivered to the airway wall during a series of bronchoscopies. It results in a prolonged reduction in airway smooth muscle mass. But, patient still needs to use their asthma-maintenance medications after the procedure.
- **Risks:** Lung collapse, bleeding, and additional breathing problems, mostly related to the bronchoscope.
- **Benefits:** Patient may use rescue inhalers less often and are able to engage strenuous physical activity than before.

General Measures in Asthmatics

- Avoid:
  - Opiates, sedatives, and tranquilizers in acutely ill patients with asthma
  - β-blockers and parasympathetic agonists in asthmatics
- Expectorants and mucolytic agents have no significant role in the management of bronchial asthma.

Assessment of Asthma Control (Table 2.19)

- Features, which suggest that asthma is under control, are listed in **Box 2.4**.
- Complications of asthma are mentioned in **Box 2.5**.

TABLE 2.19: Assessment of asthma control.

Characteristics	*Controlled asthma*	*Partly controlled asthma*	*Uncontrolled asthma*
Daytime symptoms more than twice/week	No	Any 1 or 2 characteristics present	Any 3 or more characteristics partly controlled asthma
Limitation of activities due to asthma	No		
Nocturnal symptoms/ awakening due to asthma	No		
Need for reliever/rescue medicine more than twice/week	No		

Box 2.4: Features of controlled asthma.
- Daytime symptoms develop two times or less/week
- No limitation of daily activities including exercise
- No awakening in the night due to symptoms
- Need for short-acting β-agonists twice or less/week
- No exacerbations

Box 2.5: Complications of asthma.
- Pneumonia
- Collapse of part or all of the lung
- Respiratory failure
- Status asthmaticus

Global Initiative for Asthma (GINA) Severity Grades (See Table 2.21)

Severe asthma:

- It is an asthma, which requires treatment with high-dose inhaled corticosteroids and long-acting β_2-agonist and/or leukotriene receptor antagonists for the previous year or systemic corticosteroids for ≥50% of the previous year to prevent from becoming uncontrolled asthma
- Asthma which remains uncontrolled despite this therapy

Stepwise Management of Chronic Asthma (Table 2.20 and Fig. 2.10)

Q. Write short essay/note on step care management of bronchial asthma.

All asthmatics should be educated regarding self-monitoring and correct use of inhalers.

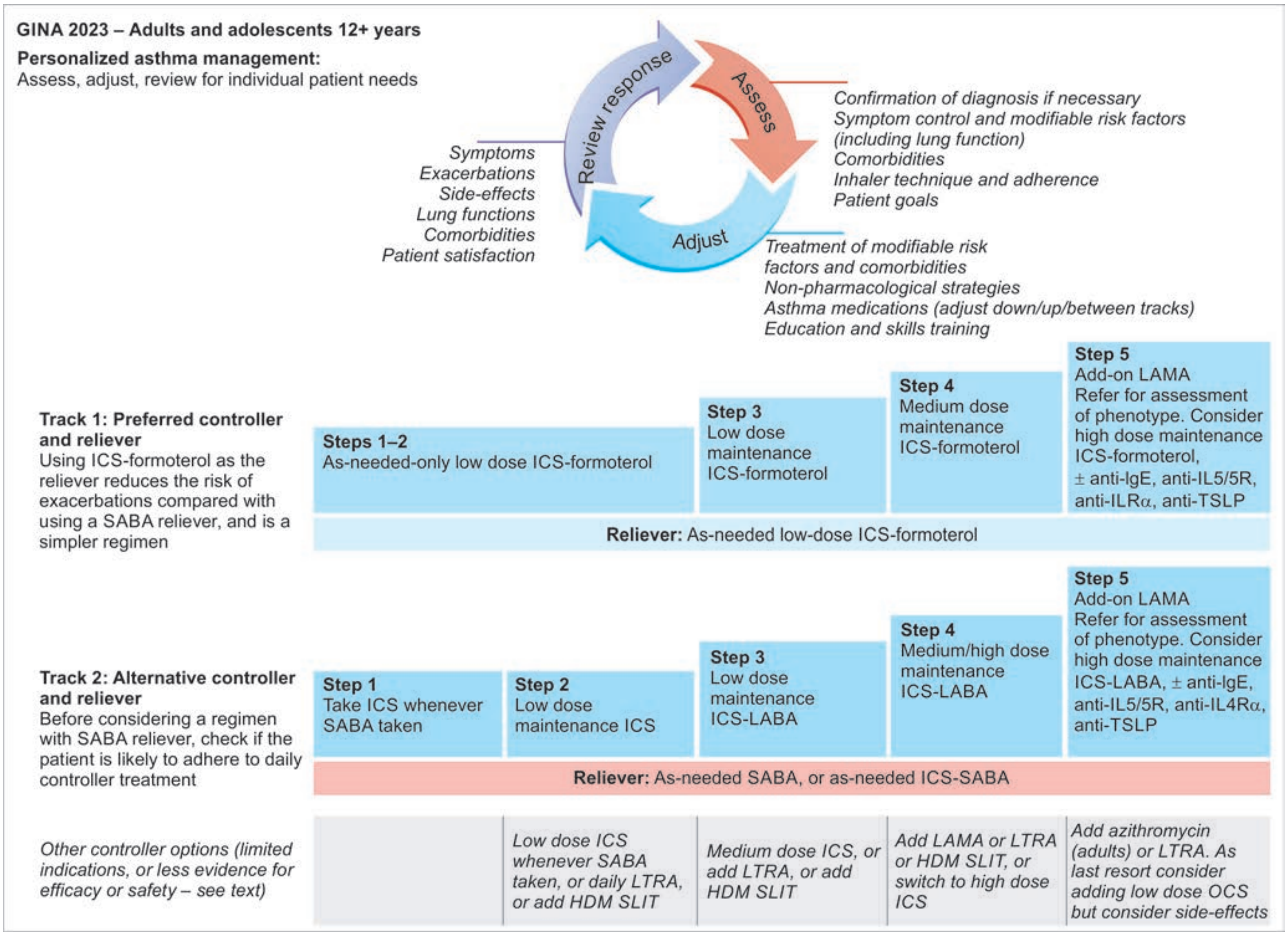


FIG. 2.10: Stepwise management of asthma in adults.
(LABA: Long-acting inhaled β_2-agonist; LTRA: Leukotriene receptor antagonists; SR: Sustained release)

TABLE 2.20: Stepwise management of asthma.

Step	***Management***
Reliever	➢ Low dose ICS + formoterol ➢ As needed SABA or as needed ICS-SABA
Step 1: (only for intermittent/less frequent symptoms)	➢ ICS whenever SABA taken ➢ Short-acting inhaled β_2-agonist as required
Step 2: Daily symptoms	**Regular inhaled preventer therapy:** ➢ Low-dose maintenance inhaled corticosteroids up to 800 µg daily ➢ As needed ICS-formoterol ➢ Leukotriene receptor antagonists (LTRA) (if patient develops side effects to inhaled corticosteroids) and SLIT (sublingual immunotherapy)

Contd...

Contd...

Step	***Management***
Step 3: Severe symptoms	**Inhaled corticosteroids and long-acting inhaled β_2-agonist:** ➢ Continue low-dose inhaled corticosteroids plus long-acting β_2-agonist (ICS-LABA), or ➢ Medium- or high-dose inhaled corticosteroids, or ➢ Low-dose inhaled corticosteroids plus leukotriene receptor antagonists (LTRA), or ➢ SLIT (sublingual immunotherapy)
Step 4: Severe symptoms uncontrolled with high-dose inhaled corticosteroids	**High-dose inhaled corticosteroid and regular bronchodilators:** ➢ Medium- or high-dose inhaled corticosteroids (up to 2,000 µg daily) plus long-acting β_2-agonist ➢ May add leukotriene receptor antagonists (LTRA) or ➢ May add LAMA or sublingual immunotherapy (SLIT) or ➢ Switch to high dose ICS
Step 5: Severe symptoms deteriorating	**Regular oral corticosteroids:** ➢ Add oral corticosteroids (prednisolone 40 mg daily) ➢ Add oral azithromycin or LTRA ➢ Consider anti-IgE treatment (omalizumab) ➢ Anti-IL5/anti-IL4 /anti-TSLP therapy

Step-down Therapy

Once asthma is controlled, the dose of inhaled (or oral) corticosteroid should be reduced to the lowest dose at which effective control of asthma is maintained.

Suggested initial controller therapy is given in **Figure 2.11**.

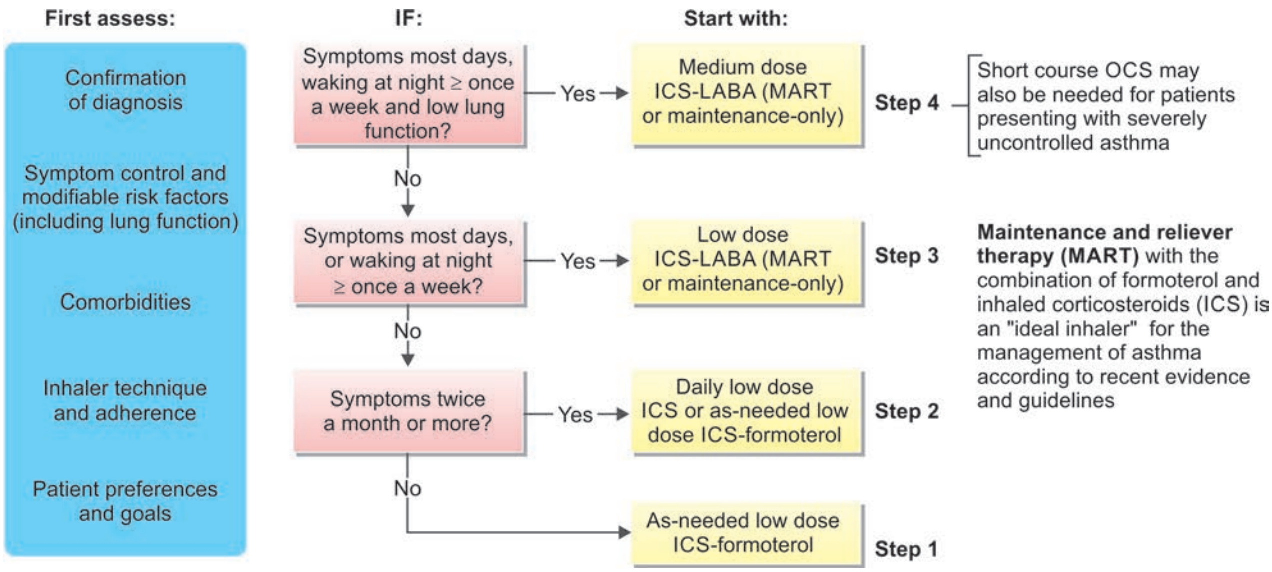


FIG. 2.11: Suggested initial controller treatment in adults and adolescents with a diagnosis of asthma.

Difficult-to-Treat Asthma

Asthma that is uncontrolled despite GINA step 4 or 5 treatment or that requires such treatment to maintain good symptom control and reduce the risk of exacerbations.

Contributory factors may include:

- Incorrect diagnosis
- Incorrect inhaler technique
- Poor adherence
- Comorbidities

Severe asthma is a subset of difficult-to-treat asthma.

It means asthma that is uncontrolled despite adherence with maximal optimized therapy and treatment of contributory factors or that worsens when high-dose treatment is decreased.

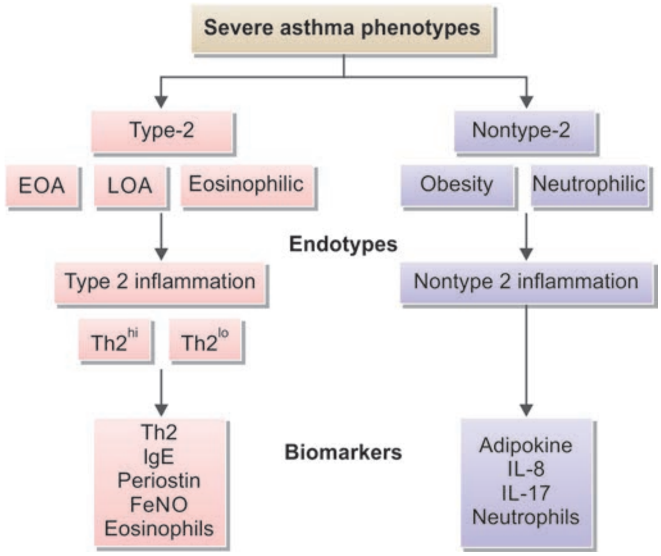
It is sometimes called "severe refractory asthma", since it is relatively refractory to high-dose inhaled therapy. However, with the advent of biologic therapies, the word "refractory" is no longer appropriate.

Phenotypes, endotypes, and biomarkers in severe asthma are shown in **Flowchart 2.3**.

Asthma Phenotype (Fig. 2.12)

- The asthma phenotypes differ in terms of genetic susceptibility, environmental risk factors, age of onset, clinical presentation, prognosis, and response to therapies; therefore, asthma is seen as a syndrome rather than a single disease.
- Asthma is divided into **phenotypes:** type 2 inflammation and non-type 2 inflammation. **Type 2 phenotype:** early onset asthma (EOA), late-onset asthma (LOA), and eosinophilic asthma; biomarkers: Th2 cytokines, IgE, periostin, FeNO (fraction of nitric oxide expired), and eosinophilia. **Non-type 2 phenotypes:** asthma associated with obesity and neutrophilic asthma; biomarkers: adipokine, IL-8 or IL-17, and neutrophilia.

FLOWCHART 2.3: Phenotypes, endotypes, and biomarkers in severe asthma.



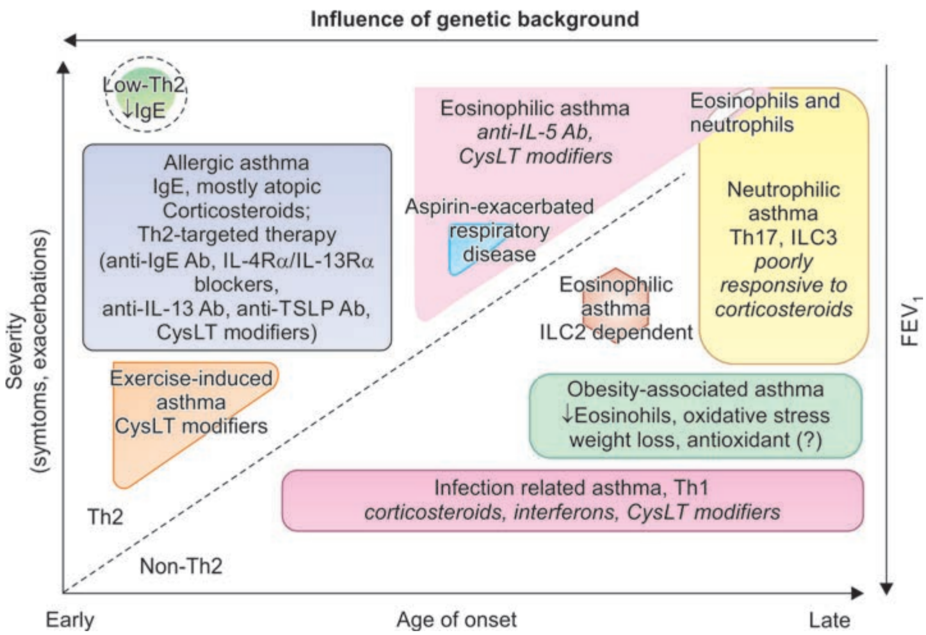


FIG. 2.12: Asthma phenotypes and pathology.

New phenotype/endotype-based therapy to treat severe asthma

Asthma phenotype	***Characterized by difficult-to-treat asthma plus***	***Targeted treatment***
Allergic asthma (IgE)	Total serum IgE = 30–700 IU/mL and demonstrated IgE-mediated hypersensitivity to a perennial allergen	Add anti-IgE biologic ➢ Omalizumab
Eosinophilic	Blood eosinophils >300 cells/uL and 2 or more exacerbations requiring OCS in past year or ≥150 cells/uL and 3 or more exacerbations requiring OCS in past year	Add anti-IL-5 biologic ➢ Mepolizumab ➢ Reslizumab ➢ Benralizumab Add anti-IL-4/IL-13 biologic ➢ Dupilumab Can also add anti-TSLP (Tezepelumab)
Neutrophilic	Sputum neutrophils in patients who do not respond to high-dose corticosteroids and do not have other type 2 markers	Consider adding a macrolide antibiotic Can also add anti-TSLP (Tezepelumab)
Airway smooth muscle (ASM) hypertrophy	Patient who does not qualify for other targeted therapies and/or has tried and failed targeted therapies for which he/she might be eligible and obstruction by bronchodilator reversibility	Consider bronchial thermoplasty, an outpatient medical procedure, in addition to regular treatment

Treatment of severe acute asthma (status asthmaticus)

Q. Write short essay/note on management of status asthmaticus.

Acute severe asthma is the term used for an exacerbation of asthma that has not been controlled by the use of standard medication.

Treatment at home

- **Give high concentrations of oxygen** (40–60%) through a mask, if available
- **Bronchodilator therapy:** Any one of the following should be given.
  - Nebulized **salbutamol** 5 mg or **terbutaline** 10 mg every 20 minutes for 3 doses
  - Salbutamol/terbutaline through metered-dose inhalers (four to eight puffs with a spacer every 20 minutes for 3 doses), followed by 4–8 puffs every 2–4 hours
- **Corticosteroids:**
  - Give IV hydrocortisone sodium succinate 200 mg
  - Give oral prednisolone 60 mg
- **Admit to hospital, if there is no response** within 1 hour or if patient becomes drowsy.

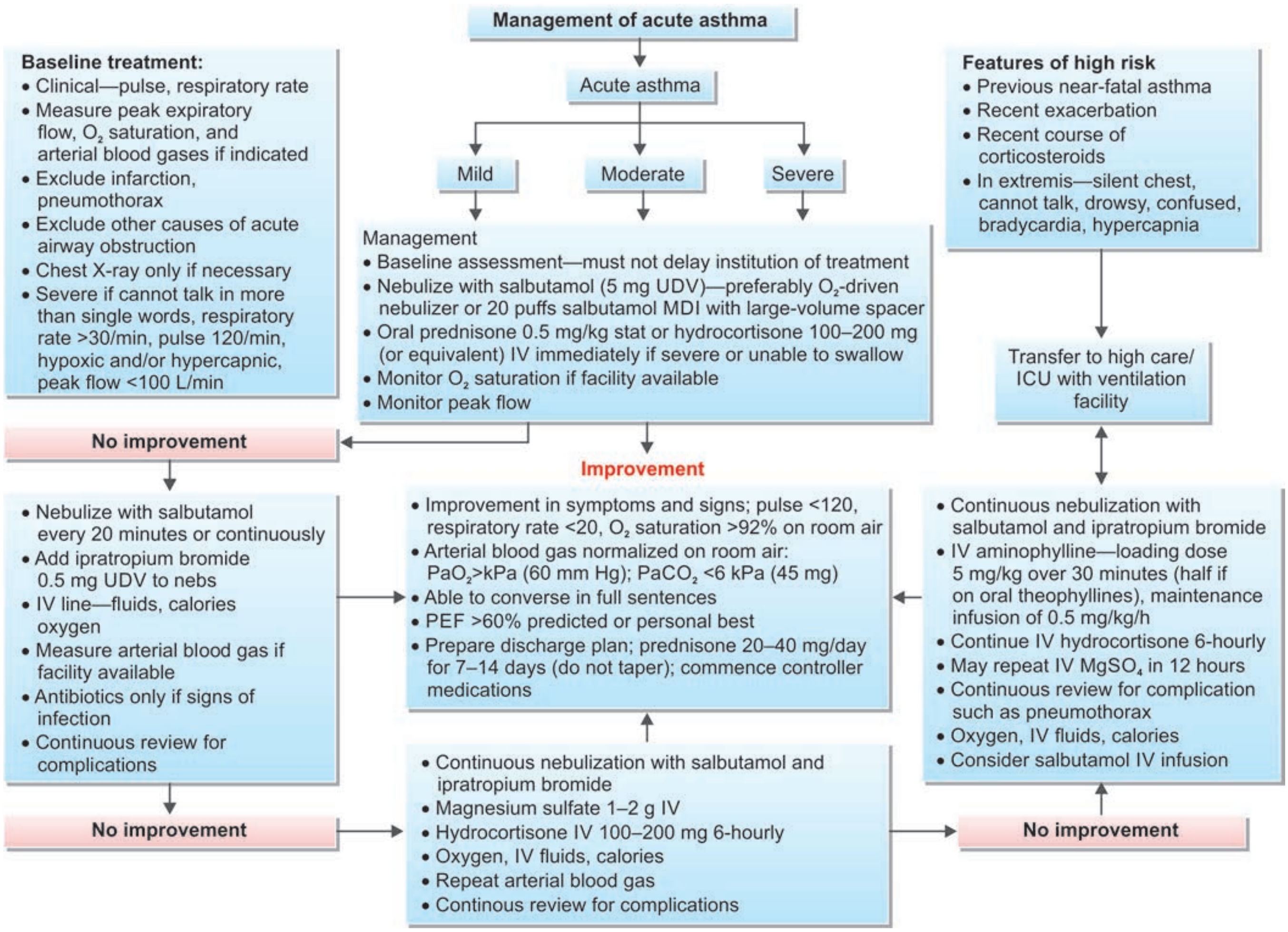
Management in hospital

Initial assessment: Take brief history, perform rapid examination of the patient, and assess the severity (**Table 2.21**). Administer high concentration of oxygen (40–60%).

Management algorithm of acute asthma

See **Flowchart 2.4**.

FLOWCHART 2.4: Management algorithm of acute asthma.



(UDV: unit dose vial)

TABLE 2.21: Assessment of severity of asthma.

	Mild	*Moderate*	*Severe*	*Respiratory arrest imminent*
Breathless	Walking Lying down possible	Taking Sitting preferred	At rest Hunched forward	
Talking in	Sentences	Phrases	Words	
Alertness	Possibly agitated	Usually agitated	Usually agitated	Drowsy or confused
Respiratory rate	Increased	Increased	Often >30/min	
Accessory muscles/ suprasternal retractions	Usually not	Usually	Usually	Paradoxical thoracoabdominal movement
Wheeze	Moderate	Loud	Usually loud	Absent
Pulse/min	<100	100–120	>120	Bradycardia
Pulsus paradoxus	Absent	May be present	Often present	Possibly absent due to muscle fatigue
PEF after bronchodilator	>80%	Approximately 60–80%	<60%	

Adapted from: O'Byme P; GINA Executive Committee. Global strategy for asthma management and prevention. National Institutes of Health. 2004. Publication No. 02-3659.

Exacerbations of asthma

Q. Write short essay on management of acute severe asthma.

Precipitating factors of exacerbations: Viral infections (most common), others include moulds, pollens, and air pollution.

Treatment of mild-to-moderate exacerbation

- β_2-agonists (via inhaler or nebulizer) every 20 minutes for 3 doses (as mentioned above)
- Oral corticosteroids, if there is no immediate response
- Patient is reassessed every hour.

Treatment of severe exacerbation

- Give **high concentration of oxygen** (40–60%)
- **Bronchodilators:**
  - Administer **nebulized (in oxygen) salbutamol** (5 mg) or terbutaline (10 mg) or levosalbutamol (1.25–2.5 mg) immediately and may be repeated after a few minutes, if there is no response.
  - **β_2-agonist (subcutaneously or intravenously)** are indicated in patients with excessive cough, too weak to inspire adequately or moribund.
  - **Terbutaline** is administered subcutaneously (0.25–0.5 mg) or intravenously (0.1–10 µg/kg/min).
  - **Epinephrine (adrenaline)** may be administered in children and young adults. Adult dose is 0.2–0.5 mg as 1:1,000 solution subcutaneously every 20 minutes.
  - **Add nebulized ipratropium bromide** 0.5 mg to nebulized salbutamol 5 mg/terbutaline 10 mg to those patients who do not respond within 15–30 minutes. It can be repeated every 20 minutes for 3 doses.
  - **Aminophylline** can be given intravenously to those patients who do not respond to nebulized bronchodilators. Give a loading dose of 5 mg/kg/h as an infusion.
- **Corticosteroids:** In severely ill patients, **hydrocortisone sodium succinate 100 mg** is administered intravenously at presentation and then repeated 4–6 hourly for 24 hours.
- **Antibiotics** are indicated only if there is respiratory infection.
- Role of magnesium sulfate either intravenously or by nebulization is not clear.
- **If no improvement** with above measures, perform **endotracheal intubation** and **mechanical ventilation**.
  - **Indications for intubation:** Cardiac or respiratory arrest, severe hypoxia (PaO_2 <60 mm Hg), hypercapnia ($PaCO_2$ > 50 mm Hg), acidosis (pH <7.3), exhaustion, or deterioration in mental status
- **NIV (noninvasive ventilation)** using continuous positive pressure of BiPAP machines and tight fitting face mask reduces the work of breathing without intubation. It is useful in assisting breathing. It is used in a cooperative and alert patient who has impending respiratory failure but does not require immediate intubation.
- Treatment with 70–80% helium with oxygen may be useful, since it reduces airway resistance and improves efficacy of bronchodilators.
- Assessment of response to treatment is done by noting the patient distress, respiratory rate, FEV_1, heart rate, presence of pulsus paradoxus, and serial arterial blood gas (ABG) studies.
- More severe cases should remain in hospital for 2–5 days with regular monitoring of oxygen saturation and peak flow rates. Bronchial thermoplasty may be beneficial for moderate-to-severe persistent asthma. This reduces the mass of airway smooth muscle, reducing bronchoconstriction.

Occupational Asthma

Q. Write a short note on occupational asthma.

Occupational asthma is relatively common.

Etiology

- It is a type of asthma caused by specific occupational sensitizer (proteins or glycopeptide). It is often associated with allergic rhinitis/conjunctivitis.
- Once the individual is sensitized, subsequent low exposure is capable of producing specific IgE antibodies and can induce asthma.
- Latency period between first exposure to sensitizer and onset of work-related symptoms can range from weeks to years.
- Triggering occupational agents include—fumes (epoxy resins and plastics), organic and chemical dusts (wood, cotton, and platinum), gases (toluene diisocyanate), animal allergens, plants and plant products (flours and cereals), and other chemicals (formaldehyde and penicillin products). **Work-exacerbated asthma** is defined as preexisting or concurrent asthma that subjectively worsens in the workplace.

Management
- Primary prevention (e.g., avoiding sensitizer agent)
- Secondary prevention (e.g., medical surveillance program)
- Tertiary prevention (e.g., appropriate treatment), i.e., treatment as per asthma severity

Hypersensitivity Pneumonitis (Extrinsic Allergic Alveolitis)

Q. Discuss the clinical features, diagnosis, and management of hypersensitivity pneumonitis (extrinsic allergic alveolitis).

Q. Write short note on farmer's lung.

Definition

Hypersensitivity pneumonitis (HP) or extrinsic allergic alveolitis is an immune-mediated lung disease that occurs secondary to inhalational exposure to a variety of antigens leading to a widespread diffuse inflammatory reaction in the alveoli and small airways (bronchioles).

Pathogenesis

It is an immune-mediated condition that develops in response to inhaled antigens that are small enough to deposit in distal airways and alveoli. It involves T_H1 and T_H17 lymphocyte subsets. Examples of hypersensitivity pneumonitis are presented in **Table 2.22**.

TABLE 2.22: Examples of hypersensitivity pneumonitis.

Disease	*Sources of antigen*	*Antigen*
Farming/Food processing		
Farmer's lung	Grain, moldy hay	Thermophilic actinomycetes, fungus
Bagassosis	Sugarcane	Thermophilic actinomycetes
Mushroom worker's lung	Mushroom	Thermophilic actinomycetes; mushroom spores
Coffee worker's lung	Coffee beans	Coffee bean dust
Malt worker's lung	Mouldy barley	*Aspergillus* species (clavatus)
Miller's lung	Infested wheat flour	*Sitophilus granarius* (wheat weevil)
Tobacco grower's lung	Tobacco	*Aspergillus* species
Birds and other animals		
Bird fancier's lung	Avian droppings (most often from pigeons and parakeets)	Protein in avian excreta, feathers
Other occupational and environmental exposures		
Humidifier fever and air-conditioner lung	Humidifiers and air conditioners (contaminated water)	*Bacillus subtilis, Aureobasidium pullulans, Candida albicans*, ameba, thermophilic actinomycetes
Woodworker's lung	Wood dust	*Alternaria* species

Clinical Features

It depends on type, intensity, and duration of exposure to offending antigen. Most of them present following months or years of continuous or intermittent inhalation of offending agent.

Clinical forms: It can present as acute, subacute, or chronic form. Most patients (80–95%) are nonsmokers.

1. **Acute form:**
   - *Onset:* Usually manifests 4–8 hours following exposure to the causative antigen and is often intense in nature.
   - *Symptoms:* Fevers, chills, cough, and malaise accompanied by dyspnea.
   - *Resolution:* Symptoms resolve within hours to days, if there is no further exposure to the causative antigen.
2. **Subacute form:** Onset of respiratory (cough, dyspnea, and cyanosis) and systemic symptoms is more gradual lasting for weeks or months.
3. **Chronic form:**
   - Presents with an even more gradual onset of symptoms than subacute form with progressive dyspnea, cough, fatigue, weight loss, and clubbing of the fingers.

- It usually occurs in patients with continuous low-dose exposure to the causative antigen.
- Clinically cannot be distinguished from pulmonary fibrosis due to other causes.

Complications

Long-standing disease leads to diffuse pulmonary interstitial fibrosis, pulmonary hypertension resulting in cor pulmonale.

Investigations

- **Chest X-ray:** Findings are nonspecific. It may show diffuse micronodular shadowing with the subsequent development of streaky shadows, particularly in the upper zones. Very advanced cases produce honeycomb lung.
- **High-resolution computed tomography (HRCT):** It shows reticular and nodular changes with ground-glass opacity.
- **Blood:** Raised ESR and neutrophilia (especially in acute form). Eosinophil counts and IgE levels are usually normal.
- **Lung function tests:** These show a restrictive ventilatory defect with normal FEV_1:FVC ratio. Diffusion capacity for carbon monoxide may be significantly impaired in chronic cases.
- **Precipitating antibodies** in the serum against offending agent indicates the evidence of exposure, but not disease.
- **Bronchoscopy with bronchoalveolar lavage (BAL):** Although not a specific, BAL shows increased T-lymphocytes and granulocytes. Most often, CD4:CD8 ratio <1 (normal is about 0.9–2.5). In sarcoidosis, it is usually above 3.5.
- **Lung biopsy:** May show chronic inflammatory cells, ill-defined noncaseating granulomas, and fibrosis in late stages.
- **Provocation test:** If about 4–10 hours after exposure to the inhaled antigen, there is respiratory or systemic findings (e.g., fever and leukocytosis); reduced diffusing capacity, diminished VC (vital capacity), or both; increased radiographic abnormalities; worsening alveolar–arterial oxygen pressure suggests a positive response.

Treatment
- **Prevention:** It should be the aim and can be achieved by identification and avoidance of the offending antigen.
- **Corticosteroids:** Because acute form is usually a self-limited disease, treatment is not necessary. However, corticosteroid has some role in subacute and chronic forms. Prednisolone in large doses (1 mg/kg/day) orally for 1–2 weeks, followed by gradual tapering over the next 4–6 weeks. It may achieve regression during the early stages but not those with established fibrosis.
- **Oxygen therapy** in severely hypoxemic individuals. Symptomatic bronchodilator therapy.

Drug-induced Asthma

Drugs Implicated

- Most commonly due to **aspirin (aspirin-induced asthma—**AIA) **followed by ibuprofen, indomethacin, naproxen, phenylbutazone, and mefenamic acid**. About 10% of asthma patients are aspirin sensitive.
- **Drugs that can cause bronchospasm:** Adenosine, prostaglandin analogs, BBs (beta-blockers), cholinergic drugs, streptomycin, pentazocine, and penicillin.
- **Nonpharmaceutical agents:** Tartrazine (coloring agent), sulfating agents (preservatives in food/medicines).

Mechanism: Decreased production of prostaglandin E_2 (PGE_2) and enhanced production of leukotriene B_4 (LTB_4).

Associations: Nasal polyps, vasomotor rhinitis, and hyperplastic rhinosinusitis **(Samter's triad)**.

Aspirin-induced asthma (AIA): About 1% of asthmatics worsen with aspirin and other COX inhibitors. Mechanism is due to polymorphism of cis-leukotriene synthase.

Two distinct forms:
1. **Cutaneous form:** Associated with urticaria and angioedema
2. **Respiratory form:** Resulting in rhinoconjunctivitis and bronchospasm

Diagnosis: Bronchial challenge with aspirin.

Drug treatment of AIA
Treat the underlying asthma and the strict avoidance of aspirin and cross-reacting NSAIDs. Acetaminophen and selective COX-2 inhibitors are safe.

Exercise-induced Asthma

- Transient airway narrowing associated with exercise, which can occur in patients with or without chronic asthma.
- **Thermal pathway:** Exercise → heat loss → rapid airway rewarming → congestion of vascular bed → airway obstruction
- **Osmotic pathway:** Exercise → water loss → hyperosmolar environment → cell shrinking → mediator release, vascular leak, and bronchospasm

Prevention/treatment of exercise-induced asthma
- Prevention of episode by the inhalation of 2 metered doses of salbutamol or terbutaline a few minutes before exercise. However, regular use may lead to loss of their effect.
- Regular use of sodium cromoglycate or leukotriene modifiers is often needed. Additional use of inhaled β2-adrenoreceptor agonists may be necessary before exercise.
- Single dose short-acting β2-agonist, sodium cromoglycate, or nedocromil sodium immediately before exercise should be used.
- Inhaled corticosteroid twice daily for 8–12 weeks reduces severity.
- If abnormal, spirometry and persistent symptom-inhaled corticosteroid with long-acting β2-agonist
- Leukotriene receptor antagonist may be used

CHRONIC OBSTRUCTIVE PULMONARY DISEASE (COPD)

Q. Write short essay/note on COPD (chronic obstructive pulmonary disease), COLD (chronic obstructive lung disease), or COAD (chronic obstructive airway disease).

Q. CT2.1 Define and classify obstructive airway disease.

Introduction

Chronic obstructive pulmonary disease is also known as chronic obstructive lung disease (COLD), chronic obstructive airway disease (COAD), chronic airflow limitation (CAL), and chronic obstructive respiratory disease (CORD).

Definition

- COPD is a **preventable and treatable pulmonary disease** associated **with** some **significant extrapulmonary effects** that may contribute to the severity in individual patients.
- Pulmonary disease is characterized by **airflow limitation,** which is **not fully reversible.**
- The airflow limitation is usually progressive and associated with an abnormal inflammatory response of the lung to various noxious particles or gases.

Conditions included under COPD: Disease is considered COPD, only if chronic airflow obstruction occurs.
- **Emphysema:** An anatomically defined condition characterized by abnormal and permanent enlargement of the airspaces distal to the terminal bronchioles. It is accompanied by destruction of the airspace walls, without obvious fibrosis (i.e., there is no fibrosis visible to the naked eye).
- **Chronic bronchitis:** It is defined as a chronic productive cough for 3 months in each of two successive years in a patient in whom other causes of chronic cough (e.g., bronchiectasis) have been excluded.
- **Small airways disease:** A condition in which small bronchioles are narrowed.

Note: **Preserved ratio impaired spirometry (PRISm)**, defined as a reduced FEV1 or FVC in the setting of preserved FEV1/FVC is seen in 10% of general poulation.

Pathogenesis

- **Major physiologic change** in the COPD is **airflow limitation**. It can develop due to both small airway obstruction and emphysema. The small airways may become narrowed by cells (hyperplasia and accumulation), mucus, and fibrosis.
- Activation of transforming growth factor-β (TGF-β) contributes to airway fibrosis, whereas absence of TGF-β may produce parenchymal inflammation and emphysema. The mechanism involved in emphysema is better understood than small airway obstruction.

CHRONIC BRONCHITIS

Q. Define chronic bronchitis. Describe the etiology, pathology, clinical features, investigations, course, prognosis, treatment, and complications of chronic bronchitis.

Incidence

- **Age and gender:** Occurs during **middle and late adult life**. It is more common in **males than in females.**
- It is more common **in smokers than in nonsmokers**. Also more often it develops in urban than in rural dwellers.

Types of Chronic Bronchitis: (1) Simple chronic bronchitis; (2) chronic mucopurulent bronchitis; (3) chronic asthmatic bronchitis; and (4) chronic obstructive bronchitis.

Q. CT2.2 Describe and discuss the epidemiology, risk factors and evolution of obstructive airway disease.

Q. CT2.6 Describe the role of the environment in the cause and exacerbation of obstructive airway disease.

Etiology

Risk Factors

See **Table 2.23**.

Smoking and COPD

- **Cigarette smoking:** It is the most important risk factor for the development of COPD. The risk of developing COPD relates to both the amount and the duration of smoking. However, only about 15–20% of smokers develop clinically significant COPD. This suggests that genetic predisposition and environmental factors play a role in the pathogenesis.
- **Second-hand smoke:** Environmental tobacco smoke that is inhaled involuntarily or passively by someone who is not smoking.
- **Environmental tobacco smoke** is generated from the **side stream** (the burning end) of a cigarette, pipe or cigar or from the exhaled **mainstream** (the smoke puffed out by smokers) of cigarettes, pipes, and cigars.
- **Abnormalities due to smoking:** Cigarette smoking is associated with a variety of abnormalities of the respiratory system that predisposes to the development of chronic bronchitis. These include:
    - Sluggishness of movement of cilia
    - Bronchoconstriction due to constriction of smooth muscle
    - Hypertrophy and hyperplasia of mucus secreting glands. **The ratio of the thickness of the mucous gland layer to** the **thickness of the bronchial wall** between the base of the surface epithelium and the inner limit of the cartilage plates is called **Reid index**. It is useful for detecting the increase in the size and number of the mucus glands. Reid index (normally 0.44 ± 0.094) is increased in chronic bronchitis (>0.51). There is a direct correlation between the value of Reid index and the volume of daily sputum production by the patient.
    - Release of proteolytic enzymes from polymorphonuclear leukocytes and release of inflammatory mediators in lungs
    - Inhibits the function of alveolar macrophages
    - Adverse effect on surfactant and favors overdistension of the lungs

TABLE 2.23: Risk factor for chronic obstructive pulmonary disease (COPD).

Environmental:
➢ Tobacco smoke
➢ Indoor air pollution. Cooking with biomass fuels
➢ Toxic industrial inhalants: Occupational dust exposure (e.g., coal dust, silica, and cadmium)
➢ Respiratory infections: Recurrent infection; HIV infection (associated with emphysema), previous tuberculosis
➢ Low-birth weight and bronchopulmonary dysplasia
➢ Lung growth: Childhood infections or maternal smoking may affect growth of lung during childhood
➢ Low socioeconomic status and antioxidant deficiency
➢ Cannabis smoking
Host factors:
➢ Genetic factors: α1-antiproteinase deficiency TGF β-1 polymorphism, *SERPINE2* gene expression
➢ Airway hyper-reactivity

Pathogenesis (Flowchart 2.5)

Major physiologic change in COPD is airflow limitation. It can result from both small airway obstruction and emphysema.

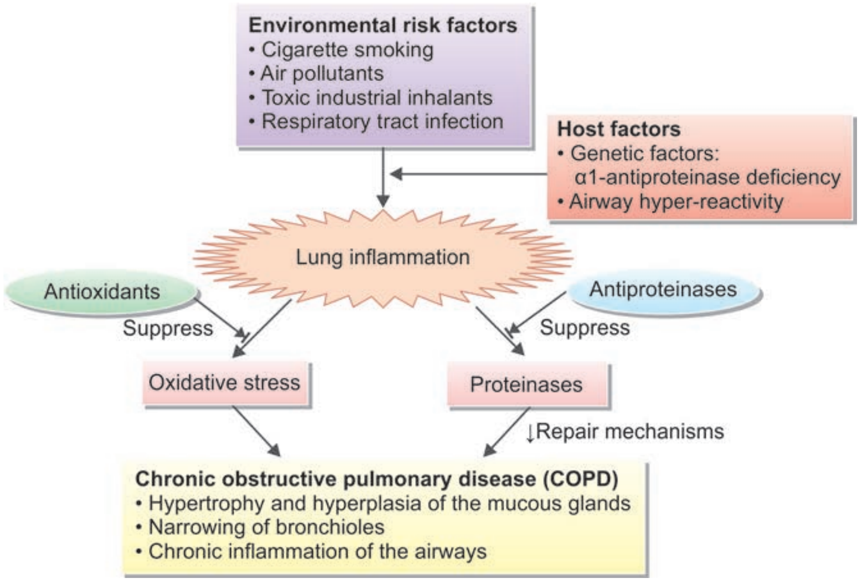
- Irritants cause inflammation → infiltration by CD8 + T-lymphocytes, macrophages, and neutrophils.
- **Hypersecretion of mucus:**
    - **Hyperplasia/hypertrophy of the submucosal glands** in l**arge airways (trachea and bronchi):** Develops as response to inhaled **environmental irritants** and **proteases released from neutrophils** (e.g., elastase and cathepsin). This leads to **hypersecretion of mucus**.
    - **Marked increase of goblet cells in small airways (small bronchi and bronchioles): They produce excessive mucus** → Mucus plugging of bronchial lumen → inflammation and fibrosis of bronchial wall → **leads to airway obstruction.**

Clinical Features

History: Most common three symptoms in COPD are impressive history of cough, sputum production, and exertional dyspnea (breathlessness).

1. **Cough:** Initially, the cough is present only in the winter seasons (often referred as "winter cough" or "smoker's cough"), especially in the mornings ("morning cough"). Later, cough increases in frequency, severity, and duration.
2. **Sputum:** Usually **scanty, mucoid** and more in the mornings. It may be occasionally blood stained (hemoptysis) or frankly purulent ("mucopurulent relapse").
3. **Breathlessness:** It is **relatively insidious in onset** and is due to airflow obstruction. It is aggravated by infection, excessive smoking, and adverse atmospheric conditions. Breathlessness severity can be assessed by the modified MRC dyspnea scale (**Table 2.24**).

FLOWCHART 2.5: Pathogenesis of chronic obstructive pulmonary disease.



Other symptoms: Fever during mucopurulent relapses, wheezing, and tightness in the chest.

Physical Signs

- Patient is usually overweight.
- In the early stages, patients are entirely normal on physical examination. At rest, there is no respiratory distress, respiratory rate is normal and accessory muscles of respiration are not acting.
- **Auscultation:** (i) Vesicular breath sounds with prolonged expiration; (ii) inspiratory and expiratory rhonchi; (iii) crepitations that either disappear or change in location and intensity after coughing; and (iv) forced expiratory time >4 seconds.

Associated comorbidities in COPD (Table 2.25): Although COPD affects the lungs, it is associated with comorbidities probably a part of a generalized systemic inflammatory process.

TABLE 2.24: Modified Medical Research Council (mMRC) scale for dyspnea.

Grade	*Features*
0	No breathless except on strenuous exercise
1	Breathless when hurrying on the level or walking up a slight hill
2	Walks slower than people of same age on the level ground because of breathlessness or has to stop for breath while walking at his own pace on the level ground
3	Stops for breath after walking 100 meters or after a few minutes on the level ground
4	Too breathless to leave the house or gets breathless while dressing or undressing

TABLE 2.25: Common comorbidities in chronic obstructive pulmonary disease (COPD).

Cardiovascular disorders: ➢ Pulmonary hypertension ➢ Right heart failure and cor pulmonale ➢ *Vascular disease:* Coronary artery disease, cerebrovascular disease, and peripheral vascular disease ➢ Systemic hypertension
Nutritional disorders: Cachexia
Musculoskeletal disorders: ➢ Muscle dysfunction ➢ Osteoporosis
Cancer: Lung cancer
Other: Sleep disorders, sexual dysfunction, diabetes, depression, anxiety, anemia, osteoporosis, peptic ulcer, and glaucoma

Investigations

Q. CT2.11 Describe, discuss and interpret pulmonary function tests in chronic obstructive airway disease.

- **Radiological examination:**
  - **Chest X-ray:** May assist in the classification of the type of COPD. There are no reliable radiographic signs that indicate the severity of airflow limitation.
    - **Features of emphysema:** Presence of bullae, paucity of parenchymal markings, or hyperlucency
    - **Features of hyperinflation:** Increased lung volumes and flattening of the diaphragm
    - **Chronic bronchitis:** No characteristic abnormality
    - Essential **to identify complications** such as cardiac failure, other complications of smoking (e.g., lung cancer)
  - **High-resolution CT scans (HRCT):** Useful in detection, characterization, and quantification of emphysema

- **Pulmonary function tests (Table 2.26):** In patients with diffuse obstructive disorders, pulmonary function tests show decreased maximal airflow rates during forced expiration, usually expressed as the forced expiratory volume at 1 second (FEV_1) over the forced ventilatory capacity (FVC).

TABLE 2.26: Pulmonary function tests findings in chronic obstructive pulmonary disease (COPD).

➤ Reduced: FEV_1, FVC, FEV_1: FVC ratio, and PEF
➤ Increased: TLC, RV, RV, and TLC
➤ Gas transfer factor for carbon monoxide (diffusion) is reduced

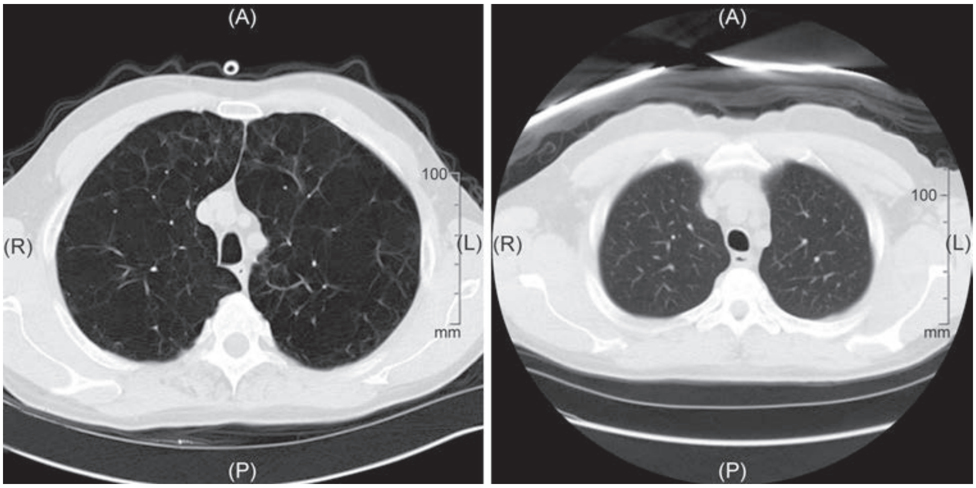


FIG. 2.13: COPD vs normal CT scan.

- **Arterial blood gases (ABG) study:** To demonstrate resting or exertional hypoxemia. It is important in the evaluation of patients with exacerbation and decide need for LTOT.
- **Measurement of lung volumes:** It assesses hyperinflation and is usually performed by using the helium dilution technique.
- **Exercise testing:** Six-minute walk test used to assess exercise tolerance, response to bronchodilator therapy, disability and effectiveness of pulmonary rehabilitation.
- Blood:
  - **Hemoglobin** level and **PCV** may be elevated due to persistent hypoxemia (secondary polycythemia). In patients with normal kidney function, an elevated serum bicarbonate may indirectly identify chronic hypercapnia.
  - **α_1-antiproteinase:** It should be assayed in younger patients with predominantly basal emphysema
- **Electrocardiography:** Often normal. In advanced cases, it may show features of right atrial and ventricular hypertrophy (tall P waves-P-Pulmonary; right bundle branch block; RSR pattern in V_1)

BODE index:
- It is a multidimensional prognostic index. It takes into account several indicators of COPD prognosis **[body mass index (BMI), obstructive ventilatory defect severity, dyspnea severity, and exercise capacity]**.
- The components are derived from measures of the body mass index (weight in kg/height in m^2), FEV_1 percent predicted, the modified Medical Research Council dyspnea, and 6-minute walk test.
- A BODE score greater than 7 is associated with a 30% 2-year mortality.
- A score of 5 to 6 is associated with 15%, 2-year mortality.
- If score is less than 5, the 2-year mortality is less than 10%.

Complications of COPD

Q. Write short note on complications of chronic bronchitis/emphysema.

- **Mucopurulent relapses:** It may develop due to secondary bacterial infection by *Streptococcus pneumoniae, Haemophilus influenzae, or Moraxella catarrhalis.* It presents with fever and increased production of purulent sputum.
- **Carbon dioxide narcosis:** Persistent retention of CO_2 (hypercarbia: high $PaCO_2$) manifests as clouding of consciousness, altered behavior, drowsiness, headache, and papilledema.
- **Respiratory failure:**
  - *Type 1 respiratory failure (low PaO_2 normal $PaCO_2$):* In mild-to-moderate COPD
  - *Type II respiratory failure:* Acute or chronic in severe COPD
- **Secondary polycythemia:** Due to hypoxemia which stimulates erythropoiesis
- **Pulmonary hypertension and right ventricular failure (cor pulmonale)**

- **Pneumonia**
- **Tuberculosis**
- **Lung cancer**
- **Pneumothorax (emphysema)**
- **Deep vein thrombosis**
- **Pulmonary embolism**

Pathogenesis of Complications (Flowchart 2.6)

Management

Q. CT2.16 **Discuss and describe therapies for OAD including bronchodilators, leukotriene inhibitors, mast cell stabilisers, theophylline, inhaled and systemic steroids, oxygen and immunotherapy.**

General measures:

- Regular exercises and management of nutritional status
- Weight loss, if the patient is obese.

Reducing exposure to noxious particles and gases that cause bronchial irritation:

- **Smoking cessation:** Complete smoking cessation and this may be aided by bupropion (a noradrenergic antidepressant) nicotine replacement therapy (by gum, transdermal patch lozenge inhaler, or nasal spray) or varenicline [partial agonist of the nicotinic acetylcholine receptor (nAChR) subtype α-4β_2].
- **Reduce smoke:** Reducing the risk from indoor and outdoor air pollution. Reduce exposure to smoke from biomass fuel, particularly among women and children.
- **Avoid:** Dusty and smoke-laden atmospheres.

Drug therapy (Table 2.27)

Used both for short-term management of exacerbations and for long-term relief of symptoms. However, none of the medications for COPD reduce the rate of decline of lung functions.

- **Bronchodilators:** They are central to the management of breathlessness.
  - **β_2-Adrenergic agonists:** The inhaled route is preferred.
    - *Mild disease:* Short-acting agents namely salbutamol 200 µg or terbutaline 500 µg 6 hourly
    - *Moderate-to-severe disease:* Long-acting agents such as salmeterol 50 µg twice daily or formoterol (12-µg powder inhaled twice daily) or indacaterol (150–300 µg daily) achieve bronchodilation and also reduce the incidence of infective exacerbations. LABAs include salmeterol, formoterol, arformoterol, indacaterol, vilanterol, and olodaterol; all are beta-2 selective.
  - **Antimuscarinic (anticholinergic) drugs:** More prolonged and greater bronchodilatation is achieved by adding ipratropium bromide (40–80 µg 6 hourly) or tiotropium bromide (18 µg once a day) or oxitropium (200 µg twice daily) in severe disease.
  - Oral long-acting theophylline or doxophylline may be beneficial in selected cases. Umeclidinium, aclidinium, and glycopyrronium are newer long-acting anticholinergic drugs available.

FLOWCHART 2.6: Pathogenesis of complications of chronic bronchitis.

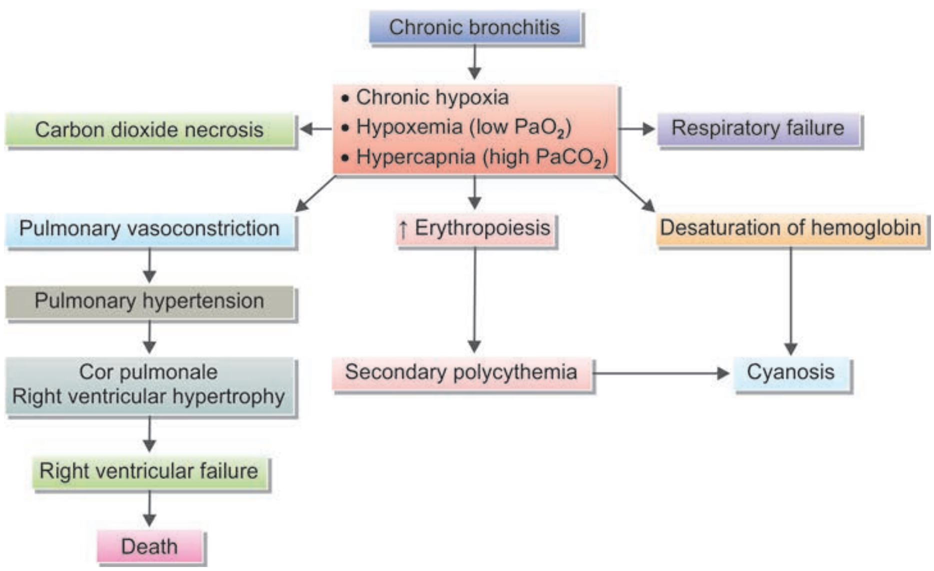


TABLE 2.27: Drug therapy in chronic bronchitis.

➢ **Beta2-agonists:** – Short-acting beta2-agonists (SABA) – Long-acting beta2-agonists (LABA)	➢ **Methylxanthines** ➢ **Inhaled corticosteroids (ICS)** ➢ **Combination long-acting beta2-agonists + corticosteroids in one inhaler**
➢ **Anticholinergics/muscarinic antagonists:** – Short-acting anticholinergics (SAMA) – Long-acting anticholinergics (LAMA)	➢ **Systemic corticosteroids** ➢ **Phosphodiesterase-4 inhibitors**
➢ **Combination short-acting beta2-agonists + anticholinergic in one inhaler**	➢ **Acebrophylline:** An airway mucoregulator and anti-inflammatory agent

- **Phosphodiesterase type 4 inhibitors:** Roflumilast is an inhibitor with anti-inflammatory properties. It may be used as an adjunct to bronchodilators. Weight loss is a significant side effect.
- **Corticosteroids:**
  - Inhaled corticosteroids (ICS) reduce the frequency and severity of exacerbations, and are used in moderately severe COPD. These include beclomethasone, budesonide, fluticasone, ciclesonide, flunisolide, and beclomethasone.
  - Oral corticosteroids are useful during exacerbations and should be avoided as a maintenance therapy because it may lead to osteoporosis and impaired skeletal muscle function.

Respiratory infections:
- **Treatment of infection:** Bacterial infection precipitates exacerbations. Azithromycin (has both anti-inflammatory and antimicrobial properties) administered daily to subjects with a history of exacerbation in the past 6 months may reduce the exacerbation. If patient develops purulent (yellow or green) sputum, oral tetracycline or ampicillin 250 mg 6 hourly or cotrimoxazole 960 mg 12 hourly for 10 days should be given. If there is no response, sputum culture and sensitivity is done and the antibiotic is changed accordingly.
- **Prevention of infection:** Patients with COPD should receive **vaccination with polyvalent pneumococcal and influenza vaccines**.

Symptomatic measures:
- **Antimucolytic agents:** They reduce viscosity of sputum and can reduce the number of acute exacerbations and total number of days of disability. Mucolytic agents including bromhexine, N-acetylcysteine carbocisteine, ambroxol, and erdosteine can be tried.
- **Antitussives:** Regular use of antitussives to control cough in stable COPD is not recommended.
- Chest physiotherapy

Pulmonary rehabilitation:
- It is an individually designed treatment program consisting of education and cardiovascular conditioning (reverse muscular and cardiovascular dysfunction).
- Program includes breathing technique, chest physiotherapy, postural drainage, activities of daily living (work simplification, energy conservation), and exercise conditioning (upper and lower extremity).

Oxygen Therapy

Q. CT2.20 **Describe and discuss the principles and use of oxygen therapy in the hospital and at home.**

Long-term domiciliary oxygen therapy (LTOT)
- **Aim of therapy:** To increase the PaO_2 to at least 8 kPa (60 mm Hg) or SaO_2 to at least 90%. It is administered through nasal cannulae for at least 15 hours per day at a low dose (2 L/min)
- **Indications:** In COPD patient with exertional hypoxemia or nocturnal hypoxemia
- Daytime PaO_2 ≤55 mm Hg at rest or oxygen saturation ≤ 88% with or without hypercapnia during a period of clinical stability OR
- Daytime PaO_2 between 56 and 59 mmHg or oxygen saturation > 88% in the presence of secondary polycythemia, nocturnal hypoxemia, peripheral edema, or evidence of pulmonary hypertension
- **Benefits:** LTOT has significant benefits and reduces mortality rates in selected patients. It decreases pulmonary hypertension and prolongs life in hypoxemic COPD patients with right heart failure. It also reduces polycythemia, pulmonary artery pressures, dyspnea, and hypoxemia during sleep and reduces nocturnal arrhythmias.

Pharmacologic therapy in COPD depending on severity (GOLD 2020) is presented in **Figure 2.14.**

Treatment of pulmonary hypertension
By long-term oxygen therapy, sildenafil, bosentan, and synthetic prostacyclin (epoprostenol)

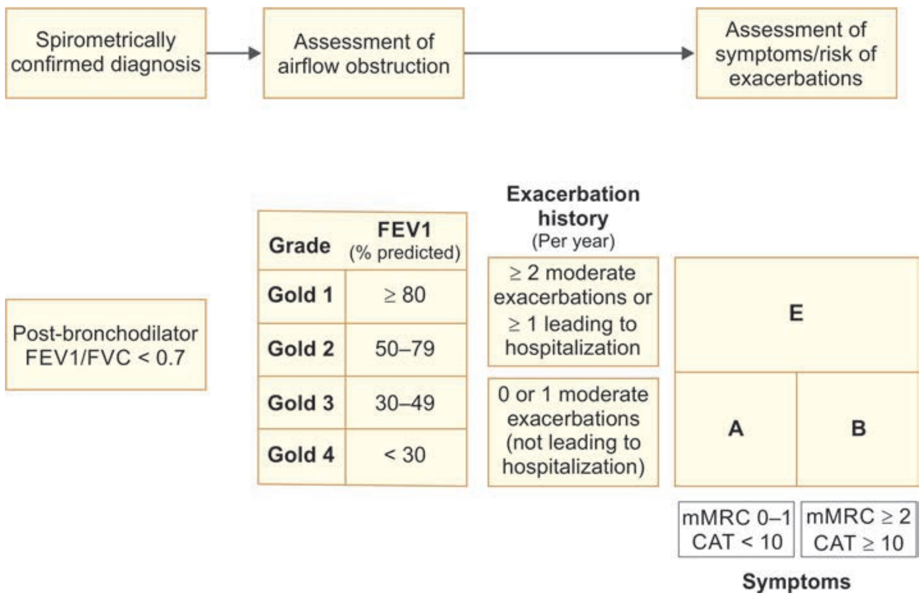


FIG. 2.14: Pharmacotherapy based on severity of chronic obstructive pulmonary disease (COPD).

Surgical treatments

Lung volume reduction surgery (LVRS) is more efficacious than medical therapy among patients with upper-lobe predominant emphysema and low-exercise capacity.

In appropriately selected patients with very severe COPD, **lung transplantation** has been shown to improve quality of life and functional capacity.

Criteria and classification of acute COPD exacerbation are given in **Box 2.6.**

Acute Exacerbations of COPD

Q. CT2.19 Develop a management plan for acute exacerbations including bronchodilators, systemic steroids, antimicrobial therapy.

- **Definition:** "An event in the natural course of COPD characterized by a change in patient's baseline dyspnea, cough, and/or sputum that is beyond normal day-to-day variations. It is acute in onset and may warrant a change in regular medication in a patient with underlying COPD."
- **Causes of exacerbation:** (1) Infection of the tracheobronchial tree and (2) air pollution. In about one-third of cases, no cause can be identified.
- **Trigging factors:** Infections by bacteria, viruses, or a change in air quality.

Box 2.6: Criteria and classification of acute COPD exacerbation.

Major criteria
- Increase in sputum volume
- Increase in sputum (generally yellow and green)
- Worsening of baseline dyspnea

Additional criteria
- Upper respiratory infection in past 5 days
- Fever of no apparent cause
- Increase in wheezing and cough
- Increase in respiration rate or heart rate 20% above baseline
- Various nonspecific signs and symptoms may accompany these findings, such as fatigue, insomnia, depression, and confusion

Degree of exacerbation
- Mild exacerbation = 1 major criterion + 1 or more additional criteria
- Moderate exacerbation = 2 major criteria
- Severe exacerbation = all 3 major criteria

Treatment of Severe Acute Exacerbations

Oxygen
- Adequate oxygenation (i.e., to achieve an oxygen saturation of 88–92%) must be assured.
- **Method of administration:** By nasal catheter or through a face mask equipped to control the inspired oxygen fraction. Venturi masks are preferred because they permit a precise fraction of inspired oxygen (FiO_2).

Bronchodilators

Nebulized short-acting β2-agonists (salbutamol 2.5 mg every 20 minutes for initial 1–2 hours) and/or anticholinergic agent (ipratropium bromide 0.5 mg) should be given. Intravenous aminophylline may be added, if the patient fails to respond to the above treatment.

Antibiotics

- **Indications:** Patients with: (1) increase in sputum purulence, sputum volume, or (2) breathlessness (dyspnea), or (3) those requiring mechanical ventilation.
- **Most common organisms during exacerbations:** S. *pneumoniae*, *H. influenzae,* and *M. catarrhalis.* Risk factors for *P. aeruginosa* infection include recent hospitalization, frequent antibiotics use and severe exacerbation.
- **Antibiotics used:**
  - **Outpatient:** Doxycycline, cotrimoxazole, or amoxicillin–clavulanate can be given. Patients older than 65 years are treated with newer fluoroquinolones (levofloxacin, gemifloxacin, and moxifloxacin).
  - **Hospitalized patients:** Intravenous antibiotics (azithromycin or fluoroquinolone or a third-generation cephalosporin-like ceftriaxone or cefotaxime)
  - **Severe exacerbations:** Third-generation cephalosporin plus a fluoroquinolone or an aminoglycoside

Corticosteroids

Intravenous or oral corticosteroids shorten the recovery time and improve lung functions (FEV_1) and hypoxemia.

Diuretics

These are given to patients with gross right ventricular failure.

Respiratory stimulants

These may be used when there is no response to the conventional agents. Doxapram in the dose of 1.5–4 mg/minute as infusion is the most often used agent. Other respiratory stimulants are almitrine, nikethamide, medroxyprogesterone, and acetazolamide.

Mechanical Ventilatory Support

- **Noninvasive positive airway pressure ventilation (NIPPV)**
  - NIPPV is by using tight-fitting facemask to deliver BiPAP.
- **Invasive (conventional) ventilation**
  - It is administered via an endotracheal tube.

Management of associated co-morbidities

It is necessary to manage co-morbidities because they are responsible for mortality and hospitalization.

GOLD staging for severity of COPD and management (Fig. 2.15)

It is used for both chronic bronchitis and emphysema.

GOLD Stage	I. Mild	II. Moderate	III. Severe	IV. Very severe
Spirometry findings	• $FEV_1/FVC < 0.70$ • FEV_1 >80% of predicted	• $FEV_1/FVC < 0.70$ • FEV_1 50–79% of predicted	- $FEV_1/FVC < 0.70$ - FEV_1 30 to < 49% of predicted	$FEV_1/FVC < 0.70$ FEV_1 30% of predicted or FEV_1 < 50% of predicted, if respiratory failure present
Management	Short-acting β_2-agonists (salbutamol, terbutaline, levosalbutamol) as needed	Long-acting β_2-agonists [salmeterol, formoterol ± anticholinergics (ipratropium, tiotropium]	Add inhaled glucocorticoids (beclomethasone, budesonide, fluticasone, ciclesonide) ± methylxanthines	Add oxygen therapy, ventilator assistance, management of right-sided heart failure, surgical options
	Avoid risk factors, influenza vaccination and use short-acting β_2-agonists to all stages →			

FIG. 2.15: GOLD staging for severity of chronic obstructive pulmonary disease (COPD) and management.

EMPHYSEMA

The word "emphysema" literally means inflation or distension with air. Emphysema can be classified depending on the organ or structure involved as:

- **Pulmonary emphysema**
- **Compensatory emphysema/hyperinflation:**
  - It is characterized by **dilation of alveoli without destruction of septal walls**. It develops as a compensatory response to loss of extensive lung substance elsewhere (e.g., removal of a diseased lung or lobe).

- **Mediastinal emphysema (Fig. 2.16):**
  - It occurs as a result of entry of air rapidly into the mediastinum following rupture of overdistended alveoli as in severe bronchial asthma, rupture of emphysematous bulla (during coughing) and rupture of esophagus.
  - Severe mediastinal emphysema can lead to cardiac tamponade.
  - *Auscultation:* It may reveal a crunching sound (mediastinal crunch).
- **Subcutaneous emphysema (Fig. 2.16):**
  - It is characterized by the **entry of air into** subcutaneous tissue.
  - *Causes:* Penetrating chest injuries, fracture of ribs of intercostals tube introduction.
  - On palpation, it produces a characteristic crepitation or crackling sensation.
  - *Treatment:* Not needed because the air will get absorbed slowly. In severe cases, subcutaneous incisions may be necessary to relieve pressure.

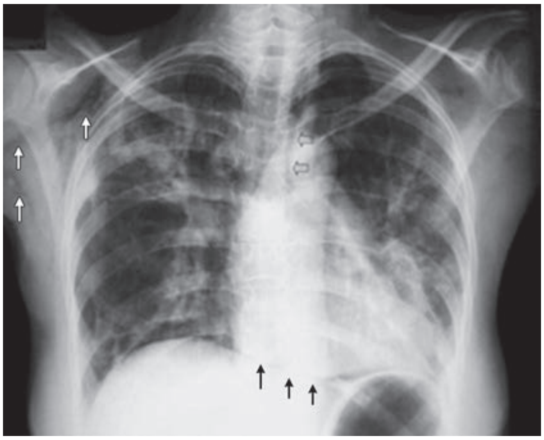


FIG. 2.16: Chest X-ray shows subcutaneous (white arrows) and mediastinal emphysema (black arrows).

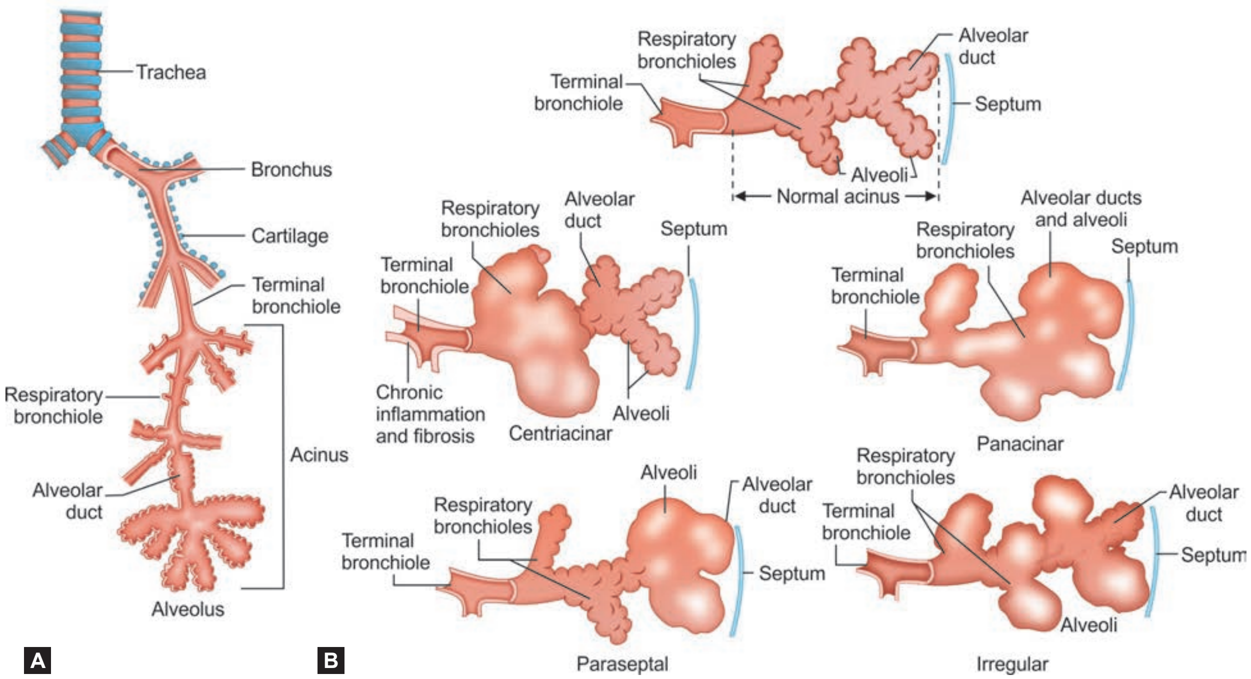
Q. Discuss the etiology, pathology, clinical features, investigations, complications, and management of pulmonary emphysema.

Definition: Emphysema (pulmonary) is a **chronic lung disease** characterized by abnormal **irreversible** (permanent) **dilatation of the airspaces distal to the terminal bronchiole**. This is associated with destruction of their walls but without obvious fibrosis.

Types of Emphysema/Classification (Figs. 2.17A and B)

Emphysema is **classified according to its anatomic distribution** (location of the lesions) within the lobule into **four major types:** (1) centriacinar, (2) panacinar, (3) paraseptal, and (4) irregular.

1. **Centriacinar (centrilobular) emphysema:**
   - Dilatation **involves the central or proximal parts of the acini** (formed by respiratory bronchioles), whereas distal alveoli are spared.
   - Common and severe in the **upper lobes**, especially in the **apical segments**



FIGS. 2.17A AND B: (A) Normal components of respiratory tree; (B) Types of emphysema.

- **Association:** It occurs in **heavy smokers** and in association with **chronic bronchitis** and **coal workers' pneumoconiosis**.

2. **Panacinar (panlobular) emphysema:**
   - All the airspaces beyond terminal bronchiole are more or less uniformly/equally dilated.
   - **Site:** More common in the lower lobes, and is usually most severe at the bases.
   - It is associated with **α_1-antitrypsin (α_1-AT) deficiency**
3. **Distal acinar (paraseptal) emphysema:**
   - Dilatation affects the distal airspace at the periphery of the lobule and the proximal portion is normal.
   - It is found near the pleura. **Dilated spaces of more than 1 cm in size** are known as **bullae,** which may rupture and cause spontaneous pneumothorax.
   - It occurs **adjacent to areas of fibrosis, scarring, or atelectasis**.
4. **Irregular (scar or cicatricial) emphysema:**
   - Acinus is irregularly involved and may be asymptomatic.
   - It is most common form of emphysema.
   - **It occurs near the scar** and is commonly found **around old healed inflammatory process such as tuberculous scars.**

Etiology and Pathogenesis (Fig. 2.18)

The **major event** in emphysema is **destruction of alveolar wall**.

- **Mechanism that checks the destruction of alveolar wall:** These include: (1) **Anti-elastases** (e.g., α_1-antitrypsin) and (2) **antioxidants**. If these two mechanisms are defective → results in (1) **protease–antiprotease imbalance** (e.g., α_1-antitrypsin deficiency) and (2) **imbalance between oxidants and antioxidants**.
  - **Unchecked inflammation and proteolysis:** It develops due to deficiency of the above protective mechanism.
- **Genetic factors**

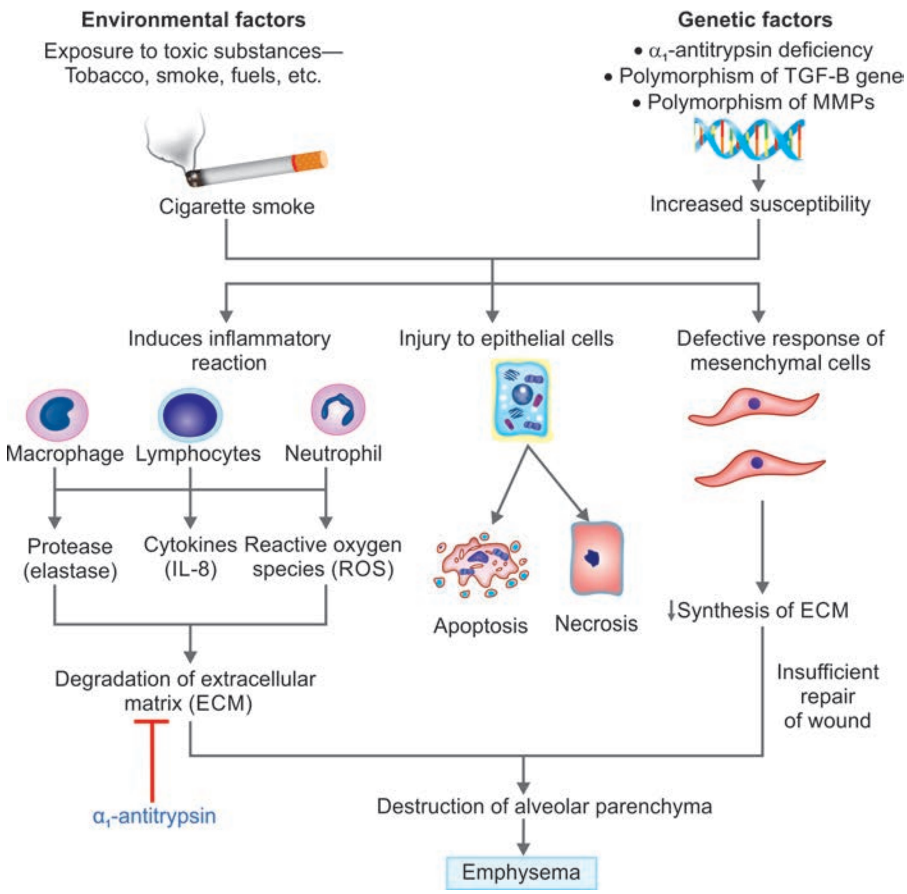


FIG. 2.18: Pathogenesis of emphysema. Exposure to environmental toxins (e.g., cigarette smoke) causes inflammatory reaction, cell death, and proteolysis of extracellular matrix (ECM). α1-antitrypsin (α1-AT) deficiency also results in increased degradation of ECM.
(IL-8: Interleukin-8; TNF: Tumor necrosis factor; TGF: Transforming growth factor)

Q. CT2.5 **Describe and discuss the genetics of alpha 1-antitrypsin deficiency in emphysema.**

Deficiency of α1-antitrypsin: It is inherited as **autosomal recessive**, which exhibits polymorphism → tendency to develop emphysema. α**1-antitrypsin is a major inhibitor of proteases** (particularly elastase). It is produced in hepatocytes by SERPINA1 gene expression. It is normally present in serum, tissue fluids, and macrophages and a balance is maintained between protease and antiproteases. During inflammation, protease (proteolytic enzyme) is secreted by neutrophils and digests the connective tissue of the lung. α_1-antitrypsin is a protease inhibitor (antiprotease), preventing this proteolytic digestion. Hence, a deficiency or absence of α_1-antitrypsin results in the proteolytic destruction of lung. These patients develop severe panacinar emphysema (LL >UL).

The alleles are designated M, Z, S with prefix of Pi (protease inhibitor). M allele is normally expressed (PiMM). S allele confers partially functional A1-AT protein. Z allele confers absent expression. PiZZ > PiSZ > PiMZ >PiMM in terms of severity of disease. Patients can develop liver cirrhosis in later stages. A1-AT replacement therapy has been tried with partial response. Lung and liver transplant is reserved for end stage disease.

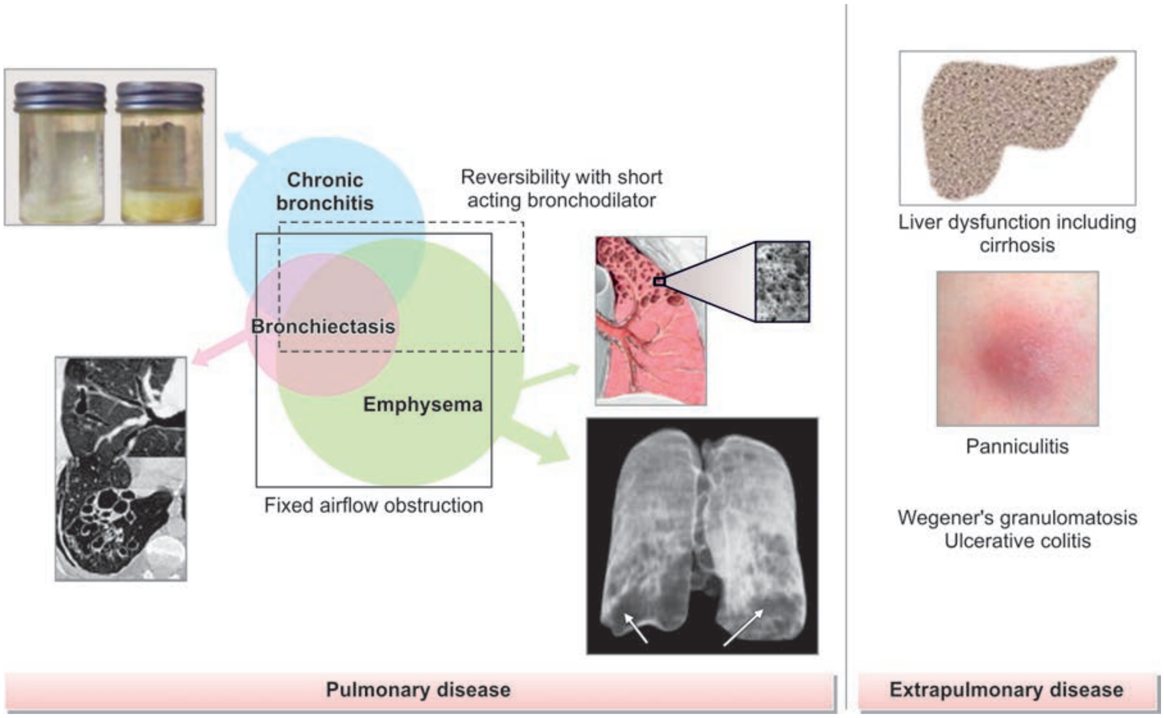


FIG. 2.19: Pathology and clinical diseases in alpha 1-antitrypsin deficiency.

Clinical Features of Emphysema

Manifestations appear late, until, at least one-third of the functioning pulmonary parenchyma is damaged.

- **Dyspnea** is the most striking feature that begins insidiously and steadily progresses, ultimately ending in breathlessness on trivial exertion and even at rest.
- **Cough and expectoration** of scanty mucoid sputum.
- **Weight loss,** weakness, anorexia, and lethargy are common with advanced disease.

Physical Findings

General: Body build is asthenic, short, and thick neck; neck veins may appear distend during expiration and collapse during inspiration. Patient leans forward, extending the arms to brace himself during sitting posture.

Respiratory

- **Inspection:**
  - Patient appears **distressed** and **tachypneic**, **hypertrophy of accessory muscles of respiration** (sternomastoid and scalene muscles), length of the trachea above the suprasternal notch is reduced, and apical impulse is invisible or feeble.
  - **During inspiration: Exaggerated tracheal descent** (Campbell's sign), excavation of the suprasternal and supraclavicular fossae, and **indrawing of the costal margins.**

- **Expiration: Prolonged through pursed lips** (purse-lip breathing) and beginning of expiration with a **grunting sound**.
- **Chest:** Cylindrical or barrel like (**barrel-shaped chest**), anteroposterior diameter of the chest is markedly increased. Whole chest is in a fixed state of full inspiration. Ribs are placed more horizontally and widely. Chest expansion diminished symmetrically. Thoracic kyphosis is exaggerated and the subcostal angle is widened.
- **Dahl sign:** Above the knee, patches of hyperpigmentation or bruising are caused by constant "tenting" position of hands or elbows.
- **Hoover's sign:** Briefly, during inspiration, a paradoxical medial movement of the chest. The "subcostal angle" is the angle between the xiphoid process and the right or left costal margin. Normally during inhalation, the chest expands laterally, increasing this angle. When the diaphragms are flattened (as in COPD), inhalation paradoxically causes the angle to decrease.
- **Harrison's sulcus:** This is a horizontal grove where the diaphragm attaches to the ribs; it is associated with chronic asthma, COPD, and rickets.

- **Percussion:** Hyper-resonant percussion note over the lungs, reduced cardiac dullness, and liver dullness are pushed down or absent. Tidal percussion is negative.
- **Auscultation:** Diminished intensity of the breath sound; breath sounds are vesicular with prolonged expiration. Scattered, faint, high-pitched, and end-expiratory rhonchi may be audible.

FIG. 2.20: Chest X-ray of emphysema. Posteroanterior (PA) view showing low set, flat diaphragm, translucent lung field, long and narrow heart ("tubular heart"), loss of peripheral vascular markings, and prominent pulmonary artery shadows at the hilum.

Investigations

- **Chest X-ray (Fig. 2.20):**
  - **PA view:** Features include low set, flat diaphragm, translucent lung field, long and narrow heart ("tubular heart"), loss of peripheral vascular markings, prominent pulmonary artery shadows at the hilum and bullae.
  - **Lateral view:** Large retrosternal translucency
- **Computed tomography:** Lower lobe affected predominantly
- **Pulmonary function tests (Table 2.28)**
- **Arterial blood gas studies:** Slightly reduced PaO_2 and normal or mildly elevated $PaCO_2$.

Complications

Emphysema progresses steadily and gradually.

1. **Pulmonary bullae:**
   - They represent inflated thin-walled spaces produced due to the rupture of alveolar walls.
   - May be single or multiple, small or large, and resembles an amulet
   - Usually located in the subpleural region along the anterior borders of lungs
   - *Complications:* A subpleural bulla may rupture producing spontaneous pneumothorax. Large bullae can interfere with pulmonary ventilation.
2. **Respiratory failure:** Type 1 and type II respiratory failure can occur.
3. **Pulmonary hypertension and right heart failure** (**cor pulmonale**)**:** These are late complications and right ventricular failure in emphysema is usually a terminal event.
4. **Severe weight loss:** Leading to emaciation.

TABLE 2.28: Pulmonary function test findings in emphysema.

Reduced: FEV_1, FVC, FEV_1: FVC ratio, and PEF	**Increased:** TLC, RV, and RV:TLC
GAS transfer factor for carbon monoxide (diffusion) is reduced	

Treatment: Treatment similar as described for COPD

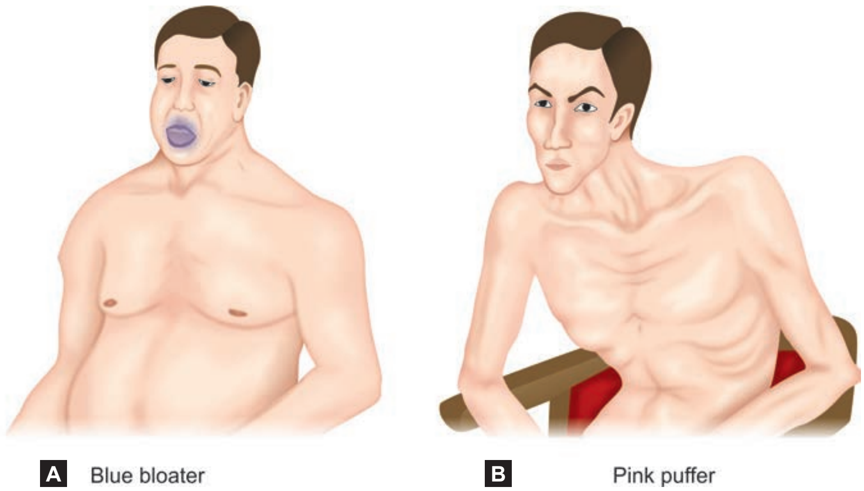
- No specific treatment for established case of emphysema. Bronchodilators and steroids may be helpful in few patients.
- **Prevention of progression:** Cessation of smoking and avoidance of occupational exposure
- **Treatment of aggravating factors and complications:** Treatment of infections, respiratory failure, and right heart failure
- **Physiotherapy**
- **Surgical therapy: Ablation of giant bullae,** lung volume reduction surgery reduced hyperinflation of one or both lungs and/or laser resection
- **Heart and lung transplantation:** In young patients with severe emphysema due to α_1-antitrypsin deficiency

Blue Bloaters (Fig. 2.21A)

Q. What are blue bloaters?

It is a distinctive clinical pattern seen in chronic bronchitis, the characteristics of which are:

- Marked/heavy cyanosis ("blue") and peripheral edema ("bloated") and secondary polycythemia
- Current evidence demonstrates that most patients have elements of both bronchitis and emphysema and by physical examination cannot reliably differentiate "blue bloaters" from "pink puffers".



FIGS. 2.21A AND B: (A) Blue bloater (in chronic bronchitis); (B) Pink puffer (in emphysema).

Pink Puffers (Fig. 2.21B)

Q. What are pink puffers?

- It is a distinctive clinical pattern seen in emphysema of lung.
- Patients are thin and noncyanotic at rest (hence "pink").
- They have marked dyspnea ("puffer") and have prominent use of accessory muscle. They develop steadily progressive dyspnea.

Differences between Emphysema and Chronic Bronchitis (Table 2.29)

Q. What are the differentiating features of emphysema and chronic bronchitis?

TABLE 2.29: Differences between emphysema and chronic bronchitis.

Feature	*Emphysema*	*Chronic bronchitis*
Clinical features		
Dyspnea	Severe	Mild to moderate
Cough	Develops after dyspnea starts	Frequent, develops before dyspnea starts
Sputum—amount and nature	Scanty and mucoid	Copious and purulent
Frequency of mucopurulent relapses	Less	More
Cyanosis	Absent	Present
Pulmonary hypertension	Late and mild	Early and severe
Right ventricular failure and respiratory failure	Late and often terminal	Repeated episodes
Mechanism of airway obstruction	Loss of elastic recoil	Decreased airway lumen due to mucus and inflammation
Investigations		
Hematocrit (PCV)	Normal	Increased
PaO_2	Normal to low "pink puffer"	Low "blue bloater"
$PaCO_2$	Normal mildly increased	High (>40)
FEV_1	Decreased	Decreased
Diffusing capacity	Reduced	Normal
Chest X-ray	Features of hyperinflation, bullae, and tubular heart	Increased bronchovascular markings and cardiomegaly
Elastic recoil	Decreased	Normal

Contd...

Contd...

Feature	***Emphysema***	***Chronic bronchitis***
Airway resistance	Normal to slightly increased	Increased
Cor pulmonale	Late and mild	Early and marked
Prognosis	Good	Poor

(FEV_1: forced expiratory volume in the first second; $PaCO_2$: partial pressure of carbon dioxide; PaO_2: partial pressure of oxygen)

Differentiating Features of Asthma and Chronic Obstructive Pulmonary Disease (Table 2.30)

Q. What are the differentiating features of asthma and chronic obstructive pulmonary disease?

TABLE 2.30: Differentiating features of asthma and chronic obstructive pulmonary disease (COPD).

Characteristics	***Bronchial asthma***	***COPD***
Age of onset	Usually children and young adults	Usually older individuals
Risk factors	Family history of allergy, exposure to allergens, and occupational sensitizers	Smoking, atmospheric pollution, occupational exposure, and α1-antitrypsin deficiency
Respiratory symptoms		
Main symptoms	Wheezing, cough, and dyspnea	Chronic dyspnea and productive cough
Nature of symptoms	Vary from time to time and even over hours and days	Usually continuous symptoms
Triggers	Exercise, dust, or exposure to allergens	Unrelated to triggers
Recovery of symptoms	Symptoms improve spontaneously or with treatment	Slowly progressive despite therapy
Comorbidities	Generally absent	Often present (cardiovascular diseases, metabolic syndrome, depression, osteoporosis, and muscle wasting)
Chest X-ray	Normal	Hyperinflation
Spirometry	Reversibility of airway obstruction and normal between symptoms	$FEV_1/FVC<0.7$ and persistent airflow limitation

(FEV_1: Forced expiratory volume in the first second; FVC: Forced vital capacity)

CIGARETTE SMOKING

Q. Write short note/essay on the various components of cigarette smoke and list the diseases caused by smoking.

Q. CT2.21 Describe discuss, and counsel patients appropriately on smoking cessation.

Cigarette smoke is a complex aerosol containing gaseous and particulate compounds.

Components of cigarette smoke: It consists of mainstream smoke and sidestream smoke.

1. *Mainstream smoke*: It is produced by inhalation of air through cigarette. It is the primary source of smoke exposure in smokers.
2. *Sidestream smoke:* It is produced from emitting of smoke between cigarette puffs and is the main source of environmental smoke or second hand smoke.

Chemical constituents of cigarette smoke: It contains about 2,000–4,000 chemical substances and more than 60 carcinogens. About 95% of the weight of the mainstream smoke is derived from 400 gaseous compounds and about 5% of the weight is made up of about 3,500 particulate components.

Tobacco addiction is due to nicotine present in cigarette smoke and is the total particulate matter responsible for carcinogenesis.

- **Carcinogens:** Tar, polycyclic aromatic hydrocarbons, benzo[a]pyrene, and nitrosamine
- **Others:**
  - Nicotine causes ganglionic stimulation and depression; tumor promotion
  - Phenol causes tumor promotion; mucosal irritation
  - Carbon monoxide impairs oxygen transport and utilization
  - Formaldehyde produces toxicity to cilia; mucosal irritation
  - Nitrogen oxides produces toxicity to cilia; mucosal irritation

Diseases caused by smoking are listed in **Table 2.31**.

TABLE 2.31: Diseases caused by smoking.

System	***Diseases produced***
General	Cancers of lung, oropharynx, esophagus, stomach, pancreas, bladder, kidney, cervix, colon, and acute myeloid leukemia
Respiratory	COPD, chronic cough, and infections
Cardiovascular	Coronary artery disease, cerebrovascular disease, peripheral artery disease, and abdominal aortic aneurysm
Reproductive	Miscarriage, prematurity, low-birth weight, ectopic pregnancy, SIDS (sudden infant death syndrome)
Gastrointestinal	Gastro-esophageal acid reflux, peptic ulcer, and Crohn's diseases
Others	Poor oral and skin health, cataract, fire-related injuries, osteoporosis, and macular degeneration

(COPD: Chronic obstructive pulmonary disease)

Strategies for Smoking Cessation

Nicotine Replacement Therapy (NRT)

LONG ACTING NRT	SHORT ACTING NRTs. Combine with nicotine patch for best effect		
Patch Wear for 24 hours at a time. Alternate sites to minimize skin irritation **Dose:** If <10 cig/d, start with 14 mg/d If >10 cig/d, start with 21 mg/d Taper down with a regimen that is easiest for the patient **Side effects:** Vivid dreams, contact dermatitis	**Mini Lozenge** Allow to slowly dissolve. Do not chew or swallow lozenge **Dose:** 2 mg/hr for patients who smoke their first cig >30 mins after awakening 4 mg/hr for patients who smoke their first cig <30 mins after awakening **Frequency:** Every 1–2 hours. Try not to wait for cravings	**Gum** Chew minimally, "park" between teeth and cheek. Rechew when tingling is gone and rotate sites. Avoid use with acidic foods **Dose:** 2 mg/hr for patients who smoke their 1st cig >30 mins after awakening 4 mg/hr for patients who smoke their 1st cig <30 mins after awakening **Frequency:** Every 1–2 hours. Try not to wait for cravings **Side effects:** GI irritation when chewed	**Nasal spray** Can be used once to each nostril every 1–2 hours **Side effects:** Can cause nostril irritation **Inhaler** Take short puffs but keep air in mouth Do not inhale

Non-nicotine Replacement Therapy		Electronic Cigarettes
Varenicline (Chantix) Nicotinic receptor partial agonist designed to decrease cravings, reduce withdrawal, and dampen nicotine-induced reward pathway **Dose:** Take one week before quit date 0.5 mg daily × 3d, 0.5 mg BID × 3d Then 1 mg daily for 3–6 months **Side effects:** Nausea, vivid dreams	**Bupropion (Wellbutrin)** Nicotinic receptor antagonist and norepinephrine and dopamine reuptake inhibitor. Designed to reduce cravings and withdrawal. **Dose:** Take one week before quit date 150 mg tablet once daily × 3 days. Then 150 mg tablet twice daily for 3–6 months **Side effects:** Lowers seizure threshold, insomnia, dry mouth	Reserve use until after FDA-approved treatments have failed Patients should avoid using traditional cigarettes with e-cigs (to minimize adding known harms to unknown harms) Before prescribing e-cigs, discuss a plan for duration of use and when to stop

FIG. 2.22: Strategies for smoking cessation.

Asthma COPD Overlap Syndrome (ACOS)

Major Criteria for ACOS:

- It is characterized by persistent airflow limitation with several features usually associated with asthma and several features usually associated with COPD.
- History or evidence of atopy (e.g., hay fever and elevated total IgE)
- Age 40 years or more
- Smoking >10 pack-years, post-bronchodilator FEV_1 <80% predicted and FEV_1/FVC <70%
- A ≥15% increase in FEV_1 or ≥12% and ≥200 mL increase in FEV_1 post- bronchodilator treatment with albuterol would be a minor criterion.

PULMONARY TUBERCULOSIS

Mycobacteria

Classification

Q. Write short note on classification of mycobacteria and give an account of disease produced by them.

Mycobacteria are classified into three groups:

1. ***Mycobacterium tuberculosis* complex** (*M. tuberculosis, M. bovis, and M. africanum*)
2. ***Mycobacterium leprae***
3. **Atypical mycobacteria** or **nontuberculous mycobacteria** (NTM) or **mycobacteria other than tuberculosis (MOTT)**

Q. Write short note on mycobacteria other than tuberculosis (MOTT)/nontuberculous mycobacteria (NTM).

- MOTT or NTM are ubiquitous in the environment. They occur in soil and water and are **not usually pathogenic** due to their lack of virulence. Therefore, their isolation from a site that is not normally sterile (e.g., sputum, skin, or urine) does not constitute proof of disease. Groups of atypical *Mycobacterium* are listed in **Table 2.32**.

- **Patients with NTM lung disease** often **have predisposing disease of the lung** (e.g., COPD, bronchiectasis, cystic fibrosis, pneumoconiosis, etc.).

- ***Mycobacterium avium* intracellulare:**
  - Also known as MAC (*Mycobacterium avium* complex)
  - Most common nontuberculous mycobacterial infection associated with AIDS
  - Symptoms include fever, swollen lymph nodes, diarrhea, fatigue, weight loss, and shortness of breath.
  - May develop into pulmonary MAC
- ***Mycobacterium marinum*** causes infections of skin and swimming pool granuloma.
- ***Mycobacterium ulcerans*** cause skin infections.
- ***Mycobacterium kansasii*** causes lung disease.

TABLE 2.32: Groups of atypical *Mycobacterium*.

Slow growing:
- **Group I** photochromogens (P): Pigment producers in the presence of light, e.g., *Mycobacterium kansasii, M. marinum, and Mycobacterium simiae*
- **Group II** scotochromogens (S): Pigment producers in the absence of light, e.g., *Mycobacterium scrofulaceum, Mycobacterium szulgai, and Mycobacterium gordonae*
- **Group III** non-photochromogens (N): Do not produce any pigment, e.g., *Mycobacterium malmoense, Mycobacterium xenopi, and M. avium* intracellulare

Fast growing:
- **Group IV** fast growers (3–5 days), e.g. *Mycobacterium fortuitum, Mycobacterium chelonae*, and *Mycobacterium abscessus*

ATS diagnostic criteria
1. **Clinical:** symptoms of cough, fatigue, weight loss
2. **Radiological:** CXR shows cavity, nodules or CT showing bronchiectasis with nodules
3. **Microbiology:** *Single* positive culture from *BAL* or ≥2 positive sputum cultures or positive lung biopsy

Combination of All three is required for a diagnosis of NTM disease
Positive AZ stain with negative CBNAAT is clue towards a NTM disease.

Treatment Considerations

- **Patient related:** Symptomatic patients, radiographic abnormalities (cavity indicates destruction).
- **Organism:** More than 200 species exist but not all require treatment. Organisms like *M. szulgai, M. celatum, M. kansasii, M. abscessus, M. avium* are associated with disease.
- **Goals of therapy:** Complete cure is aimed in organism like *M. kansasii*, MAC. Bacteriologic conversion is aimed in some organisms.
- **Species of NTM** helps guide the duration and goals of therapy.
- Antimicrobial susceptibility testing against **amikacin, macrolides (azithromycin/clarithromycin)** is mandatory before starting therapy since presence of *macrolide resistance changes therapy.*

M. abscessus group includes (*abscessus, massiliense, chelonae*). Abscessus subspecies contain an inducible erm gene that can induce macrolide resistance despite initial in-vitro sensitivity. Hence crucial to test for inducible resistance. Masiliense subspecies contains truncated erm gene thereby producing favorable macrolide response.

Duration of treatment is 12 months after culture becomes negative.

Therapeutic options in atypical mycobacterial infections (**Table 2.33**).

TABLE 2.33: Therapeutic options in atypical mycobacterial infections.

Atypical mycobacteria	***Therapeutic options***
MAC (Mycobacterium avium complex)	Nodule/bronchiectasis – Rifampicin + ethambutol + azithromycin thrice weekly Cavitory disease: REA – daily + IV amikacin/ streptomycin (thrice weekly) If macrolide resistant: RE + other (Cfz, mfx, bdq) + IV amikacin
M. abscessus	Erm resistance present: Initital phase (2 months) : ≥1 Oral drugs (linezolid/ Bdq/omadacycline) + IV amikacin + IV drug (Imipenem/tigecycline/tedizolid) Continuation: 2 oral drug + inhaled amikacin
M. xenopi	Rifampin + ethambutol + INH
M. kansasii	Rifampin + ethambutol should be treated for at least 18 months
M. malmoense	Rifampin or ethambutol
M. marinum	Rifampin or clarithromycin + ethambutol 2–3 months. Often resistant to isoniazid. Treatment with other antibiotics should be for at least 2 months
Rapid growers	Doxycycline, amikacin, imipenem, quinolones, sulfonamides, cefoxitin, and clarithromycin
Mycobacterium chelonae	Clarithromycin in combination with another agent, sometimes surgical excision is the best approach

Characteristics of Mycobacteria

- It is an aerobic, slender, and rod-shaped bacterium. It measures 2–10 μm in length.
- It has a high-lipid content in the cell wall, which makes it difficult to stain, but **once stained resists decolorization by acids and alcohol**. Hence, it is termed as acid-fast bacilli (AFB), because once stained by carbol fuchsin (present

in **Ziehl–Neelsen stain**), it is not decolorized by acid and alcohol. Acid-fast organisms and structures are listed in **Table 2.34**.

TABLE 2.34: Acid-fast organisms and structures.

Acid-fast organism	*Other acid-fast structures*
➢ *Mycobacteria* ➢ *Nocardia* ➢ *Isospora* ➢ *Cryptospora* ➢ *Microsporidia* ➢ *Rhodococcus* ➢ *Legionella*	➢ Head of human sperm ➢ Embryophore of *T. saginata* ➢ Hooklets of *E. granulosus* ➢ Keratin

TUBERCULOSIS

Tuberculosis (also called Koch's disease) is a **communicable, chronic granulomatous disease** caused by ***Mycobacterium tuberculosis***.

Tuberculosis (TB) is caused by four main mycobacterial species collectively termed *Mycobacterium tuberculosis complex (MTb)*: (1) *Mycobacterium tuberculosis* (reservoir human), (2) *Mycobacterium bovis* (reservoir cattle), (3) *Mycobacterium africanum, and* (4) *Mycobacterium microti. These* are obligate aerobes and facultative intracellular pathogens, which usually infect mononuclear phagocytes.

- **Majority of tuberculosis are due to *Mycobacterium tuberculosis* hominis** (human strain). The source of infection is patients suffering from active open case of tuberculosis.
- **Oropharyngeal and intestinal tuberculosis can be due to drinking milk contaminated by *M. bovis*** (bovine strain) from infected cows. Routine pasteurization has almost eliminated this source of infection.
- ***M. avium and intracellulare* are nonpathogenic to normal individuals. They cause infection in patients suffering from AIDS.**

Epidemiology

Q. CT1.1 Describe and discuss the epidemiology of tuberculosis and its impact on the work, life and economy of India.

- Tuberculosis is common in India. High incidence of tuberculosis is observed with poverty, overcrowding, and chronic debilitating illness.
- In 2019, 1.4 million people died from TB, including nearly 208,000 people who were coinfected with HIV.
- In 2019, 87% of new TB cases occurred in the 30 high TB burden countries. Eight countries accounted for two-thirds of the new TB cases: India, Indonesia, China, Philippines, Pakistan, Nigeria, Bangladesh, and South Africa.
- **Diseases associated with increased risk** of tuberculosis include diabetes mellitus, Hodgkin lymphoma, malnutrition, immunosuppression, alcoholism, chronic lung disease (e.g., silicosis), and chronic renal failure. Alcohol use disorder and tobacco smoking increase the risk of TB disease by a factor of 3.3 and 1.6, respectively.
- **HIV is the most important risk factor.** People who are infected with HIV are 18 times more likely to develop active TB.
- A global total of 206,030 people with multidrug- or rifampicin-resistant TB (MDR/RR-TB) were detected and notified in 2019, a 10% increase from 186,883 in 2018. About half of the global burden of MDR-TB is in 3 countries—India, China, and the Russian Federation.

Determinants of Virulence

- **Three genes:** (1) 1 kagG—encodes catalase, (2) rpo V-signs factor (initiates transcription of many enzymes), and (3) erp—encodes a protein required for multiplication.
- *NRAMP-1* gene: ***NRAMP1* is a transmembrane protein** (a product of *NRAMP1* gene), which **inhibits microbial growth** and it determines the susceptibility to tuberculosis. In individuals with **polymorphisms in the *NRAMP1* (n**atural **r**esistance-**a**ssociated **m**acrophage **p**rotein 1**) gene**, **tuberculosis may progress due to the absence of an effective immune response**.

Mode of Transmission

- **Inhalation:** It is the **most common mode** of transmission. **Source of organisms is an active open case of tuberculosis to a susceptible individual.** Infection spreads by inhalation of respiratory droplet from other infected patients.

- **Ingestion:** Tuberculosis may be transmitted by drinking nonpasteurized milk from infected cows contaminated with *M. bovis*. It causes oropharyngeal and intestinal tuberculosis. Nowadays, the ingestion mode of transmission occurs when a patient with open case of tuberculosis swallows the infected sputum resulting in tuberculosis of intestine.
- **Inoculation:** It is **extremely rare** and may develop during postmortem examination through cuts resulting from handling tuberculous infected organs.

Primary Tuberculosis

Q. CT1.2 Describe and discuss the microbiology of tubercle bacillus, mode of transmission, pathogenesis, clinical evolution and natural history of pulmonary and extrapulmonary forms (including lymph node, bone and CNS).

Q. Write short essay/note on primary complex of Ranke and Ghon complex/features of primary tuberculosis.

Initial infection that occurs on first exposure to the organism (*Mycobacterium tuberculosis*) **in an unsensitized** (previously unexposed tuberculin-negative) **individual** is known as primary tuberculosis. First infection of the lung caused by the tubercle bacillus is termed as primary pulmonary tuberculosis.

Primary infection usually occurs during childhood. Many patients give a history of contact with a case of active pulmonary tuberculosis. About 5% of newly infected people develop clinically significant disease. Source of the organism is always exogenous.

Sites of Primary Tuberculosis

Lung, intestine, tonsil, and skin (very rare).

Primary tuberculosis of lung

It is the **most common site** of primary tuberculosis. Primary pulmonary tuberculosis develops when the bacillus is inhaled and lodged in the alveoli of the lung.

Ghon lesion/focus

Following inhalation, tubercle bacilli reach distal airspaces.

- **Site of deposit:** Lower part of the upper lobe or upper part of the lower lobe near the pleural surface (subpleural) is the usual sites of deposit.
- **Ghon focus:** About 2–4 weeks after the infection, a circumscribed **gray-white area of** about **1- to 1.5-cm develops in the lung** is known as the Ghon focus → the center of which undergoes caseous necrosis.
- **Regional lymphadenitis:** Tubercle bacilli (free or within macrophages) are carried along the lymphatics to the regional draining nodes → which often show caseous necrosis.

Ghon complex

It is the combination of subpleural parenchymal lung lesion (**Ghon focus**) and **regional lymph node** involvement.

Fate of Ghon complex

- **Healing:** In **majority** (about 95%), cell-mediated immunity controls the infection and primary tuberculosis **heals**.
  - The **hallmark of healing is fibrosis.** Ghon complex may undergo progressive fibrosis and calcification. It is radiologically detected as a small calcified nodule **(Ranke complex)** in caseous material and very rarely undergoes ossification.
  - In the majority of individuals infected by *Mycobacterium,* the immune system contains the infection and the patient develops cell-mediated immune memory to the bacteria. This is termed as **latent tuberculosis.**
- **Spread:** Lymphatic and hematogenous spread to other organs or parts of the body occurs during the first few weeks.
  - **Progressive pulmonary tuberculosis:** In few patients, primary lesion in the lung may progress from the beginning (progressive pulmonary tuberculosis or progressive primary pulmonary tuberculosis).
  - **Bronchial spread:** Tuberculous lymph node may rupture/ulcerate through the bronchial wall and discharge caseous material into the bronchial lumen. This results in spread of infection to the related lobe or segment through bronchi.
  - **Hematogenous spread:** In some patients, the tubercle bacilli may enter the blood and produce tuberculous lesions in different parts of the body. The hematogenous spread can be of two types:
    - **Acute form:** It is more likely to occur in infants or young children and results in **miliary tuberculosis** or tuberculous meningitis.
    - **Chronic form:** It is characterized by tuberculosis in the lungs, bones, joints, liver, and kidneys. These lesions may develop months or even years after primary infection. The infection in these secondary foci may remain dormant for years.

- **Lymphatic spread:** In some cases, the infection may be carried by lymphatics from mediastinal lymph nodes to pleura or pericardium resulting in tuberculous pleurisy with effusion or tuberculous pericarditis with effusion.

Other Sites of Primary Tuberculosis

- **Intestine:** Primary focus always involves the small intestine (usually ileal region) and is associated with mesenteric lymphadenitis.
- **Tonsils:** Primary focus in the pharynx and tonsil with cervical lymph node enlargement
- **Skin:** Primary focus in the skin associated with regional lymphadenopathy

Bronchial Complications

- **Middle lobe syndrome:** Enlarged mediastinal lymph nodes of primary complex may compress a bronchus causing collapse of the lung. Compression of middle lobe bronchus may lead to collapse–consolidation and bronchiectatic changes. This may be present later as the **"middle lobe syndrome or Brock's syndrome"**.
- An unusual phenomenon in primary tuberculosis is obstruction of bronchus by lymph nodes referred to as **epituberculosis** resulting in some clinical signs, e.g., localized wheeze or bronchial breathing.
- **Obstructive emphysema:** Rarely, the compression of bronchus may result in a valve action with air trapping. This leads to obstructive emphysema.
- **Broncholith:** Calcification in a Ghon focus or regional lymph node may be extruded into a bronchus leading to "broncholith" or present as hemoptysis.

Clinical Features: Pulmonary Disease

Primary pulmonary TB

Symptoms:

- Majority are asymptomatic.
- A few patients may present with self-limiting febrile illness, which may lasts no more than 7–14 days. It occurs at the time of tuberculin conversion.
- Clinical disease only occurs, if there is progressive infection. If the infection is severe or the host resistance is low, child may present with reduced appetite and failure to gain weight. Slight dry cough may be occasionally present.

Physical signs:

- Majority of patients do not reveal any abnormal physical signs.
- **General features:** If the lesion is severe or extensive, signs of general debility may be present. Child is thin, pale, and fretful with less glossy hair and less elastic skin.
- **Respiratory system:** Usually no abnormal physical signs detected in the chest. Sometimes, few crepitations may be heard over lung parenchyma involved by the primary complex. More extensive physical signs in the chest may be produced when there are complications. Rarely, pleural effusion can be seen.
- **Erythema nodosum:** It may accompany primary pulmonary tuberculosis. They are bluish-red, raised, and tender; skin lesions are commonly seen on the shins and less commonly on the thighs. In few, it may be associated with fever and polyarthralgia.

Diagnosis

- **History of contact:** With a case of active tuberculosis.
- **Tuberculin test:** It is very valuable in children. A positive test in a previously nonsensitized/immunized child strongly indicates the disease. A negative test makes the diagnosis of tuberculosis very unlikely.
- **Chest radiograph:** Primary complex may appear as a peripheral parenchymal lung lesion and an enlarged hilar lymph node.
  - **In children:** Enlargement of (e.g., hilar) lymph node of the primary complex is more prominent than the pulmonary component.
  - **In adults:** Pulmonary component (peripheral parenchymal lesion) of the complex is more obvious than the lymph node component.
- **Bacteriological examination:**
  - Sputum examination—for AFB is preferable
  - Alternatively, three laryngeal swabs or fasting gastric washings can be examined
  - Detection of the tubercle bacilli either in the direct smear or culture confirms the diagnosis.

Postprimary (Secondary) Tuberculosis

Q. Describe the etiology, pathogenesis, pathology, clinical features, complications, and diagnosis of postprimary tuberculosis/secondary tuberculosis.

(Synonyms: Postprimary tuberculosis or reactivation tuberculosis)

- **Tuberculosis** developing **in a previously sensitized** individual (by earlier exposure) is known as secondary tuberculosis.
- It may develop shortly or after many years following primary tuberculosis, when resistance host is reduced.

Source of Infection

- **Endogenous:** Most common source is **reactivation of a latent infection.**
  - Direct progression of a primary tuberculous lesion
  - Reactivation of a dormant primary lesion
  - Hematogenous spread to the lungs
- **Exogenous:** Rarely exogenous new infection (reinfection)

Any location may be involved in secondary tuberculosis, but lungs are by far the most common site.

Morphology

Gross

- **Site:** In the lungs, postprimary (secondary) tuberculosis usually involves the **apex of the upper lobes** of one or both lungs, **within 1 to 2 cm of the apical pleura**. It commonly involves apical and posterior segments of the upper lobe or apical segment of the lower lobe. This predilection may be due to good ventilation, decreased blood and lymphatic supply of these regions in the erect posture, and the oxygen tension that favors survival of the strictly aerobic tubercle bacilli.
- **Appearance: Initially small focus (**less than **2 cm in diameter) of consolidation**, sharply circumscribed, firm, and **gray–white to yellow** in color. The central caseated liquefied material of a tuberculous primary lesion may be discharged into a bronchus and forms a tuberculous cavity in the lung.
- Regional lymph node involvement is not as prominent as that seen in primary tuberculosis.

Fate of Secondary Tuberculosis (Fig. 2.23)

Healing: In immunocompetent individuals, localized and apical focus may **heal with fibrosis and calcification but** rarely ossification.

Progress: It may occur along several different pathways.

- **Progressive pulmonary tuberculosis:** It occurs mainly in the elderly and immunosuppressed. Apical lesion may expand into surrounding lung and may erode into bronchi and vessels.
  - **Erosion into bronchi:** It leads to release of the central area of caseous necrosis → resulting in a ragged, irregular **apical cavity** surrounded by fibrous tissue. This produces an **important source of infection**, because when the patient coughs, sputum contains bacteria.
  - **Erosion of blood vessels:** It may result in hemoptysis.
- **Spread of infection:** If the treatment is inadequate or if host defenses are impaired, the infection may spread via: (1) airways, (2) lymphatics, or (3) blood vessels.
  - **Local/direct spread:** Tuberculosis can directly spread to the surrounding tissue. In the lung, local spread to the pleura may result in serous **pleural effusions**, **tuberculous empyema**, or **obliterative fibrous pleuritis**.
  - **Spread through bronchi/airways:** It may produce tuberculous pneumonia.
  - **Spread along mucosal lining:** Spread through lymphatic channels or along the mucosal lining from mycobacteria present in the expectorated infectious material may lead to **endobronchial**, **endotracheal**, and **laryngeal tuberculosis**. In the past, intestinal tuberculosis was due to drinking of contaminated unpasteurized milk. However, this is rare following pasteurization of milk. Nowadays, it is caused by the **swallowing of coughed-up infective material** in patients with open case of advanced pulmonary tuberculosis. It **mainly develops in the ileum.**
  - **Lymphatic spread:** Spread through lymphatic channels mainly reach regional lymph nodes. It may also cause disseminated disease.

    Miliary pulmonary disease: It is the disseminated form of tuberculosis. If the dissemination is only limited to the lungs, it is termed as miliary pulmonary disease.

    Lymphadenitis: It is most frequent presentation of extrapulmonary tuberculosis and usually occurs in the cervical region (“scrofula”).

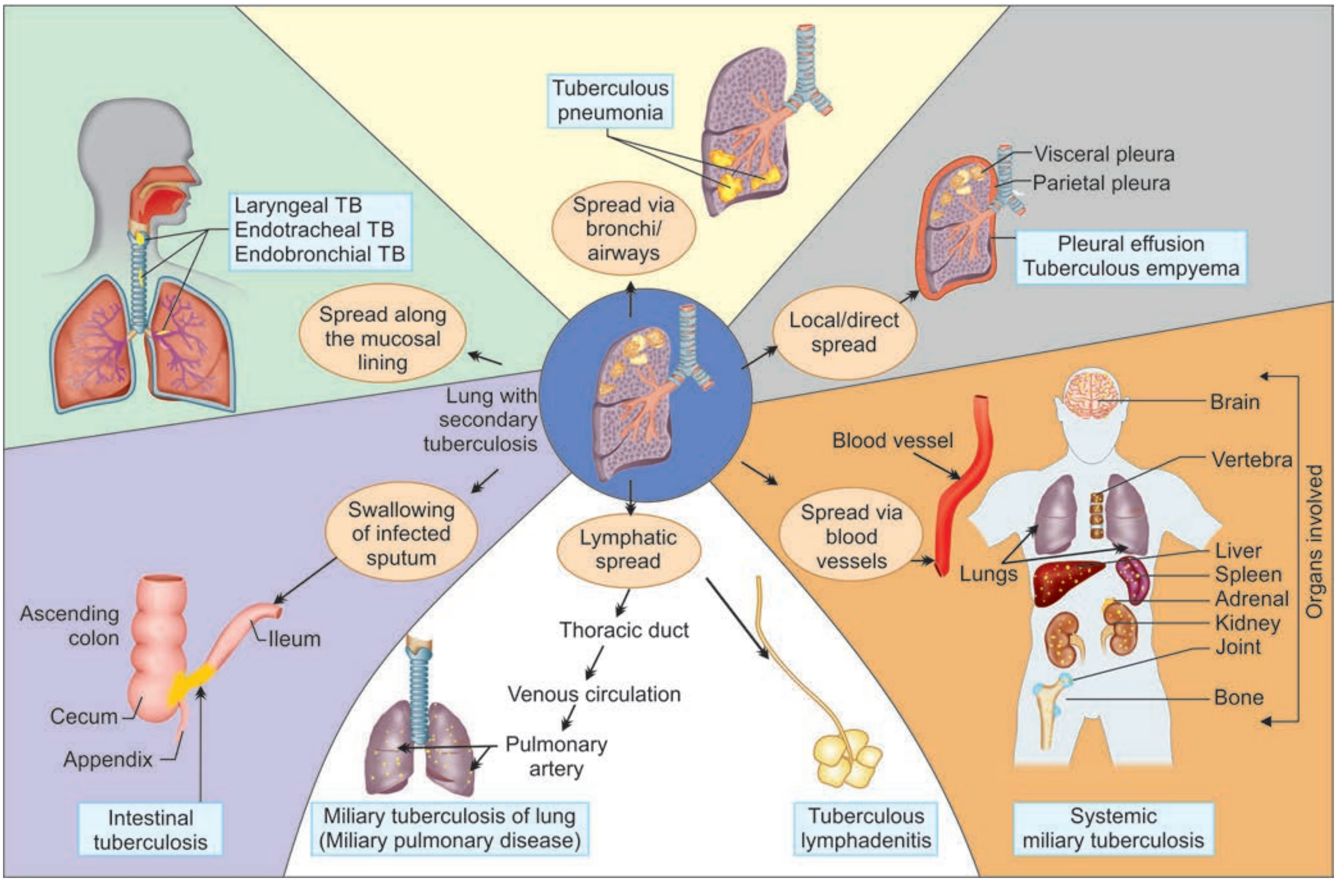


FIG. 2.23: Progression and complications of secondary tuberculosis (TB) of lung.

- **Spread via blood vessels**
  Systemic miliary tuberculosis occurs when tubercle bacilli disseminate through the systemic arterial system. Miliary tuberculosis most commonly involves liver, bone marrow, spleen, adrenals, meninges, kidneys, fallopian tubes, and epididymis. Tuberculosis can involve any organ *except nail, hair, and enamel*.
  Isolated organ tuberculosis: Dissemination of tubercle bacilli through blood may seed any organ or tissue → resulting in isolated organ tuberculosis. Commonly involved organs are:
  - Meninges (tuberculous meningitis)
  - Kidneys (renal tuberculosis)
  - Adrenals
  - *Bones (osteomyelitis):* When it involves the vertebrae, the disease is referred to as **Pott's disease**. Paraspinal "cold" abscesses may track along tissue planes and present as an abdominal or pelvic mass.
  - Fallopian tubes (salpingitis)/orchitis.

Clinical Features

Q. Write short note on clinical features of fibrocavitary tuberculosis.

Symptoms of pulmonary tuberculosis:

- Localized secondary tuberculosis may remain asymptomatic.
- Many patients are symptom free, and may be detected on routine radiography.
- Onset is usually insidious or gradual, with symptoms developing slowly over weeks or months.
- **Nonspecific:** Malaise, anorexia, loss of appetite and weight, and tiredness
- **Low-grade fever:** It is **remittent** (appearing late each afternoon and then subsiding—commonly known as **evening rise of temperature**) and **night sweats**.
- *Others:* Amenorrhea

Respiratory symptoms:

- **Chronic cough:** It is the most consistent symptom. If a patient has cough of more than 3 weeks, he/she should be investigated for pulmonary tuberculosis.
- **Hemoptysis:** It is a classical symptom. **Hemoptysis** is present in 50% of cases of pulmonary tuberculosis.
- **Sputum:** It may be mucoid, purulent, or blood stained. Classical sputum is described as "**nummular**".
- **Pain in the chest:** Pain may be due to pleurisy, intercostals myalgia, or cough fracture.
- **Unresolved pneumonia:** It may be another mode of presentation.
- **Breathlessness** may be a feature observed in advanced and extensive disease or due to pleural effusion.
- Localized wheeze may be observed due to local ulceration and narrowing of a major bronchus.
- Recurrent cold may be also a presenting symptom.
- Presentation due to complications: Very occasionally, it may present with one of the complications, which are given in **Table 2.35**.

TABLE 2.35: Complications of pulmonary tuberculosis.

Pulmonary	***Nonpulmonary***
Exudative pleural effusion/empyema Spontaneous pneumothorax Massive hemoptysis pulmonary or bronchial arteritis and thrombosis, bronchial artery dilatation, and Rasmussen aneurysm) Cor pulmonale Persistence of cavities even after treatment Pulmonary fibrosis/emphysema *Infection of cavities*: ➢ Atypical mycobacterial infection ➢ Aspergillus → aspergilloma ➢ Lung/pleural calcification **Airway lesions:** These include bronchiectasis, tracheobronchial stenosis, and broncholithiasis Bronchopleural fistula Bronchogenic carcinoma	Empyema necessitans Spread of tuberculosis to other organs (especially Addison's) Laryngitis *Following swallowing of infected sputum:* ➢ Enteritis ➢ Anorectal disease ➢ Amyloidosis ➢ Poncet's polyarthritis **Mediastinal lesions:** These include lymph node calcification and extranodal extension, esophagomediastinal or esophagobronchial fistula, constrictive pericarditis, and fibrosing mediastinitis Venous thromboembolism

Physical signs:

- Fever, tachycardia, and tachypnea
- Pallor and cachexia may be seen in advanced stages of the disease.
- Clubbing of finger is unusual.

Chest:

- Often, there are no abnormal signs detected.
- **Fine crepitations:** Most common sign is fine crepitations in the upper part (apices) of one or both lungs. They are better heard particularly on taking a deep breath after coughing **(post-tussive crepitations)**.
- Classical physical signs of consolidation (dullness to percussion), cavitation, fibrosis, bronchiectasis, pleural effusion, or pneumothorax may be present.
- **Cavernous bronchial breathing** with **post-tussive suction** may be heard, if there is a superficial collapsible cavity.
- There may be bronchial breathing in the upper part and localized wheeze due to local tuberculous bronchitis or pressure by a lymph node on a bronchus may be heard. Chronic tuberculosis, when accompanied with fibrosis, may show evidence of volume loss and mediastinal shift.

Extrapulmonary manifestations depend on the organ/system involved.

Conditions/diseases that favor reactivation/reinfection of tuberculosis are presented in **Table 2.36**.

TABLE 2.36: Condition/diseases that favor reactivation/reinfection of tuberculosis.

➢ Immunosuppression: HIV, anti-tumor necrosis factor (TNF) therapy, high-dose corticosteroids, and cytotoxic agents ➢ Malnutrition ➢ Diabetes mellitus ➢ Hemophilia ➢ Chronic kidney disease	➢ Silicosis ➢ Malignancies (e.g., lymphoma and leukemia) ➢ Gastrointestinal disease associated with malnutrition (gastrectomy, jejunoileal bypass, cancer of the pancreas, and malabsorption) ➢ Deficiency of vitamin D or A

Investigations

Q. CT1.9 Order and interpret diagnostic tests based on the clinical presentation including: CBC, Chest X-ray PA view, Mantoux, sputum culture and sensitivity, pleural fluid examination and culture, HIV testing.

Presence of an unexplained cough for more than 2–3 weeks, particularly in regions where TB is prevalent, or typical chest X-ray changes, should prompt for further investigation.

Blood examination:

- *Anemia:* Moderate degree
- *White cell count:* Usually normal or below normal
- *ESR:* Usually raised
- *Other findings:*
  - *Serum electrolytes*: Hyponatremia and hyperkalemia may be observed in severe disease or adrenalitis
  - *Liver function tests:* Occasionally may be impaired.

Radiological examination:
For practical purposes, a normal chest radiograph excludes the diagnosis of pulmonary tuberculosis (**Fig. 2.24**).

Radiological Features of Pulmonary Tuberculosis (Table 2.37)

Q. Write short note on the radiological features of pulmonary tuberculosis.

For all practical purposes, a normal chest radiograph excludes the diagnosis of pulmonary tuberculosis.

- **Radiological findings:** It shows ill-defined opacification in one or both of the upper lobes. As the disease progresses features of consolidation, collapse and cavitation develop to varying degrees (**Fig. 2.24**). It is often difficult to distinguish between active from quiescent form of tuberculosis on radiological criteria alone, but the presence of a miliary pattern or cavitation favors active disease.
- In extensive disease, collapse may cause significant displacement of the trachea and mediastinum. Occasionally, a caseous lymph node may drain into an adjoining bronchus, resulting in tuberculous pneumonia.
- **CT chest:** It may be useful in evaluating parenchymal and lymph node lesions. It may show **tree in bud appearance** (**Fig. 2.25**).
- ^{18}F-FDG PET scans and ^{11}C-choline PET scans may be done in few selected patients.

TABLE 2.37: Radiological features of tuberculosis of lung.

Radiological shadows that strongly suggest tuberculosis	*Radiological shadows that may be due to tuberculosis*
➢ Patchy or nodular shadows in the upper zone (on one or both sides) ➢ Cavitation (especially if more than one) ➢ Calcified lesion ➢ Pleural effusion/thickening	➢ Oval or round single shadow (tuberculoma) ➢ Hilar and mediastinal shadows (due to enlarged lymph nodes) ➢ Diffuse small nodular shadows (miliary tuberculosis)

Sputum examination

- **Direct microscopic examination of sputum:** It remains the most important first step investigation in pulmonary TB. Earlier, three specimens of sputum should be examined and if two of these smears are positive, diagnosis of TB is certain.
- **Revised WHO definition of a new sputum smear-positive case of pulmonary tuberculosis:** Presence of at least one acid-fast bacillus in at least one sputum sample in countries with a well-functioning external quality-assurance system. Presently, WHO recommends that the number of sputum specimens to be examined for screening of tuberculosis cases can be **reduced from three to two,** in places (i) where a well-functioning external quality-assurance system exists, (ii) where the workload is very high, and (iii) human resources are scarce.
- **Stain:** Rapid identification of the presence of tubercle bacilli by immediate stains is essential and should be done within 24 hours. The most effective stains are the Ziehl–Neelsen and rhodamine–auramine. Auramine–rhodamine staining is more sensitive (though less specific) than Ziehl–Neelsen.
- Grading of smears (RNTCP)

Bacilli observed	*Report*	*Grade*
10 or more AFB/OIF	Positive	3+
1-10 AFB/OIF	Positive	2+
10-99 bacilli/100 OIF	Positive	1+
1-9 AFB/100 OIF	Doubtful (repeat)	Scanty
No AFB/100 OIF	Negative	0

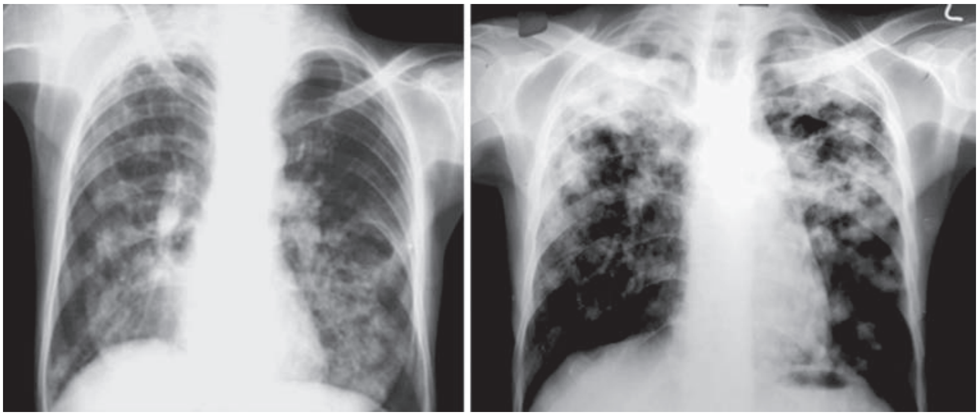


FIG. 2.24: Chest X-rays of tuberculosis showing bilateral infiltrates and thin-walled cavities.

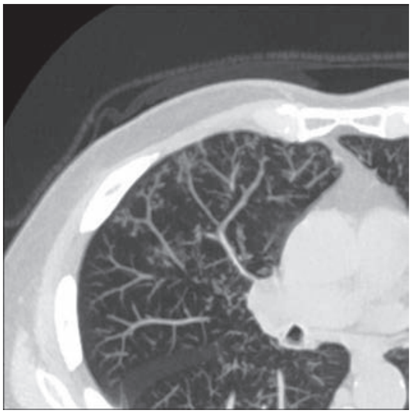


FIG. 2.25: Tree in bud appearance of tuberculosis on computed tomography (CT).

Culture of sputum

- A **positive sputum smear is sufficient for the presumptive diagnosis of TB** but definitive diagnosis requires culture of tubercle bacillus. Smear-negative sputum should also be cultured.
- **Culture medias:** It may be—
  - Liquid/broth culture (Middle brook 7H12) or the nonradiometric mycobacteria growth indicator tube (MGIT): Faster growth (1–3 weeks) occurs in liquid media. *The BACTEC radiometric growth detection method detects mycobacterial growth by measuring the liberation of ^{14}CO, following metabolism of ^{14}C-labeled substrate present in the medium. The growth can be detected in 4–8 days.*
  - Solid media (Löwenstein–Jensen slopes): MTB grows slowly and may take between 4 and 6 weeks.
- **Drug sensitivity testing:** It should be done in selected cases. It is important in patients with a previous history of TB, treatment failure or chronic disease, and in those who are resident in or have visited an area of high prevalence of resistance, or who are HIV-positive. Using liquid culture in the presence of antimycobacterial drugs (usually first line therapy initially) establishes the drug sensitivity for that strain and usually takes approximately 3 weeks).
- **Nucleic acid amplification (NAA) tests:** The **Amplified *Mycobacterium tuberculosis* direct (MTD) test and the GeneXpert MTB/RIF test**. Cartridge-based nucleic acid amplification test (CBNAAT) is more sensitive than smear but less sensitive than culture, as few as 1–10 organisms/mL may give a positive result. Resistance to rifampin can be detected by Xpert MTB/RIF or MTBDR plus, resistance to isoniazid can be detected by MTBDR plus **(Fig. 2.26)**.

Line probe assay (LPA)

- Immobilization of the multiple probes of interest on the nitrocellulose strip followed by application of amplicons to the strip that determines which probe hybridizes the amplicon.
- LPA is a rapid technique based on polymerase chain reaction (PCR) that is used to detect *Mycobacterium tuberculosis* (MTB) complex as well as drug sensitivity to rifampicin (RPM) and isoniazid (INH)
- It can detect resistance to second line drugs too.

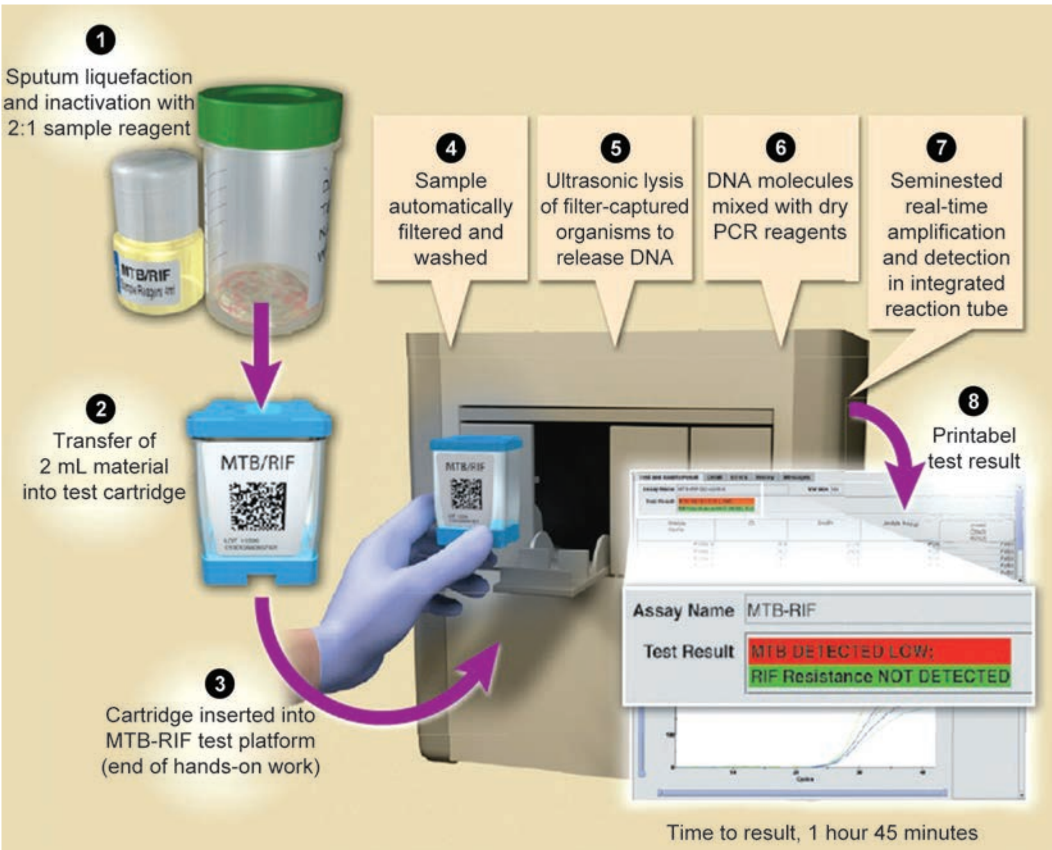


FIG. 2.26: Process of CBNAAT.

Other investigations

- Laryngeal swab, early morning gastric lavage, and bronchoalveolar lavage samples can be used for detecting AFB.

- **Tuberculin test:** It is used for diagnosis but is less valuable. However, this test may be negative in patients with active tuberculosis associated with malnutrition or other diseases. It may be positive in patients without active tuberculosis. Strongly positive test favors tuberculosis, whereas a negative test does not exclude tuberculosis.

Q. Write short note on newer methods of diagnosis of tuberculosis.

- **Interferon gamma release assays (IGRAs):**
  - IGRAs are *in-vitro* tests of cellular immunity. These assays measure cell-mediated immune response by quantifying interferon gamma (IFN γ) released by T cells in response to stimulation by *Mycobacterium tuberculosis*-specific antigens. These specific antigens include early secretory antigenic target-6 (ESAT-6) and culture filtrate protein-10 (CFP10).
  - The test does not differentiate between active and latent infection. This test requires high cost and trained personnel.
- **Other methods of diagnosis:**
  - **MGIT (mycobacteria growth indicator tube) method:** In this method, growth is detected by a nonradioactive detection system using fluorochromes for detection and drug screening.
  - Identification by mycolic acids using high-pressure liquid chromatography
  - *ELISA testing for IgM and IgA:* It is commonly used but has low specificity.
  - Mycobacterial-specific phages (reporter phages) to detect luciferase gene. It can be used to detect drug-resistant isolates.

Causes of Hemoptysis in Pulmonary Tuberculosis

Q. List the causes of hemoptysis in pulmonary tuberculosis.

- **Hemoptysis from a pulmonary cavity**
  - **Rasmussen's aneurysm:** Blood vessels traversing a tuberculous cavity can undergo changes due to inflammatory and necrosis. Over a period of time, these vessels may develop aneurysmal dilatation (Rasmussen's aneurysm). These aneurysms may rupture resulting in hemoptysis.
  - **Allergic response of vessel:** Occasionally, intense allergic response to antigens of tubercle bacilli can damage the walls of the blood vessels in and around the tuberculous cavities leading to hemoptysis.
- **Hemoptysis from endobronchial tuberculosis:**
  - **Tuberculosis of endobronchial region** may be surrounded by vessels with small aneurysmal dilatation. Rupture of these aneurysms can produce hemoptysis.
  - Occasionally, sloughing of the part of the granuloma may result in hemoptysis.
- **Hemoptysis as a sequel of pulmonary tuberculosis:**
  - **Open-healed cavities:** A tuberculous cavity may persist as a sequelae following chemotherapy, which are designated as "open-healed cavities"/INH cysts. The aneurysmal dilations of vessels may also persist in these open-healed cavities, which can rupture producing hemoptysis.
  - **Post-tuberculous bronchiectasis:** The upper lobe bronchiectasis is common sequelae of pulmonary tuberculosis. This may be characterized by repeated attacks of hemoptysis without sputum production (bronchiectasis sicca or dry bronchiectasis).
  - **Broncholith:** Calcification in a primary/Gohn focus or lymph node may be extruded into a bronchus as a "broncholith" and can cause hemoptysis. Hemoptysis may also result from the broncholith eroding through blood vessels.
  - **Aspergilloma:** Treated and healed tuberculous cavities may sometimes remain open and can be infected by the fungus *Aspergillus fumigatus*. This may produce a fungal ball (aspergilloma) in the cavity and can present as severe hemoptysis.
- Hemoptysis due to scar carcinoma

Tuberculin Skin Test

Q. CT1.7 Perform and interpret a PPD (Mantoux) and describe and discuss the indications and pitfalls of the test.

First infection with mycobacteria leads to development of delayed hypersensitivity to *M. tuberculosis* antigens (tuberculin) and this is detected by the tuberculin skin test.

Tuberculins: There are two commonly used tuberculins in present use:

1. PPD-S has been adopted as the international standard for PPD of mammalian tuberculin.
2. PPD-RT23 is widely used in epidemiological studies through- out the world.

Mantoux test (Fig. 2.27): It is ideal to begin the test with 5 IU PPD-S or 1 or 2 IU PPD-RT23.

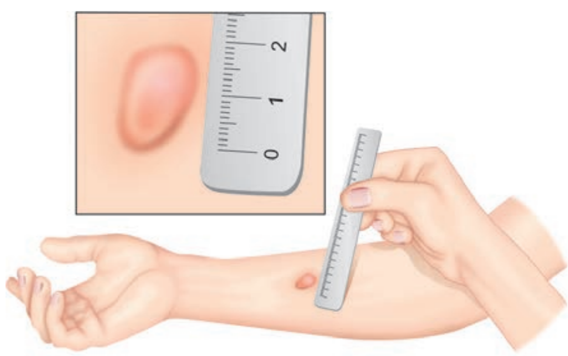


FIG. 2.27: Mantoux test.

Method

- Select an area of skin at the junction of the mid and upper thirds of flexure surface of the left forearm.
- Skin is cleaned with soap and water and allowed to dry.
- Using a tuberculin syringe and an intradermal needle, inject 0.1 mL of the tuberculin solution strictly intradermally. It should result in papule in the skin measuring 5–6 mm in diameter.

Reading and Interpreting the Result

- The test is read after 48–72 hours.
- If a reaction has taken place, there will be an area of erythema (redness) and an area of induration (thickening) of the skin. Measure the diameter of induration across the transverse axis of the arm. The reaction is considered positive, if an area of **induration** of the skin **of 10 mm diameter or more** at the site of injection of PPD. The amount of erythema (redness) present is not important. Induration ≥5 mm is considered as positive in patients with HIV infection (or risk factors for HIV infection, but unknown status), recent close contact to person with known active TB, patients with chest X-ray consistent with prior TB, patients with organ transplants, and other immunosuppressed patient.

Significance

- **Positive tuberculin test: Indicates T-cell-mediated immunity** to mycobacterial antigens. A strongly positive test is particularly valuable in children, especially very young children and favors the diagnosis of tuberculosis.
- **False-negative reactions:** If the diameter of induration is below 10 mm, the test is considered negative. But, a negative test does not exclude tuberculosis. It is seen in certain viral infections, sarcoidosis, malnutrition, Hodgkin lymphoma, immunosuppression, and overwhelming active tuberculous disease, HIV infection, measles, chickenpox, glandular fever (infectious mononucleosis), cancer, corticosteroids, and similar drugs.
- **False-positive reactions:** It is seen in infection by **atypical mycobacteria** or prior **vaccination with BCG** (*Bacillus Calmette–Guerin*) or lymphoma. Most infants immunized with BCG at birth have a negative tuberculin test by 1–2 years. In infants immunized after 1 year, the tuberculin reaction often remains positive for some years. It may also be positive in NTM infections.

Latent Tuberculosis

Q. Describe latent tuberculosis infection (LTBI).

- In the majority of individuals infected by *Mycobacterium tuberculosis*, the immune system contains the infection and the patient develops cell-mediated immune memory to the bacteria. These individuals do not currently have active tuberculosis disease. This is termed latent tuberculosis.
- Individuals with latent tuberculosis are at risk of progression to active tuberculosis. About 5–10% is the lifetime risk of progression. The increased risk of progression from latent tuberculosis to active tuberculosis is during the first 2 years after infection. Groups of individuals at high-risk of tuberculosis infection are listed in **Table 2.38**.

Groups at increased risk of progression to active tuberculosis are listed in **Table 2.39**.

TABLE 2.38: Groups of individuals at high-risk of tuberculosis infection.

- Employees working at long-term care facilities, hospitals, and medical laboratories
- Individuals having close contact with patients with active tuberculosis
- Residents and employees of congregate living facilities (e.g., prison and jails, nursing homes, hospitals, and homeless shelters)

TABLE 2.39: Groups at increased risk of progression to active tuberculosis.

- Children younger than 5 years of age
- History of tuberculous infection:
  - Individuals infected with *Mycobacterium tuberculosis* within the past 2 years
  - Past history of untreated or inadequately treated tuberculosis
- Associated conditions:
  - Individuals with HIV infection
  - Silicosis
  - IV drug users
  - Immunocompromised conditions
  - Long-term use of corticosteroids or other immunosuppressants (including anti-TNF-α)
  - Chronic renal failure
  - Diabetes mellitus
  - Malignancy

Screening for Latent Tuberculosis

- Tuberculin skin test
- T-cell IGRAs

Prophylaxis (to Prevent Development of Active Tuberculosis)

WHO and CDC recommend 3-4 months regimens compared to longer 6-9 months regimens as they are more effective and have better adherence. The following regimens are approved by WHO,CDC and preferred. Preferred Regimens 1.3HR: Isoniazid (300) + Rifampicin (600) daily for 3 months

(Total 90 doses) 2.4R: Rifampicin (600 mg) daily for 4 months (Total 120 doses) 3.3HP: Isoniazid (15 mg/kg) + Rifapentine (900 mg) once weekly for 3 months (total 12 doses).

Alternative regimen: 6H/9H = Isoniazid daily or twice weekly for 6–9 months.

Antituberculous Drugs (ATDS)

Q. CT1.14 **Describe and discuss the pharmacology of various anti-tuberculous agents, their indications, contraindications, interactions and adverse reactions.**

- **Bactericidal action:** It is the capacity of antituberculous drugs to rapidly kill large number of actively metabolizing bacilli. Most of the antituberculous drugs (except thiacetazone and PAS) are bactericidal. Isoniazid is the most potent bactericidal. Ethambutol is bacteriostatic at low doses and bactericidal at high doses.
- **Sterilizing action:** It is the capacity of antituberculous drugs to kill special populations of slowly or intermittently metabolizing semi-dormant bacilli (so-called "persisters"), e.g., rifampicin and pyrazinamide.

Q. Write short essay/note on:
- **Management of pulmonary tuberculosis, antituberculous drugs, and their dosages in adults.**
- **List the first- and second-line antituberculous drugs. Explain the rationale for using a multidrug regime.**
- **Bactericidal drugs used in the treatment of tuberculosis.**

Classification of Antituberculous Drugs

See **Table 2.40**.

TABLE 2.40: Classification of antitubercular drugs.

First-line antituberculous drugs	
➢ Isoniazid (H) ➢ Rifampin (R) ➢ Pyrazinamide (Z)	➢ Ethambutol (E) ➢ Streptomycin (S)
Second-line antitubercular drugs	
➢ Thiacetazone (Tzn) ➢ Para-aminosalicylic acid (PAS) ➢ Ethionamide (Etm) ➢ Cycloserine (Cys) ➢ Kanamycin (Kmc) ➢ Amikacin (Am) ➢ Capreomycin (Cpr)	➢ Newer antitubercular drugs (bedaquiline, delamanid) ➢ Levofloxacin/Moxifloxacin ➢ Moxifloxacin ➢ Clarithromycin ➢ Azithromycin ➢ Rifabutin

First-line Antituberculous Drugs

Q. Write short note on:
- **First-line antituberculous drugs.**
- **Modes of action of first-line drugs.**

1. **Isoniazid (INH):** It is primarily tuberculocidal drug.
   Mechanism of action: Inhibition of mycolic acid cell wall synthesis via O_2-dependent pathways (e.g., catalase–peroxidase reaction). Bactericidal against rapidly multiplying and bacteriostatic against resting bacilli. **Active against both extracellular and intracellular organisms.** Resistance occurs spontaneously in 1 in 10^5 bacilli.
   Pharmacokinetics:
   - Excreted in urine: Decrease dose, if creatinine clearance <30 mL/min
   - Slow versus rapid acetylators ($t_{1/2}$).

Q. Write short note on toxic effects of INH.

   Adverse effects:
   a. **Peripheral neuropathy:** More common is slow acetylators, diabetic, alcoholics, and malnourished patients
      - Prevention: Pyridoxine 10 mg/day
      - INH neurotoxicity is treated by pyridoxine 100 mg/day. Convulsions should be treated by IV pyridoxine 100 mg.
      - *Other TB-related drugs that cause peripheral neuropathy:* Pyridoxine, ethambutol, and cycloserine
   b. **Hepatitis:**
      - *Increased risk*: Age >35 years, alcohol abuse, rapid acetylators, co-administration of rifampin, pyrazinamide, HIV infection, chronic hepatitis B, pregnant females, and immediate postpartum (3 months).
      - Dose related and is reversible on stopping the drug.
   c. **Others:** Idiosyncratic reactions, SLE, gynecomastia acne, rash, anemia psychosis, memory impairment, and optic neuritis (atrophy).
2. **Rifampicin:** Semisynthetic derivative of rifamycin B obtained from *Streptomyces mediterranei*.
   Mechanism of action: Inhibition DNA-dependent RNA synthesis. **Bactericidal against both extracellular and intercellular organisms.**
   Adverse effects: Patients have discolored (orange) body secretions, hepatitis, thrombocytopenic purpura, respiratory syndrome, shock, renal failure (azotemia). Minor effects are:

- **Cutaneous (red man) syndrome:** Flushing, pruritus, rash (especially face and scalp). Exfoliative dermatitis is more frequent in HIV-positive TB patients.
- Influenza-like (Flu) syndrome
- **Abdominal syndrome:** Pseudomembranous colitis (especially rifabutin)

Newer rifampicin-related antitubercular agents.

Q. Write short essay/note on newer rifampicin-related antitubercular agents.

Rifabutin:

- **Actions:** It is related to rifampin. It is active against rifampicin-resistant *M. tuberculosis*; more active than rifampicin against *M. avium* intracellulare complex/NTM; longer t½; extent of absorption remains unchanged with food; recommended instead of rifampicin in patients HIV patients.
- **Dose:** It is recommended for tuberculosis in HIV-infected patients who are on protease inhibitors. Dose is 150 mg/day.
- **Adverse effects:** GI distress, rash, myalgias and insomnia, flu-like syndrome, anterior uveitis, leukopenia, skin discoloration, and hepatitis. Patients also have discolored (orange) body secretions.

Rifapentine:

- **Features:** It is lipophilic and has longer duration of action. Mycobacteria resistant to rifampicin are also resistant to this drug. It may be used in the treatment of pulmonary tuberculosis in place of rifampicin. Higher likelihood of relapse, but lower risk of adverse effects and less frequent administration than with rifampicin.
- **Dose:** 600 mg once or twice a week
- **Side effects:** Similar to rifampicin.

Q. Write short essay/note on uses of rifampin.

Uses of rifampin are listed in **Box 2.7**.

Box 2.7: Uses of rifampin.

- Tuberculosis
- Leprosy
- Prophylaxis of *meningococcal* and *H. influenza* meningitis and carrier state
- Second or third choice of drug for MRSA, diphtheroids, and *Legionella* infections
- Combination of doxycycline and rifampin as first line drugs in Brucellosis

3. **Pyrazinamide:**
   - Chemically similar to INH
   - **Mechanism of action:** Inhibition of mycolic acid cell wall synthesis and resembles INH. Bactericidal to slowly metabolizing bacilli in phagosome/granuloma. Most effective in acidic pH (<6.0)
   - **Adverse effects:**
     - **Hepatotoxicity:** Dose dependent
     - Arthralgias and polyarthralgias (especially shoulders) are common. Arthralgias are not related to the serum uric acid level.
     - Hyperuricemia is common due to inhibition of uric acid secretion by kidney. Development of new-onset gout is rare, but pre-existing gout may be exacerbated.
4. **Ethambutol:**

Q. Write short essay/note on ethambutol.

- **Mechanism of action** (MOA): It inhibits arabinose (arabinosyl transferase) involved in arabinogalactan synthesis. Bacteriostatic
- Excreted in urine (dose reduction required for patients with creatinine clearance <50 mL/min).

Adverse effects:

a. **Retrobulbar neuritis:** Dose-dependent. Usually occurs after many months of treatment. Manifests with reduced visual acuity, central scotoma, disturbance of *red–green* discrimination (loss of ability to see green). Permanent blindness may develop, if not discontinued.
b. **Others:** Hyperuricemia and peripheral sensory neuropathy.

5. **Streptomycin:**
   - It is an aminoglycoside, bactericidal antibiotic derived from *Streptomyces griseus.*

Adverse effects

a. **Ototoxicity, cochlear and vestibular damage, deafness, and neuromuscular blockage:** Dosage should be reduced, if headache, vomiting, vertigo, and tinnitus occur. Avoid in young children.
b. **Renal damage-nephrotoxicity** (nonoliguric renal failure): Dosage must be reduced to half immediately, if (1) urinary output falls, (2) if albuminuria occurs, or (3) if tubular casts are detected in the urine.
c. **Others:** Rare and include hemolytic anemia, aplastic anemia, agranulocytosis, thrombocytopenia, and lupoid reactions

Mode of Action of First-line Antituberculous Drugs

In a tuberculous lesion (particularly in a cavitary lesion), mycobacteria exist in several foci (**Table 2.41**).

TABLE 2.41: Tuberculous foci and the drugs acting on them.

Tuberculous foci	*Drugs acting on them*
Extracellular, in alkaline medium	Streptomycin
Rapidly metabolizing mycobacteria (in a cavity)	Rifampicin
Less actively multiplying bacilli in acidic and closed lesions	Isoniazid
Dormant bacilli (that cause a relapse)	Pyrazinamide

Second-line Antituberculous Drugs

Q. **Write short note on second-line antituberculous drugs.**

- **Ethionamide:** It is structurally related to INH and acts by inhibiting mycolic acid synthesis. It is effective against bacilli, resistant to other drugs, and is effective in infections due to atypical mycobacteria. It is effective against both intracellular and extracellular organisms.
- **Cycloserine:** It is mainly bacteriostatic and acts by inhibiting the synthesis of the bacterial cell wall. It is effective against bacilli resistant to INH or streptomycin and against atypical mycobacteria. Antitubercular activity is less than that of these two drugs.
- **Fluoroquinolones:** Ciprofloxacin, ofloxacin, levofloxacin, moxifloxacin, and gatifloxacin are active against *M. tuberculosis*, even in cases resistant to other drugs. Given orally or IV. It is useful in treating infections resistant to standard drugs and in relapse cases.
- **Capreomycin:** It is bactericidal and its mechanism of action, pharmacokinetics, and adverse reactions are similar to those of streptomycin. Administer with caution in presence of renal impairment.
- **Kanamycin and amikacin:** Both are bactericidal and are active against bacilli resistant to streptomycin, INH, and cycloserine.
- **Macrolides:** Newer macrolides, azithromycin and clarithromycin, also have action against tubercular bacilli. They are used to treat typical mycobacterial infection as well as in relapse cases.

Newer Antituberculous Drugs

- **β-lactams** (imipenem, amoxicillin-clavulanic acid), linezolid, clofazimine, clarithromycin, dapsone, and metronidazole have been used rarely for the treatment of multidrug-resistant (MDR) tuberculosis. However, their roles are not well established.
- **Bedaquiline:** It is a diarylquinoline class of antibiotics that selectively targets the proton pump of ATP synthesis, leading to inadequate ATP synthesis (necessary for bacterial metabolism). It is a new drug used for MDR tuberculosis.
- **Delamanid and pretomanid:** These are nitroimidazole class of antibiotics that inhibit the synthesis of mycolic acids (components of the cell envelope of *M. tuberculosis*). It is still on clinical trials.
- **Sutezolid:** It is oxazolidinone class of antibiotics. It prevents the initiation of protein synthesis by binding to 23s RNA in the 50s ribosomal subunit of bacteria.
- **SQ109, a 1,2-ethylenediamine:** It is an analog of ethambutol.
- **Benzothiazinones:** They inhibit the synthesis of decaprenyl-phospho-arabinose (the precursor of the arabinan) in the mycobacterial cell wall.

Side Effects of the Commonly Used Antituberculous Drugs (Table 2.42)

Q. **Write short notes on side effects of the commonly used antituberculous drugs/rifampicin/INH/ethambutol/streptomycin.**

TABLE 2.42: Side effects of the commonly used antituberculous drugs.

	Adverse reactions	
Drug (daily dosages)	*Major*	*Less common (rare)*
Isoniazid (H) (5–10 mg/kg)	➢ Hepatitis ➢ Peripheral neuropathy (preventable and treatable with pyridoxine) ➢ Cutaneous hypersensitivity	Giddiness, seizures, optic neuritis, mental symptoms, hemolytic anemia, aplastic anemia, agranulocytosis, lupoid reactions, arthralgia, and gynecomastia
Rifampicin (R) (10 mg/kg)	Febrile reactions ("flu" syndrome; more common with intermittent therapy), hepatitis, cutaneous reactions, and gastrointestinal disturbances	Shortness of breath, shock, hemolytic anemia, interstitial nephritis, and thrombocytopenia
Pyrazinamide (Z) (20 mg/kg)	Anorexia, nausea, flushing, hepatitis, gastrointestinal disturbance, and hyperuricemia	Hepatitis (dose related), vomiting, arthralgia, cutaneous hypersensitivity, and gout
Ethambutol (E) (15 mg/kg)	Retrobulbar neuritis (dose related) and arthralgia	Peripheral neuropathy and rash

Contd...

Contd...

Drug (daily dosages)	Adverse reactions	
	Major	**Less common (rare)**
Streptomycin (S) and other aminoglycosides (15–20 mg/kg)	8th nerve damage, cutaneous hypersensitivity, giddiness, numbness, and tinnitus	Vertigo, ataxia, deafness, hypokalemia, renal damage, aplastic anemia, and agranulocytosis
Ethionamide (Etm) (10–20 mg/kg)	Anorexia and vomiting	Serious neurologic reactions and hepatitis
Cycloserine (Cys) (10–20 mg/kg)	Headache and somnolence	Psychosis, seizures, and peripheral neuropathy
Quinolones (7.5–15 mg/kg)	GI intolerance and skin rashes	Phototoxicity (with sparfloxacin), dizziness, headache, and insomnia
Thiacetazone (Tzn) (2.5 mg/kg)	Gastrointestinal reactions, cutaneous hypersensitivity, vertigo, and conjunctivitis	Hepatitis, erythema multiforme, exfoliative dermatitis, hemolytic anemia
Para-aminosalicylic acid (PAS) (8–12 g/day)	Gastrointestinal reactions, hepatitis, cutaneous hypersensitivity, and hypokalemia	Acute renal failure, hemolytic anemia, thrombocytopenia, and hypothyroidism

Antituberculous Chemotherapy

Q. Write short essay/note on:
- **Short-course chemotherapy and its advantages.**
- **Discuss regimen of antituberculous chemotherapy.**

Global targets of detecting 70% of infectious cases and curing 85% of those detected. Goals of antituberculous drug therapy are listed in **Box 2.8**.

Box 2.8: Goals of antituberculous therapy.

1. Kill the dividing bacilli
2. Kill the persisting bacilli
3. Prevent emergence of resistance

Short-course Chemotherapy

- Short-course chemotherapy (SCC) is regimens of 6–9-month duration, which are highly effective and widely accepted as the treatment of choice for tuberculosis.
- All the short course regimens have two phases: An initial intensive (bactericidal) phase and a continuation (sterilizing) phase.
  - **Initial phase:** It lasts for 2–3 months and aimed to rapidly kill majority of mycobacteria. The symptoms resolve, sputum becomes negative, and the patient becomes noninfectious.
  - **Continuation phase:** It lasts for 4–6 months during which the remaining bacilli are eliminated so that relapse does not occur.

Advantages of short course of chemotherapy are mentioned in **Box 2.9**.

Box 2.9: Advantages of short-course chemotherapy.

- Easy to take and produces minimal upsets in patients
- Patient will recover more quickly
- Sputum becomes negative more quickly. About 85% at 2 months
- Low relapse rate. Even if relapse occurs, the tubercle bacilli remain sensitive and the same treatment can be given repeated

WHO (2009) Definitions

Q. CT1.17 Define criteria for the cure of tuberculosis; describe and recognize the features of drug resistant tuberculosis, prevention and therapeutic regimens.

- **New case:** A patient who has never had been treated for TB or not had anti-TB drugs for less than 1 month. New case may have positive or negative bacteriology and may have tuberculosis at any anatomical site.
- **Previously treated case:** It is defined as a newly registered episode of TB in a patient who has received one month or more of anti-TB drugs in the past. A culture and drug sensitivity test should be done before starting treatment. It is also referred to as "retreatment cases" and forms a heterogeneous group composed of several subcategories:
  - *Relapse*: A patient who has been previously treated for TB and was declared cured and is now been diagnosed bacteriologically positive tuberculous case (either a true relapse or a new episode of TB caused by reinfection).
- **Treatment after failure:** A patient who have been previously treated for TB and whose treatment failed at the end of course of treatment. The sputum smear or culture is positive at 5 months or later during treatment. This includes patients who have a multidrug-resistance strain at any point of time during the treatment.
- **Treatment after loss to follow-up:** Patients have previously been treated for TB and were declared lost to follow-up at the end of their most recent course of treatment (previously known as treatment after default patients) and now have bacteriologically positive tuberculosis.
- **Other previously treated patients:** All cases, which do not fit the above definitions. It includes patients who are sputum smear-positive at the end of a retreatment regimen (previously defined as chronic cases) and who may be resistant to the first-line drugs.

Categories of Diagnosis and Treatment

Treatment regimen for tuberculosis is presented in **Table 2.43.**

Diagnostic algorithm for pulmonary TB is shown in **Flowchart 2.7**.

Severe Tuberculosis

- **Miliary, disseminated TB** is considered to be severe.
- **Forms of extrapulmonary tuberculosis** (EPTB) classified as severe include: Meningeal, pericardial, peritoneal, bilateral or extensive pleural effusive, spinal, intestinal, and genitourinary.

Less severe: Lymph node, pleural effusion (unilateral), bone (excluding spine), peripheral joint, and skin tuberculosis.

Directly Observed Treatment, Short Course

Q. Write short essay/note on directly observed treatment, short course (DOTS) in tuberculosis.

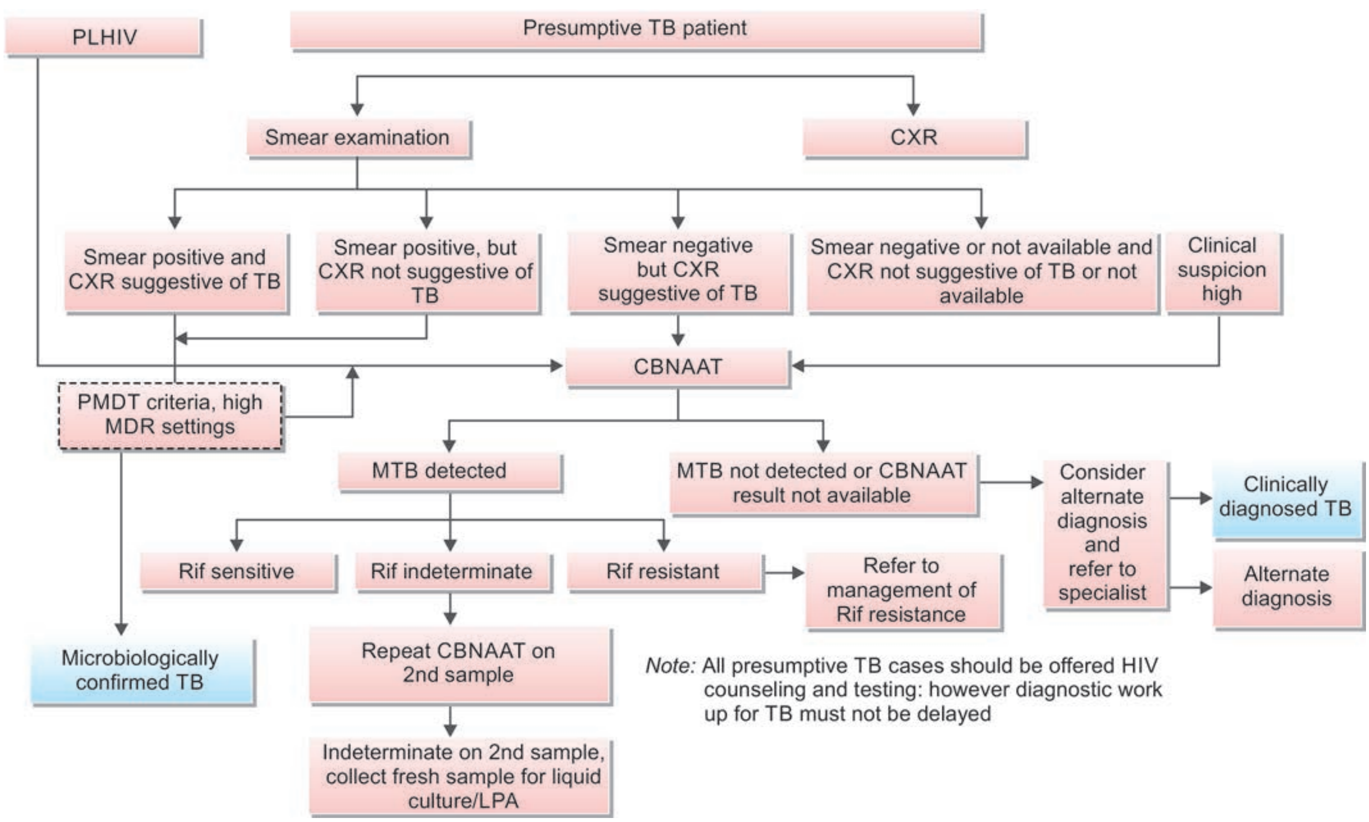
- Directly observed treatment, short course is an **intermittent method** of administering potent antituberculous regimens to a patient with tuberculosis **under direct supervision**.

TABLE 2.43: Treatment regimen for tuberculosis—all regimens are daily regimens.

A. DOTS (Directly observed treatment, short course)	***Regimen***
For new TB cases	2 months (HRZE) + 4 months (HRE) **Duration: 6 months**
For previously treated cases: Includes previously treated sputum smear-positive pulmonary TB: ➢ Relapse ➢ Treatment after interruption ➢ Treatment failure	2 months (HRZES) followed by 1 month (HRZE), followed by 5 months (HRE) **Duration: 8 months**
Management of extrapulmonary TB	The CP in both new and previously treated cases may be extended 3–6 months in certain TB such as CNS, skeletal, disseminated TB, and so on based on clinical decision of the treating physicians
B. DOTS PLUS	
1. **MDR TB:** Includes MDR and chronic TB cases (still sputum-positive after supervised retreatment)	6–9 months (KM LVX, ETO CS, Z, and E) followed by 18 months (LVX, ETO, CS, and E) **Duration: 24–27 months**
2. **XDR TB**	6–12 months intensive phase followed by 18 months continuation phase (capreomycin, PAS, moxifloxacin, clofazimine, linezolid, amoxicillin/clavulanate, clarithromycin, thiacetazone) **Duration: 24–30 months**

Note: Since 2017, the thrice weekly regimen has been changed daily regimen.
(H: Isoniazid; R: Rifampin; Z: Pyrazinamide; E: Ethambutol; S: Streptomycin; KM: Kanamycin; LVX: Levofloxacin; ETO: Ethionamide; CS: Cycloserine; PLHIV: people living with HIV)

FLOWCHART 2.7: Diagnostic algorithm for pulmonary TB.



- Daily chemotherapy is excellent. However, it is expensive and not possible to supervise therapy and hence compliance rate is low and relapse rate is high.
- **DOTS is a five-point program** (WHO), which can effectively control TB:
  1. Political and administrative support.
  2. Diagnosis in patients attending health facilities is by microscopic examination of sputum.
  3. Good antituberculous drugs are given for short course.
  4. Directly observed treatment, which is easily accessible, acceptable, and accountable.
  5. Systematic monitoring and accountability.

Monitoring the Treatment and Treatment Results

Q. Write short essay/note on monitoring the progress of treatment and the assessment of treatment result.

Main method of monitoring is bacteriological examination. Other methods include radiological assessment, ESR, and body weight changes.

- **Bacteriological examination:** Bacteriological assessment can be made by examining sputum smear and sputum culture.
  a. **Serial sputum smear examinations:** These help to assess the progress and ultimate result.
    - Sputum smear microscopy should be **performed at completion of the intensive phase** of treatment.
    - **New patients:**
      - **If sputum smear is positive at the end of the intensive phase (2 months), repeat the sputum smear at the end of the 3rd month**.
      - **If** the **smear is positive at the end of 3 months**, perform sputum culture and drug susceptibility testing (DST).
    - **Previously treated patients:**
      - If the smear is positive at the end of the intensive phase (3 months), perform sputum culture and DST.
    - **Favorable response:** If sputum bacillary count steadily decreases. In most patients, sputum will be negative in 1 month and almost all in 2 months and all in 3 months. However, sputum culture may be positive for another 1 or 2 months. Finally, both smears and cultures will be negative.
    - **Indications of treatment failure:**
      - Persistence of bacilli (no response)
      - **Fall and rise phenomenon:** Initial decline followed by a steady rise in sputum positivity
      - Relapse: Initial decline and sputum negativity, followed much later by sputum positivity
  b. **Culture:** Confirms the diagnosis and is indicated only in selected cases
    - **Radiological assessment** by chest radiographs: Serial radiographs can assess progress and determine final result. Improvement in the serial radiographs is a favorable response. However, radiological assessment may be misleading many times and radiological improvement may be associated with persistence of tubercle bacilli in the sputum.
    - **Erythrocyte sedimentation rate (ESR):** It is not a very satisfactory method of assessing the progress or activity of disease. However, a reduction in ESR can be a favorable response.
    - **Body weight changes** are also not very reliable. However, a body weight gain may be a favorable response.

Drug-Resistant TB

Q. PH1.45 Describe the dugs used in MDR and XDR tuberculosis.

Definition: Drug-resistant TB is defined by the **presence of resistance to any first-line antituberculous drug/agent.** Diagnosis is challenging (especially in developing countries) and although cure may be possible, it needs prolonged treatment with less effective, more toxic, and more expensive drugs.

- **Primary drug resistance and initial drug resistance:**
  - **Primary drug resistance**
    - It develops in **patients who have not received any antituberculous chemotherapy before or received it for less than 1 month**.
    - **Cause: Infection by drug-resistant organism** from another patient with secondary resistance due to inadequate chemotherapy.
  - **Initial drug resistance:** When it is impossible to obtain a reliable history of previous chemotherapy from a new drug-resistant patient, it is better to term it has initial drug resistance. This covers both true primary and undisclosed acquired resistance.

- **Secondary or acquired drug resistance: Resistance to one or more antituberculous drugs,** usually due to incorrect chemotherapy.
- **Natural drug resistance:** Mycobacterial strains, which have never been exposed to any antibacterial drug are called "wild strains." Thus, natural drug resistant strain is a wild strain resistant to a particular drug without ever having any contact with it. Thus, neither the patient with naturally resistant bacilli nor his source of infection has had chemotherapy in the past.
- **Transient drug resistance:** A positive culture of bacilli may be resistant to one or more (rarely two) drugs of a regimen during the course of successful chemotherapy. This may be due to resistant organisms for unknown reasons outlived the sensitive part of the bacterial population. Transient resistance does not need any change of treatment, since it does not results in treatment failure.
- **Mono-resistance (MR):** A TB patient whose biological specimen is resistant to one first-line anti-TB drug only.
- **Isoniazid resistance TB:** A TB patient, whose biological specimen is resistance to isoniazid and susceptibility to rifampicin has been confirmed.
- **Polyresistance (PDR):** A TB patient whose biological specimen is resistant to more than one first-line anti-TB drug, other than both INH and rifampicin.
- **Rifampicin resistance (RR):** Resistance to rifampicin detected by phenotypic or genotypic methods with or without resistant to other ATD excluding INH. Patient with RR should be managed as if they are in MDR-TB case.
- **Pre-XDR:** Resistance to INH and Rifampicin and either a fluoroquinolone or a second-line injectable agent but not both.
- **Extensive drug resistance (XDR):** Resistance to any FQ (fluoroquinolone) + atleast one SLID (2nd line injectable) in addition to multidrug-resistance.

Q. Write short essay on multidrug-resistant (MDR) tuberculosis, its diagnosis, and management.

- **Multiple drug resistance or multidrug-resistant TB:**
  - Multidrug-resistant tuberculosis (MDR-TB) is a form of TB that is **resistance to at least both of INH and rifampicin**, with or without other drug resistance. Hence, a patient should not be classified as multidrug resistant disease, if the patient has an infection with a bacterium susceptible to rifampicin but resistant to many other drugs.
  - MDR-TB can rarely be observed in new cases. It is more common in individuals in re-treatment cases (prior history of TB, particularly if treatment has been inadequate, and those with HIV infection). It is a man-made phenomenon.
  - Chronic cases and MDR-TB cases are not synonymous. Chronic patients probably have MDR-TB because they have previously received at least two full courses of treatment with essential antituberculous drugs.
- **Extensive drug-resistance TB (XDR-TB):**
  - Extensively drug resistance TB is a form of TB that is resistant to at least four of the core anti-TB drugs. These drugs include most important (core) anti-TB drugs—(1) isoniazid, and (2) rifampicin, and (3) injectable second-line aminoglycoside drugs (amikacin, capreomycin, or kanamycin) + (4) fluoroquinolone (such as ofloxacin or moxifloxacin).
- **Totally drug-resistant TB or (extremely XXDR, TDR):**
  - Totally drug-resistant tuberculosis (TDR-TB) is a form of TB strains that shows in vitro resistance to all first- and second-line drugs tested (isoniazid, rifampicin, streptomycin, ethambutol, pyrazinamide, ethionamide, para-aminosalicylic acid, cycloserine, ofloxacin, amikacin, ciprofloxacin, capreomycin, and kanamycin).
  - It was first reported in 2003 from Italy. In early January 2012, twelve cases had been diagnosed in Mumbai and in all cases, the strain of TB was resistant to all first- and second-line antitubercular drugs.

Factors contributing to the emergence of drug-resistant TB are listed in **Table 2.44**.

Q. Write short essay/note on common causes of drug resistance.

Suspicion of Drug Resistant TB

- A close contact of drug resistant TB case
- All retreatment cases
- Extensive disease at start of treatment
- Extrapulmonary TB not responding to standard ATT regime
- Treatment failures
- No sputum conversion after initial 2 months of ATT
- All HIV patients with TB

TABLE 2.44: Factors contributing to the emergence of drug-resistant tuberculosis (TB).

➢ Drug shortages	➢ Infection due to organisms with primary resistance
➢ Use of poor-quality drugs	➢ Transmission of drug-resistant strains
➢ Inadequate treatment regimen	➢ Prior antituberculosis treatment
➢ Inadequate supervision of therapy	➢ Treatment failure (smear-positive at 5 months)
➢ Poor absorption of drugs	➢ Lack of good laboratory facilities to monitor drug susceptibility
➢ Development of adverse drug reactions	➢ Genetic factors
➢ Inadequate duration of treatment	

Drug Susceptibility Testing (DST)

Q. Write short essay/note on various methods for performing drug susceptibility tests for *M. tuberculosis*.

Two methods available for determining drug resistance.

1. Rapid molecular methods (Drug resistance testing-**DRT)**
   - **Xpert MTB/Rif:** Catridge based NAAT. Detects DNA specific to *M. tuberculosis* complex and mutations in rpoB gene associated with rifampicin resistance. Turn around time 2 hours.
   - **Xpert MTB/XDR:** Detects mutation associated with INH, FQ, SLI (Amikacin, kanamycin, capreomycin, ethionamide). Results available within 90 mins.
   - **Truelab real time PCR:** Chip based, real-time PCR based NAAT for Tb detection and rifampicin resistance. Results obtained in 1 hour.
   - **Line probe assay:** Utilizes PCR, reverse hybridization technology for detection of resistance. FL-LPA (First line LPA) detects *rpoB gene (*R resistance), *KatG and InhA promoter* (H and ethionamide resistance). SL-LPA (2nd line LPA) detects mutation in *gyrA and gyrB (*FQ resistance), *rrs and eis (*low level kanamycin resistance).

   Interpretation of gene resistance
   - *rpoB:* R is not effective
   - *KatG:* High level resistance to INH = High dose INH ineffective
   - *InhA:* Low level resistance to INH = High dose INH effective. Ethionamide *ineffective*
   - *gyrA,B:* Levofloxacin resistance and low level moxiflox resistance = Lfx ineffective. High dose Mfx can be tried
   - *rrs:* SLI (2nd line injectable resistance) = amikacin, kanamycin, capreomycin ineffective
   - *eis:* Resistance inferred or detected for SLI = kanamycin ineffective. Amikacin, capreomycin *effective.*
2. Growth based drug susceptibility testing **(DST)**

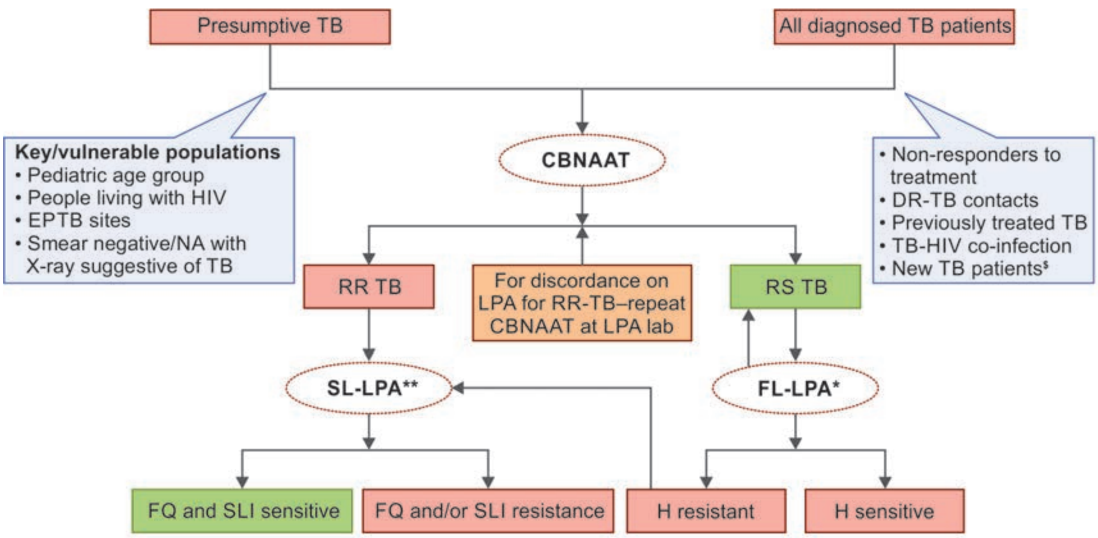
   WHO recommends DST for first-line and second-line anti-TB drugs to detect MDR-TB and XDR-TB, respectively.

   Various methods of DST are:
   - **Lowenstein Jensen (L-J) culture:** For drug sensitivity testing, it takes 6–8 weeks time.
   - **Radiometric methods (BACTEC radiometric method):** Results obtained within 10 days of inoculation.
   - **Mycobacteria growth indicator tube (MGIT) system:** Rapid, nonradioactive method.
   - **Nitrate reduction assays:** Based on the capacity of *M. tuberculosis* to reduce nitrate to nitrite.
   - **Ligase chain reaction (LCR):** Enzyme DNA ligase functions as a link two strands of DNA.
   - **Luciferase reporter assay:** Useful for rapid determination of drug resistance.
   - **PCR-based sequencing:** Detects mutations responsible for drug resistance.
   - **Line probe assays:** Detects drug resistant (particularly rifampicin resistant) bacilli.

Management of Multidrug-resistant Tuberculosis (Flowchart 2.8)

Q. Write short essay/note on multidrug-resistant (MDR) tuberculosis, its diagnosis, and its management (drugs used).

FLOWCHART 2.8: Drug-resistant TB—diagnostic algorithm.



Note: *Offer molecular testing for H mono/poly resistance to TB patients prioritized by risk as per the available lab capacity
**LC DST (Mfx 2.0, Km, Cm, Lzd) will be done only for patients with any resistance on baseline SL-LPA, DST to Z, Cfz, Bdq and Dlm would be considered for policy in future, whenever available, standardized and WHO endorsed.
$States to advance in phased manner as per PMDT scale up plan for universal DST based on lab capacity and policy on use of diagnostics

Q. Write short essay/note on management of multidrug- resistant (MDR) tuberculosis.

Principles of Management of MDR-TB tuberculosis (Flowchart 2.9)

Current guidelines have tried to implement an all oral shorter tolerable regimen for MDR/RR-TB. Resistance assays (Xpert/Rif and LPA) should be performed before treating MDR.

Classifications of drugs for MDR-TB is as follows:

- **Group A:** FQ/Bedaquiline (Bdq)/Linezolid (Lzd)
- **Group B:** Clofazimine (cfz), cycloserine (cs), terizidone (tzd)
- **Group C:** Z/E/S or Amikacin/delamanid (Dlm)/PAS/ or imipenem/meropenem

After confirming diagnosis of MDR/RR-TB eligibility for *Short course MDR-regimen* is to be assessed. Exclusion for short course include:

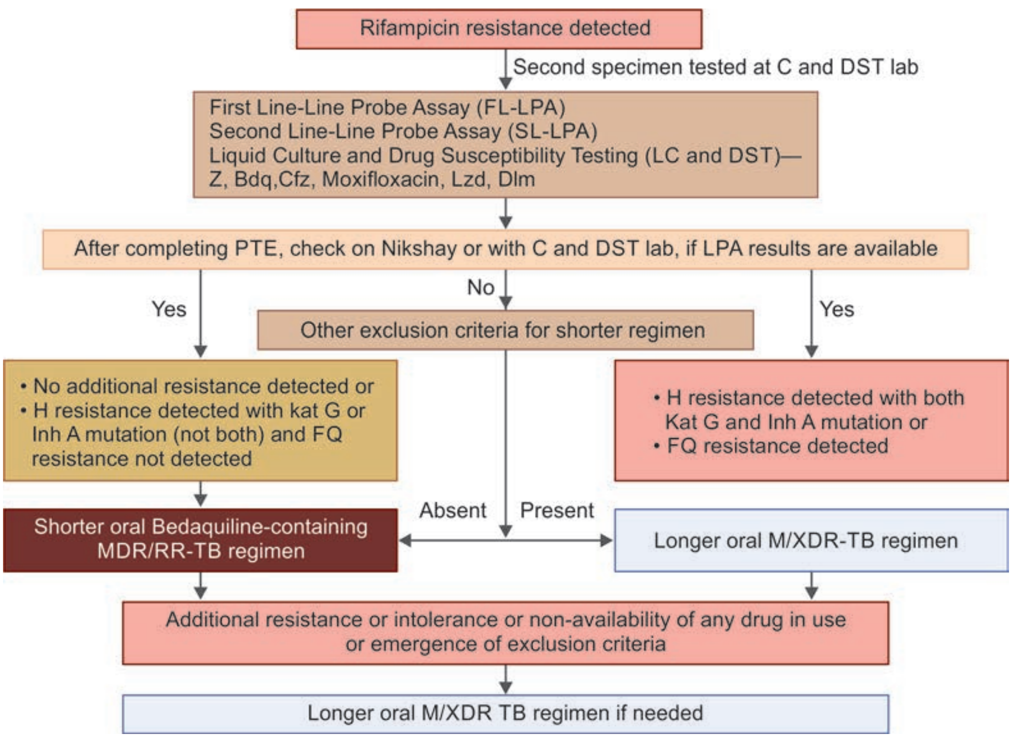
- Children <5 years
- Pregnancy/lactation
- Severe pulmonary TB (B/L cavity) or severe EPTB (miliary/CNS TB)
- >1 month exposure to FQ/Bdq/Cfz
- FQ resistance or INH resistance (both KatG and InhA)
- Intolerance to MDR regimen.

Short course regimen: Bdq minimum 6 months. Also known as *Bangladeshi regimen*. Total duration 9–11 months of therapy.

- **Intensive phase:** (4–6 months) Bdq^{6m} Lfx, Cfz, Z, E, H^{h} (high dose INH) and Eto
  Note: Sputum to be repeated after 4 months if negative continuation phase (CP) to be started.
- **Continuation phase:** (5 months) Lfx, Cfz, Z, E

Long course MDR regimen (18–20 months) Bdq, Lfx, Lzd, Cfz, Cs (all from group A, one or both from group B, add group C if group A, B drugs cannot be used).

FLOWCHART 2.9: Treatment algorithm for rifampicin resistance tuberculosis (MDR/RR-TB).



Indications for Treatment with Steroids

Q. Write short essay/note on the indications of corticosteroids use in the management of tuberculosis.

- Severely ill patients, e.g., TB meningitis with decreased consciousness, neurological defects, or spinal block or severe pulmonary TB.
- Severe hypersensitivity reaction to anti-TB drugs.

- To prevent exudation, its organization and stricture formation:
  - TB pericarditis with effusion or constriction
  - Large TB pleural effusion with severe symptoms
  - Meningeal tuberculosis
  - Renal tract TB to prevent ureteric scarring
  - TB laryngitis with life-threatening airway obstruction
- Hypoadrenalism (tuberculosis of adrenal glands)
- Massive lymph node enlargement with pressure effects
- In AIDS patient for TB-IRIS.

EXTRAPULMONARY TUBERCULOSIS

Q. Write short essay/note on extrapulmonary tuberculosis.

The term extrapulmonary tuberculosis (EPTB) is used for occurrence of tuberculosis at body sites other than the lung. However, when an extrapulmonary focus is present in a patient with pulmonary tuberculosis, such patients are categorized as pulmonary tuberculosis.

EPTB constitutes 15–20% of all cases of TB immunocompetent patients and in HIV-positive patients, EPTB accounts for more than 50% of all cases of TB.

Sites of EPTB: In order of frequency, these include lymph nodes (mediastinal and/or hilar, cervical), pleura (effusions without radiographic abnormalities in the lungs), genitourinary tract, bones and joints, meninges, peritoneum, and pericardium. However, virtually all organ systems may be affected **except nail, hair and enamel.**

Lymph Node TB

- Most common site for EPTB is lymph node. Extrathoracic nodes are more commonly involved than intrathoracic or mediastinal. Usually, this presents as a firm, painless (nontender) enlargement of a posterior cervical or supraclavicular node (a condition historically referred to as **scrofula**). Lymph nodes are usually discrete in early disease but develop into a matted nontender mass over time. The portal of entry is through the tonsils.
- The overlying skin is frequently indurated or there can be sinus tract formation with draining caseous material, but characteristically there is no erythema (**cold abscess** formation). The diagnosis is established by fine-needle aspiration biopsy. TB lymphadenitis is seen in nearly 35% of extrapulmonary TB cases.
- Antituberculous drugs are highly effective for lymph node tuberculosis. **Scrofuloderma** is a mycobacterial infection of the skin caused by direct extension of tuberculosis into the skin from underlying structures or by contact exposure to tuberculosis.

Tuberculous Osteomyelitis

Tuberculous osteomyelitis is **usually solitary** but in patients with acquired immunodeficiency syndrome, it is frequently multifocal. It tends to be **more destructive and resistant to control** than pyogenic osteomyelitis.

- **Age:** Usually **adolescents or young adults** in developing countries.
- **Source of infection:** Pulmonary or extrapulmonary tuberculosis.
- **Predisposing factors:** Diabetes, elderly, immune compromised states, and general debility.
- **Route of infection:**
  - *Blood-borne:* Usually blood-borne infection, which is from a focus of **active pulmonary or extrapulmonary disease**.
  - *Direct extension:* From lung into a rib and tracheobronchial nodes into adjacent vertebrae.
- **Sites:**
  - **Spine** (thoracic and lumbar vertebrae) commonly known as **Pott's disease.** The **infection breaks through intervertebral disks** to involve multiple vertebrae and **extends down into the soft tissues** forming abscesses (**cold abscess–psoas abscess**).
  - Knees and hips
- **Clinical course:**
  - Low-grade fever with evening rise of temperature
  - Pain on motion and localized tenderness
  - Weight loss
- **Complications:**
  - **Spine**
    - **Destruction of vertebrae:** Causes severe **scoliosis or kyphosis** and neurologic deficits due to spinal cord and nerve compression.

- **Psoas abscess:** Infection from spine may rupture into the soft tissue anteriorly and pus and necrotic debris may drain along the spinal ligaments and form a **cold abscess, i.e., an abscess lacking acute inflammation. Psoas abscess** is the condition in which infection from lower lumbar vertebrae dissects along the pelvis, and appears as a draining sinus of the skin in the inguinal region. It may be the first manifestation of tuberculous spondylitis.
  - **Tuberculous arthritis**
  - **Sinus tract formation**
  - **Amyloidosis**

TABLE 2.45: Extrapulmonary sites of tuberculosis and their presentation.

Extrapulmonary site	*Presentation*
Pleural	Pleural effusion and pleuritis
Lymph nodes	Tuberculous lymphadenopathy (including mediastinal) nonhealing sinuses
Skeletal system	Tuberculous osteomyelitis, cold abscess, vertebral tuberculosis, and pyarthrosis
Nervous system	Tuberculous meningitis tuberculous arteritis and cerebral tuberculoma
Gastrointestinal	Ulcerations of the tongue, intestinal tuberculosis, and tuberculous peritonitis
Pericardium	Pericardial effusion and tamponade, constrictive pericarditis
Genitourinary	Renal tuberculosis, salpingitis, tubal abscess, and tuberculous epididymitis
Miscellaneous	Addison's disease (tuberculous adrenalitis), skin tuberculosis (Scrofuloderma, lupus vulgaris, tuberculids), phlyctenular keratoconjunctivitis, choroiditis, iritis, and erythema nodosum

Tuberculous Meningitis

Refer **Chapter 16**.

Gastrointestinal Tuberculosis

- TB can affect any part of the bowel.
- Upper gastrointestinal tract involvement is rare.

Intestinal Tuberculosis

Discussed in Chapter 5.

Extrapulmonary sites of tuberculosis and their presentation are summarized in **Table 2.45** and **Figure 2.28**.

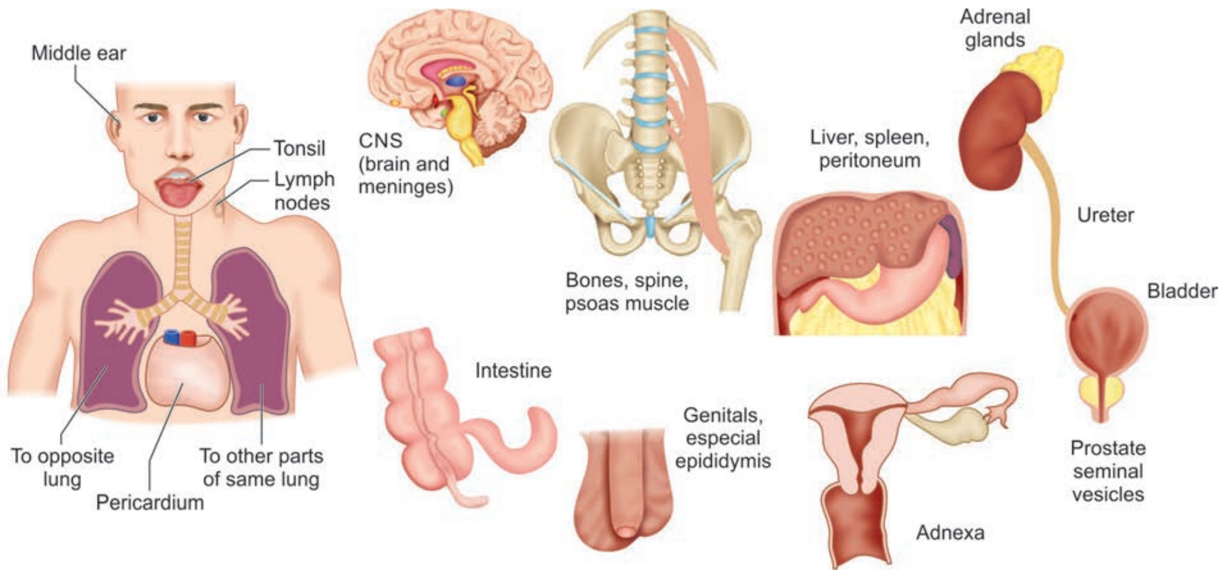


FIG. 2.28: Various extrapulmonary sites of tuberculosis.
(CNS: Central nervous system)

Miliary or Disseminated TB

Q. Discuss the pathogenesis, types, clinical features, diagnosis, and management of miliary tuberculosis in adults.

Miliary TB is the disseminated form of tuberculosis. The lesions are usually yellowish granulomas 1–2 mm in diameter. These lesions resemble millet seeds (hence termed miliary).

Route of Spread

Miliary tuberculosis results from widespread hematogenous dissemination of tubercle bacilli. The tubercle bacilli may enter the bloodstream by either hematogenous or lymphatic route.

Types

Q. Write short essay/note on miliary tuberculosis/nonreactive miliary tuberculosis/disseminated nonreactive tuberculosis.

Clinically, the miliary TB patients may be divided into three different types: (1) Classical (acute) miliary tuberculosis, (2) cryptic (obscure) miliary tuberculosis, and (3) nonreactive miliary tuberculosis.

1. **Classical (acute) miliary tuberculosis:**
   - **Age:** Can occur at any age, but more commonly affects children and young adults.
   - **Onset:** Sudden or gradual. Usually present with insidious onset of fever, malaise, and weight loss over weeks.
   - **Other symptoms:**
     - Systemic symptoms include high-grade fever, drenching night sweats, and progressive pallor.
     - Cough and breathlessness are occasionally present.
   - **Signs:**
     - There may not be any abnormal physical signs in the lungs. Widespread crepitations may be heard late in the disease.
     - Hepatosplenomegaly may be seen.
     - **Choroidal tubercles** on ophthalmoscopy—diagnostic.
2. **Cryptic (obscure) miliary tuberculosis**

Q. Write short essay/note on cryptic miliary tuberculosis.

   - **Age:** Usually in the elderly
   - **Symptoms:**
     - Prolonged low-grade pyrexia common presenting manifestation
     - Lassitude, weight loss, and general debility
   - **Signs:**
     - Hepatosplenomegaly may occur
     - Chest is usually normal
     - Choroidal tubercles are rare.
3. **Nonreactive miliary tuberculosis:**
   - Rare, usually develops in elderly with disease reactivation
   - Acute severe form of tuberculous septicemia, resulting in necrotic lesions without granulomatous reaction containing numerous bacilli.
   - Patients are extremely ill and die rapidly.

Diagnosis

- **Chest radiograph (Fig. 2.29):** If it shows the characteristic miliary shadows (miliary mottling), it is virtually diagnostic. Miliary mottling appears as diffuse small shadows of 1–2 mm diameter and evenly distributed throughout both lung fields. Upper zones are always involved. The early lesions may be difficult to appreciate. These lesions are better visualized by: (1) an over penetrated (dark) radiograph and a bright light behind the outer rib spaces, (2) lateral chest film, (3) underpenetrated anteroposterior radiograph, or (4) high-resolution CT of chest.
- **Positron emission tomography CT (PET-CT):** Using radiopharmaceutical ^{18}F labeled 2-deoxy-D-glucose (FDG) may show “hot” spots. It can determine the activity of lesion, guide biopsy, and detect occult foci.
- **Sputum** smear is usually negative but bronchoalveolar lavage and bronchial biopsy are likely to be positive.
- **Culture:** Confirmation of diagnosis should be done by culture of sputum, urine, or bone marrow.

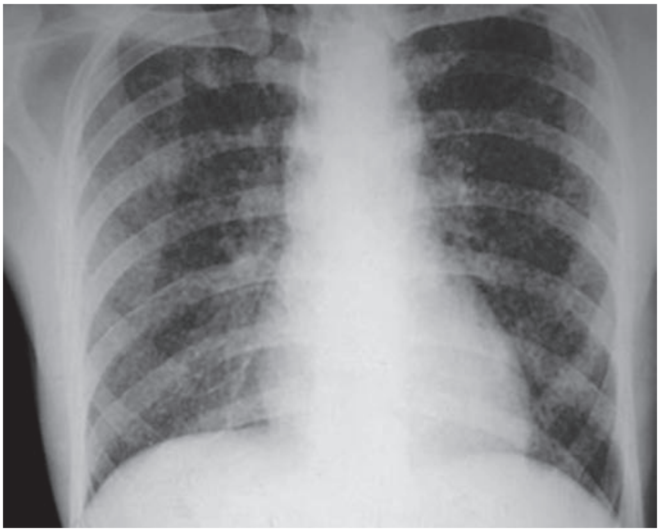


FIG. 2.29: Chest X-ray showing miliary mottling.

- **Hematological abnormalities:** These include anemia, leukopenia, neutrophilic leukocytosis, and leukemoid reaction. Rarely DIC can develop.
- Elevation of alkaline phosphatase and other liver enzymes may be observed in patient with severe liver involvement.
- Hyponatremia may develop in about 50% cases.
- **Bone marrow biopsy:** May show miliary tubercles/bacilli on histology. Part of the specimen should be sent for culture for tubercle bacilli.
- **Liver biopsy:** May show miliary tubercles.
- **Tuberculin test:** Only of limited value in miliary tuberculosis.

Q. Write short essay/note on management of miliary tuberculosis.

Management
- **Acute and cryptic miliary tuberculosis:** Standard antituberculous chemotherapy. In severely ill patients, prednisolone is given along with chemotherapy. It reduces life-threatening toxicity and gives time for antituberculous drugs to act.
- **If the diagnosis is not proved** (e.g., cryptic miliary tuberculosis): Therapeutic trial of antituberculous chemotherapy.

Tuberculous Pleural Effusion

Q. Discuss the etiology, pathogenesis, clinical features, investigations, complications, and management of tuberculous pleural effusion.

Q. Write short note on tuberculous pleural effusion and its laboratory diagnosis.

- **Age group:** Tuberculous pleural effusion usually occurs in **younger individuals**.
- Underlying pulmonary tuberculosis: **Only one-third** of patients show simultaneous pulmonary tuberculosis.

Pathogenesis

Involvement of pleura by *M. tuberculosis* may occur by various routes namely via lymphatics, bloodstream, or by direct extension.

- **Isolated pleural effusion** usually due to **recent primary infection**. The collection of fluid in the pleural space represents a hypersensitivity response to mycobacterial antigens.
- Pleural disease may also develop from contiguous parenchymal spread from **postprimary/secondary tuberculosis of lung**.
- **Rupture of subpleural caseous focus** into the pleural cavity. This produces delayed hypersensitivity reaction to tuberculous protein. It causes increased permeability of pleural capillaries, mild lymphatic block by fibrosis. Cultures of the pleural fluid from most of these patients with tuberculous pleural effusions are negative (because it is a hypersensitivity response).
- **Sequelae:** If left untreated, the pleura may become thick and fibrotic (pleural fibrosis) and pleural adhesions may develop. Pleural adhesion causes restrictive ventilator dysfunction. Early treatment is necessary to prevent these sequelae.

Clinical Features, Investigations, and Management

Refer Section on pleural effusion later.

Pleural fluid analysis

Q. Write a note on pleural fluid findings in tuberculous pleural effusion.

- **Color:** Fluid is usually straw/amber colored, but sometimes hemorrhagic.
- **Exudative in nature:** Characteristically fluid is an exudate, with a high protein content (>3 g/dL) >50% of that in serum (usually 4–6 g/dL) a normal to low glucose concentration, a pH of –7.3 (occasionally <7.2) and raised white blood cells (usually 500–6,000/μL).
- **Cells: Predominant lymphocytosis**. Neutrophils may predominate in the early stage (less than 2 weeks), but lymphocytosis is the typical finding later. Mesothelial cells are usually rare or absent. If the pleural fluid shows more than 10% eosinophils, diagnosis of tuberculous effusion is unlikely unless the patient has a pneumothorax or had previously undergone thoracocentesis.
- **Smear for AFB:** Smears prepared from the centrifuged deposit may rarely show the tubercle bacilli (<10% of immunocompetent cases).
- **Culture for *M. tuberculosis*:** Positive in approximately 25–50% of patients and are more common among postprimary/secondary cases.
- **Determination of the pleural concentration of adenosine deaminase (ADA):** It is a useful screening test, and TB can be excluded if the value is very low.

Q. Write short note on adenosine deaminase (ADA).

- It is a T-lymphocyte enzyme.
- In majority of cases, the levels of **ADA in the pleural fluid are elevated** (>40 IU/L) and is probably due to increased activity of T lymphocytes (CD4+) in the pleural fluid.
- Other causes with high ADA include rheumatoid arthritis, lymphoma, chronic lymphatic leukemia, empyema, and mesothelioma. However, because the incidence of tuberculosis far exceeds than any other cause of a lymphocytic pleural effusion (like in India), high ADA level has a predictive value.

- Specificity of raised levels of ADA in diagnosing tuberculous effusion is nearly 0.83 and the reported sensitivity is 77–100%. Specificity increases when pleural fluid lymphocytes/polymorph ratio greater than 3.
- ADA is not useful in HIV patients with TB.
- There are two isoenzymes of ADA namely ADA1 and ADA2. ADA1 isoenzyme is found in all cells and they are high in lymphocytes and monocytes. ADA2 isoenzyme is present only in monocytes. In **tuberculous pleural effusion, ADA2 isoenzyme** is mainly responsible for high ADA concentration.

Interferon-gamma (INF-γ): Produced by lymphocytes specifically sensitized to PPD. Its level above 140 pg/mL is suggestive of TB and is elevated irrespective of immune status. It is more expensive than ADA.

Other tests on pleural fluid: These include raised LDH, raised lysozymes, marked elevation in the levels of soluble interleukin-2 (IL-2) receptors, and PCR for DNA of *M. tuberculosis*. Nucleic acid amplification technology has low sensitivity.

Pleural biopsy: Closed pleural biopsy shows noncaseating granulomas in 80% of patients. It should be stained with Z-N stain and cultured for mycobacteria. Diagnostic yield increases to 90% with pleural biopsy and biopsy cultures for AFB.

Management
- **Therapeutic aspiration of pleural fluid:** May be required in patients with severe symptoms.
- **Antituberculous chemotherapy**
- **Corticosteroids:** May reduce the symptoms of toxemia. However, they do not reduce the incidence of pleural fibrosis. Prednisolone is administered in the dose of 0.75 mg/kg/day for up to 4 weeks with gradual reduction over an additional 2–4 weeks.

BCG Vaccination

Q. Write short essay/note on BCG vaccination (Bacillus Calmette–Guerin vaccine).

BCG (Bacillus Calmette-Guerin) is a freeze-dried, live attenuated (lost its virulence) vaccine derived from *M. bovis*. In India, Danish strain 1331 is being used for BCG vaccine production.

Procedure: 0.1 mL of the reconstituted vaccine is injected intradermally at the junction of the upper and middle thirds of the left upper arm.

Contraindications: Generalized eczema and hypogammaglobulinemia, and immunodeficiency resulting from treatment with antimetabolites, irradiation or systemic corticosteroids.

Advantages: BCG vaccination reduces the risk of miliary/disseminated tuberculosis (TB) and tuberculous meningitis in children. Its efficacy in adults is very variable.

Complications: Secondary infection with local abscess formation, enlargement of regional lymph nodes, cold abscess of draining lymph nodes, local lupoid reactions (rare), erythema nodosum, urticaria, and disseminated BCG infection (rare).

Tuberculosis Chemoprophylaxis—Isoniazid Preventive Therapy

Q. Write a short note on tuberculosis chemoprophylaxis.

- **Purpose:** To prevent progression of latent tuberculous infection to active disease.
- **Types:**
  - *Primary or infection prophylaxis:* Drug is given to individuals who have not been infected in order to prevent development of disease (e.g., breastfed infants of sputum-positive mother).
  - *Secondary or disease prophylaxis:* Drug is given to prevent development of disease in individuals already infected.

Drugs Used

- **Isoniazid** (H) at the dose of 5 mg/kg/day (not exceeding 300 mg/day) for 6–12 months is used for chemoprophylaxis.
- *Alternate option:*
  - Isoniazid plus rifampicin (10 mg/kg daily) for 3 months, or
  - Isoniazid in the dose of 5 mg/kg (adults 900 mg) plus rifampicin at a dose of 10 mg/kg twice weekly for 3 months.
- **Precaution:** Exclude active TB by history, physical examination, chest radiograph and, if necessary, by other tests before starting chemoprophylaxis.

Indications

- Close contacts of open case of TB who show recent Mantoux conversion.
- Close contacts children aged 5 years or below with strongly positive Mantoux and a TB patient in the family.
- Breastfed neonates/infants of sputum-positive mothers.
- Newly infected patients as shown by recent change in tuberculin test from negative to positive.
- Patients with old inactive disease who are assessed to have received inadequate treatment.
- Certain diseases in which TB is more likely to develop. These include HIV infection, leukemia, Hodgkin's disease, prolonged treatment with prednisolone, severe diabetes mellitus, and patients on antimalignancy drugs. In India, if tuberculin test is ≥10 mm. [WHO recommends that people living with HIV (PLHIV) who are unlikely to have active TB should receive at least 6 months of isoniazid preventive therapy (IPT) as part of a comprehensive package of HIV care].

Risks

- **Isoniazid resistance** can occur especially when preventive therapy with isoniazid is inadvertently given to individuals with subclinical or unrecognized TB.
- Hepatotoxicity

SUPPURATIVE LUNG DISEASE

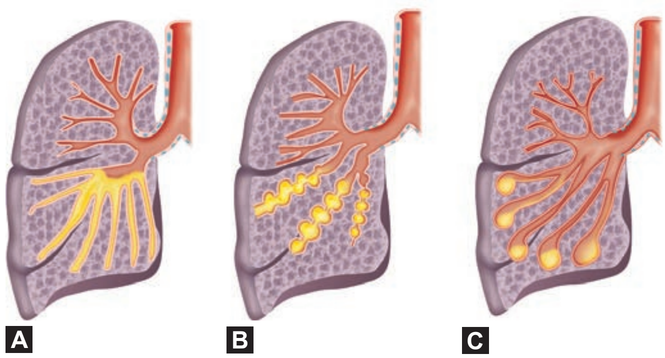
Bronchiectasis

Q. Define bronchiectasis. Describe the etiopathogenesis, classification, clinical features, investigations, complications, and management of bronchiectasis.

Definition: Bronchiectasis is defined as an **irreversible** (permanent), **abnormal dilation of the cartilage-containing airways bronchi or bronchioles.**

Classification

A. **According to the shape of the bronchial dilation (Figs. 2.30A to C)** based on the bronchographic appearance (Reid's classification)
   - **Tubular (cylindrical):** Characterized by smooth dilation of the bronchi. It is the most common form.
   - **Varicose (bulbous):** In which the bronchi are dilated with multiple indentations.
   - **Cystic (saccular/balloon appearance):** In which dilated bronchi terminate in blind ending sacs.



FIGS. 2.30A TO C: Morphological types of bronchiectasis: (A) Cylindrical (tubular) type; (B) Varicose (bulbous) type; (C) Saccular.

B. **According to the extent of involvement**
   - **Diffuse (generalized) bronchiectasis: Characterized by** widespread bronchiectatic changes throughout the lung. It is **usually bilateral** and commonly affects the **lower lobes**. **Left lobe is more commonly involved than the right**. It is most severe in the distal bronchi and bronchioles.
   - **Focal (localized) bronchiectasis:** Bronchiectatic change is **restricted to a** localized area of the lung **(single segment of the lung)** and usually occurs in association with obstruction of the airway (parenchymal tumor or aspiration of foreign bodies).

C. **According to the underlying disease/mechanism**
   - Congenital/acquired
   - Cystic fibrosis (CF) associated and noncystic fibrosis bronchiectasis
   - Associated with post fibrosis: Traction bronchiectasis
   - Without much expectorant: Dry bronchiectasis.

Etiology

Q. Write a short essay/note on causes of bronchiectasis.

The dilatation of bronchi and bronchioles is caused by **destruction of the muscle and elastic tissue of bronchial wall**. It represents a secondary disorder as the end stage of many unrelated disorders. It may be divided into obstructive and nonobstructive (postinflammatory). It may also be divided into congenital and acquired **(Table 2.46)**.

TABLE 2.46: Causes of bronchiectasis.

➢ **Idiopathic** ➢ **Postinfectious:** Bacterial pneumonia, viral pneumonia in childhood. Mycobacterial infection, aspergillosis, **Swyer-James** (secondary to childhood bronchiolitis obliterans) ➢ **Impaired immunity:** – **Primary immunodeficiency (PI):** CVID, SCID, hypogammaglobulinemia, chronic granulomatous disease, hyper IgE ➢ **Genetic/congenital:** CF, primary ciliary dyskinesia (PCD), α1-antitrypsin deficiency – **Bronchial tree malformation** (tracheomegaly—Mounier Kuhn, cartilage deficiency: William-Campbell, *pulmonary sequestration,* bronchial atresia)	➢ **Allergic/autoimmune:** – ABPA, asthma, COPD – CTD: RA, SLE, Sjogrens, relapsing polychondritis – Inflammatory bowel disease (IBD) ➢ **Obstruction:** Foreign body, anatomical abnormality, tumor ➢ **Chronic lung disease:** Sarcoidosis, interstitial pneumonias, post-transplant ➢ **Inflammation:** Aspiration, radiation induced, toxic inhalation (chlorine, ammonia, smoke) ➢ **Malignancy:** Primary lung—bronchogenic carcinoma, bronchial carcinoid, thymoma (*d/t hypogammaglobulinemia)* ➢ **Miscellaneous:** Young, yellow nail syndrome

Proximal/central bronchiectasis: Bronchiectasis involving primary (main bronchi) and secondary (lobar bronchi):

- **Allergic bronchopulmonary aspergillosis (ABPA)**
- **Brock's syndrome/middle lobe syndrome:** Primary TB/foreign body/tumor compressing main bronchus.
- **Lady Windermere syndrome (LWS):** These women have the habit of voluntarily suppressing cough. It results in inability to clear the secretions from the right middle lobe and lingual leading to infection and later bronchiectasis.
- **Congenital syndromes: Kartagener's syndrome, yellow nails syndrome, Chandra–Khetarpal syndrome** (immunodeficiency associated with levocardia, bronchiectasis, and paranasal sinus anomalies), **Young's syndrome, cystic fibrosis, Chédiak–Higashi syndrome.**

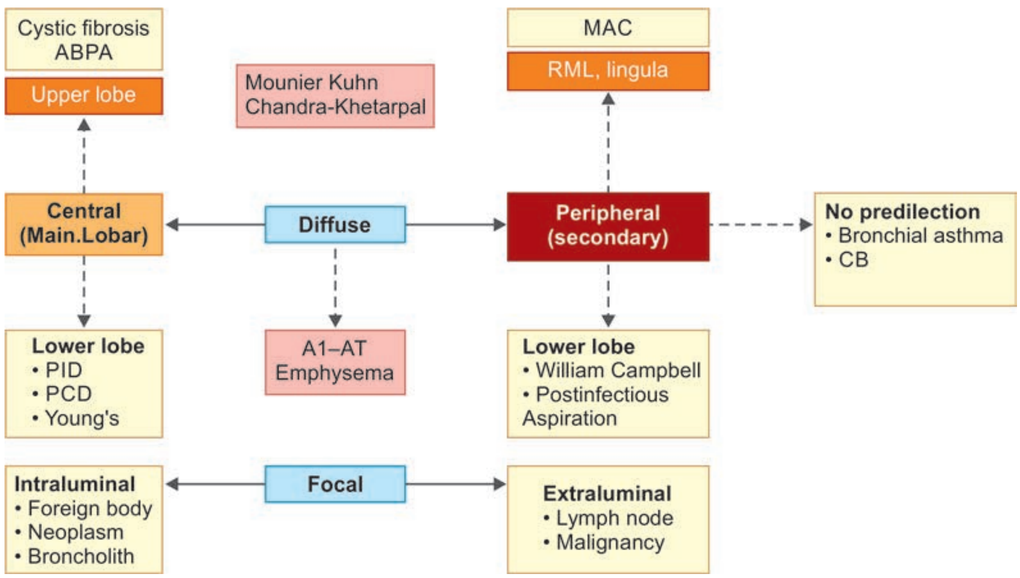


FIG. 2.31: Etiology and types of bronchiectasis.

(ABPA: Allergic bronchopulmonary aspergillosis; PID: Primary immunodeficiency; PCD: Primary ciliary dyskinesia; MAC: *Mycobacterium avium* complex; CB: Constrictive bronchiolitis)

Pathogenesis

Theories of bronchiectasis

- **Pressure of secretion theory (obstruction):** Secretions cause mechanical obstruction and obstruction **impairs clearing mechanisms** of the lung → results in **accumulation of secretions** distal to the obstruction → **leads to secondary infection** → inflammation → weakens and dilates airway.
- **Infection theory: Chronic persistent** (recurrent) **necrotizing infection** and inflammation in the bronchi or bronchioles → increased bronchial secretion → **obstruction** of airways by secretions → inflammation and fibrosis of the airway walls → weakening and dilatation of airways.
- **Traction theory:** Traction of the bronchi walls secondary to fibrosis/scarring.
- **Atelectasis theory:** Negative intrapleural pressure resulting in collapse and bronchial dilatation.
- **All of the above leads to "Vicious Cycle" Bronchiectasis proposed by Cole** which comprises of *Initial Insult resulting in* neutrophilic inflammation → Airway destruction → Abnormal mucus clearance → Mucus stasis → Bacterial colonization resulting in further *inflammation.*

Clinical Features

Q. Write a short essay/note on clinical features of bronchiectasis.

- **Severe persistent (chronic) productive cough: It is the most common symptom.** Cough is chronic, daily, and persistent. **Paroxysm of cough develops when the patient rises in the morning** because the postural changes drain the collections of pus and secretions into the bronchi. Sputum production varies with posture.
- **Sputum:** It is **foul-smelling** (due to anaerobic infections), thick, copious, tenacious, and continuously purulent, sometimes bloody.
- **Hemoptysis:** Streaks of blood is common with exacerbations of infection and is commonly recurrent. Rarely massive hemoptysis occurs. Hemoptysis occurs due to rupture of the thin-walled blood vessels present on the walls of dilated bronchi.
- **Pleuritic (chest) pain:** It may be caused due to infection of pleura, or due to segmental collapse caused by retained secretions.
- **Infective exacerbation:** Increased sputum volume with fever, malaise, and anorexia are precipitated by upper respiratory tract infections.
- **General debility:** In severe/widespread bronchiectasis, the patient presents with difficulty maintaining weight, anorexia, exertional breathlessness/dyspnea, wheezing, and orthopnea.
- **Bronchiectasis sicca/dry bronchiectasis:** Occasionally, the patient is asymptomatic or has nonproductive cough. It is termed bronchiectasis sicca and commonly follows TB of upper lobe. Only manifestation will be hemoptysis.
- Situs inversus is found in 50% cases of ciliary dyskinesia.

Physical findings

General examination: It may reveal anemia, **pandigital clubbing** (7% cases), fever, weight loss, night sweat, weakness, **halitosis** (may accompany purulent sputum) and sinusitis. Signs and symptoms of lung infection, such as fever may not be present.

Respiratory system

- Nasal polyps and signs of chronic sinusitis may be present.
- Signs may be unilateral, but are usually bilateral and basal. In dry bronchiectasis, no abnormal physical signs may be found.
- **Auscultation:** Reveals **crackles** and wheezing. Presence of large amounts of secretion is responsible for the characteristic **"bilateral, coarse, leathery crepitations"** of bronchiectasis which may be palpable (tactile fremitus).

Investigations

- **Blood:** Anemia, raised ESR and leukocytosis indicating suppuration. ABG studies may show respiratory alkalosis or hypoxemia.
- **Sputum examination:**
  - If sputum is collected in a conical flask and allowed to stand, it forms three layers (**"three-layered sputum"**), top mucoid layer, middle mucopurulent layer, and purulent layer at the bottom.
  - Stain the sputum by Gram's stain, Ziehl–Neelsen stain for acid-fast bacilli
  - *Culture and sensitivity:* Culture usually grows organism in the normal nasopharyngeal flora or *Pseudomonas*.
  - *Organisms associated with high mortality: Burkholderia cepacia, Stenotrophomonas maltophilia*
- **Chest radiograph (Fig. 2.32):** It lacks sensitivity. Signs on CXR include the identification of parallel linear densities, tram-track opacities, or ring shadows reflecting thickened and abnormally dilated bronchial walls.
- **Chest computed tomography (CT):** More specific and sensitive, and is the imaging modality of choice for confirmation for bronchiectasis. High-resolution CT **(HRCT) findings (Fig. 2.33)** are:
  - *Specific criteria:*
    - Thickened, dilated airways (parallel "tram tracks" or as the "signet ring sign"). Internal diameter of the bronchus is minimum 1.5 times more than that of the nearby vessel.

FIG. 2.32: Chest X-ray (CXR) showing bilateral bronchiectasis with dextrocardia with situs inversus.

- ♦ Absence of bronchial tapering in the periphery of the chest (presence of tubular structures within 1 cm from the pleural surface)
  - *Other findings:* Inspissated secretions (e.g., the tree-in-bud pattern) or cysts arising from the bronchial wall (in cystic bronchiectasis).
  - May suggest the etiology of bronchiectasis (e.g., proximal bronchiectasis suggests ABPA).
- HRCT is the investigation of choice
- Sensitivity: 82–97%
- False positive and negative rates of 1 and 2%
- 1–2 mm cuts at 10 mm intervals, reducing to 5 mm intervals at areas of particular interest
- The two main features are bronchial dilatation and bronchial wall thickening (*See* table on right):
  1. A bronchus is said to be dilated if its internal diameter is greater than that of the accompanying pulmonary artery (bronchoarterial ratio >1)
  2. Bronchial wall thickening is said to be present if the thickness of the wall is atleast equal to the diameter of the adjacent pulmonary artery branch

FIG. 2.33: High-resolution CT (HRCT) image of bronchiectasis.

Direct signs	*Indirect signs*
➢ Bronchial dilatation	➢ Bronchial wall thickening
➢ Signet ring sign	➢ Fluid or mucus filled bronchi
➢ Tram tracts sign	➢ Mosaic perfusion
➢ Varicose appearance	➢ Centrilobular nodules
➢ Air filled cysts	➢ Atelectasis/consolidation
➢ Lack of tapering > 2 cm distal to bifurcation	➢ Air trapping on expiratory scan
➢ Visibility of the peripheral airways thickening	

- **Sinus X-rays:** About 30% of patients have rhinosinusitis.
- **Bronchoscopy:** Does not establish the diagnosis.
  - *Indications:* (1) to identify the source of secretions, (2) to identify the site of bleeding in patients with hemoptysis, (3) therapeutically to remove secretions, and (4) localized bronchiectasis.
- **Bronchography:** Rarely indicated.
- **Pulmonary function tests:** It may detect mild to moderate airflow obstruction, but a restrictive pattern evolves with advanced disease.
- **Urine examination:** In advanced and chronic cases, proteinuria may develop due to renal amyloidosis.
- **Electrocardiogram:** Usually normal, but right ventricular hypertrophy may be detected when cor pulmonale develops.
- **Sweat electrolytes:** Measurement of sodium and chloride concentrations in sweat is useful in cystic fibrosis.
- **Serum immunoglobulins:** Up to 10% of adults with bronchiectasis have antibody class or subclass deficiency (mainly IgA). Its estimation is also useful when primary hypogammaglobulinemia is suspected.
- **Patients suspected of ciliary dysfunction syndrome:** Assessment of ciliary function may be done by several ways:
  - *Mucociliary clearance (nasal clearance of saccharin):* Measures the time taken for a small pellet of saccharin placed in the anterior chamber of the nose to reach the pharynx, where patient can taste it. Normally, it should be <30 minutes. A prolongation of this time (>60 minutes) is found in patients with ciliary dysfunction.
  - *Measurement of ciliary beat frequency:* Assessed by using biopsies taken from the nose.
  - *Electron microscopy:* It can detect structural abnormalities of cilia.
  - Study of the sperms.

TABLE 2.47: Complications of bronchiectasis.

➢ Hemoptysis	➢ Metastatic abscesses (e.g., brain abscess)
➢ Pneumonia	➢ Generalized edema (100 mL sputum/4–5 g protein)—protein loosing pneumopathy
➢ Lung abscess	➢ Generalized amyloidosis
➢ Empyema	➢ Aspergilloma
➢ Cor pulmonale	➢ Respiratory failure
➢ Septicemia	➢ Microbial resistance to antibiotics
➢ Meningitis	
➢ Osteomyelitis	

Complications of Bronchiectasis (Table 2.47)

Prognostic scores: Bronchiectasis severity score (BSI), FEV1, age, chronic colonization, extension, and dyspnea (FACED).

Severity scoring

FACED score includes:
- **F:** FEV1
- **A:** Age
- **C:** *Pseudomonas aeruginosa* colonization
- **E:** Extent of bronchiectasis
- **D:** Dyspnea
- Each variable is scored as 0,1 or 2
- The 5 years mortality in mild (0–2), moderate (3–4) and severe (5–7) disease is 4%, 25% and 56%

BSI score:
- Includes age, BMI, FEV1, previous hospitalization, exacerbation frequency, colonization status, radiological appearances
- The score was designed to predict future exacerbations, hospitalizations, health status and death over 4 years

Both scores have similar results regarding severity and prognosis. *Pseudomonas*, chronic pulmonary aspergillosis in pre-existing bronchiectasis lead to early decline in lung function.

Q. Write a short essay/note on complications of bronchiectasis.

Management: Principle of management rests in working on different arms of the *Vicious Cycle*

Goals of treatment of bronchiectasis is presented in **Box 2.10**.

Improvements in secretion clearance and bronchial hygiene

- Bronchial hygiene so as to reduce the microbial load within the airways and minimize the risk of repeated infections.
- Many methods are used to increase secretion clearance in bronchiectasis. These include chest physiotherapy (e.g., postural drainage), hydration and mucolytic administration, aerosolization of bronchodilators, and hyperosmolar agents (e.g., hypertonic saline), oscillatory PEEP device (ACAPELLA).
- **Postural drainage:** Postural drainage is valuable and consists of adopting a position in which the affected lobe(s) to be drained is uppermost. Patients must be trained by physiotherapists and this should be performed at least three times daily for 5–10 minutes. Lying over the side of the bed with head and thorax down is effective in most patients. Gentle mechanical chest percussion through hand clapping to the chest helps to dislodge the sputum.
- Bronchoscopic removal of inspissated secretions is rarely necessary.

Box 2.10: Goals of treatment in bronchiectasis.

- Treatment of the underlying cause/disorder
- Improvements in secretion clearance and bronchial hygiene
- Antibiotics to control active infections
- Anti-inflammatory therapy
- Reversal of airflow obstruction
- Surgery

Treatable causes:

- Obstructed airway on CT secondary to foreign body/compression improved with bronchoscopy
- **Cystic fibrosis:** CFTR modulators (Ivacaftor, lumicaftor)
- **Immunoglobulin deficiency:** Immunoglobulin replacement
- **ABPA:** Itraconazole with oral corticosteroids

Antibiotic therapy:

Oral/nebulized antibiotics: If ≥3 exacerbations per year. Exacerbation is defined if 3 out of 5 are present.

- A deterioration in cough and sputum volume or consistency for at least 48 hours
- Increase in sputum purulence
- Breathlessness or exercise intolerance
- Fatigue or malaise
- Hemoptysis for at least 48 hours
- Determination by a clinician that a change in bronchiectasis treatment is needed.

Suppressive antibiotics

- After resolution of an acute infection in patients with recurrences, the use of suppressive antibiotics may minimize the microbial load and reduce the frequency of exacerbations.
- Inhaled antibiotics are safe and effective, e.g., tobramycin, gentamicin
- *Monitoring of* QTc, GI effects, drug-drug interactions recommended with the above.

Eradication therapy: Recommended for Pseudomonas aeruginosa

Oral/IV antibiotic (sensitive antibiotic usually ciprofloxacin/amikacin) for 2 weeks followed by 3 months of nebulized antibiotic is recommended.

Treatment for exacerbation

- Choice of the antibiotic depends on the results of culture and sensitivity of sputum.
- If no specific pathogen is identified and the patient is not seriously ill, oral agents like amoxicillin, ampicillin, cotrimoxazole, tetracycline, one of the fluoroquinolones or a fixed combination of amoxicillin and clavulanic acid are recommended.
- More seriously ill patients with pneumonitis require parenteral antibiotics.
- **Duration of therapy:** Usually a 7 to 10-day course is sufficient. Few patients may need prolonged therapy for several weeks.

Anti-inflammatory therapy

- Control of the inflammatory response may be of benefit in bronchiectasis.
- Inhaled or oral steroids can worsen outcomes in non-CF bronchiectasis and not routinely recommended.
- Macrolides inhibit *Quorum sensing by P.aeruginosa* and has immunomodulatory effect.
- Macrolides should NOT be used until nontuberculous mycobacterial (NTM) infection is ruled out

Reversal of airflow obstruction

- Bronchodilators (β-adrenoreceptor agonists, anticholinergics) improve obstruction and help in clearance of secretion. They are useful in patients with demonstrable airflow limitation.
- Inhaled corticosteroids may be useful in some patients.

Surgical treatment

- It can be considered only in refractory cases.
- The procedure involved is excision of bronchiectatic areas. It is usually done in cases where the bronchiectasis is restricted to a single lobe or segment on CT.
- Indications of surgery:
  - Children or young adults with localized lesions who fail to respond to medical treatment.
  - Recurrent hemoptysis
  - Recurrent localized pneumonias
- Lung transplantation is considered in patients with advanced disease and respiratory failure.
- **Treatment of the hemoptysis:** Bed rest and antibiotics. Blood transfusion is given if necessary. Occasionally, fiberoptic bronchoscopy is needed to detect the source of bleeding. If the hemoptysis continues, embolization of bronchial artery is the treatment of choice. Surgical resection may be needed if embolization fails.

Newer Therapies

1. **Brensocatib:** Oral inhibitor of dipeptidyl peptidase 1 (DPP1) involved in activation of neutrophilic enzymes.
   - It delays time to exacerbation and can be used in prevention
2. **Biologicals:** Use of benralizumab/mepolizumab have been tried in type 2 inflammation phenotype
3. **Phage therapy:** Phage therapy has been used in difficult to treat *Pseudomonas, Mycobacterium abscessus* masiliense infection.

Other Measures

- **General management:** Graded exercise, routine deep breathing, and maintenance of good nutrition
- Vaccination
- Promptly treat episodes of sinusitis.
- Treat complicated ABPA with prednisolone and itraconazole.
- Mucolytic dornase (DNase) is recommended in cystic fibrosis-related bronchiectasis. It reduces viscosity of sputum by breaking down DNA released from neutrophils.

Management of bronchiectasis is summarized in **Flowchart 2.10**.

Pseudobronchiectasis

Q. Write a short note on pseudobronchiectasis.

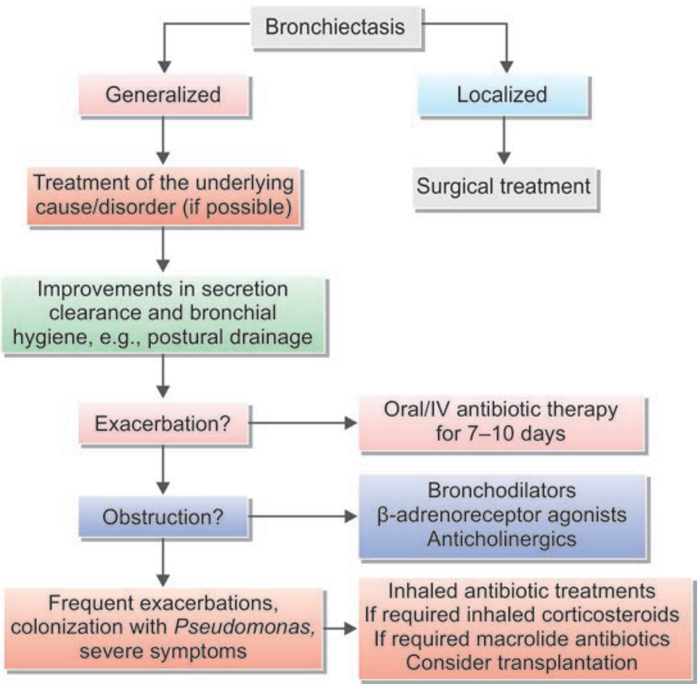
Pseudobronchiectasis (functional bronchiectasis) is characterized by dilated bronchi and is reversible. It is a common in patients with pneumonia of any cause. But re-expansion of the collapsed lung in atelectasis and regeneration of the mucosa in tracheobronchitis leads to reversal of the bronchographic findings. This reversible dilatation of bronchi is termed as pseudobronchiectasis.

Postobstructive Bronchiectasis

Q. Write a short note on postobstructive bronchiectasis.

- It is bronchiectasis that develops distal to a bronchial obstruction.
- **Causes:** Partial or total obstruction of the bronchial lumen due to endobronchial tumors, foreign body aspiration, mucus plugs, enlarged hilar lymph nodes or tumor masses, and bronchostenosis (due to endobronchial TB).

FLOWCHART 2.10: Management of bronchiectasis.



Bronchiectasis Sicca (Dry Bronchiectasis)

Q. Write a short note on bronchiectasis sicca (dry bronchiectasis).

- Usually, bronchiectasis presents with copious sputum.
- **Bronchiectasis sicca** is a condition where bronchiectasis presents with repeated episodes of hemoptysis without sputum production.
- It usually occurs in bronchiectasis of upper lobe following TB.

Atelectasis

Q. Write a short note on atelectasis and signs of lung collapse.

- Atelectasis refers either to **incomplete expansion of the lungs** (neonatal atelectasis) or to the **complete collapse** of previously inflated lung parenchyma.
- It produces areas of relatively airless pulmonary parenchyma.

Classification

Main types are acquired atelectasis:

A. Obstructive atelectasis (absorption atelectasis)

- Most common type
- **Mechanism:** Complete obstruction (intrabronchial) of an airway causes obstruction of communication between the alveoli and major airways. This leads to absorption of air from the dependent alveoli → diminished lung volume → shifting of the mediastinum toward the atelectatic lung.
- **Causes:** Due to complete obstruction (intrabronchial) of an airway.
  - Exogenous, e.g., foreign body aspiration, or recurrent aspiration of either gastric or oral contents due to a swallowing disorder
  - Endogenous, e.g., excessive secretions (e.g., mucus plugs) or exudates within smaller bronchi (e.g., in bronchial asthma, chronic bronchitis, bronchiectasis, and postoperative states), bronchial tumors.

B. Nonobstructive atelectasis

- **Compression atelectasis:**
  - Develops due to compression of lung.
  - *Causes:* It develops from any space-occupying lesion of the thorax (e.g., tumors, cysts, enlarged lymph nodes, and cardiomegaly). It can also occur with chest wall defects (e.g., scoliosis), neuromuscular diseases, and compression by emphysematous bulla.
  - In compression atelectasis, the mediastinum shifts away from the affected lung.
- **Relaxation or passive atelectasis:**
  - Contact between visceral and parietal pleura is lost resulting in passive atelectasis of lung.
  - *Causes:* Significant volumes of fluid (transudate, exudate or blood) or air (*pneumothorax*) accumulation within the pleural cavity.
- **Fibrotic or cicatrization or contraction atelectasis:** It occurs when focal or generalized pulmonary or pleural fibrosis prevents full expansion of lung.

C. Atelectasis due to surfactant deficiency or dysfunction

- Surfactant deficiency or dysfunction causes increased alveolar surface tension and failure to maintain small airway patency.
- **Causes:** ARDS (particularly in preterm neonates and meconium aspiration), and pneumonia in elderly.

Clinical Features

- Depends on the underlying cause, the degree of volume loss within the lung, and rate of volume loss.
- No symptoms may develop when atelectasis develops slowly.
- Chronic atelectasis may be a nidus of chronic purulent infection. This may damage bronchial wall leading to bronchiectasis.
- **Physical examination:**
  - Reduced chest movement during breathing on the involved region
  - Deviation of trachea and apex beat toward affected side
  - Dullness over the involved region
  - Absence of breath sounds over involved region. No added sounds.

Investigations

- Chest radiograph **(Fig. 2.34)** and CT chest findings
  - Homogenous opacification of the **atelectatic region**
  - Displacement of fissure
  - Loss of volume and shift of trachea and mediastinum
- Bronchoscopy to identify obstructive lesions and is also useful in removing the mucus plug.
- Arterial blood gas may reveal hypoxemia. Hypocapnia may develop due to tachypnea.

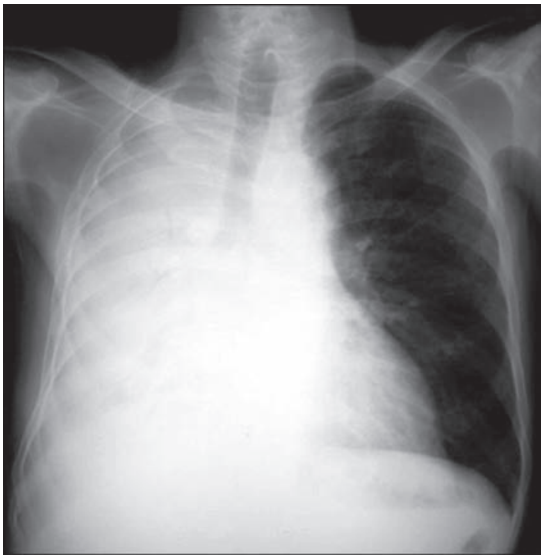


FIG. 2.34: Chest X-ray shows collapse of the right lung.

Treatment of collapse
- Emergency bronchoscopy and removal of mucus plug, foreign body
- Treatment of underlying cause

Middle Lobe Bronchiectasis (Middle Lobe Syndrome or Brock's Syndrome)

Q. Write a short note on middle lobe bronchiectasis (middle lobe syndrome or Brock's syndrome).

- It usually develops as a sequel of primary pulmonary TB of lung. It can be also seen with lymphoma and other causes of mediastinal lymphadenopathy.
- It is a type of postobstructive bronchiectasis usually due to obstruction of the middle lobe bronchus by tuberculous lymph nodes.
- This syndrome is characterized by recurrent atelectasis of the right middle lobe in the absence of any endobronchial lesion. Recurrent episodes of atelectasis result in bronchiectasis and fibrosis of right middle lobe.

When a middle-aged woman presents with MAC infection following right middle lobe or lingular lobe bronchiectasis, the patient is diagnosed to have LWS. This appears to be more common in thin females, who have never smoked, and who have no underlying pulmonary disease. Voluntary suppression of cough (hence, fastidious) in these women is hypothesized to cause reduced clearance of secretions from the right middle lobe and lingular segments, which have long and narrow bronchi with acute angulations, thus predisposing the patients to MAC infection.

Ciliary Dysfunction Syndromes or Primary Ciliary Dyskinesia

Q. Enumerate the clinical features, diagnosis, and treatment of ciliary dysfunction syndromes (ciliary dyskinesia syndromes).

Q. Write a short note on primary ciliary dyskinesia (PCD)/Kartagener's syndrome, and Young's syndrome.

- These are a group of genetic disorders characterized by dysfunction of cilia of the respiratory tract epithelium, sperms, and other cells. Dysfunction of cilia causes impairment of mucociliary clearance, left-right body asymmetry, and impaired sperm motility.
- Majority are transmitted as an autosomal recessive disorder.
- **Kartagener or immotile cilia syndrome:** It is an **autosomal recessive** syndrome. It is one of the ciliary dysfunction syndromes with **lack of ciliary function** (due to absence of inner or outer dynein arms of cilia) and causes retention of secretions and recurrent infections. It comprises of recurrent **sinusitis**, **bronchiectasis**, **dextrocardia** (with or without situs inversus), and **male infertility** (sperm dysmotility).
- **Young's syndrome:** Patients develop bronchiectasis, sinusitis, and obstructive azoospermia. The cause is not known.

TABLE 2.48: Clinical features of primary ciliary dyskinesia.

System involved	*Clinical features*
Lung	Respiratory distress (in neonates), recurrent infections, and bronchiectasis
Nasal sinus	Chronic sinusitis
Ear	Otitis media, cholesteatoma, hearing loss
Genital tract	Male infertility
Organ laterality	Situs inversus totalis, polysplenia or asplenia
Cardiovascular system	Complex congenital heart disease, vascular anomalies
Central nervous system	Hydrocephalus (rare)

Clinical Features of Primary Ciliary Dyskinesia

See **Table 2.48.**

Diagnosis

- Screening tests
  - **PICADAR** (**P**r**I**mary **C**ili**A**ry **D**yskinesi**A** **R**ule): Diagnostic prediction tool. Score ≥5 signification (Sn 90, Sp 75)
  - Exhaled nasal nitric oxide: Low (<100 mL/min)
  - **Mucociliary clearance (nasal clearance of saccharin):** Tests for ciliary function.
- **Electron microscopy:** It can detect structural abnormalities of respiratory cilia in samples of nasal or airway mucosa. It is characterized by defects in the outer or inner dyenin arms of the cilia.
- **Genetic study:** Shows mutation in PCD genes (*DNAI1, DNAH5* gene)

Treatment
- No definite treatment.
- Treat bronchiectasis.
- Avoid cough suppressants because cough is the only intact mechanism for mucociliary clearance in these patients.

Suppurative Pneumonias

Q. Write a short essay on suppurative pneumonias and necrotizing pneumonias.

Suppurative pneumonia, aspiration pneumonia, and pulmonary abscess are conditions which have overlap in their etiology and clinical features. Suppurative pneumonia is a type of pneumonic consolidation characterized by destruction of the lung parenchyma by the inflammatory process. Though histologically there is formation of microabscess, the term "lung/pulmonary abscess" is usually used for lesions having a large localized collection of pus, or a cavity lined by chronic inflammatory tissue. The etiological factors are same for both suppurative pneumonia and lung/pulmonary abscess.

Lung Abscess

Q. Define lung abscess. Describe the etiology, clinical features, investigations, diagnosis, complications, and management of lung abscess.

Definition: Lung (pulmonary) abscess is defined as a severe, **local suppurative process within the lung** associated with cavity formation. It is characterized by necrotic area of lung parenchyma containing **pus accompanied by the destruction of lung tissue. Necrotizing pneumonia:** Often used to describe similar pathologic process with multiple small (<2 cm) cavities in contiguous are as of the lung.

Classification

- **Duration of symptoms** prior to diagnosis: Acute <1 month, and chronic >1 month
- **Primary or secondary.**

Etiology (Table 2.49)

- **Infectious causes:** Any pathogen can produce lung abscess.
- Noninfectious causes.

Pathogenesis

Lung abscess may be **primary or secondary**.

A. **Secondary lung abscess:** Develops as a complication of several conditions.
   - **Complication of necrotizing pneumonia:** The microbes usually associated are: *Staphylococcus aureus*, *Klebsiella pneumoniae*, and the serotype type 3 *Pneumococcus.*
   - **Pulmonary TB:** Important cause of lung abscess.
   - **Septic embolism:** The source of embolus may be from thrombophlebitis in any part of the systemic venous circulation or from the vegetations of infective bacterial endocarditis on the right side of the heart. The embolus may be trapped in the lung causing multiple pyemic abscesses.

TABLE 2.49: Causes of lung abscess.

A. Infectious causes
1. Bacteria – **Usual:** Mouth flora anaerobes, most frequently isolated anaerobes: *Peptostreptococcus, Fusobacterium nucleatum, Prevotella melaninogenica* – **Less common:** *Staphylococcus aureus, Streptococcus pyogenes, Pseudomonas aeruginosa, Klebsiella pneumoniae, Streptococcus pneumoniae*, gram-negative bacilli, such as *Escherichia coli, Haemophilus influenzae type B, Legionella, Nocardia asteroides, Burkholderia pseudomallei, Burkholderia mallei, Rhodococcus sp.* Mixed infections occur when lung abscess develops due to inhalation of foreign material – **Mycobacteria:** *Mycobacterium tuberculosis, M. avium complex, M. kansasii*, other mycobacteria
2. Fungi: *Aspergillus* spp., *Histoplasma capsulatum, Pneumocystis jirovecii, Coccidioides immitis, Blastocystis hominis, Cryptococcus*
3. Parasites: *Entamoeba histolytica, Paragonimus westermani, Strongyloides stercoralis* (postobstructive)
B. Noninfectious causes
1. Neoplasms: Primary lung cancer, metastatic carcinoma, lymphoma **2. Pulmonary infarction:** Due to bland embolus (may be secondarily infected in <5%) **3. Septic embolism:** Tricuspid endocarditis due to *S. aureus* and others (typically with positive blood cultures), jugular venous septic phlebitis due to ***Fusobacterium necrophorum* (Lemierre's syndrome)** **4. Vasculitis:** Wegener's granulomatosis, rheumatoid lung nodule **5. Developmental:** Pulmonary sequestration **6. Airway disease:** Bullae, blebs, or cystic bronchiectasis (usually thin-walled) **7. Other:** Sarcoidosis, transdiaphragmatic bowel herniation giving appearance of cavity with air fluid level

- **Bronchial obstruction:** By a **bronchial cancer** (primary or secondary) or foreign body (postobstructive pneumonia) → secondary infection
- **Miscellaneous:**
  - Direct penetrating trauma to the lungs
  - Spread of infections from a neighboring organ (e.g., suppuration in the esophagus, spine, subphrenic space, or pleural cavity)
  - Spread from an amebic liver abscess
  - Secondary infection of cavitary malignancy, pulmonary infarct.

B. **Primary cryptogenic lung abscesses:** This type of lung abscess has no apparent cause and most of them develop as consequence of **aspiration of infected material**. About 80% of lung abscess is primary (50% of these associated with putrid sputum). Bacteria responsible are usually the mixed anaerobes found in the oropharynx or nasopharynx (aspiration abscess). Pre-dominant organisms found in aspiration abscess include anerobic organisms, streptococci and *Haemophilus influenzae.* Preexisting sources of infection for aspiration are sinusitis, dental sepsis, gingivitis, periodontal infection, etc.
- **Aspiration of infective material:** Depression of cough reflex favors aspiration from **infected nasal sinuses or tonsils** or periodontitis gingivitis. It may occur during alcoholic stupor, general anesthesia, sleep, epilepsy, coma, head injury or neurological disease-causing loss of consciousness.
- **Aspiration of gastric contents:** Occurs in a chalasia cardia, carcinoma of esophagus, hiatushernia, and gastro-esophageal reflux disease (chemical pneumonitis—Mendelson's syndrome).

C. Acute (<6 weeks) vs Chronic (>6 weeks)

D. Bronchogenic (Inhalation/aspiration) vs. Hematologic (bloodstream spread from other sites)

Features of aspiration abscess:
- **Site and number:** Abscess due to **aspiration** is more common on the **right** lung (because of the more vertical right main bronchus) and is most often **single.** RLL (apical) >RUL (postsegment) > LL (apical)
- **Segment involved:** Aspiration abscess cavities occur in those bronchopulmonary segments that are most dependent at the time of aspiration.
  - Aspiration in supine position → produces abscess in posterior segment of the upper lobes or superior segments of the lower lobes.
  - Aspiration in the upright position → produces abscess in the basilar segments (bilateral).
- About one-third of patients develop empyema due to direct extension.
- Amebic lung abscess typically occurs in right lower lobe due to direct extension of liver abscess through the diaphragm.

Clinical Features

Q. Write short essay on diagnosis of acute lung abscess.

Lung abscess may present either as an acute (symptoms <1 month) or chronic (symptoms >1 month).

- **Acute:**
  - Majority present acutely with dry **cough**, high-grade **fever**, chills, rigors, and pleuritic chest pain.
  - After a few days, when the abscess ruptures into a patent bronchus, the patient suddenly starts expectorating **large amounts of foul-smelling purulent or sanguineous sputum.** The sputum may often be blood-tinged and expectoration **varies with posture.**
- **Chronic:** Lung abscess secondary to aspiration often presents as chronic, insidious in onset with low-grade fever, malaise, weight loss, anorexia, and a deep-seated chest **pain**/discomfort.

Physical findings

General examination: Anemia, fever, **clubbing of the fingers and toes** (may develop rapidly), **halitosis,** and oronasal sepsis.

Respiratory system examination

- **Early stages:** May be normal
- **Later:**
  - **Signs of consolidation:** Dullness of percussion, increased vocal fremitus and vocal resonance, bronchial breathing, crepitations, and pleural rub.
  - **Signs of cavitation:** Once the abscess opens into a bronchus, signs of cavitation like cavernous or amphoric bronchial breathing and coarse post-tussive crepitations are heard on auscultation.

Investigations

- **Blood:** Normocytic anemia and/or raised inflammatory markers (ESR/CRP), leukocytosis, and raised ESR.
- **Sputum:**
  - Gram's stain, Ziehl–Neelsen staining for acid-fast bacilli
  - *Culture and sensitivity:* For aerobic and anaerobic
  - *Cytological examination:* For malignant cells.
- **Chest radiograph (Fig. 2.35):** It reveals radiolucency in an opaque area of consolidation. The wall of the abscess cavity completely surrounds the radiolucent area. An air-fluid level may be seen in the abscess cavity.
- **CT scan of thorax:** It shows lung abscess.
- **Bronchoscopy:** Indicated (1) to exclude malignancy, (2) to obtain specimens for studies, and (3) for removal of secretions or endobronchial obstructions.

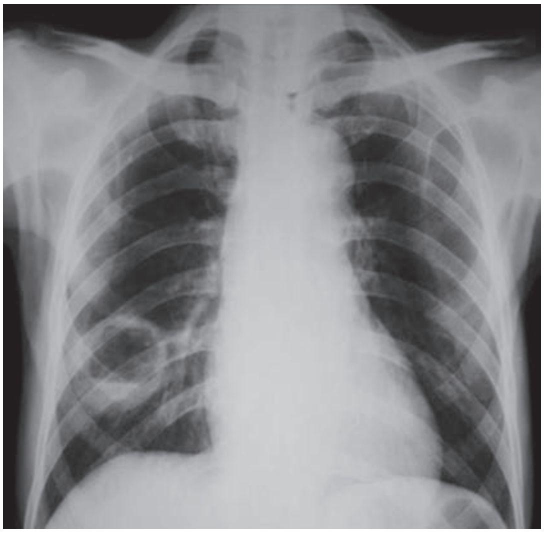


FIG. 2.35: Chest X-ray shows lung abscess in right lower zone.

Complications (Table 2.50)

Q. Write a short essay on management of acute lung abscess/ management of lung abscess.

Treatment

- **Postural drainage** (refer bronchiectasis) and chest physiotherapy.
- **Oral** chlorhexidine (periodontal disease commonly associated)
- Adequate **nutrition**
- **Antibiotic therapy:** Choice of drug depends on culture and sensitivity result. Broad guidelines include:
  - **Oral treatment:** Majority respond to oral treatment with ampicillin 500 mg four times daily or cotrimoxazole 960 mg twice daily or clindamycin 300 mg thrice daily.
  - **Anaerobic bacterial infection**, e.g., patients with foul-smelling sputum, oral metronidazole 400 mg 8 hourly should be combined with the above oral treatment. It should not be used alone.
  - **Nocardia:** TMP-SMX
  - **Parenteral antibiotic therapy:** Required in seriously ill patients and consists of beta-lactamase inhibitor (e.g., ampicillin-sulbactam 3 g intravenously every 6 hours) or a carbapenem (e.g., imipenem, meropenem) with clindamycin or metronidazole
- **Duration of antibiotic therapy:** Usually given for 4–6 weeks. Antibiotic treatment should be continued until the chest radiograph has shown either the resolution of lung abscess or the presence of a small stable lesion. There is a risk of relapse with shorter antibiotic regimen.
  - No response by day 10 of antibiotics must prompt evaluation for resistant organism or airway obstruction
- **Percutaneous drainage:** Beneficial in patients who are ineligible for extensive surgeries or critically ill
  - Ongoing sepsis despite antibiotics
  - Progressively enlarging in size
  - Failure to wean from mechanical ventilator
  - Contamination of opposite lung
- **Resectional surgery:** Indicated in few in cases. These include:
  - Massive hemoptysis
  - Lung abscess associated with symptomatic bronchiectasis
  - Lung abscess associated with localized malignancy
  - Persistent lung abscess cavity after 4–6 weeks of antibiotic
  - Complications (empyema /bronchopleural fistula)

The surgical procedure is either lobectomy or pneumonectomy. Consider lobectomy/pneumonectomy with large cavities (>6 cm), resistant organisms like *Pseudomonas*, obstructing neoplasm or massive hemorrhage.

TABLE 2.50: Complications of lung abscess.

Extension of the infection into the pleural cavity: Leading to empyema/pneumothorax/pyopneumothorax/bronchopleural fistula/pleural effusion/pleurocutaneous fistula
Hemorrhage into the abscess cavity
Hemoptysis
Septic emboli may cause metastatic brain abscesses or meningitis
Secondary amyloidosis (type AA)
Aspergilloma
Residual fibrosis and bronchiectasis

Differential diagnosis: Empyema and necrotizing pneumonia
Lung abscess: Round cavity on CXR, acute angle with chest wall, claw sign on CT
Empyema: Lenticular/biconvex on CXR, obtuse angle with chest wall, split pleura sign on CT (separation of parietal, visceral pleura)

Cystic Fibrosis

Q. Write a short note on complications of cystic fibrosis.

Cystic fibrosis is a fatal multisystem genetic disorder because of abnormal ion transport function causing inability to adequately hydrate mucus.

Genetics and Pathogenesis

It is transmitted as an autosomal recessive disorder and characterized by mutation in a gene on the long arm of chromosome 7. This gene codes for a chloride channel known as cystic fibrosis transmembrane conductance regulator (*CFTR*). This influences salt and water movement across epithelial cell membranes.

CFTR protein:
- Normally present in epithelia and functions as cAMP-regulated chloride ion channel and as inhibitor of Na^+ channels.
- Mutation in *CFTR* gene causes intracellular degradation of CFTR. Thus, epithelial membranes are unable to secrete chloride ion in response to cAMP-mediated signals.

Types of epithelia affected:
- **Volume-absorbing epithelia:** Airways and distal intestinal epithelium.
  - In cystic fibrosis, **Na^+ absorption increased and chloride ion secretion is decreased.** This leads to reduced volume of periciliary fluid (with relative dehydration of the airway epithelium), thickening of mucus, adhesion and failure to clear mucus from the airway lumen. Mucus stasis and mucus hypoxia predispose to chronic bacterial infection (favors *Pseudomonas* growth) and ciliary dysfunction. It can lead to bronchiectasis.
- **Salt-absorbing epithelia:** In the sweat duct epithelium, cystic fibrosis is associated with increased sodium and chloride content in sweat. Aquagenic wrinkling of the palms (wrinkling and nodules) that develop after several minutes of immersion in water is quite characteristic.
- **Volume secretary epithelia:** Epithelium of proximal intestine and pancreas
  - *Pancreas:* Failure of Cl-HCO_3 exchanger to secrete Na^+, HCO_3 and water → enzymes retained → steatorrhea, azotorrhea, and pancreatic destruction.
  - *Intestine:* Reduced bicarbonate secretion → low pH in duodenum → thick intestinal mucus → predisposition to obstruction.
  - *Hepatobiliary system:* Retention of biliary secretion, focal biliary cirrhosis, bile duct proliferation, chronic cholecystitis, and cholelithiasis.

Clinical Features

- Most patients present in infancy. Earliest presentation is **meconium ileus**. Pancreatic and intestinal manifestations occur early.
- Pulmonary manifestations occur late. After the neonatal period, maximum morbidity and mortality is due to pulmonary disease.

Respiratory tract
- **Upper respiratory tract:** Chronic sinusitis, rhinorrhea, and nasal polyps
- **Lower respiratory tract:**
  - Earliest functional abnormality is small airways disease and earliest symptom is cough. Final expression is bronchiectasis. Earliest and most severe changes in **right upper lobe**.
  - Persistent cough → viscous, purulent sputum. Intermittent exacerbations → increased cough, increased sputum volume, decrements in pulmonary function, weight loss. As exacerbations become more frequent, lung function deteriorates → eventually, respiratory failure.
  - *Pathogens:* In newly diagnosed patients include *H. influenzae* and *S. aureus* and in established diseases *Pseudomonas aeruginosa* (mucoid form).
  - *Chest X-ray:* Earliest manifestation is hyperinflation later bronchiectatic changes.
  - *Pulmonary function tests:* Obstructive pattern with partial bronchodilator response.
  - *Complications:* Pneumothorax, hemoptysis, clubbing, respiratory failure, cor pulmonale.

Gastrointestinal tract
- Most common is **exocrine pancreatic insufficiency:** Steatorrhea, azotorrhea → consequent malnutrition. Recognized only when secretion of amylase and lipase falls below 90%.

- Endocrine pancreatic insufficiency occurs in 10% much later.
- *Others:* Intestinal obstruction, appendicitis, recurring acute or chronic pancreatitis.
- Increased incidence of GI malignancy.

Genitourinary tract

- Late onset of puberty in both males and females.
- 95% azoospermic, 20% women infertile, 90% completed pregnancies produce viable infants; breastfeeding normal.
- Retardation of bone age, heatstroke.

Classes of mutation in cystic fibrosis

Normal	***Class I***	***Class II***	***Class III***	***Class IV***	***Class V***	***Class VI***
Defect	No synthesis	Processing	Gating	Conductance	↓production	Defective stability
Functional abnormality	No protein synthesis	Folding defect	Channel opening defect	Ion transport defect	Decreased protein synthesis	Decreased half life of protein synthesized
Mutation	G542X	F508	G551D	A117H	A455E	4326 Del
Type	Nonsense frameshift	Missense AA deletion	Missense (AA change)		Splicing defect	Missense (AA change)
Therapy	Read through agents	Corrector + Potentiators	Potentiators	Potentiators	Amplifiers NMD inhibitors Splicing modulators	Stabilizers
Drugs	Ivacaftor	Lumacaftor + Ivacaftor	Elexacaftor + Tezacaftor + Iva caftor		Ivacaftor	-

Diagnosis

- **Aquagenic wrinkling:** Screening test in children. Wrinkling of palms occur <3 mins on immersion in water.
- **Diagnosis** requires **1 clinical presentation** (any 1) + Evidence of **CFTR dysfunction**
  - *Clinical presentation:* (a) Positive newborn screening test, (b) Signs/symptoms of CF and (c) Family history
  - *CFTR dysfunction*
    - Sweat chloride (>60 mEq/L)
    - **CFTR genetic analysis:** 2 CF causing CFTR mutations
    - **CFTR physiologic testing:** Dysfunction positive.
- One or more characteristic clinical features + abnormal sweat chloride ions or nasal bioelectrical response [nasal transepithelial potential difference (TEPD)]. TEPD is the voltage across an epithelium, and is the sum of the membrane potentials for the outer and inner cell membranes. This test measures the salt (sodium and chloride) transport in and out of the cells in the nose in response to different salt solutions. The way nasal cells respond to the changing salt solutions can be used to make a diagnosis of cystic fibrosis.
- **Other disorders raised sweat chloride ions:** Ectodermal dysplasia, glycogen storage disorders, adrenal sufficiency, mucopolysaccharidoses, acute respiratory disorders (group, epiglottitis, and viral pneumonias), chronic respiratory disorders (alpha-1 antitrypsin deficiency, bronchopulmonary dysplasia).

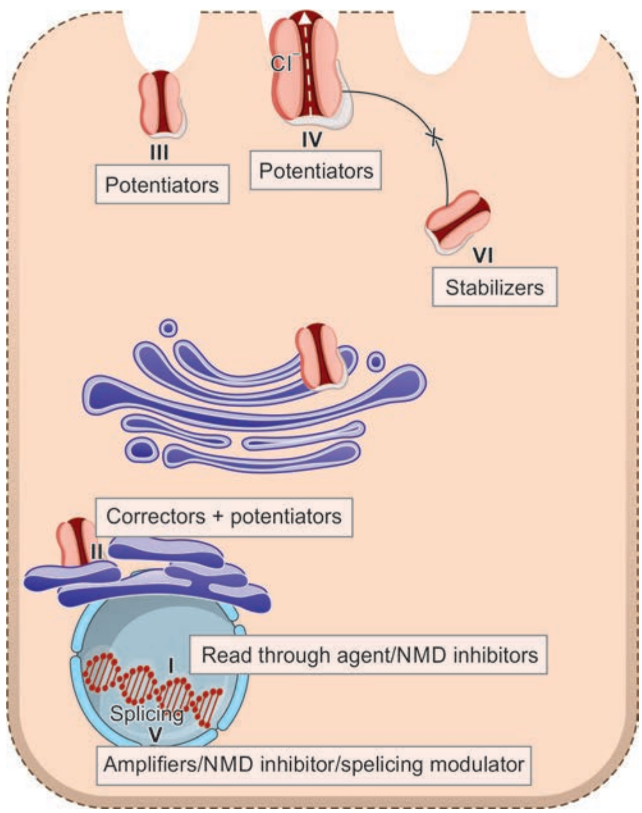


FIG. 2.36: Mutations in cystic fibrosis.

- Pilocarpine iontophoresis
- Sweat chloride ions >70 mEq/L distinguishing feature
- **Nasal TEPD:** Increased and on amiloride application → loss of this potential difference and with β-agonist → no response.
- DNA testing (for mutation) not useful because >1,000 mutations exist.

Treatment

- **Lung disease**
  - *Enzyme therapy:* Elexacaftor+tezacaftor+ivacaftor (triple therapy) preferred in ΔF508 mutations with age >12 years. Triple combination preferred to single agent. They reduce decline in FEV1
  - *Clear secretions:* By breathing exercises, flutter valves, chest percussion, and recombinant human DNase, oscillatory PEEP device (ACAPELLA)
  - *Infection:* Long courses of culture and sensitivity guided antibiotic therapy; higher doses required; oral/IV/aerosolized
  - *Inhaled β-agonists/anticholinergics:* Short-term benefit. O_2 and medical management are temporary measures
  - Long-term high-dose NSAID (some patients)
  - *Pulmonary rehabilitation:* Reduces exacerbations, improve airway clearance.
  - The only effective therapy for respiratory failure in cystic fibrosis (CF): Lung transplantation.
- **Gastrointestinal**
  - Pancreatic enzyme replacement (microsphere formulation)
  - Replacement of fat-soluble vitamins
  - Treatment of acute obstruction: Enema of hypertonic radiocontrast material (megalodiatrizoate).
- **Reproductive:** Assisted reproductive technology.

PLEURAL EFFUSION

Normal Composition of Pleural Fluid (Table 2.51)

Q. Discuss the causes/etiology, clinical features, investigations, radiological findings, diagnosis, pleural fluid analysis, complications, and management/treatment of pleural effusion.

Definitions

- **Pleural effusion:** Excessive accumulation of serous fluid within the pleural cavity/space (between parietal pleura and visceral pleura). It can be detected on X-ray when ≥300 mL of fluid is accumulated and clinically, when a minimum of 500 mL is present.
- **Empyema:** Accumulation of purulent fluid (frank pus) within the pleural cavity/space.
- **Hydrothorax:** Passive transudation of fluid into the pleural cavity. It occurs in congestive heart failure, nephrotic syndrome, cirrhosis of liver, severe malnutrition, etc.
- **Hemothorax:** Accumulation of blood within the pleural cavity/space.
- **Chylothorax:** Accumulation of chyle within the pleural cavity/space.

TABLE 2.51: Normal composition of pleural fluid.

Feature	*Normal value*
➢ Volume	0.1–0.2 mL/kg
➢ Cells	1,000–5,000/mm^3
– Mesothelial cells	3–70%
– Monocytes	30–70%
– Lymphocytes	2–30%
– Granulocytes	~ 10%
– Eosinophils	0%
➢ Protein	1–2 g/dL
➢ Albumin	50–70%
➢ Glucose	~ plasma level
➢ Lactate dehydrogenase (LDH)	<50% plasma level

Mechanism of Pleural Effusion

- Increased hydrostatic pressure (e.g., left ventricular failure)
- Decreased oncotic pressure in microcirculation (e.g., hypoalbuminemia)
- Decrease in pleural pressure (e.g., atelectasis)
- Increased permeability of microcirculation (e.g., pneumonia)
- Impaired lymphatic drainage from pleural space (e.g., malignancy)
- Movement of fluid from abdomen to pleural space (e.g., cirrhosis).

TABLE 2.52: Classification and causes of pleural effusion.

Types of effusion	*Causes*	
Transudative effusion	➢ Cardiac failure ➢ Hypoproteinemia (e.g., nephrotic syndrome, cirrhosis of liver, severe malnutrition) ➢ Constrictive pericarditis	➢ Hypothyroidism ➢ Meigs' syndrome (benign ovarian tumors with ascites and pleural effusion) ➢ Peritoneal dialysis
Exudative effusion	➢ Tuberculosis ➢ Bacterial pneumonia ➢ Malignancy ➢ Pulmonary infarction ➢ Autoimmune diseases (e.g., rheumatoid arthritis, systemic lupus erythematosus) ➢ Acute pancreatitis ➢ Postmyocardial infarction syndrome (Dressler's syndrome)	➢ Drug-induced effusion ➢ Benign asbestos-related effusion ➢ Intra-abdominal abscess ➢ Meigs' syndrome (can be transudative as well) ➢ Ruptured amebic liver abscess, chylous pleural effusion ➢ Acute rheumatic fever

Classification and Causes

See **Table 2.52**

Q. Write a short essay/note on:

- **Causes of pleural effusion.**
- **Causes and diagnosis of exudative pleural effusion.**
- **Causes of transudative pleural effusion.**

Q. Write a short essay/note on causes of left-sided pleural effusion.

Causes of left-sided pleural effusion is listed in **Box 2.11**.

Box 2.11: Causes of left-sided pleural effusion.

- Pancreatitis
- Pericardial inflammation
- Rupture of esophagus
- Left-sided subdiaphragmatic abscess
- Hepatic hydrothorax (20%)
- Thoracic duct involvement above D_3 level

Clinical Features

- Symptoms (pain on inspiration and coughing) and signs of pleurisy (a pleural rub): They often precede the development of a pleural effusion.
- **Breathlessness:** It may be the only symptom and its severity depends on the size and rate of accumulation of fluid.

Physical Findings in the Chest

Q. Write a short essay/note on signs of pleural effusion.

- **Inspection: Tachypnea.** Shift of trachea to opposite side.
- **Palpation:**
  - **Shift of trachea and mediastinum** (shift of apex beat) to the opposite side
  - **Reduced chest movements** on the affected side, bulging of the intercostal spaces, fullness of the affected chest, and markedly reduced vocal fremitus
  - **Measurements:** Diminished chest expansion, increase in the size of the affected hemithorax and an increase in spinoscapular distance.
- **Percussion:**
  - **Stony dullness** over the fluid. Upper level of the dullness is highest laterally in the axilla, and is lower anteriorly and posteriorly (**Ellis-S-shaped curve**).
  - **Small effusions:** When there is a small effusion, it may be detected as follows:
    - **Left-sided** small pleural effusion may be detected only by the obliteration of Traube's space on percussion.
    - **Right-sided** small effusion may be detectable only by tidal percussion.
  - **Moderate to large effusions:** Percussion reveals a triangular area of dullness or impaired note over the back of chest on the contralateral side or opposite side of the effusion. It may be due to shift of the posterior mediastinum to the opposite side by effusion.
    Causes of pleural effusion without tracheal mediastinal shift are listed in **Box 2.12**.

Box 2.12: Causes of pleural effusion without trachea/mediastinal shift.

- Minimal effusion
- Loculated effusion
- Bilateral effusion
- Effusion with underlying collapse/fibrosis
- Effusion with fixed mediastinum (malignancy/fibrosis)

- **Auscultation:**
  - **Breath sounds: Intensity is markedly diminished** or absent over the fluid.
  - **Vocal resonance: Markedly diminished** over the fluid.
  - **Bronchial breathing or crackles** the level of a pleural effusion
  - **Occasional findings:** Egophony and enhanced breath sounds can often be appreciated at the superior border of the effusion because of underlying atelectatic lung tissue.

Q. Write a short note on Grocco's sign.

Grocco's sign (Grocco's triangle; paravertebral triangle of dullness): It is a triangular area of paravertebral dullness on the side opposite a pleural effusion. Boundaries of Grocco's triangle are:

- **Medial boundary:** Midspinal line from the upper level of effusion down to the level of the 10th thoracic vertebra.
- **Lower boundary:** A horizontal line of about 3–7 cm extending laterally from the 10th thoracic vertebra, along the lower limit of lung resonance.
- **Lateral boundary:** A curved line connecting the above two lines.

Investigations

Q. Write a short note on radiological findings/X-ray features in pleural effusion.

- **Radiological investigations**
  - *75 mL of pleural fluid:* Subpulmonic space without spillover can obliterate the posterior costophrenic sulcus.
  - *Upright chest radiograph*
    - 175 mL is required to obscure the lateral costophrenic sulcus.
    - 500 mL of fluid will obscure the diaphragmatic contour.

- ♦ 1,000 mL of effusion reaches the level of the 4th anterior rib.
- ♦ **Decubitus radiographs and CT scans:** Less than 10 mL (even as little as 2 mL) can be identified.
- ■ Small effusions are thinner than 1.5 cm, moderate effusions are 1.5–4.5 cm thick, and large effusions exceed 4.5 cm.
- ■ Effusions thicker than 1 cm are usually large enough for sampling by thoracentesis, since at least 200 mL of liquid is already present.
- ■ A significant pleural effusion is large enough to produce a pleural fluid strip >10 mm wide on lateral decubitus radiographic views. Radiological features of pleural effusion in an erect chest film **(Fig. 2.37)** are as follows:
  - ♦ **Shift of mediastinum** to the opposite side
  - ♦ **Obliteration of costophrenic angle**
  - ♦ **Opacity:** Dense uniform opacity in the lower and lateral part on the involved side. Upper border of the opacity is concave upward and is highest laterally.

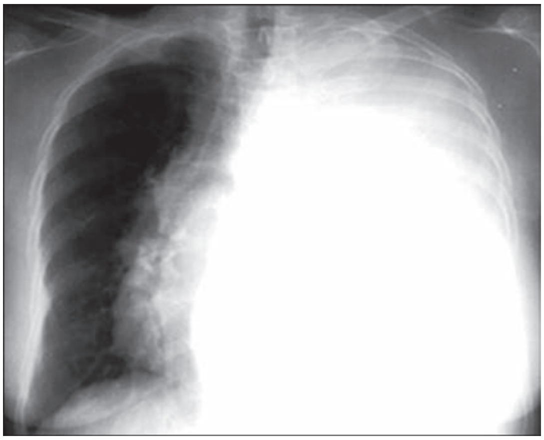


FIG. 2.37: Chest X-ray shows left-sided massive pleural effusion.

- ■ *Interlobar effusion:* Wider than normal interlobar fissure.
- ■ *Encysted interlobar effusion:* Round opacity resembling solitary pulmonary nodule (phantom tumor).
- **Ultrasonography**
  - ■ More accurate than plain chest X-ray for detection of pleural effusion and it can detect as little as 5 mL of effusion.
  - ■ **Interpretation**
    - ♦ **Transudate versus exudate:** A clear hypoechoic space favors transudate and the presence of moving floating densities suggests an exudate.
    - ♦ Presence of septation favors an evolving empyema or resolving hemothorax.
    - ♦ Useful in differentiating loculated pleural effusion from pleural tumor or pleural thickening.
    - ♦ Useful for locating an effusion prior to aspiration and biopsy.
    - ♦ Presence of static air bronchograms, plankton sign suggest complicated parapneumonic effusion
    - ♦ Detection of solid pleural abnormalities may suggest pleural malignancy. However, a CT scanning is indicated where malignancy is suspected.
- **Pleural aspiration and fluid analysis.**

Pleural Fluid Analysis

Q. How will you differentiate transudative pleural effusions from exudative pleural effusions?

Q. Write a short essay/note on criteria for distinguishing pleural transudate from exudate.

Q. Write a short essay/note on pleural fluid analysis in a typical pleural effusion.

Differentiation of transudative from exudative effusion **(Table 2.53)**.

TABLE 2.53: Differentiation of transudative from exudative effusion.

Characteristics	*Transudative effusion*	*Exudative effusion*
Cause and mechanism	Noninflammatory process. Ultrafiltrate of plasma, due to increased hydrostatic pressure or decreased serum oncotic pressure with normal vascular permeability	Inflammation process and is rich in proteins due to increased vascular permeability
Appearance	Clear, serous	Cloudy/purulent/hemorrhagic/chylous
Color	Straw yellow	Yellow to red
Specific gravity	<1.018	>1.018
Protein		
➢ Absolute value	Low, <2 g/dL, mainly albumin	High, >2 g/dL
➢ Pleural fluid: serum ratio	<0.5	>0.5
Clot	Absent	Clots spontaneously because of high fibrinogen
Leukocytes		
➢ Total leukocytes	<1,000/mm³	>1,000/mm³
➢ Type of cells	>50% lymphocytes or mononuclear cells	>50% lymphocytes (tuberculosis, malignancy)
Differential leukocytes	mesothelial cells	>50% polymorphs (acute inflammation)
➢ Erythrocytes	<500/mm³	Variable

Contd...

Contd...

Characteristics	***Transudative effusion***	***Exudative effusion***
Bacteria	Absent	Usually present
Lactate dehydrogenase (LDH)		
➢ Absolute value	<200 IU/L	200 IU/L
➢ Pleural fluid LDH: Serum LDH ratio	<0.6	>0.6
Glucose	>60 mg/dL (usually same as in blood)	<60 mg/dL (variable)

Interpretation of Pleural Fluid Parameters (Table 2.54)

Light's criteria

TABLE 2.54: Interpretation of pleural fluid parameters.

Parameter	***Interpretation***
Appearance of pleural fluid	Putrid odor (anaerobic empyema), food particles (esophageal rupture), bile-stained (chylothorax/biliary fistula), milky (chylothorax/pseudochylothorax), anchovy sauce-like fluid (ruptured amebic abscess)
Pleural fluid glucose concentration	
➢ Low glucose concentration (<60 mg/dL)	Suggests empyema, malignancy or tuberculosis
➢ Very low glucose concentration (<15 mg/dL)	Empyema, rheumatoid effusions
Pleural fluid eosinophilia (>10% of all cells)	May be observed in resolving infections, pneumothorax, hydropneumothorax, hemothorax and asbestos-related pleural effusion, dantrolene, bromocriptine, nitrofurantoin, paragonimiasis or Churg–Strauss syndrome
Pleural fluid erythrocyte counts >100,000/mm^3	Most often in malignancy or pulmonary infarction/embolism, but may result from a traumatic tap
Low pH of pleural fluid <7.2	Complicated parapneumonic effusion, esophageal rupture, rheumatoid pleuritis, tuberculous pleuritis, malignant pleural disease, hemothorax, systemic acidosis, paragonimiasis, lupus pleuritis, urinothorax
Raised pleural fluid amylase	Pancreatic diseases and esophageal rupture. However, routine amylase estimation is not recommended unless the clinical features suggest either of the two diseases
Pleural fluid antinuclear antibody titers or rheumatoid factor	No diagnostic significance and is not indicated in most cases
Mesothelial cells	Absent: Tuberculosis Markedly increased: Pulmonary embolism

Q. Write a short essay/note on Light's criteria for distinguishing pleural transudate from exudate.

- Light's criteria **(Box 2.13)** are used to differentiate exudative from transudative pleural effusion by measuring the lactate dehydrogenase (LDH) and protein levels in the pleural fluid. Exudative pleural effusions must meet at least one of the following criteria, whereas transudative pleural effusions meet none.
- **Precautions:**
  - **Low specificity:** These criteria are highly sensitive for exudative effusions but have lower specificity (i.e., few transudative effusions will be classified as exudative using these criteria).
- In general Light's criteria, occasionally misidentify a transudative effusion as an exudative effusion as in cardiac failure with diuretic therapy (25%).

Patient suspected to have transudative effusion, but meets Light's criteria for an exudative effusion, serum-pleural fluid albumin gradient (SPAG) or serum-pleural protein gradient is calculated **(Roth's criteria).**
- Serum-effusion albumin gradient of >1.2 g/dL—transudative. SPAG is more sensitive and preferred
- Serum-effusion protein gradient >3.1 g/dL—transudative

Box 2.13: Modified Light's criteria for distinguishing pleural transudate from exudate.

Modified Light's criteria

Pleural fluid is an exudate if one or more of the following criteria are met:

Pleural fluid
- Serum protein ratio >0.5
- Serum LDH ratio >0.6
- LDH >2/3 upper limit of normal serum LDH
- Protein >30 g/L

If only one of the above criteria is met, then calculate the fluid to serum albumin gradient
- If the albumin gradient >12 g/L, consider a transudate

Sensitivity and specificity of tests to distinguish exudative from transudative effusions

	Sensitivity for exudates (%)	***Specificity for exudates (%)***
Light's criteria	98	83
Pleural-fluid cholesterol level >60 mg/dL	54	92
Pleural-fluid cholesterol level >43 mg/dL	75	80
Ratio pleural-fluid cholesterol/serum cholesterol >0.3	89	81
Serum albumin level minus pleural fluid albumin level <1.2 g/L	87	92

(LDH: lactate dehydrogenase)

Source: Modified with permission from Light RW. Pleural effusion. Engl J Med. 2002;346:1971-7.

Causes of Lymphocytic Pleural Effusion (Table 2.55)

- **Pleural biopsy is indicated in undiagnosed cases.**
- **Other investigations in pleural effusion**
  - *Pleural fluid tumor markers:* Carcinoembryonic antigen (CEA), cancer antigen 125 (CA 125), cancer antigen15–3 (CA 15–3) and cytokeratin 19 fragments (CYFRA). They are not used in routine investigations of pleural effusion.
  - *Blood examination:* Total and differential leukocyte counts, ESR, proteins, sugar, LDH, amylase, rheumatoid factor and antinuclear factor.
  - *Sputum examination:* For tubercle bacilli and malignant cells.
  - Mantoux test
  - *Repeat radiograph:* When effusion is massive, a repeat radiograph after removal of a large volume of fluid may reveal an underlying parenchymal lesion.
  - Biopsy or fine-needle aspiration of scalene lymph nodes
  - Bronchoscopy and biopsy
  - Thoracoscopy and biopsy.

TABLE 2.55: Causes of lymphocytic pleural effusion.

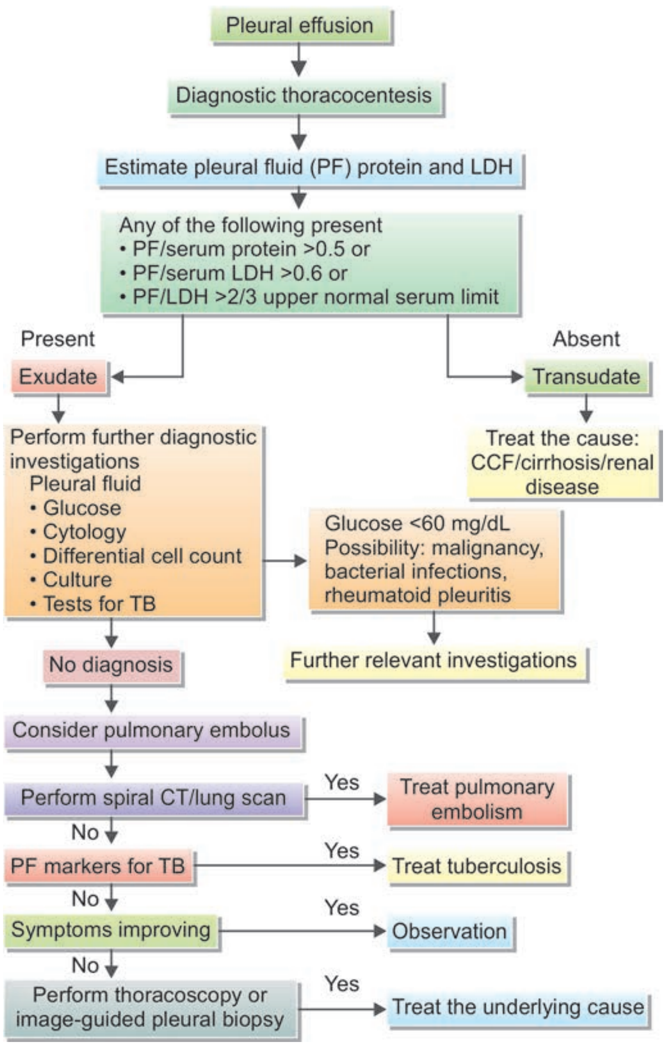
Malignancy ➢ Primary lung cancers ➢ Secondaries	Long-standing congestive cardiac failure
Tuberculosis	Effusion associated with rheumatoid arthritis
Lymphoma	Chylothorax, uremic pleuritis

Management of pleural effusion
- **Treat the underlying cause** (e.g., heart failure, pneumonia, pulmonary embolism or subphrenic abscess).
- **Therapeutic aspiration:** It may be necessary to relieve breathlessness/dyspnea. At one sitting not >1 L should be removed because it is associated with a small risk of re-expansion pulmonary edema. If there is rapid re-accumulation of fluid insertion of chest tube is necessary.
- **Drainage:** If the fluid is purulent (empyema).
- **Pleurodesis:** For malignant effusion.

Approach to the Diagnosis of Pleural Effusions

See **Flowchart 2.11.**

FLOWCHART 2.11: Approach to the diagnosis of pleural effusions.



Re-expansion Pulmonary Edema

Q. Write a short essay/note on re-expansion pulmonary edema.

- It occurs in small number of patients when the lung undergoes rapid inflation of lung (e.g., evacuation of >1 L of air/fluid) after a prolonged period of collapse (usually >3 days) from either pleural effusion or pneumothorax.
- Edema usually involves the entire re-expanded lung. Occasionally, it may involve a single lobe or the contralateral lung, or bilateral.

Mechanism: Exact mechanism is not known. Following may be implicated in its development.

- **Reactive oxygen species:** Ventilation and reperfusion of a previously collapsed lung may cause an inflammatory response, with generation of reactive oxygen species and superoxide radicals, which leads to increased capillary permeability.
- **Increase pulmonary hydrostatic pressure:** It may be due to increased venous return.
- **Pressure-induced mechanical disruption** of alveolar capillaries
- Reduced levels of functional surfactant
- Increased pressure across the capillary-alveolar membrane due to bronchial obstruction
- Alteration of lymphatic clearance.

Clinical Features

- **Onset:** Symptoms usually develop within 24 hours (majority within 1–2 hours after lung re-expansion).
- **Symptoms:** Chest discomfort, persistent severe cough, dyspnea and frothy sputum.

- **Signs:** Tachypnea, tachycardia, and crackles on the affected side of the lung.
- **Chest radiograph:** Features of pulmonary edema.
- Mortality rate is about 20%.

Treatment

- **Supportive treatment:** Most patients recover completely within 5–7 days with supportive treatment (includes oxygen administration in mild cases to mechanical ventilation in severe cases). Patient is advised to lie on unaffected side.
- **Others:** Use of diuretics, bronchodilators, prostaglandin analogs (e.g., misoprostol), ibuprofen, and steroids is controversial.

Prevention of Re-expansion Pulmonary Edema

- Restricting the drainage of pleural fluid to <1 L.
- Avoid routine use negative suction in pneumothorax/effusion
- Large volumes of fluid can be drained safely if pleural pressures are monitored. If the patient complains of vague chest pressure during thoracentesis, it suggests a sudden drop in intrapleural pressure, and the thoracentesis should be stopped.

Subpulmonic Effusion

Q. Write a short essay/note on subpulmonic effusion. How will you diagnose it?

- It is a small pleural effusion which is seen underneath the lung. Its detection is extremely difficult.
    - Left-sided subpulmonic effusion may show obliteration of Traube's semilunar space of tympany.
    - Right-sided subpulmonic effusion may be suspected with an abnormal tidal percussion.
- Chest radiograph:
    - Posteroanterior view in the erect posture
        - Mild elevation of hemidiaphragm
        - Lateral displacement and slight flattering of the dome of the diaphragm
        - Wide density between the gastric air shadow and upper border of the diaphragm in left-sided subpulmonic effusion
    - Lateral decubitus view with the affected side down: Pleural fluid layering out along the lateral chest wall
- **Ultrasonography:** Reveals subpulmonic effusion with more certainty.

Hemorrhagic Pleural Effusion

Q. Write a short note on common causes of hemorrhagic pleural effusion.

Causes of hemorrhagic pleural effusion **(Table 2.56)**

TABLE 2.56: Causes of hemorrhagic pleural effusion.

➢ Malignancy	➢ Tuberculosis (very rare)
➢ Pulmonary infarction	➢ Postcardiac injury effusion
➢ Asbestos-related pleural effusion	

Q. Write a short note on hemothorax.

Causes of hemothorax

- Trauma
- Rupture of a blood vessel
- Rupture of a tumor

Differentiation between hemorrhagic pleural effusion and hemothorax:

- Differentiation of hemothorax from hemorrhagic pleural effusions can be done by performing a hematocrit on the pleural fluid.
- If the pleural fluid hematocrit is >50% of the patient's peripheral blood hematocrit, it is considered to be diagnostic of a hemothorax.

Malignant Pleural Effusion

Q. Write a short note on malignant pleural effusion.

Effusion secondary to metastatic disease is called malignant pleural effusion.

Causes

- Malignant pleural effusions may be either due to a primary malignancy of pleura (e.g., mesothelioma) or secondary invasion of pleura by primary malignancy elsewhere in the body (more common). Most common malignancy-associated effusions occur with cancer of lung (35%), breast cancers (25%), and lymphomas (10%).
- **Mechanism:**
    - **Local effects:** Malignant effusion develops usually due to the local effects of the tumor, e.g., lymphatic obstruction, bronchial obstruction with pneumonia or atelectasis.
    - **Systemic effects** of tumor can also produce effusion.

Causes of dyspnea in patients with malignancy **(*Table 2.57*)**

TABLE 2.57: Causes of dyspnea in patients with malignancy.

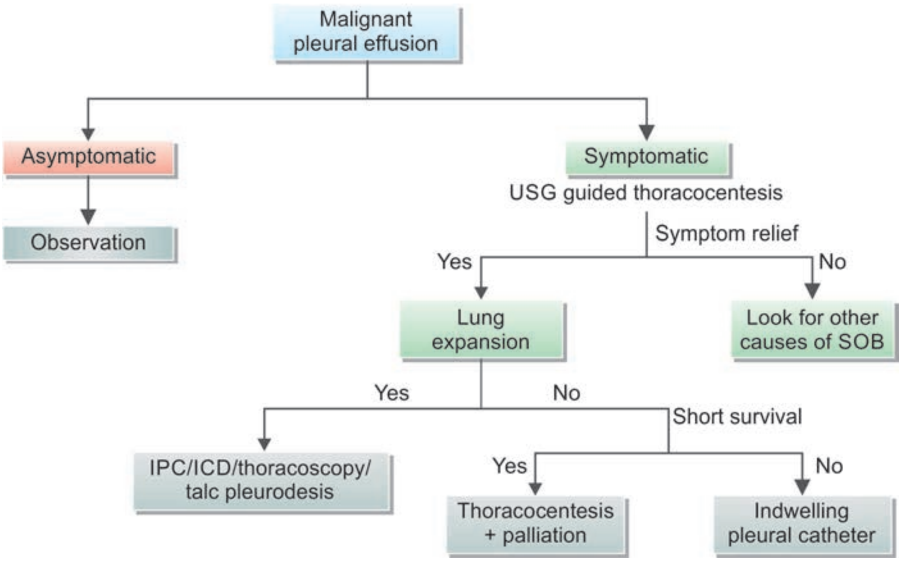
➢ Pleural effusion	➢ Underlying lung disease
➢ Direct effects of a tumor on the lung	➢ Pulmonary vascular disease (e.g., pulmonary emboli)
➢ Airway obstruction	➢ Lymphangitis carcinomatosis
➢ Chest wall invasion	➢ Radiation pneumonitis
➢ Pericardial disease	➢ Cardiac compromise

Characteristic features of malignant pleural effusion:

- Effusion is usually unilateral, massive and often symptomatic.
- Rapid reaccumulation of fluid after aspiration
- **Pleural fluid:**
  - Often hemorrhagic, with a high erythrocyte count (>100,000/mm^3)
  - Shows the characteristics of an exudates such as high protein content (>3.0 g/dL), high LDH levels (>200 IU/L), low glucose levels (<60 mg/dL), and a total leukocyte count exceeding 1,000/mm^3
  - Predominant cells are lymphocytes (>50%).
  - Cytological examination may show malignant cells.
- *Closed pleural biopsy:* It may reveal malignant tumor (in about 40% of cases).

Treatment (Flowchart 2.12)

FLOWCHART 2.12: Treatment of malignant pleural effusion.



(SOB: Shortness of breath; IPC: Indwelling pleural catheter; ICD: Intercostal drain)

Parapneumonic Effusion (Synpneumonic Effusion)

Q. Write a short note on parapneumonic effusion (synpneumonic effusion).

Parapneumonic effusion is a pleural effusion developing as a complicating pneumonia or lung abscess or bronchiectasis.
Synpneumonic effusion is used to describe a sterile parapneumonic effusion in general.
Empyema refers to a grossly purulent pleural effusion.

- More frequently associated with bacterial pneumonias (especially gram-negative and pneumococcal). Effusion in these cases is large and consists of predominant polymorphonuclear leukocytes.
- Less frequent with viral pneumonias and this effusion is small and predominant shows lymphocytes.

Common Organisms

- **Community acquired:** GPC (*S. viridans, Pneumococcus*, MSSA), GNR (*Enterobacteriaceae, Pseudomonas*), anerobes. Mortality—17%
- **Hospital acquired:** GPC (*S. aureus, Enterococcus, Viridians*), GNR (*Enterobacteriaceae, Pseudomonas*), anerobes. Mortality—47%

Clinical Presentation

- **Acute illness:** Aerobic bacterial pneumonia with pleural effusion presents with an acute illness with fever, chest pain. sputum, and leukocytosis.
- **Subacute illness:** Anaerobic infections present with a subacute illness with weight loss, leukocytosis, mild anemia, and predisposing factor that has resulted in aspiration pneumonia.

Categories of Parapneumonic Effusion (Table 2.58)

TABLE 2.58: Parapneumonic pleural effusion and empyema: Light's classification and corresponding treatment.

Category	*Type*	*Characteristics*	*Treatment*
1.	Nonsignificant	<1 cm thick on an ipsilateral decubitus view. Thoracocentesis not required	Antibiotic
2.	Typical parapneumonic	>1 cm thick, glucose >40 mg/dL pH > 7.20, negative Gram's stain and culture	Antibiotic + consider therapeutic thoracentesis
3.	Borderline complicated	pH 7–7.20 or LDH >1,000 Negative Gram's stain and culture	Antibiotic + pleural drainage tube + consider fibrinolytics
4.	Simple complicated	pH <7.0, positive Gram's stain or culture. Not loculated, no pus	Antibiotic + pleural drainage tube + fibrinolytics
5.	Complex complicated	pH <7.0, positive Gram's stain or culture. Multiloculated	Antibiotics + pleural drainage tube + fibrinolytics + consider VAT
6.	Simple empyema	Frank pus. Single loculation or free-flowing fluid	Antibiotics + pleural drainage tube + fibrinolytics + consider VAT
7.	Complex empyema	Frank pus. Multiple loculations. Often requires decortication	Antibiotics + pleural drainage tube + fibrinolytics + VAT + other surgical procedures if VAT fails

(LDH: Lactate dehydrogenase; VAT: Video-assisted thoracoscopy)

Simplified Classification and Stages of Parapneumonic Effusion (Table 2.59)

TABLE 2.59: Simplified classification and stages of parapneumonic effusion.

	Stage 1	*Stage II*	*Stage III*
	Exudative	Fibropurulent	Organising
pH	>7.20	<7.20	
Glucose	>40	<40	
LDH	<1000	>1000	
Microbilogy	Negative c/s staining	Pus + or Organism+	

Investigations

Presence of pleural fluid can be demonstrated with a lateral decubitus radiograph, computed tomography (CT) of the chest, or ultrasound.

Newer biomarkers in pleural fluid

- **Pleural CRP:** High negative predictive value in parapneumonic effusion
- **Pleural procalcitonin:** Routine use not recommended. May be used for infection in malignant pleural effusion
- **Pleural calprotectin:** Low in infection, elevated in malignancy, pulmonary embolism
- **Pleural soluble urokinase plasminogen activator receptor (SuPAR):** This enzyme plays an important role in development of pleural loculations. >35: Tube thoracostomy, >65: Surgical intervention

Treatment

- When there is **mild** parapneumonic effusion, patient may be **observed** and a repeat tap is done if it persists or increases.
- Pleural fluid should be directly inoculated into culture bottle (increases positivity rate 40→ 60%)
- Pleural LDH/ADA >15.5 differentiates parapneumonic effusion from empyema.
- **Antibiotics:** Duration 2–6 weeks. Organisms to be covered anerobes, Gram+ cocci, Gram -ve rods (GNR)
  - Community acquired: Ceftriaxone + Metronidazole
  - Hospital acquired: Additional *Pseudomonas* + MRSA coverage

  (Piperecillin-Tazobactum + vancomycin) OR (linezolid + Piperecillin-Tazobactum) OR (vancomycin + carbapenem)
  (Vancomycin + metronidazole + anti-*pseudomonas* (3rd/4th gen cephalosporins – cefepime/ceftazidime)
  Switch to oral antibiotic once clinical improvement noted (no fevers, ↓CRP, radiography)
- If the free fluid separates the lung from the chest wall by >1 cm, a **therapeutic thoracentesis** should be done.
- Indications for **more invasive procedure,** i.e., intercostal tube drainage are as following:
  - Loculated pleural fluid
  - Pleural fluid pH <7.20
  - Pleural fluid glucose <3.3 mmol/L (<60 mg/dL)
  - Positive Gram's stain or culture of the pleural fluid
  - Presence of gross pus in the pleural space.
- If there is recurrence after the initial therapeutic thoracentesis and if any of above factors is present, a repeat thoracentesis should be done.
- If the fluid cannot be completely removed with the therapeutic thoracentesis, insert a chest tube and instill a fibrinolytic agent (e.g., tissue plasminogen activator 10 mg) and deoxyribonuclease (5 mg) or performing a thoracoscopy and release the adhesions.

- **Intrapleural enzyme therapy (IET):** Most effective in the early fibrinolytic stage (e.g., streptokinase, streptodornase, urokinase, and tPA). They break down loculations and enhance clearance.
  - **Indications: (**1) Occluded small-bore catheter, (2) multiloculated pleural space, and (3) as a trial before committing the patient to surgery.
  - Persistent clinical (fever, ↑HR), biochemical (<50% CRP fall), radiological (persistent collection) features at 48 hours post ICD insertion mandates intrapleural enzyme therapy.
  - Administration of tPA (10 mg) and DNAse (5 mg) together twice daily for 3 days is preferred.
- When to refer for surgery?
  - Ongoing sepsis, clinical worsening, persistent radiological features by day 3
  - Nondraining collection
- **Decortication** is performed when the above measures are not effective.
- **Treatment of underlying cause.**

Chylous Pleural Effusion (Chylothorax)

Q. Write a short note on chylous pleural effusion (chylothorax)/causes of milky pleural fluid.

- **Definition:** It is the accumulation of chyle in the pleural cavity.
- **Mechanism:** A chylothorax results from leakage of chyle from the thoracic duct into the pleural space.
- **Causes:** Most common cause is trauma (most frequently during thoracic surgery). It may also result from lymphomas, lung cancer with mediastinal spread, mediastinal fibrosis, and tumors.
- **Presentation:** Dyspnea and a large pleural effusion on chest X-ray.
- **Pleural fluid findings:**
  - Appears milky and shows the characteristics of exudates **(Fig. 2.38)**.
  - Total triglyceride level is >110 mg/dL. The cholesterol level is very low. Sudan III staining shows fat globules.
  - Occasionally, empyema can be so turbid to be confused with chyle. Centrifugation of fluid leaves a clear supernatant fluid in empyema compared to persistence of milky appearance in chylous effusion. In starved patients, chyle may not be milky in appearance.
- If the trauma is not the cause of chylothorax, a contrast-enhanced CT scan of mediastinal and a lymphangiogram should be performed.

FIG. 2.38: Gross appearance of pleural aspirate in chylothorax.

Treatment

Treatment of choice is insertion of a chest tube and use of octreotide. Long-term tube drainage may result in malnutrition and immune deficiency. Therefore, if patient does not respond, a pleuroperitoneal shunt may be tried. Alternatively, ligation of the thoracic duct and percutaneous transabdominal thoracic duct blockage may be useful.

Pseudochylous Pleural Effusion (Pseudochylothorax)

Q. Write a short essay/note on pseudochylous pleural effusion (pseudochylothorax).

- Rare condition in which the pleural fluid appearance is similar to that of chylous effusion (milky).
- **Causes:** Long-standing benign pleural effusion (e.g., tuberculous effusion, rheumatoid effusion, etc.).
- **Pleural fluid:** Appears milky. Cholesterol level of the fluid is very high (>200 mg/dL) which is responsible for the milky appearance of fluid. Cholesterol crystals can be demonstrated and Sudan III staining does not show fat globules.
- **Differences between pseudochylous thorax and chylothorax (Table 2.60).**

TABLE 2.60: Differences between pseudochylous thorax and chylothorax.

Features	*Pseudochylothorax*	*Chylothorax*
Triglycerides	<0.56 mol/L (50 mg/dL)	>1.24 mmol/L (110 mg/dL)
Cholesterol	>5.18 mmol/L (200 mg/dL)	<5.18 mmol/L (200 mg/dL)
Cholesterol crystals	Often present	Absent
Chylomicrons	Absent	Present
Etiology	Tuberculosis, rheumatoid arthritis, poorly treated empyema	Neoplasm (lymphoma, metastatic carcinoma), trauma (operative, penetrating injuries), miscellaneous (tuberculosis, sarcoidosis, cirrhosis, lymphangioleiomyomatosis, obstruction of central veins, amyloidosis)

Pleural Effusions in HIV Infection

- A pleural effusion is seen in 7–27% of patients with HIV.
- **Leading causes are:** Kaposi sarcoma, parapneumonic effusion, TB, lymphoma, *Pneumocystis jirovecii* pneumonia.

Empyema Thoracis

Q. Discuss the etiology, clinical features, investigations, complications, and management of empyema thoracis.

Definition

- Empyema thoracis is defined as collection of pus in the pleural space/cavity. It may be as thin as serous fluid or so thick that it is impossible to aspirate, even with a wide-bore needle. Microscopically, it shows numerous neutrophil leukocytes.
- It usually involves whole pleural space/cavity and unilateral. If only a part of the pleural space is involved, it is termed encysted or loculated empyema.

Etiology

- **Spread of infection from neighboring structures:** For example, bacterial pneumonias, bronchiectasis, lung abscess, rupture of subphrenic abscess through the diaphragm, esophageal perforation, and infection of hemothorax following trauma or surgery.
- **From distant source:** For example, bacteremia.
- **Direct infection from external source:** For example, penetrating chest injury, chest tube placement, and thoracic surgery.
  Common organisms: *Pneumococcus, Streptococcus, Staphylococcus, Pseudomonas, Mycobacterium tuberculosis, H. influenzae*, and anaerobes.

Clinical Features (Table 2.61)

Q. Write a short answer on:
- **Clinical features of empyema.**
- **Physical sign of right-sided empyema.**

TABLE 2.61: Clinical features of empyema.

Systemic features:
➢ High-grade, remittent fever
➢ Chills, rigors, sweating, malaise, and weight loss
Respiratory symptoms:
Pleuritic chest pain, breathlessness, and dry cough. Copious, purulent sputum develops when the empyema ruptures into a bronchus (bronchopleural fistula)
Physical examination:
Clinical signs of pleural effusion plus intercostals tenderness, digital clubbing, and edema of the chest wall

Investigations

- **Blood:** Polymorphonuclear leukocytosis, high CRP (C-reactive protein).
- **Chest X-ray:** Findings may be similar to those of pleural effusion. However, pleural adhesions may produce a "D"-shaped shadow against the inside of the chest wall **(Fig. 2.39)**. If air is present along with pus (pyopneumothorax), a horizontal "fluid level" mark is detected at the air/liquid interface.
- **Ultrasound:** It shows the position of the fluid, the extent of pleural thickening and whether fluid is collected as a single locule or multiloculated.
- **CT:** It gives information regarding the pleura, underlying lung parenchyma, and patency of the major bronchi. Ultrasound or CT is used to detect the optimal site for aspiration.
- **Aspiration of empyema (Table 2.62)**

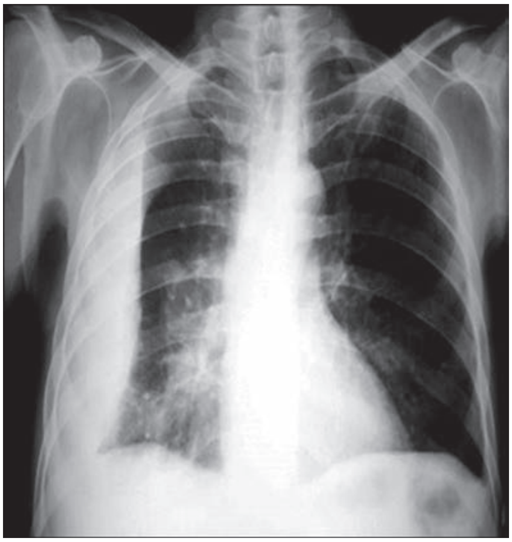


FIG. 2.39: Chest X-ray of loculated effusion—"D" sign.

Complications

See **Table 2.63.**

TABLE 2.62: Characteristic features of aspirated pus from empyema.

Consistency	Vary from very thin to very thick
Microscopy	Numerous polymorphonuclear leukocytes
Gram's stain	May show positive organism
Culture and sensitivity	To identify the causative organism
Tubercle bacilli	May be found, if it is due to *M. tuberculosis*

TABLE 2.63: Complications of empyema.

➢ **Empyema necessitans** is the formation of a subcutaneous abscess or sinus that develops when the pus track through the chest wall	➢ Pleural fibrosis ➢ Secondary amyloidosis
➢ **Bronchopleural fistula and pyopneumothorax** due to rupture of pus into a bronchus	➢ Metastatic brain abscess

Q. Write a short essay/note on the management on nontuberculous empyema.

Management of Empyema Thoracis

A. Nontuberculous empyema

- **Acute**
  - **Antibiotics:** It should be given depending on the culture and sensitivity report.
  - **Drainage of pus:** The pus in the pleural cavity should be drained.
    - **Aspiration:** Can be attempted in early and small loculated collections
    - **Tube drainage** (Pigtail/small bore ICD (12-14F) is recommended in all patients for definitive management.
    - **Intrapleural fibrinolysis:** Recommended if incomplete drainage or no clinical response by 48–72 hours of tube drainage.
    - **Limited thoracotomy:** If tube drainage fails or pus is thick or loculated, limited thoracotomy is performed. It involves resection of a small segment of the rib, clearing the empyema cavity breaking down any adhesions and introducing a wide-bore tube (for prolonged drainage).
    - **Surgery: VATS** guided drainage recommended as 1st line. If patient cannot tolerate single lung ventilation open thoracotomy is preferred.
- **Chronic:** Characterized by thickened pleura forming pleural peel compressing the lung, volume loss, mediastinal shift.
  - **Surgery:** Aimed as stripping off the pleural thickening (peel) which is termed *decortication*.
    - Open thoracotomy is initial preferred line and VATS performed if thoracotomy fails
    - Alternatives include removal of peel and open communication with atmosphere (thoracic window). In case of poor lung expansion muscle flaps, pleural tenting or thoracoplasty is performed as rescue measure

B. Tuberculous empyema

- **Antituberculous chemotherapy**
- **Repeated aspiration:** Through a wide-bore needle or tube drainage.
- Rarely, surgical treatment may be required.

PNEUMOTHORAX

Q. Describe the types, etiology, clinical features, investigations, and management of pneumothorax.

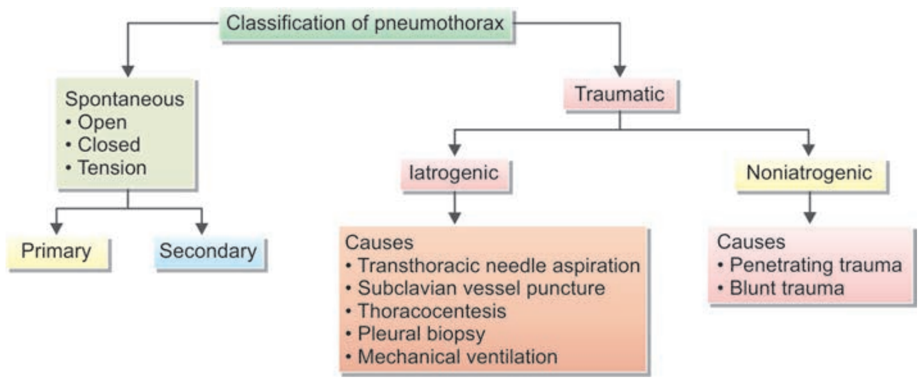
Q. Describe etiology, clinical features, investigations, and treatment of tension pneumothorax.

- **Definition:** Presence of air/gas in the pleural cavity/space is known as pneumothorax.
- Pneumothorax may be localized (if there is a prior disease-causing adhesion of visceral pleura to parietal pleura), or generalized (if there are no pleural adhesions).

Classification

See **Flowchart 2.13**

FLOWCHART 2.13: Classification of pneumothorax.



Etiology

Spontaneous pneumothorax

- **Primary (simple) spontaneous pneumothorax** occurs in the absence (no evidence) of overt lung disease.
  - It occurs in individual without any underlying lung disease or any trauma.
  - *Age:* Commonly occurs between the age group of 20–40 years.
  - *Risk factors:* Smoking, tall stature, and the presence of apical subpleural blebs, mostly familial.
  - *Pleural porosity:* Replacement of mesothelial lining to inflammatory cell lining increases porosity and inflammation
  - Subsequent recurrence rate after 1st episode is 20%, 2nd episode is 60%, 3rd episode is 83%
- **Secondary spontaneous pneumothorax** occurs in the presence of an underlying lung disease.
  - **Causes:** Most common causes are **COPD** (chronic bronchitis and emphysema) **and** cavitary active pulmonary **TB. It occurs** due to rupture of emphysematous bullae and subpleural TB focus. Other causes include bronchial asthma, suppurative diseases of lung and pleura, cystic fibrosis, and *P. jirovecii* pneumonia.

Traumatic pneumothorax results from penetrating or nonpenetrating injuries to the chest.

- **Iatrogenic:** Following diagnostic or therapeutic interventions. These include transthoracic and transbronchial needle aspiration/biopsy (24%), subclavian vessel puncture (22%), thoracocentesis (22%), pleural biopsy (8%), and mechanical ventilation (7%).
- **Noniatrogenic:** Blunt and penetrating injuries to the chest wall, bronchi, lung or esophagus.

Clinical Features

Q. Write a short note on clinical features of acute pneumothorax.

- A small pneumothorax may be asymptomatic without any abnormal physical signs in the chest.
- Most common symptoms are **sudden onset unilateral pleuritic chest pain** and **breathlessness** (dyspnea).
- **Severity depends on:** (1) extent of lung collapse and (2) amount of pre-existing lung disease.
- **Tension pneumothorax:** Distressed with rapid labored respiration, cyanosis, marked tachycardia, and profuse diaphoresis.

Physical Signs

General examination: Patient will be cyanosed, tachypneic, peripheral pulses may be feeble and hypotension may be present.

Respiratory system

- **Inspection and palpation**
  - **Accessory muscles of respiration in action**, **trachea and mediastinal** (apex beat) **shift to the opposite side.**
  - **On the affected side:** Fullness of the chest, diminished chest movements, increase in the size and diminished expansion of the hemithorax, increased spinoscapular distance, and markedly diminished vocal fremitus. Subcutaneous emphysema may be present.
- **Percussion:** Hyper-resonant note over the affected hemithorax.
- **Auscultation:**
  - **On the affected side:** Markedly diminished/absent of breath sounds and vocal resonance, absence of adventitious sounds. Open pneumothorax with a bronchopleural fistula, there may be **amphoric bronchial breathing.**

FIG. 2.40: Chest X-ray shows right-sided pneumothorax.

Investigations

Radiological findings on chest radiograph **(Fig. 2.40)**: **Standard erect chest X-ray in inspiration** is recommended for the initial diagnosis of pneumothorax rather than expiratory films. Following features are observed:

- Sharply defined edge of the deflated lung.
- Complete translucency and absence of bronchovascular markings (no lung markings) in the area between the edge of the lung and chest wall. A clear **visceral pleural line/collapsed lung margin** can be seen.
- Mediastinal shift/displacement to the opposite side.
  It may also reveal the presence or absence of pleural fluid, complicating empyema or underlying lung lesion. CT is used in difficult cases.

CT scanning is done if accurate size estimates are required.

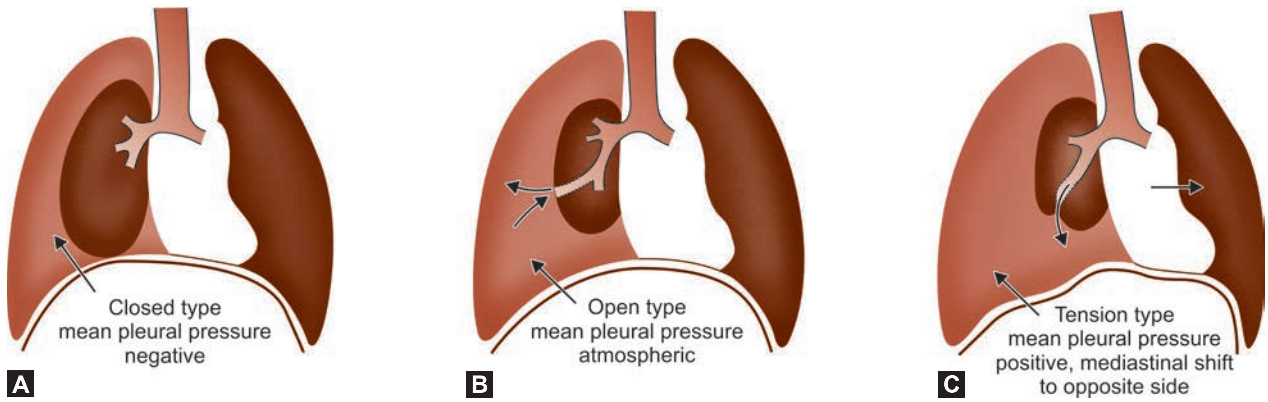
- Recommended only in difficult cases such as patients in whom the lungs are obscured by overlying surgical emphysema.
- To differentiate a pneumothorax from suspected bulla in complex cystic lung disease.
  USG signs of pneumothorax: (1) loss of lung sliding, (2) loss of comet tails, (3) loss of seashore sign (M mode), and (4) stratosphere sign or barcode sign (M mode).

Types of Spontaneous Pneumothorax (Figs. 2.41A to C)

There are three types namely: (1) closed spontaneous pneumothorax, (2) open spontaneous pneumothorax, and (3) tension (valvular) pneumothorax. Differences between closed, open, and tension pneumothorax are listed in **Table 2.64**.

TABLE 2.64: Types of spontaneous pneumothorax (Figs. 2.37A to C).

Closed pneumothorax	***Open pneumothorax***	***Tension pneumothorax***
The pleural tear is **sealed**	The pleural tear is **open**	The pleural tear acts as a **ball and valve** mechanism
The pleural cavity pressure is less than the atmospheric pressure	The pleural cavity pressure is equal to the atmospheric pressure	The pleural cavity pressure is more than the atmospheric pressure



FIGS. 2.41A TO C: Types of spontaneous pneumothorax. (A) Closed type; (B) Open type; (C) Tension (valvular) type.

Closed Spontaneous Pneumothorax

Pneumothorax where the communication between pleural space and the lung seals off and does not reopen. The mean pleural pressure remains negative and air can neither enter nor leave the pleural cavity/space. The trapped air is slowly and spontaneously reabsorbed and the lung re-expands completely in 2–4 weeks. Infection of the pleural cavity is uncommon.
Clinical features: Trivial breathlessness that gradually abates over a few days.

Open Spontaneous Pneumothorax

- Pneumothorax where the communication between bronchus and pleura does not seal off and remains patent and air continues to pass freely between the bronchial tree and pleural space. It results in a bronchopleural fistula.
- Free flow of air through the bronchopleural fistula results in intrapleural pressure (normally negative) same as that of atmospheric pressure throughout the respiratory cycle. This prevents the re-expansion of the collapsed lung. Development of bronchopleural fistula facilitates the spread of infection into the pleural space causing empyema.
- Open pneumothorax usually develops secondary to rupture of an emphysematous bulla, a small pleural bleb, a tuberculous cavity or a lung abscess into the pleural cavity/space.

Clinical features: Presents with breathlessness that does not improve. If there is infection of pleural space, fever and systemic features are observed. The physical signs are those of hydropneumothorax (air and fluid in the pleural space).

Tension (Valvular) Pneumothorax

Q. Write a short essay/note on clinical features and signs of tension pneumothorax.

In tension pneumothorax, the pressure in the pleural space is positive throughout the respiratory cycle.

- Pneumothorax where communication between pleura and lung persists. This is due to formation of a valvular mechanism (one-way valve through) in which air is sucked into the pleural space during inspiration (coughing, sneezing and straining) but not expelled during expiration. Large quantity of air gets "trapped" in the pleural space/cavity and raises the intrapleural pressure much higher than the atmospheric pressure.
- The high intrapleural pressure causes compression of the underlying lung, shifts the mediastinum to the opposite side with consequent compression of the opposite lung also. It also decreases venous return to the heart by compressing the vena cava resulting in reduced cardiac output.

Clinical features: Rapidly progressive breathlessness, central cyanosis, rapid thread pulse, and signs of peripheral circulatory failure. Signs of pneumothorax are present.

Treatment

- Tension pneumothorax should be treated as an acute medical emergency.
- **Emergency treatment:** Insertion of a large-bore needle into the pleural space through the 5th intercostal space in axilla (ATLS). The diagnosis is confirmed, if large amounts of air escape through the inserted needle. The needle should be left in place till a **thoracostomy** tube can be inserted. Cover the open end of the needle with a glove finger. Other methods:
  - Insertion of wide-bore plastic cannula. The opposite end is attached to long rubber tubing, the end of which is placed underwater in a bottle.
  - Introduction of an intercostals catheter connected to a water-seal drainage system.
  - If above methods cannot be performed, simple stab on chest wall to release pressure.

Recurrent Spontaneous Pneumothorax

- After primary spontaneous pneumothorax, recurrence occurs within a year in about 25% of patients.
- Recurrent pneumothorax is common with emphysematous bullae. The recurrence usually occurs on the same side. It can also occur with lymphangioleiomyomatosis (LAM).

Treatment
- Obliteration of the pleural space by artificial pleurodesis. This can be accomplished by intrapleural instillation of an irritant like tetracycline hydrochloride of talc powder.
- Pleurodesis is recommended for all patients following a second pneumothorax. Pleurodesis can be performed with chemicals (talc, doxycycline, povidone iodine) or surgically by pleural abrasion or parietal pleurectomy at thoracotomy or thoracoscopy.

Catamenial Pneumothorax

Q. Write a short note on catamenial pneumothorax.

- Rare condition occurring in females above the age of 25–30 years.
- It presents with repeated attacks of spontaneous pneumothorax usually on the right side, in association with menstruation. Attacks usually occur within 72 hrs before or after the onset of menstruation. Hemoptysis may also develop. Most frequently associated with endometriosis of thorax.
- It occurs due to **pleural endometriosis** that erode the visceral pleura and peritoneal air that enters via **diaphragmatic fenestrations**.

Treatment:
Surgery + Medical management
Surgery: VATS—remove endometrial deposits, repair diaphragm fenestrations and pleurodesis
Medical Mx: OCP/Danazol/GnRH agonist

Clicking Pneumothorax

Q. Write a short note clicking pneumothorax.

In clicking pneumothorax, a small left-sided pneumothorax gets localized in front of the pericardium. This produces alteration of the heart sounds so that the sound becomes loud and resonant ("clicking").

Complications of Pneumothorax

Q. Write a short note on causes of pyopneumothorax.

- **Pyopneumothorax:** It is caused by aspiration or intercostal chest tube insertion (iatrogenic). It may also result from necrotic pneumonia, lung abscess, or caseous pneumonia. Its causes are listed in **Box 2.14**.
- Hydropneumothorax
- **Hemopneumothorax:** Bleeding in pleural space and commonly caused due to rupture of vessels in adhesions. When lung re-expands, bleeding will stop. If bleeding persists, surgical ligation may be required.
- Mediastinal and subcutaneous emphysema

Box 2.14: Causes of pyopneumothorax.
- Complication of pneumothorax
- Thoracocentesis
- Trauma to thorax
- Bronchopleural fistula
- Esophagopleural fistula

Q. Write a short note on management of acute pneumothorax.

Treatment of pneumothorax
- **Goals:** (1) To promote lung expansion, (2) to eliminate the pathogenesis, and (3) to decrease recurrence of pneumothorax.
- **Treatment options according to:** (1) classification of pneumothorax, (2) pathogenesis, (3) pneumothorax frequency, (4) the extension of lung collapse, (5) severity of disease, and (6) complication and concomitant underlying diseases.
- Normal resolution of pneumothorax is **1–2%/day.** Average resolution occurs in 6 weeks. Hence stable patients can follow-up after 6 weeks

1. **Primary spontaneous pneumothorax (PSP):**
   - **Asymptomatic or size < 2 cm:** Observation + high flow oxygen (100%)
   - **Symptomatic or size >2 cm:** Needle aspiration, ICD (if failed aspiration)
   - **Persistent air leak >96 hrs:** Surgery (VATS preferred)
   - **Recurrent pneumothorax (≥2):** Pleurodesis
2. **Secondary spontaneous pneumothorax (SSP):**
   - **Asymptomatic/size<1 cm:** Observation + high flow oxygen (reduces partial pressure of nitrogen thereby increasing pressure gradient between pleural capillary and pleural cavity.
   - **Asymptomatic/size 1–2 cm:** Needle aspiration
   - **Symptomatic/size >2 cm:** Intercostal drainage tube (ICD) connected to underwater seal.
   - **Pleurodesis:** In patients with COPD, cystic fibrosis, Birt Hogg Dube syndrome or significant decompensation due to pneumothorax in the first episode to prevent recurrence **(BTS 2022)**
   - **Note:** Smokers aged >50 years even in the absence of COPD or lung disease are treated like SSP

Air travel can be advised only **After 7 days** of complete **radiological resolution** of pneumothorax

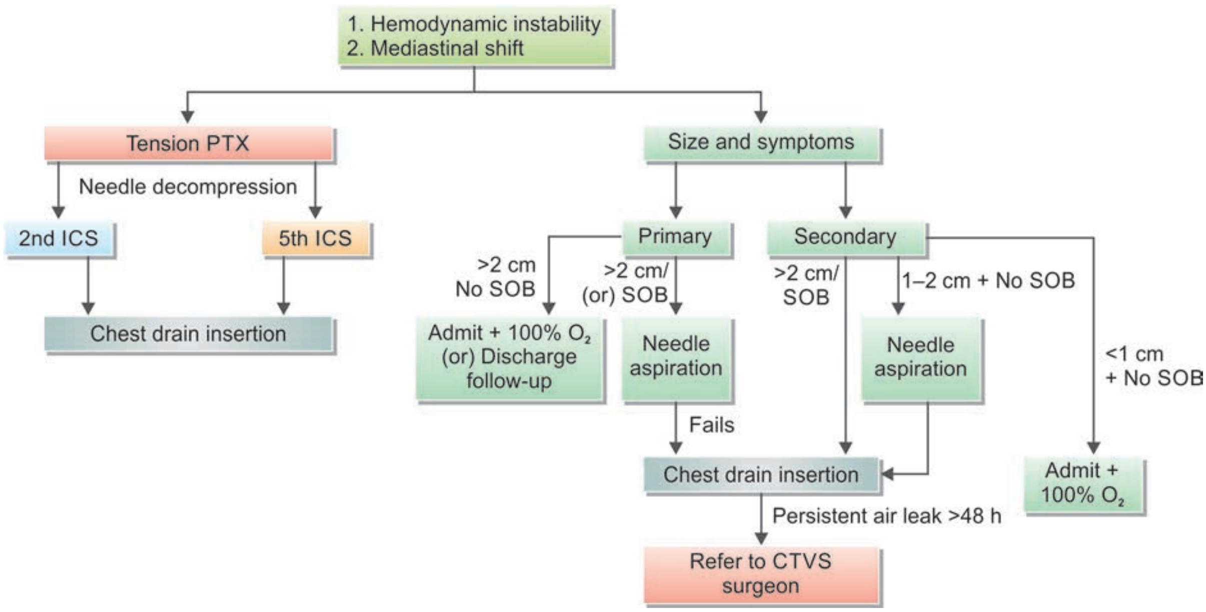
Q. Write a short note on indications for pleurodesis.

- **Goals:** (1) to prevent pneumothorax recurrence and (2) to produce inflammation of pleura and adhesions.
- **Indications:** Persistent air leak and repeated pneumothorax, bilateral pneumothoraces, those complicated with bullae, occupations where pneumothorax should not occur (e.g., drivers, pilots).
- **Sclerosing agents: Talc (Best), Tetracycline, minocycline, doxycycline and erythromycin.**
- The instillation of sclerosing agents into the pleural space should lead to an aseptic inflammation with dense adhesions.
  - Other types pleurodesis:
- Biological pleurodesis using ***Corynebacterium parvum.***
- Physical pleurodesis **mechanical abrasions.**

Surgical treatment

- **Indication:** No response to medical treatment, persistence of air leak, hemopneumothorax, bilateral pneumothoraces, recurrent pneumothorax, tension pneumothorax failed to drain, thicken pleura making lung unable to re-expansion, and multiple blebs or bullae.
- **Open thoracotomy and pleurectomy** remain the procedure with the lowest recurrence rate for difficult or recurrent pneumothorax.
- Minimally invasive procedures, thoracoscopy (VATS), pleural abrasion, and surgical talc pleurodesis are all effective alternative strategies.

FLOWCHART 2.14: Treatment approach to pneumothorax.



(PTX: Pneumothorax; SOB: Shortness of breath)

Hydropneumothorax

Q. Write a short note on the physical signs of hydropneumothorax.

Similar findings as pneumothorax except the following findings:

- **Percussion note** is **hyper-resonant over the upper air-containing part** and **stony dull over the lower fluid-containing part.**

➢ **Straight line dullness. Shifting dullness** can be elicited.	➢ **Coin-test** is positive over the upper air-containing part.
➢ **Amphoric bronchial breathing** in case of bronchopleural fistula.	➢ **Succussion splash** can be elicited on the affected side.

PNEUMONIA

Q. IM3.1 Define, discuss, describe and distinguish community acquired pneumonia, nosocomial pneumonia and aspiration pneumonia.

Definition: Pneumonia is as an acute respiratory illness, defined as inflammation with exudative solidification of the lung parenchyma. It causes the alveoli to be filled with inflammatory exudates and usually results in consolidation (solidification) of lung.

- **Pathological definition:** Infection of the alveoli, distal airway, and interstitium of the lung. Characterized by increased weight, replacement of the normal sponginess by the consolidation. Alveoli filled by the WBC, RBC, and fibrin.
- **Clinical definition:** Constellation of symptoms and signs (fever, chills, cough, pleural chest pain, sputum, bronchial breathing, egophony, crackles, wheeze, and pleural friction rub) with at least one opacity on chest X-ray PA view.

Classification of Pneumonia

Pneumonias can be classified in different ways (**Table 2.65**).

1. ***Classification depending on the anatomic distribution***
   a. **Lobar pneumonias (alveolar or air space pneumonia):** The organism causes inflammatory exudates involving many contiguous alveoli (e.g., pneumococcal pneumonia). This radiologically appears as nonsegmental consolidation.
   b. **Bronchopneumonia:** Inflammation involving conducting airways, especially terminal and respiratory bronchioles, and the surrounding alveoli (e.g., staphylococcal pneumonia).
   c. **Interstitial pneumonia:** The inflammation is confined to interalveolar septa. X-ray chest gives a reticular pattern (e.g., *Mycoplasma pneumoniae*, *P. jirovecii*, and viruses).
2. ***Etiological classification***
   a. **Primary pneumonia:** It is caused by a specific pathogenic organism and there is no pre-existing abnormality of the respiratory system. The causative organisms are listed in **Table 2.66**.

Q. Write a short essay on list of bacteria causing pneumonia.

   b. **Secondary pneumonia (including aspiration pneumonia):** discussed earlier
   c. **Suppurative pneumonia (necrotizing pneumonia):** discussed earlier

Q. IM3.2 Discuss and describe the etiologies of various kinds of pneumonia and their microbiology depending on the setting and immune status of the host.

3. **Clinical setting in which the infection occurs (if no pathogen can be isolated):** Pneumonia is classified by the setting in which the person has contracted their infection.
   d. **Community setting or community-acquired pneumonia (CAP):** It is defined as an acute pulmonary infection in a patient who is not hospitalized or living in a long-term care facility 14 days or more before presentation and does not meet the criteria for health care-associated pneumonia (HCAP). This category includes both immunocompetent and immunocompromised patients as causative agents are almost similar in both the conditions.
   e. **Nosocomial pneumonia or hospital acquired pneumonia**
      - **Definition:** Hospital-acquired pneumonias (HAPs) are pulmonary infections acquired in the course of a hospital stay (development of pneumonia after >48 hours of hospitalization).
      - Most of this pneumonia occurs outside intensive care units. However, the highest risk is observed in patients on mechanical ventilation [ventilator-associated pneumonia (VAP)]
      - **Etiology:** Predisposing factors include severe underlying disease, immunosuppression, prolonged antibiotic therapy, patients on mechanical ventilation or invasive access devices such as intravascular catheters. The HAPs are serious and may be life-threatening. The various causative organisms causing HAP are shown in **Table 2.67**.
4. **Pneumonia in immune compromised host***:* This is a type of pneumonia found in patient whose immune system is compromised, through either genetic defect, immunosuppressive medication, or acquired immunodeficiency such as HIV infection and malignancies.
   - It may be caused by classical organisms, atypical organisms, *M. tuberculosis* or *P. jirovecii*.
   - Usually, symptoms are more than the signs.

   f. **Health care-associated pneumonia:** Currently this definition is no longer used.
      - **Definition:** Hospital-acquired pneumonias are pulmonary infections acquired in the course of a hospital stay. It occurs:
        - Within 90 days of a 2-day or longer hospitalization. In a nursing home or long-term care residence.

TABLE 2.65: Classification of pneumonia.

1. Classification depending on the anatomic distribution
   - Lobar pneumonia
   - Bronchopneumonia
   - Interstitial pneumonia
2. Etiological classification
   - Primary
   - Secondary
   - Suppurative
3. Clinical setting in which the infection occurs (if no pathogen can be isolated)
   - Community-acquired acute pneumonia
   - Community-acquired atypical pneumonia
   - Nosocomial pneumonia or hospital-acquired pneumonia
   - Pneumonia in immunocompromised host
   - Health care-associated pneumonia

TABLE 2.66: List of organisms causing primary pneumonia.

Common	*Less common*
➢ *Streptococcus pneumoniae (most common)* ➢ *Haemophilus influenzae* ➢ *Moraxella catarrhalis* ➢ *Staphylococcus aureus* ➢ *Legionella pneumophila* ➢ *Mycoplasma pneumoniae*	➢ *Enterobacteriaceae (Klebsiella pneumoniae)* and *Pseudomonas* spp. ➢ *Streptococcus pyogenes* ➢ *Pseudomonas aeruginosa* ➢ *Coxiella burnetii* (Q fever) ➢ *Chlamydia* spp. *(C. pneumoniae, C. psittaci, and C. trachomatis)* ➢ Viruses: Respiratory syncytial virus, H1N1 influenza virus, seasonal influenza virus, parainfluenza virus, and human metapneumovirus (children); influenza A and B (adults); adenovirus (military recruits) coronavirus producing severe acute respiratory syndrome (SARS) ➢ *Actinomyces israelii*

TABLE 2.67: Various etiological agents causing hospital-acquired pneumonia.

Gram-negative rods, Enterobacteriaceae (*Klebsiella* spp., *Serratia marcescens*, *Escherichia coli*) and *Pseudomonas* spp.
Staphylococcus aureus (penicillin and usually methicillin-resistant

- Within 30 days of receiving intravenous antibacterial therapy, chemotherapy, or wound care or after a hospital or hemodialysis clinic visit; or in any patient in contact with a multidrug-resistant pathogen.
- Current studies show treatment with broad-spectrum antibiotics in all patients in the absence of specific risk factors for MDR organisms had worse outcomes.

Noninfective Pneumonias

- **Lipid/Lipoid pneumonia:**
  - Aspiration of fatty/oily material into lungs.
  - Decreasing order of severity of manifestations: Mineral oil > animal oil > vegetable oil (because vegetable oil to some extent, animal oils can be hydrolyzed in the body).
  - Liquid paraffin causes the most cases of lipoid pneumonia.
  - Chronic persistence of the oil in the lung can produce fibrosis/paraffinomas (paraffin granulomas).
- **Radiation pneumonitis**
  - Dose >25 Gy. Risk depends on radiation dose and volume of lung irradiated.
  - Early phase: Cough, fever, chest X-ray infiltrate.
  - Late phase (3–6 weeks): Dyspnea
  - Lung fibrosis: With excessive dose/large lung volume irradiation.
  - Treatment: Glucocorticoids improve symptoms, but have no ultimate effect on the development of fibrosis.
  - Recall pneumonitis: Pneumonitis due to chemotherapy, within the distribution of a previous radiotherapy field (the radiation sensitizes the lung tissue to the toxic effects of chemotherapy).
- **Chemical pneumonitis**
  - If aspirated fluid: pH <2.5 and volume >0.3 mL/kg result in chemical pneumonia (Mendelson's syndrome)—ARDS/secondary bacterial infection.
  - If gastric pH alkaline: Colonization of gastric mucosa by enteric gram-negative bacilli—pneumonitis.
  - Poor orodental hygiene with gross aspiration and lung abscess.
  - Pathogenesis: Most common mechanism by which pathogenic organisms reach the lower respiratory tract through microaspiration.
  - Others: Hematogenous/inhalation (droplet nuclei/aerosols)/contiguous spread.

Community-acquired Pneumonia

Q. Write an essay on community-acquired pneumonia—its definition, etiology/causative organisms, clinical features, investigations, diagnosis, complications, and treatment/management.

Q. Write an essay on streptococcal pneumonia—its etiopathogenesis, clinical features, complications, and management.

Q. Write a short essay on etiology/organisms causing community-acquired pneumonia.

- Community-acquired pneumonia affects all ages, but is commoner at extremes of age.
- Most cases are spread by droplet infection.
- CAP may occur in previously healthy individuals. However, several factors may impair the effectiveness of local defenses and predispose to CAP **(Table 2.68)**.
- Pneumonia can be classified either according to the organism responsible for infection or anatomical distribution of infection.
- *Streptococcus pneumoniae (Pneumococcus)* is the most common cause. Viral infections are important causes of CAP in children.

TABLE 2.68: Predisposing conditions for community-acquired pneumonia.

- Extremes of age
- Upper respiratory tract infections
- Comorbidities: For example, congestive heart failure, diabetes, chronic kidney disease, recent influenza infection, and malnutrition
- Cigarette smoking
- Alcohol
- Corticosteroid therapy
- Congenital or acquired immune deficiencies, e.g., HIV
- Decreased or absent splenic function, e.g., sickle cell disease or postsplenectomy (risk for infection with encapsulated bacteria)
- Other respiratory conditions: Cystic fibrosis, bronchiectasis, COPD, obstructing lesion (endoluminal cancer, inhaled foreign body)
- Indoor air pollution

Q. IM3.3 Discuss and describe the pathogenesis, presentation, natural history and complications of pneumonia.

- Depending upon the anatomical distribution of infection, it may be classified as lobar pneumonia (localized with the whole of one or more lobes affected) or bronchopneumonia (diffuse in which lobules of the lung are mainly affected, often due to infection centered on the bronchi and bronchioles).

- Potential causative agents in CAP: Bacteria, fungi, viruses, and protozoa.
- Causative agent and salient features of community-acquired acute pneumonia are presented in **Table 2.69**.

Q. Write a short essay on etiology/organisms causing community-acquired pneumonia.

TABLE 2.69: Causative agent and salient features of community-acquired acute pneumonia.

Microorganism	*Features*
Streptococcus pneumoniae or Pneumococcus	Most common cause
Haemophilus influenzae	Most common bacterial cause in COPD
Moraxella catarrhalis	Elderly
Staphylococcus aureus	Secondary bacterial pneumonia following viral respiratory illnesses
Enterobacteriaceae (Klebsiella pneumoniae)	In debilitated and malnourished people
Pseudomonas aeruginosa	Common in patients with neutropenia
Legionella pneumophila	Organ transplant recipients

Types of Presentations of Community-acquired Pneumonia

Two types of presentations of CAP are presented in **Table 2.70**.

TABLE 2.70: Types of presentations of community-acquired pneumonia (CAP).

Classical	*Atypical*
➢ Sudden onset of CAP ➢ High-fever, shaking chills ➢ Pleuritic chest pain, sudden onset of breathlessness ➢ Productive cough ➢ Rusty sputum, blood tinge ➢ Poor general condition ➢ High mortality up to 20% in patients with bacteremia ➢ Cause: *S. pneumoniae*	➢ Gradual and insidious onset ➢ Low-grade fever ➢ Dry cough, no blood tinge ➢ Low mortality 1–2%; except in cases of legionellosis ➢ Cause: *Mycoplasma, Chlamydia, Legionella, Rickettsia*, viruses ➢ Systemic manifestations present

Clinical Features of Community-acquired Pneumonia

- The clinical features vary according to the immune status of the patient and the infecting agent.
- **Systemic features:** Pneumonia (especially lobar pneumonia), usually presents as an acute illness. Sudden onset of fever, rigors, shivering, and malaise are predominant symptoms and delirium may be present. The appetite is lost and there may be headache. Fever can be as high as 39.5–40°C. Swinging fevers often indicates empyema.
- **Pulmonary symptoms**
  - *Cough:* First, it is characteristically short, painful and dry. Later, it is productive with expectoration of mucopurulent sputum. Characteristically rusty sputum may be seen in patients with *S. pneumoniae* (pneumococcal pneumonia) and the occasional hemoptysis can occur.
  - *Chest pain:* It is commonly pleuritic chest pain and may be a presenting feature. It is due to inflammation of the pleura. A pleural rub may be heard. Occasionally, pain may be referred to the shoulder or anterior abdominal wall. Upper abdominal tenderness is sometimes apparent in patients.
  - *Breathlessness:* The alveoli become filled inflammatory exudate impairs gas exchange producing breathlessness.
  - *Other features:* CAP can present with confusion or nonspecific symptoms in the elderly. When symptoms have been present for several weeks or have failed to respond to standard antibiotics, the possibility of TB should always be considered.

TABLE 2.71: Extrapulmonary features of community-acquired pneumonia.

Extrapulmonary symptoms	*Infectious agent*
Myalgia, arthralgia, and malaise	*Legionella* and *Mycoplasma*
Myocarditis and pericarditis	*Mycoplasma pneumoniae*
Headache, abdominal pain, diarrhea, and vomiting	*Legionella pneumoniae*
Labial herpes simplex reactivation	Pneumococcal pneumonia
Skin rashes: Erythema multiforme and erythema nodosum	*Mycoplasma pneumoniae*

- **Extrapulmonary features (Table 2.71):** They are more common in certain infections and sometimes give a clinical clue to the etiology.

Examination

- **Fever:** About 80% are febrile, although this finding is frequently absent in older patients. This is a helpful diagnostic clue if present.
- Respiratory and pulse rate may be raised and the blood pressure low and this may be the most sensitive sign in the elderly.
- **Tachycardia** is common. However, relative bradycardia is a characteristic feature of Legionnaires' pneumonia. There may be delirium.
- Oxygen saturation on air may be low, and the patient cyanosed and distressed.
- **Chest examination:**
  - Chest signs vary, depending on the phase of the inflammatory response.
  - *During consolidation phase:* Lung is typically dull to percussion and, as conduction of sound is enhanced, auscultation reveals tubular bronchial breath sounds over areas of consolidated lung bronchophony and whispering pectoriloquy. Coarse crackles are heard throughout on auscultation, due to consolidation of the lung parenchyma.

Q. IM3.11 Describe and enumerate the indications for further testing including HRCT, Viral cultures, PCR and specialized testing.

Investigations in Community-acquired Pneumonia (Table 2.72)

TABLE 2.72: Investigations in community-acquired pneumonia (CAP).

Investigation	*Significance*
Complete blood count	
➢ Very high (>20 × 10^9/L) or low (<4 × 10^9/L) WBC count	➢ Marker of severity. In viral and atypical pneumonias, total leukocyte count is often <5,000/mm^3
➢ Neutrophilic leukocytosis >15 × 10^9/L	➢ Suggests bacterial pneumonia
➢ Hemolytic anemia	➢ Occasionally complicates *Mycoplasma*
Urea and electrolytes	
➢ Urea >7 mmol/L (~20 mg/dL)	➢ Marker of severity
➢ Hyponatremia	➢ Marker of severity may occur in patients with Legionnaires' disease
Liver function tests	
➢ Abnormal transaminitis, raised bilirubin	➢ When basal pneumonia inflames liver, or in atypical pneumonia
➢ Hypoalbuminemia	➢ Marker of severity
Erythrocyte sedimentation rate/ C-reactive protein	➢ Nonspecifically elevated
Blood culture	➢ Bacteremia is a marker of severity. Causative organism may be grown (e.g., pneumococcal pneumonia). However, blood cultures are recommended only in hospitalized patients
Serological and antigen detection tests	➢ Pneumococcal antigens can be detected in the serum or urine in pneumococcal pneumonia ➢ Acute and convalescent titers for *Mycoplasma*, *Chlamydia*, *Legionella*, and viral infections
Cold agglutinins	Positive in 50% of patients with *Mycoplasma*
Arterial blood gases	Measure when SaO_2 <93% or when severe clinical features to assess ventilatory failure or acidosis
HIV testing	Since pneumonia is a common in previously undiagnosed HIV infection, a test should be offered to all patients with pneumonia
Sputum: It can be distinguished from saliva by microscopic examination. Sputum contains alveolar macrophages	Gram's stain, culture, antimicrobial sensitivity testing and Ziehl–Neelsen staining
Oropharynx swab	PCR for *Mycoplasma pneumoniae* and other atypical pathogens
Urine	Pneumococcal and/or *Legionella* antigen. Hematuria may occur in patients with Legionnaires' disease
Chest X-ray	Essential for the confirmation of diagnoses, follow-up and detection of complications like parapneumonic effusion and empyema
➢ Lobar pneumonia	➢ Patchy opacification evolves into homogeneous consolidation of affected lobe ➢ Air bronchogram (air-filled bronchi appear lucent against consolidated lung tissue) may be present
➢ Bronchopneumonia	Patchy and segmental shadowing
➢ Complications	Parapneumonic effusion, intrapulmonary abscess or empyema
➢ *Staphylococcus aureus*	Multilobar shadowing, cavitation, pneumatoceles, and abscesses
➢ *Mycoplasma*	Usually, one lobe is involved but infection can be bilateral and extensive
➢ *Legionella*	There is lobar and then multilobar shadowing, with the occasional small pleural effusion. Cavitation is rare
Aspiration	➢ Percutaneous transtracheal aspiration of secretions ➢ Percutaneous transthoracic needle aspiration, preferable under CT guidance
Fiberoptic bronchoscopy with BAL and brushings	Gram's stain, AFB stain, culture, and cytology
Biopsy	A transbronchial biopsy of the lung tissue for culture and histopathology may be done in selected cases. Diagnostic open-lung biopsy, which carries a high-risk, is reserved for selected patients
Pleural fluid	Aspirate and culture when present in more than trivial amounts, preferably with ultrasound guidance

Q. Write a short essay on radiological findings in pneumonias.

Complications of Pneumonia

Q. Write a short essay on complications of pneumonia.

General Complications (Box 2.15)

Local complications

- Lung abscess
- **Organization:** Delayed and incomplete resolution can cause **ingrowth of granulation tissue into the alveolar exudate.**

Box 2.15: General complications of pneumonia.

- Respiratory failure ARDS
- **Bacteremic dissemination (bacteremia):** It **can cause:**
  - Endocarditis (heart valves)
  - Pericarditis (pericardium)
  - Meningitis (meninges)
  - Suppurative arthritis (joint)
  - Metastatic abscesses in kidneys or spleen
  - Sepsis—multisystem failure

The intra-alveolar plugs of granulation tissue are known as organizing pneumonia. Gradually, **increased alveolar fibrosis leads to a shrunken and firm lobe and is called as cornification**.

- **Spread of infection to the pleural cavity:** It may result in:
  - Pleuritis
  - Parapneumonic pleural effusion
  - *Pyothorax:* Which may lead to fibrothorax.
  - *Pneumothorax*, especially in *S. pneumoniae* due to pneumatocele rupture.

Q. IM3.12, 3.13 Select, describe and prescribe based on the most likely etiology, an appropriate empirical antimicrobial based on the pharmacology and antimicrobial spectrum.

Management Guidelines for CAP

- Rational use of microbiology laboratory.
- Pathogen-directed antimicrobial therapy whenever possible.
- Prompt initiation of antibiotic therapy.
- Decision to hospitalize based on prognostic criteria CURB 65

Severity: The need to hospitalize (severity) a patient is commonly assessed by CURB 65 or the CRB 65 score **(Table 2.73)**. The CRB 65 score is used in the community where the serum urea level is not usually available. Other severity score available is Pneumonia Severity Index (PSI), which combines several clinical and laboratory features, and comorbid conditions.

General Management (Treatment) of Pneumonia

- Check the airway, breathing, and circulation. The most important aspects are: oxygenation, fluid balance, and antibiotic therapy.
- **Oxygen:** Oxygen is indicated in all patients with tachypnea, hypoxemia, hypotension or acidosis. The aim is to maintain saturations between 94 and 98%. In patients with known COPD, high concentrations (35% or more), preferably humidified oxygen should be used to maintain a saturation between 88 and 92%. If hypoxia continues or patient develops increasing hypercapnia, ventilate the patient mechanically.
- **Intravenous fluids:** These are required in patients with severe illness, older patients and those who are vomiting. Treat shock (hypotensive showing any evidence of volume depletion) with intravenous fluids initially. Otherwise, an adequate oral intake of fluid is enough.

New Treatment Paradigm

- **Hit Hard Early with Antibiotics → De-escalate.**
  1. ***Antibiotics:*** Administer antibiotics as soon the diagnosis of CAP is established preferably within 4 hours of presentation in hospital and treatment should not be delayed while investigations are awaited. Prompt administration of antibiotics improves the outcome. Empiric regimens **(Table 2.74)**

     Duration of therapy: Minimum of 5 days (usually 7–10 days). Afebrile for at least 48–72 hours. Duration is 10–14 days for patients with *Mycoplasma* and *Chlamydia pneumoniae*. Patients initially treated with intravenous antibiotics can be switched to oral agents when they become febrile. Longer duration of therapy is needed if initial therapy was not active against the identified pathogen or complicated by extrapulmonary infection.
  2. ***Mild analgesics for pleuritic pain:*** Pleural pain may prevent the patient from breathing normally and coughing efficiently resulting in sputum retention, atelectasis or secondary infection. Hence, it is relieved by simple analgesia such as paracetamol, codeine or NSAIDs.

Q. IM3.15 Describe and enumerate the indications for hospitalisation in patients with pneumonia.

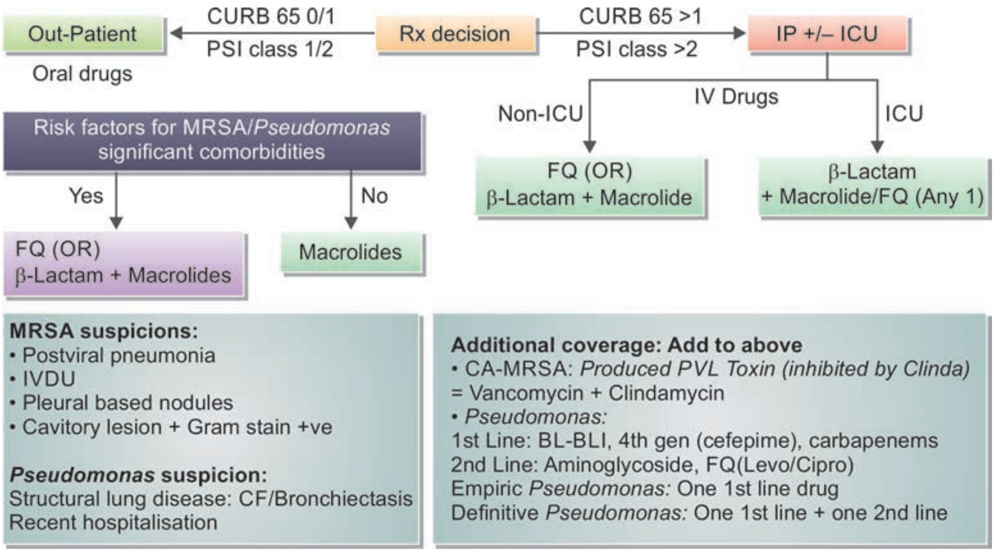
TABLE 2.73: CURB 65 rule.

Confusion: New mental confusion
Urea >7 mmol/L (>20 mg/dL)
Respiratory rate >30 breaths/minute
Blood pressure: Diastolic BP <60 mm Hg or systolic blood pressure <90 mm Hg
Age ≥**65** years of age (1 point for each)
Group 1: 0 or 1 of the above—mortality low—1.5%. Likely suitable for treatment at home
Group 2: 2 of the above—mortality—9.2%. Hospitalization for treatment
Group 3: 3 or more of the above—mortality—22%. Likely requires admission to ICU

TABLE 2.74: Empiric regimens for pneumonia.

Setting	*Therapeutic options*
Ambulatory, not requiring hospitalization, age under 60 years	Oral macrolide (erythromycin or azithromycin)
Ambulatory, not requiring hospitalization, comorbidity or age over 60 years	Oral β-lactam/β-lactamase inhibitor + macrolide **or** oral antipneumococcal fluoroquinolone
Requiring hospitalization	β-lactam (cefoperazone or ceftriaxone) + macrolide or antipneumococcal fluoroquinolone
Aspiration pneumonia requiring hospitalization	β-lactam/β-lactamase inhibitor alone (ampicillin/sulbactam, piperacillin/tazobactam)

FLOWCHART 2.15: Treatment of pneumonia.



(FQ: Fluoroquinolone; PSI: Pneumonia severity index; CF: Cystic fibrosis; IVDU: Intravenous drug user; CA-MRSA: Community-acquired methicillin-resistant *Staphylococcus aureus*)

Antibiotic treatment for community-acquired pneumonia has been shown in **Table 2.75.**

TABLE 2.75: Antibiotic treatment for community-acquired pneumonia (CAP).

Antibiotic	*Dosage, route, frequency, and duration*
Doxycycline	100–200 mg PO/IV BID for 7–10 days
Azithromycin	500 mg OD IV—3 days + 500 mg OD PO for 7–10 days
Clarithromycin	250–500 mg BID PO for 7–14 days
Telithromycin	800 mg PO OD for 7–10 days
Levofloxacin	750 mg PO/IV OD for 5 days
Gatifloxacin	400 mg PO or IV OD for 5–7 days
Moxifloxacin	400 mg PO or IV OD for 5–7 days
Gemifloxacin	320 mg PO OD for 5–7 days
Amoxiclav	2 g of amoxicillin + 125 mg of clavulanic acid PO BID for 7–10 days
Ceftriaxone	2 g IV BID for 5–7 days
Ertapenem	1 g OD IV or IM for 7–14 days

(PO: per oral; IV: intravenous; IM: intramuscular; OD: once daily; BID: twice daily)

Unresolved/Slow-resolving Pneumonia

- Patient is considered to have responded if: (1) fever declines within 72 hours, (2) temperature normalizes within 5 days, and (3) respiratory signs (tachypnea) return to normal. The expected time course for resolution is controversial.
  - In 1975, Hendin defined slowly resolving as **pulmonary consolidation persisting >21 days.**
  - In 1991, Kirtland and Winterbauer defined slowly-resolving CAP in immunocompetent patients based upon radiographic criteria. **>50% clearing by 2 weeks** or > **complete clearing at 4 weeks**.
- **Nonresolving pneumonia** is defined as a clinical syndrome in which focal infiltrates begin with some clinical association of acute pulmonary infection and despite a minimum of 10 days of antibiotic therapy patients either do not improve or worsen or radiographic opacities fail to resolve within 12 weeks.
- **Progressive pneumonia:** Increase in radiographic abnormalities and clinical deterioration during first 72 hours of treatment.
- Causes of unresolved/slow-resolving pneumonia have been shown in **Table 2.76**.

Recurrent pneumonia: Many conditions cause recurrent pneumonias **(Table 2.77)**.

Prevention of Further Episodes of Pneumonia

- **Cessation smoking**
- **Influenza vaccine** (Flu shot—October through February): It offers 90% protection and reduces mortality by 80%.
- **Pneumococcal vaccine** (Pneumonia shot): It protects against 23 types of Pneumococci. About 70% of individuals have pneumococci in the respiratory tract. Vaccination reduces mortality.

Pneumococcal Pneumonia (Lobar Pneumonia)

Q. Discuss the etiology, clinical features, investigations, diagnosis, complications, and treatment/management of pneumococcal pneumonia (lobar pneumonia/*S. pneumoniae*).

Definition: Lobar pneumonia is characterized by **diffuse inflammation** affecting the **part or entire lobe**, **usually the lower lobes.**

Etiology

- **Causative organism:** Most common form of pneumonia and is caused by *S. pneumoniae* (e.g., *Pneumococcus*, a gram-positive, lancet-shaped *Diplococcus*).
- **Mode of infection:** By droplet infection.

Risk factors for pneumococcal pneumonia

- **Age:** Younger than 2 years or older than 65 years.
- Strongest independent risk factor for invasive pneumococcal disease: Cigarette smoking
- **Strongest independent risk factor for CAP: Alcoholism**
  (ALPS: Alcoholism, leukopenia, pneumococcal sepsis: Mortality rate 80%)
- Poverty and overcrowding
- **Lowered systemic resistance of the host:** It may be due to:
  - *Chronic diseases:* Diabetes mellitus, severe liver disease, and chronic lung disease
  - *Immunological deficiency:* Defects in innate immunity and humoral immunodeficiency (complement or immunoglobulin), defects in cell-mediated immunity (congenital and acquired), HIV infection.
  - Treatment with immunosuppressive agents
  - *Leukopenia*
  - Asplenia or hyposplenia
- Impaired local defense mechanisms
  - *Loss or suppression of the cough reflex:* For example, **coma**, **anesthesia**, **drugs**, **chest pain**, or neuromuscular disorders. It may lead to aspiration of gastric contents into the lung.
  - *Damage or injury to the mucociliary apparatus:* It may be due to **cigarette smoke, viral diseases, inhalation of hot or corrosive gases**, or genetic defects of ciliary function (e.g., the immotile cilia syndrome).
  - *Accumulation of secretions:* Cystic fibrosis and bronchial obstruction.
  - *Interference with the phagocytic or bactericidal action of alveolar macrophages:* It may be due to alcohol intoxication, tobacco smoke, anoxia, or oxygen intoxication.
  - Antecedent influenza
  - Pulmonary congestion and edema.

TABLE 2.76: Causes of unresolved/slow-resolving pneumonia.

Incorrect microbiological diagnosis (e.g., tuberculosis instead of classical organisms) or incomplete antimicrobial treatment	➢ Underlying antibiotic resistance ➢ Inadequate dose/duration ➢ Nonadherence ➢ Malabsorption
Complication of CAP (community-acquired pneumonia)	➢ Parapneumonic pleural effusion (exudative), empyema, lung abscess
Underlying neoplastic lesion or other lung disease	➢ Bronchial obstruction causing partial or complete obstruction, bronchoalveolar cell carcinoma, bronchiectasis
Alternative diagnosis	➢ Pulmonary thromboembolic disease, cryptogenic organizing pneumonia, eosinophilic pneumonia, pulmonary hemorrhage
Host factors	➢ Age especially >50 years ➢ Comorbid illness: Diabetes, COPD (chronic obstructive pulmonary disease) ➢ Others (e.g., alcoholism, immunosuppressive/cytotoxic therapy)
Superinfection	➢ Fungi, *Mycobacterium tuberculosis*
Defects in defense	➢ For example, impaired cough (sedatives, neuromuscular illness, stroke), impaired mucociliary transport (chronic bronchitis), immune deficiency states—primary and secondary (B-cell and T-cell)

TABLE 2.77: Conditions causing recurrent pneumonias.

➢ Bronchial obstruction (e.g., foreign body, bronchial stenosis, compression of bronchus, endobronchial lesions)
➢ Bronchiectasis
➢ Immunocompromised state
➢ Sequestration of lung
➢ Ciliary dyskinesia
➢ Multiple myeloma and other lymphoreticular malignancies

Clinical Features

- **Sudden onset of high fever**, **chills** and rigors, **coughs**, and vomiting. Fever is usually high grade (39–40°C). Convulsions may occur in children.
- Cough is initially short, painful and dry, but soon becomes productive with **mucopurulent sputum.** Rust-colored sputum (**"rusty" sputum**) is characteristic but occasionally may be frankly blood stained.
- Nonspecific symptoms include loss of appetite, headache, and pains in the body and limbs.
- Localized pleuritic chest pain develops at an early stage due to fibrinosuppurative pleuritis. It may be referred to the shoulder or abdominal wall. It may be companied by pleural friction rub. Breathing is rapid and shallow due to pleuritic pain.
- Other features are tachycardia, hot and dry skin, herpes labialis, and flushed face.

Physical signs in the chest

- **First 2 days:** The physical signs are minimal and include diminished respiratory movements, slight impairment of percussion note, and pleural rub. In early stages, numerous fine crepitations are audible.
- **After 2 days:** Frank signs of consolidation appear **(Box 2.16)**.

Investigations

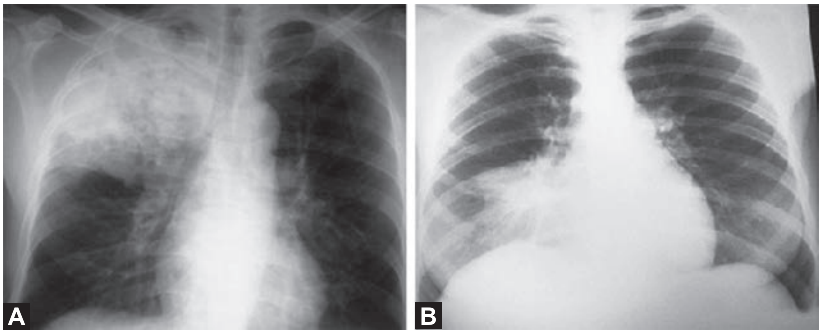
- **Blood**
  - Severe neutrophil leukocytosis
  - Blood culture may show *S. pneumoniae.*
- **Sputum**
  - Gram's staining of the sputum may show pneumococci which appear as gram-positive, lancet-shaped diplococci.
  - Sputum culture may show *S. pneumoniae.*
  - Assays on sputum are based on detecting nucleic acids for pneumococci.
- **Serological tests:** They can detect pneumococcal antigen in serum, urine, and sputum. Detection of C-polysaccharide (part of pneumococcal cell wall) in urine by an immunochromatography assay is quite sensitive. Urinary antigen remains positive for weeks after onset of severe pneumococcal pneumonia. However, it is often negative in mild pneumococcal pneumonia.
- **Chest radiograph (Figs. 2.42A and B)**
  - Involved lobe or segment appears homogeneous radiopaque with air bronchograms.
  - Associated parapneumonic effusion or empyema can also be detected.
- **Others:** In rare instances, fiberoptic, bronchoscopic aspiration or transthoracic needle aspiration may be necessary.

Box 2.16: Signs of consolidation.

- Diminished respiratory movements
- Dull percussion note
- No mediastinal shift
- Markedly increased vocal fremitus and vocal resonance
- High-pitched tubular bronchial breathing
- Bronchophony, egophony and whispering pectoriloquy may be present
- During resolution coarse crepitations
- If parapneumonic effusion develops, additional signs of pleural effusion

TABLE 2.78: Complications of lobar pneumonia.

Pulmonary ➢ Delayed/incomplete resolution ➢ Spread to other lobes (rare)	**Cardiovascular** ➢ Acute circulatory failure ➢ Acute pericarditis ➢ Endocarditis (rare)
Pleural: Spread of infection to the pleural cavity: ➢ Sterile pleural effusion ➢ Empyema ➢ Pyothorax	**Neurological** ➢ Mental confusion ➢ Meningism ➢ Meningitis (rare)



FIGS. 2.42A AND B: Chest X-ray. (A) Shows right upper lobe pneumonia; (B) Shows right middle lobe pneumonia.

Complications (Table 2.78)

Q. Write a short essay on complications of lobar pneumonia.

Treatment

General measures

- Administration of oxygen in high concentration to all hypoxemic patients.
- Treatment of pleuritic pain with mild analgesics like paracetamol and NSAIDs. However, few may require pethidine 50–100 mg or morphine 10–15 mg intramuscularly or intravenously.

Antibiotic therapy (Discussed earlier)

Vaccine: Two types:

1. Pneumococcal polysaccharide vaccine (PPSV 23): It consists of 23 common capsular serotypes that produce invasive pneumococcal disease (IPD).
   – Advantages: Inexpensive, high coverage for IPD in elderly (60–70%), cost effective in elderly
   – Disadvantages: Induce IgM (T-cell *independent* response, hence no memory cell activation and antibody response last 3–5 years. It does not prevent against pneumococcal disease burden
2. Polysaccharide-protein conjugate pneumococcal vaccine (pneumococcal conjugate vaccine—PCV13): It targets seven serotypes responsible for most of pneumococcal infections in children and adults.
   – *Advantages:* It produces *T-cell dependent response.* High efficacy against pneumococcal pneumonia (CAPITA study), IPD in children, reduces nasopharyngeal carriage. Hence, it is used mainly in children, elderly and high-risk patients.
   – *Disadvantages:* Expensive, small serotype coverage for IPD in elderly (30–40%)

3. Dosing schedule
   - Age ≥65 years
     - No previous pneumococcal vaccine: PCV13 followed by PPSV23. Minimum duration between two vaccines is 1 year except immunocompromised patients (cochlear implant, CSF leak, asplenia, sickle cell disease, CKD, HIV, malignancy, post-transplant)—8 weeks
   - Age <65 years
     - PPSV23 recommended in Cochlear implant, CSF leaks, asplenia, CKD, immunocompromised (HIV, malignancy, post-transplant)
     - Initial PCV13 followed by PPSV 23. If patient received PPSV23 at <65 years of age, 1 year duration between two vaccines. Minimum duration between two PPSV23 doses is 5 years.

Indications for vaccination

See **Table 2.79**.

TABLE 2.79: Indications for vaccination.

➢ Age 65 years
➢ Patients with risk factors: Congestive heart failure, asthma, COPD, diabetes mellitus, chronic liver disease, and alcoholism
➢ Splenectomized patients including patients with sickle cell anemia
➢ Persons living in long-term care facilities
➢ Immunocompromised persons

Staphylococcal Pneumonia

Q. Write a short essay/note on staphylococcal pneumonia.

- It produces bronchopneumonia that is characterized by **widespread** focal/**patchy areas of acute suppurative inflammation.**
- They are centered on bronchioles and bronchi with subsequent spread to surrounding alveoli. The involved alveoli show consolidation.
- **Causative agent:** *S. aureus*. Methicillin-resistant *S. aureus* (MRSA) is an important pathogen in nosocomial pneumonia. Recently, community-acquired MRSA (CA-MRSA) infections (skin and soft tissue infections, and necrotizing pneumonia) developing in previously healthy persons have emerged as a serious clinical condition.
- **Predisposing factors:** Common following influenza, in debilitated patients in hospital, and patient with cystic fibrosis.
- **Characteristic features:**
  - Abscess formation is very common. These abscesses are multiple and often bilateral.
  - Abscesses may rupture into pleura leading to pneumothorax or pyopneumothorax.
- Chest radiograph shows bronchopneumonia. It is often bilateral, with multiple thin-walled cyst-like lesions (pneumatoceles).
- Sputum smear shows gram-positive *S. aureus* in clumps.

Treatment
- Drug of choice is penicillin.
- **Methicillin-sensitive *S. aureus:*** Oxacillin or flucloxacillin.
- **Methicillin-resistant *S. aureus:*** Vancomycin and teicoplanin.
- Ceftaroline and ceftobiprole, new 5th-generation cephalosporins.
- Drugs used in treatment of resistant *Staphylococcus* (discussed in Chapter 3).

Complications

Toxic shock syndrome (TSS).

Klebsiella Pneumoniae (Friedlander's Pneumonia)

Q. Write a short note on diagnosis and treatment of *Klebsiella pneumoniae* (Friedlander's pneumonia).

- **Causative agent:** *K. pneumoniae* (Friedlander's bacillus).
- **Predisposing factors:** Common in alcoholics and diabetics.
- **Characteristics:**
  - Severe illness with a high-mortality rate.
  - Massive consolidation of one or more lobes. The **upper lobes are** most often affected.
  - Abscess formation and pleural effusion are common.

Investigations

- **Sputum:**
  - It may be viscid, jelly-like and blood stained **(red-current-jelly sputum)**. Sometimes it may be purulent or rusty.
  - Sputum smear shows gram-negative *K. pneumoniae*. *K. pneumoniae* can be cultures from the sputum.

Treatment: Antibiotic therapy
- Gentamicin, ceftazidime or ciprofloxacin for 2–3 weeks.
- **Severe cases:** Piperacillin + tazobactam or meropenem.
- **Extended-spectrum β-lactamases (ESBL) producing organisms:** Meropenem (or imipenem + cilastatin), amikacin, and tigecycline. Polymyxin B for highly resistant strains.

- **Chest radiograph:**
  - It shows air space pneumonia, usually in one of the upper lobes, with abscess formation and pleural effusion.
  - *Bulging interlobar fissure* is a characteristic finding.

Atypical Pneumonias

Q. Discuss the etiology (organisms causing atypical pneumonias), clinical features, investigations, and treatment of atypical pneumonias.

- **Causes:** *M. pneumoniae, L. pneumophila, Coxiella pneumoniae, C. burnetii,* viruses (influenza, adenovirus, RSV, measles, VZV, and CMV).
- Evolve much more slowly than bacterial pneumonias.
- Symptoms > signs
- They tend to have a slower onset, often with more prominent extrapulmonary symptoms and complications.
- Suspicion of atypical pathogens as the cause of pneumonia should be thought if patient has three or more of the parameters listed in **Table 2.80**.

TABLE 2.80: Parameters that favor the diagnosis of atypical pneumonias.
➢ Age 60 years
➢ Absence of any underlying comorbid condition
➢ Paroxysmal cough
➢ No expectoration
➢ Few clinical signs on examination of chest
If total WBC count 10,000/mm^3 is added to the above five parameters, then presence of four features indicates a strong likelihood of atypical pneumonia.

Mycoplasma Pneumoniae

Q. Discuss the clinical features, laboratory diagnosis, and treatment of pneumonias caused by *M. pneumoniae.*

Cause: *M. pneumoniae* which lacks cell wall. It is most often associated with atypical pneumonia.

Clinical Presentation

- Onset is insidious ranging from several days to a week.
- Constitutional symptoms include headache exacerbated by cough, sore throat, malaise, and myalgias. Cough is dry, paroxysmal, and worse at night.
- Clinical course is usually mild and self-limited.

Complications

- **Pulmonary complications:** Pleural effusion, empyema, pneumothorax, and respiratory distress syndrome.
- Infection is severe in sickle cell disease and other HbS-related hemoglobinopathies (functional asplenia). Especially prone to digital necrosis in patients with high titer of cold antibodies.
- **Extrapulmonary complications:**
  - *Ear pain:* Bullous myringitis
  - *Erythema multiforme:* Target/iris lesion, Stevens–Johnson (SJ) syndrome
  - Digital necrosis (especially in patients with sickle cell disease)
  - *Neurological:* Encephalitis, cerebellar ataxia, GBS (Guillain–Barré syndrome), transverse myelitis, peripheral neuropathy.
  - *Hematological:* Hemolytic anemia and coagulopathy
    - ♦ *M. pneumoniae* evokes immunoglobulin M (IgM) antibodies which agglutinate human RBCs at 4°C.
  - *Others:* Myocarditis, pericarditis, and pancreatitis.

Laboratory Findings and Diagnosis

- Mild leukocytosis can be found.
- Fourfold increase in antibody titers.
- Cold agglutinins are nonspecific but helpful in diagnosis.

Treatment of *Mycoplasma pneumoniae infection*: Macrolides (azithromycin) is the drug of choice, followed by doxycycline and fluoroquinolones.

Chlamydia Pneumoniae

- **Cause:** *Chlamydophila (Chlamydia) pneumoniae* which is an obligate intracellular organism. Another species of *Chlamydophila* [i.e., *Chlamydia psittaci* can also cause pneumonia. It usually infects birds but occasionally infects humans (psittacosis)].
- **Mode of transmission:** Through respiratory secretions.
- **Incubation period:** Several weeks.
- **Predisposing factor:** Old age with comorbid diseases.

- **Clinical presentation:** Sore throat, headache, low-grade fever, and cough that can persist for months.
- **Chest radiographs:** Show less-extensive infiltrates than those with other causes of pneumonia.

Treatment: Doxycycline is the drug of choice.

Legionella Pneumonia or Legionnaires' Disease

Q. Write a short essay/note on *Legionella pneumonia* (Legionnaires' disease).

- ***Pontiac fever:*** Acute self-limiting febrile illness.
- **Legionnaires' disease:** Pneumonia.
- Most common species causing human infections: *L. pneumophila*
- Most common serogroup causing CAP: 1.
- Most common serogroup causing nosocomial pneumonia: 6
- **Mode of transmission:** Through water droplets originated from infected **humidifier cooling systems,** and from stagnant water in cisterns and shower heads.

Clinical Features

- Fever, chills, and cough with scanty mucoid sputum. Gastrointestinal symptoms like nausea, abdominal pain, and diarrhea are common. **Mental confusion or delirium** may also develop.
- **Uncommon features:** Myocarditis, pericarditis and prosthetic valve endocarditis, glomerulonephritis, pancreatitis, and peritonitis.

Investigations

- **Chest X-ray:** Shows parenchymal lesions progressing from patchy to lobar consolidation. Pleural effusion may be observed in more than one-third patients. Complete clearing of lung infiltrates may take 1–4 months.
- **Blood:** Relative lymphopenia, very high ESR, hyponatremia, proteinuria, and microscopic hematuria.
- **Culture:** Definitive and most sensitive method, requires 3–5 days.
- **Gram's stain:** Many PMNs, no organisms-suggestive (usual picture with all atypical pneumonias).
- **False +ve AFB stain:** *Legionella micdadei* also known as Pittsburgh pneumonia agent.
- **Direct fluorescent antibody (DFA) test:** Sensitive and specific.
- **Antibody:** Fourfold rise in titer from acute to convalescent stage is diagnostic. Serology has only epidemiologic significance.
- **Urinary antigen:** Second to culture in terms of both sensitive and specificity.

Treatment: With any one of the following antibacterials: azithromycin, erythromycin, clarithromycin, telithromycin, doxycycline or an extended-spectrum fluoroquinolone for 7–10 days.

Prognosis: Among the atypical pneumonia, Legionnaires' disease presents with most severe clinical course and can progress more severely if the infection is not treated appropriately and early.

Actinomycosis

Q. Write a short note/essay on *actinomycosis*.

- **Cause:** *Actinomyces israelii,* an anaerobic organism present in the oral cavity as a commensal. When local defenses are impaired, it can produce disease.
- **Forms:**
  - *Oral-cervicofacial actinomycosis:* Present as soft tissue swelling, abscess, and discharging sinuses.
  - *Abdominal actinomycosis:* Presents with discharging sinuses, abdominal mass or abscess.
  - *Pulmonary actinomycosis:* Causes widespread suppurative pneumonia, empyema (often bilateral), and persistent discharging chest wall sinuses.
- Pus obtained from the sinuses show **"sulfur grains" (**inhibit phagocytosis).

Treatment: Antibiotic therapy with benzylpenicillin 18–24 million units intravenously (four divided doses) for 4–6 weeks. This is followed by oral penicillin for 6–12 months. If penicillin allergy Clindamycin or doxycycline or macrolides are indicated. **No role for antibiogram testing.**

Acute Bronchopneumonia

Q. Write a short note on acute bronchopneumonia.

- **Bronchopneumonia** is characterized by **patchy** (scattered solid foci) **area of consolidation in the same or several lobes** of the lung.

- It is a type of secondary pneumonia, invariable preceded by bronchial infection. Common in patients with chronic bronchitis.
- Bronchopneumonia is characterized by widespread focal/**patchy areas** of **acute suppurative inflammation.** They are **centered on bronchioles and bronchi** with subsequent spread to surrounding alveoli. The involved alveoli show **consolidation.** The **consolidated areas are larger and more numerous in lower lobes** (because of the tendency of secretions to gravitate into the lower lobes) and **frequently bilateral.**
- **Predisposing factors:**
  - *In children:* As a complication of **measles or whooping cough.**
  - *In adults:* As a complication of **acute bronchitis or influenza**.
  - *In elderly or debilitated patients:* As **hypostatic pneumonia.**
- Viral pneumonias and "atypical" pneumonias may also present as bronchopneumonia.

Clinical Features

- High fever, severe cough with purulent expectoration, breathlessness, tachypnea, tachycardia, and central cyanosis.
- **Physical signs in the chest:** Early-stage signs of acute bronchitis, later-stage numerous crepitations.
- **Investigations:**
  - *Chest X-ray:* Bilateral mottled opacities, predominantly in the lower zones **(Fig. 2.43)**.
  - *Blood:* Neutrophilic leukocytosis common.

Treatment: Antibiotic therapy.
- **Mild cases:** Ampicillin or cotrimoxazole given orally is effective.
- **Serious cases:** Third-generation cephalosporin along with a macrolide.

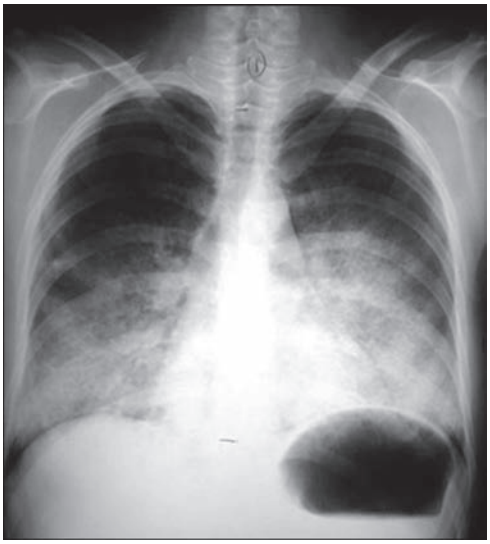


FIG. 2.43: Chest X-ray of bronchopneumonia.

Aspiration Pneumonia

Q. Write a short note on aspiration pneumonia.

- It develops due to abnormal entry of fluid, particulate exogenous substances or endogenous secretions into the lower airways.
- **Types:** (1) Chemical aspiration pneumonia and (2) bacterial aspiration pneumonia.

Predisposing Factors

- **Acute aspiration of gastric contents into the lungs:** Reduced consciousness, as a complication of anesthesia.
- Mechanical disruption of the glottis closure or cardiac sphincter due to tracheostomy, endotracheal intubation, bronchoscopy, upper endoscopy, and nasogastric feeding.
- **Disorders of the upper gastrointestinal tract:** Esophageal disease, surgery involving the upper airways or esophagus, and gastric reflux.
- Dysphagia from neurologic deficits.
- **Miscellaneous conditions:** Protracted vomiting, large volume tube feedings, and feeding gastrostomy.

Chemical Aspiration Pneumonia

- Develops due to the aspiration of substances that are toxic to the lower airways, independent of bacterial infection.
- Chemical pneumonitis associated with the aspiration of gastric acid as a complication of anesthesia particularly during pregnancy **(Mendelson's syndrome).** It can produce an extremely severe and sometimes fatal illness because of the intense destructiveness of gastric acid.

Clinical Features

- Abrupt onset of low-grade fever and dyspnea.
- Physical examination reveals cyanosis and diffuse crepitations.
- **Chest radiograph:** Infiltrates involving dependent pulmonary segments that usually develop within 2 hours of aspiration.

Treatment
- **Tracheal suction:** To clear fluids and particulate matter.
- **Support of respiration:** Mechanical ventilation may be needed.
- **Corticosteroids:** Use controversial.
- **Antibiotics:** In acute events.

Bacterial Aspiration Pneumonia

- Most common type of aspiration pneumonia caused by bacteria that normally reside in the upper airways or stomach.

- **Predisposing factors:** Hospitalized patients with depressed gag reflex, impaired swallowing, or nasogastric or endotracheal tube. Elderly individuals and patients with impaired consciousness (e.g., drug, alcohol, stroke).

Clinical Features

- Highly variable and depend upon the bacteria involved and status of the host.
- Most cases are due to anaerobic bacteria that normally reside in the gingival crevices.
- **Usual features** of pneumonia such as cough, fever, purulent sputum, and dyspnea. The process evolves over a period of several days or weeks instead of hours in usual pneumonia.
- **Other features:** Weight loss and anemia.
- **Oral examination:** May reveal periodontal disease.

Diagnosis

- **Sputum:** Putrid discharge is considered as diagnostic of anaerobic infection.
- **Chest radiograph:** Involvement of dependent pulmonary segments by aspiration. Lower lobes are involved when aspiration occurs in upright position and the superior segments of the lower lobes or posterior segment of the upper lobes when aspiration occurs in recumbent position.

Treatment

No role for routine anaerobic coverage: Routine anerobic coverage was associated with increased VAP, mortality.

- **Penicillin:** However, about 25% of cases are caused by penicillinase-producing anaerobic bacteria.
- Clindamycin (600 mg 8 hourly IV followed by 300 mg 6 hourly orally) is the drug of choice for anaerobic infections above the diaphragm (e.g., pulmonary infections).
- Other drugs include amoxicillin-clavulanate or newer fluoroquinolones (levofloxacin and gemifloxacin).
- Metronidazole alone should not be used as a monotherapy because of failure rate of about 50%.

Hospital-acquired Pneumonia (Nosocomial Pneumonia)

Q. Write a short essay on hospital-acquired pneumonia (nosocomial pneumonia) and its management.

- Hospital-acquired pneumonia or nosocomial pneumonia is a new episode of pneumonia developing in hospital in a patient who is beyond 2 days (>48 hours) of their initial admission to hospital, which was not incubating at the time of admission. Most common resistant organism causing HAP is *S. aureus.*
- Second most common (urinary infection being first) hospital-acquired infection (HAI) and the leading cause of HAI-associated death. The HAPs are serious and may be life-threatening.
- **Ventilator-associated pneumonia:** It is a subcategory of nosocomial pneumonia that occurs in patients who have been on ventilator support for any reason. Pneumonia is termed as VAP if it occurs after 48 hours after endotracheal intubation for mechanical ventilation, but within 72 hours of start of ventilation. Most common mechanism of nosocomial pneumonia is aspiration and most common organism is *S. aureus.* The most common multidrug-resistant gram-negative bacilli (MDR- GNB) causing HAP/VAP: *P. aeruginosa.*
- **Early-onset HAP and VAP:** Occurring within the first 4 days of hospitalization, usually carry a better prognosis, and are more likely to be caused by antibiotic-sensitive bacteria.
- Late-onset HAP and VAP (5 days or more) are more likely to be caused by MDR pathogens, and are associated with increased patient mortality and morbidity.

Criteria

- At least two of three clinical features (fever >38°C, leukocytosis or leukopenia, and purulent secretions), a positive culture of sputum or tracheal aspirate, and a new lung infiltrate.
- *When fever, leukocytosis, purulent sputum, and a positive culture of a sputum or tracheal aspirate are present without a new lung infiltrate, the diagnosis of nosocomial tracheobronchitis should be considered.*

Etiology (Table 2.81)

The organisms causing hospital-acquired pneumonias are different from those causing CAP.

TABLE 2.81: Various etiological agents causing hospital-acquired pneumonia.

➢ **Early onset:** (within 4 days of hospitalization): *S. aureus*, EGNB, *S. pneumoniae* and *H. influenzae*
➢ **Late onset:** (>4 days) *S. aureus, Pseudomonas, Enterobacter, Klebsiella, Acinetobacter*

Risk factors for nosocomial pneumonia: Severe underlying diseases, malnutrition, uremia, alcoholism, cigarette smoking, immunosuppression, increasing age, nasogastric tube (lesser risk with orogastric tube), endotracheal tube, decreased level of consciousness, prior AMA use, decreased gastric acidity, and upper abdominal surgery.

Risk factors for VAP: Reintubation, supine positioning, enteral nutrition, heavy sedation, paralytic agents, H_2 antagonists, and antacids.

Clinical Features

The diagnosis should be considered in any hospitalized or ventilated patient who develops:
- Purulent sputum (or endotracheal secretions)
- New infiltrate on chest radiograph
- Unexplained increase in oxygen requirement
- Core temperature of more than 38.3°C
- Leukocytosis or leukopenia

Investigations

- Patients with hospital-acquired pneumonia should have blood cultures, and microbiological confirmation should be sought whenever possible. Vigorous attempts must be made to obtain respiratory secretions for identification of the organism and determining its sensitivity to antibiotics. In mechanically ventilated patients, bronchoscopy-directed protected brush specimens, bronchoalveolar lavage (BAL) or endotracheal aspirates may be obtained.
- Full blood count (FBC), urea and electrolytes (U&E), erythrocyte sedimentation rate (ESR) and C-reactive protein (CRP), and arterial blood gas analysis and a chest X-ray are performed.

Management/Treatment
- The principles of management are similar to those for CAP. These include adequate oxygenation, appropriate fluid balance, and antibiotics.
- In early-onset HAP:
  - Patients who have not received previous antibiotics can be treated with co-amoxiclav or cefuroxime.
  - If the patient has received a course of recent antibiotics are treated with piperacillin/tazobactam or a third-generation cephalosporin
- In late-onset HAP:
  - The choice of antibiotics must cover the gram-negative bacteria, *S. aureus* (including MRSA) and anaerobes.
  - Antipseudomonal cover by a carbapenem (meropenem) or a third-generation cephalosporin combined with an aminoglycoside.
  - MRSA cover may be provided by glycopeptides (vancomycin or linezolid).
- *Acinetobacter baumannii* is usually sensitive to carbapenems. Emergence of MDR strains have been increasingly noted in the ICU which require combination of Sulbactum, Minocycline or polypetide (polymyxin/colistin) and high dose meropenem (2 g IV over 3 hours extended infusion). Recent data show no superiority of addition of meropenem.

Prevention of Ventilator-associated Pneumonia

See **Table 2.82.**

TABLE 2.82: Prevention of ventilator-associated pneumonia.

- Good hygiene: Rigorous handwashing by caregivers and sterility of any equipment used
- Minimize the risk of aspiration
- Semirecumbent positioning of the patient to prevent aspiration
- Avoidance of gastric distention
- Continuous subglottic suctioning
- Adequate nutritional support
- Early removal of endotracheal and nasogastric tubes
- Oral antiseptic (chlorhexidine 2%) may be used to decontaminate the upper airway
- Use of sucralfate (for stress-ulcer prophylaxis)
- Avoidance of prophylactic antibiotic

Influenza

Q. Discuss the etiology, clinical features, investigations, complications, and management of influenza.

Definition

Influenza is an acute systemic viral infection caused by influenza viruses that primarily affects the upper and/or lower respiratory tract.

Etiology

The influenza virus is a spherical or filamentous enveloped RNA virus.

Influenza A Virus

- It is responsible for pandemics and epidemics.
- **Antigenic variation and influenza outbreaks and pandemics:** The most extensive and severe outbreaks of influenza are due to influenza A viruses. This may be because H and N antigens of these viruses undergo periodic antigenic variation. Immunity following viral infection is type-specific and of short duration.
- **Antigenic/genetic shift:** Major antigenic variations (switch in the hemagglutinin or neuraminidase antigen generates. New influenza A subtypes) are called antigenic/genetic shift. Antigenic shift results in a virus of different antigenic character to which most or all of the population is susceptible and is the basis for influenza pandemics. For example, an antigenic shift involving both the hemagglutinin and the neuraminidase occurred in 1957 resulting in severe pandemic. It was due to shift of predominant influenza A virus subtype from H1N1 to H2N2.
- **Antigenic/genetic drifts:** These are due to minor antigenic variations. Minor antigenic variation periodically involve the hemagglutinin or the neuraminidase component or both. Antigenic drift results in viruses that are different

enough from preceding strains and enables the virus to evade previously acquired immunity and contributes to yearly seasonal epidemics. Influenza viruses also infect other species (e.g., birds and pigs) and these animal strains may infect humans and spread from person to person.

Clinical features

- **Incubation period:** 1–3 days
- **Clinical manifestations:** Influenza is respiratory illness characterized by systemic symptoms, such as sudden onset of headache, fever, chills, myalgia, nausea, vomiting and malaise. The respiratory signs and symptoms include harsh unproductive cough, and sore throat.
- **Physical findings:** Usually minimal in uncomplicated cases. Pharynx may be unremarkable despite a severe sore throat and chest is usually clear.
- In uncomplicated cases, the symptoms usually subside within 3–5 days.

Investigations

- **Blood:** Leukopenia
- **Throat swab or nasopharyngeal aspirate or sputum:** Virus may be detected in throat swabs or nasopharyngeal aspirate or sputum.
- **Reverse-transcriptase polymerase chain reaction (RT-PCR):** Most sensitive and specific technique for detection of influenza viruses.
- **Rapid influenza diagnostic tests (RIDTs):** Detect influenza virus antigens by immunologic or enzymatic techniques.

Complications

- **Pulmonary complications:** Primary influenza viral pneumonia, secondary bacterial pneumonia (most often due to superinfection with *S. pneumoniae*, *S. Aureus,* or other bacteria), mixed viral and bacterial pneumonia, exacerbation of underlying asthma and COPD.
- **Extrapulmonary complications:**
  - Myositis, myocarditis, pericarditis, worsening of underlying congestive heart failure and coronary artery disease
  - Neurological complications: Reye's syndrome in children, encephalitis, or transverse myelitis
  - Postinfluenzal asthenia and depression

Management

- Bed rest till fever subsides.
- Paracetamol 0.5–1 g every 4–6 hourly.
- Avoid aspirin, particularly in adolescents and children, because of its association with Reye's syndrome.
- **Specific antiviral therapy—administration of neuraminidase inhibitor:** Oral oseltamivir (75 mg twice daily) or inhaled zanamivir (10 mg twice daily) for 5 days reduces the severity of symptoms and duration of illness if given within 48 hours of symptom onset of both influenza A and B. Amantadine (100 mg twice a day) and rimantadine (100 mg twice a day) which are useful only for influenza A.
- Specific treatment of complications.
- Oseltamivir did not reduce rate of hospitalization in outpatients, however was associated with increased GI effects in recent metanalysis.

Prevention

Vaccination

- **Indications:** Annual seasonal vaccination is advised to elderly, individuals with chronic medical illnesses (which increases the risk of the complications of influenza), such as chronic cardiopulmonary diseases or immunocompromised or renal disease and their healthcare workers. Specific vaccination gives about 70% protection.
- **Types of vaccine:** Two types of vaccines against seasonal influenza are available, namely inactivated (killed-injectable) and live attenuated vaccines (nasal spray). Both vaccines contain three strains of influenza: an H3N2 virus, an H1N1 (seasonal) virus, and an influenza B virus. Vaccine composition is changed yearly to cover the "predicted" seasonal strains, but vaccination may fail when a new pandemic strain of virus emerges.

Chemoprophylaxis: Antiviral drugs may be used as chemoprophylaxis against influenza. However, they may lead to development of resistance.

Chemoprophylaxis with oseltamivir or zanamivir are helpful against influenza A and B; amantadine and rimantadine are useful for influenza A.

Swine Flu (H1N1 Influenza)

Refer **Chapter 3.**

Avian Influenza

Q. Write short essay on avian influenza.

- Avian influenza is caused by **avian influenza A viruses** and is primary public health concern of the 21st century.
- Avian influenza A viruses caused sporadic avian influenza and small outbreaks in humans, usually after **direct contact with birds** (most commonly poultry). There is no sustained person-to-person transmission.

Clinical Features

- **History of exposure** to infected birds.
- **Incubation period:** Usually 2–5 days after exposure (longer than with human influenza infection).
- **Symptoms:** Fever (at least 100.4°F), cough and myalgia. Watery diarrhea is fairly common. Usually complicated by viral pneumonia causing increasing respiratory distress and cyanosis. Ventilatory support is often necessary and mortality rate may be as high as 50%.

Investigations

- **Blood:** Leukopenia
- **Liver function tests:** Raised hepatic aminotransferase, lactate dehydrogenase, and creatine kinase
- **Chest X-ray:** Infiltrates seen 7 days after the onset of fever.
- **Diagnosis:**
  - Isolation of virus or detection of H5-specific RNA. Pharyngeal swabs are preferred than nasal swabs, because of greater titers in the throat and lower respiratory tract.
  - Reverse transcriptase polymerase chain reaction assays are more sensitive than commercial rapid antigen tests.

Prevention: Whenever there is an outbreak of avian influenza, avoid live animal markets and poultry farms.

Treatment

Drug treatment: Same as H1N1 influenza (*refer* **Chapter 3**)

Non-pharmacologic approaches

- Influenza surveillance: For early warning, travel restrictions, quarantine, use of N 95 particulate respiratory masks, and communication networking.
- Proper poultry handling and personal hygiene (such as handwashing) and minimizing contact with birds during an outbreak.

Severe Acute Respiratory Syndrome

Q. Write a short essay/note on severe acute respiratory syndrome and its laboratory findings.

An extraordinary outbreak of the coronavirus-associated disease known as severe acute respiratory syndrome (SARS) was described for the first time in 2002-2003 from Southern China and quickly spread to several countries within less than a year. A death rate of nearly 10% was reported. Though, presently it has not been reported from any country, there is possibility of either human or animal reservoirs of the virus and this may lead to return of SARS.

Etiology

SARS is caused by a previously unknown novel coronavirus (SARS-CoV). It is not closely related to any of the previously described coronaviruses.

Mode of spread: By close person-to-person contact via droplet transmission, or fomite.

Clinical Features

- **Incubation period:** 2–7 days (range 1–14 days).
- **Symptoms:** Begins as a systemic disease with persistent fever, chills/rigor, malaise, headache, and myalgias followed by nonproductive/dry cough and dyspnea within 1–2 days. About 25% have diarrhea and about 20% develop ARDS over a period of 3 weeks.
- Older individuals may present with decrease in general well-being, poor feeding, fall without the typical febrile response.

Laboratory Findings

Blood

- Lymphopenia due to the destruction of both $CD4^+$ and $CD8^+$ T-cells. Thrombocytopenia may occur as the disease progresses.

- Raised serum levels of aminotransferases, creatine kinase, and lactate dehydrogenase
- **Laboratory features of low-grade disseminated intravascular coagulation (DIC):** Thrombocytopenia, prolonged activated partial thromboplastin time and raised D-dimer.

Chest X-ray

- Most frequently peripheral and lower zone or interstitial infiltrates show patchy areas of consolidation. There will no cavitation, hilar lymphadenopathy, or pleural effusion.
- Patchy areas may progress to diffuse involvement.
- Spontaneous pneumomediastinum may occur in few.
- Chest X-ray may be normal in about 25%.

Diagnosis

- **Detection of SARS-CoV:** A rapid diagnosis of SARS-CoV infection by reverse transcription PCR (RT-PCR). The samples use includes:
  - *During early phase:* Respiratory tract (e.g., nasopharyngeal aspirate) samples and plasma
  - *During later phase:* Urine and stool specimen
- Quantitative measurement of SARS-CoV RNA in blood with RT-PCR technique
- Viral culture of respiratory tract samples

Treatment

- No specific therapy
- **Supportive care:** It is the mainstay in therapy and includes maintenance of fluid and electrolyte balance, oxygenation and, if necessary, ventilation with proper protection of healthcare workers.
- High-dose methylprednisolone (0.5 g daily) is given if pneumonia or hypoxemia develops. However, its benefit remains to be established.

Middle East Respiratory Syndrome

Q. Write short note on Middle East respiratory syndrome.

- Caused by novel coronavirus, namely Middle East respiratory syndrome (MERS)-CoV
- Most cases have been reported in Arab Peninsula and neighboring countries with more than 70% from Saudi Arabia.
- **Mode of transmission:** Human-to-human transmission can occur although spread may not be efficient. Unlike SARS coronavirus, MERS-CoV does not preferentially infect healthcare workers.

Source of infection: Camels are major reservoir host for MERS-CoV and an animal source of infection in humans.

Incubation period: 2–14 days (mean 5.2 days)

- The virus causes more severe disease in individuals who are old, with weakened immune systems, with chronic diseases (e.g., cancer, chronic lung disease, and diabetes).

Clinical features: Typical MERS symptoms include fever, cough, and shortness of breath. Pneumonia is common, but not always present. Gastrointestinal symptoms, including diarrhea, have also been reported.

- About 36% of reported patients with MERS died mostly due to ARDS. Renal failure also can develop.
- **Investigations:** Leukopenia, thrombocytopenia, raised elevated liver enzymes, LDH, and creatinine kinase.

Treatment

Neither vaccine nor specific treatment is currently available. Treatment is supportive and based on the patient's clinical condition.

DIFFUSE PARENCHYMAL LUNG DISEASE/INTERSTITIAL LUNG DISEASE

Q. Discuss the etiology and approach to diffuse parenchymal lung disease/interstitial lung diseases.

Definition: Diffuse parenchymal lung diseases [DPLD, also referred to as **interstitial lung diseases (**ILDs)] are a **heterogeneous group of conditions affecting the parenchyma of** lung and alveoli.

- **Diffuse:** Refers to the nonspecific **radiological patterns**
- **Lung parenchyma** includes the alveoli, the alveolar epithelium, the capillary endothelium, the spaces between those structures (interstitium) as well as the perivascular and lymphatic tissues.
- **ILD-A misnomer:**
  - Pathophysiologic processes are not restricted to the interstitium.
  - All of the several cellular and soluble constituents that make up the gas exchange units and others
  - Bronchiolar lumen
  - Terminal bronchioles

- Pulmonary parenchyma beyond the gas exchange units
- Pleura
- Lymphatic
- Lymph nodes

- **Interstitium of the lung:** It is defined as continuum of loose connective tissue throughout the lung and composed of three subdivisions:
  1. **Bronchovascular** (axial), surrounding the bronchi, arteries, and veins from the lung root to the level of the respiratory bronchiole.
  2. **Parenchymal** (acinar), situated between the alveolar and capillary basement membranes
  3. **Subpleural**, situated beneath the pleura, as well as in the interlobular septae
     - Interstitium of the lung is not normally visible radiographically. It becomes visible only when its volume increases.
     - They are noninfectious and nonmalignant and usually chronic diseases that diffusely involve the lungs.
     - Diffuse parenchymal lung diseases can progress fibrosis in the interstitium (interstitial pulmonary fibrosis) of the lung.

TABLE 2.83: Major causes of interstitial lung disease (ILD)/diffuse parenchymal lung diseases (DPLD).

ILD/DPLD due to known causes	*ILD/DPLD due to unknown causes*
Nongranulomatous interstitial inflammation (predominant inflammation and fibrosis)	
➢ Drugs: Antibiotics, gold, amiodarone, D-penicillamine, nitrofurantoin, Chemotherapeutic (e.g., busulfan, bleomycin, methotrexate, CCNU) ➢ Asbestos ➢ Fumes and gases ➢ Radiation ➢ Aspiration pneumonitis ➢ Residual of acute respiratory distress syndrome ➢ Smoking related: Pulmonary Langerhans cell histiocytosis, respiratory bronchiolitis interstitial lung disease and desquamative interstitial pneumonia ➢ Acute eosinophilic pneumonia	➢ Idiopathic interstitial pneumonias (IIPs) ➢ Idiopathic pulmonary fibrosis ➢ Pulmonary autoimmune rheumatic diseases (e.g., systemic lupus erythematosus, rheumatoid arthritis) ➢ Pulmonary alveolar proteinosis ➢ Eosinophilic pneumonias ➢ Lymphangioleiomyomatosis (LAM) ➢ Diffuse pulmonary hemorrhage ➢ Goodpasture syndrome ➢ Idiopathic pulmonary hemosiderosis ➢ Lymphocytic interstitial pneumonia (seen in HIV)
Granulomatous interstitial inflammation	
➢ Hypersensitivity pneumonitis (organic dusts) ➢ Inorganic dusts: For example, silicosis, berylliosis	➢ Sarcoidosis ➢ Pulmonary Langerhans cell histiocytosis ➢ Granulomatous lung disease with vasculitis (e.g., Wegener, Churg–Strauss)

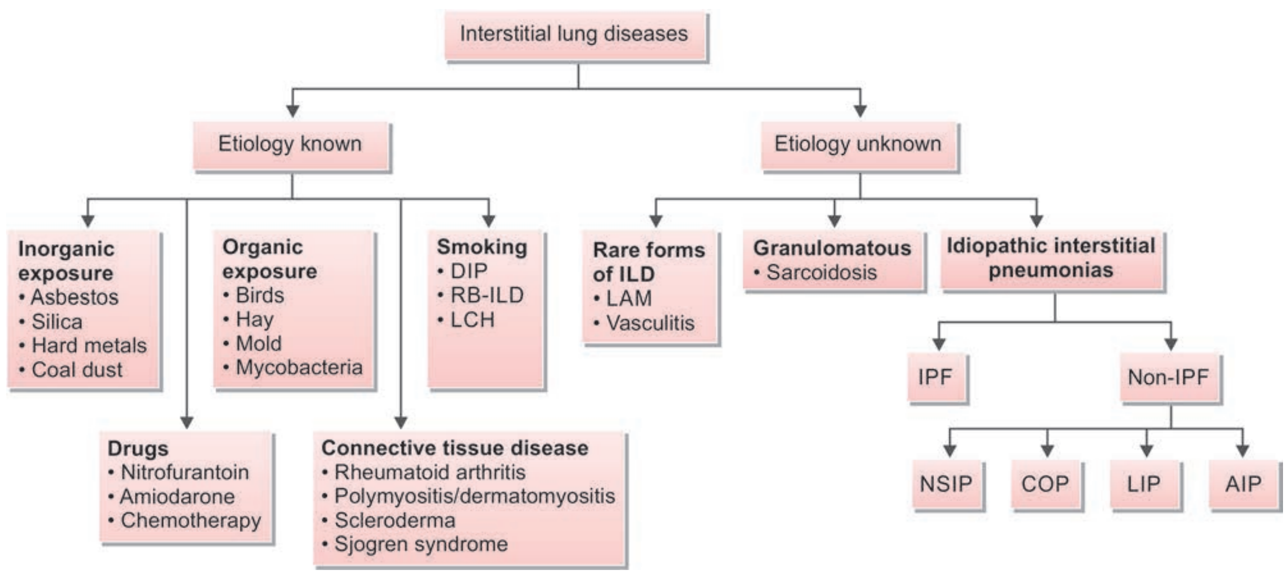
Etiology (Table 2.83)

Q. Write short essay/note on causes/etiology of interstitial lung disease.

Many diseases can produce interstitial lung disease. Based on the major underlying histopathology, ILDs can be divided into two groups: (1) those associated with predominant inflammation and fibrosis and (2) those with a predominantly granulomatous reaction in interstitial or vascular.

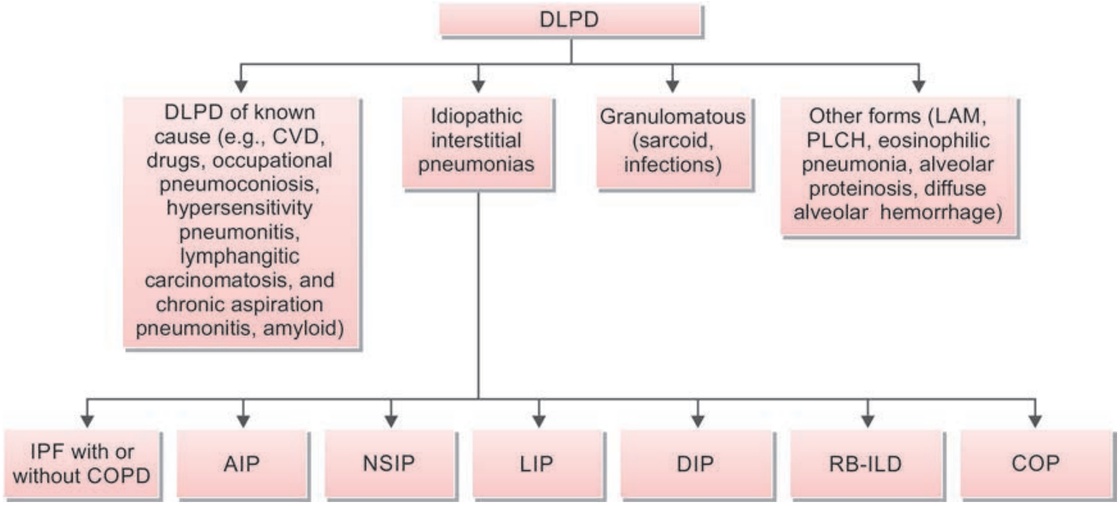
Flowchart 2.16 shows classification of ILD and **Flowchart 2.17** shows classification of DLPD.

FLOWCHART 2.16: ILD classification.



(LAM: Lymphangioleiomyomatosis; IPF: Idiopathic pulmonary fibrosis; AIP: Acute interstitial pneumonia; NSIP: Nonspecific interstitial pneumonia; LIP: Lymphocytic interstitial pneumonia; DIP: Desquamative interstitial pneumonia; RB-ILD: Respiratory bronchiolitis interstitial lung disease; COP: Cryptogenic organizing pneumonia; LCH: Langerhans cell histiocytosis)

FLOWCHART 2.17: Classification of DLPD.

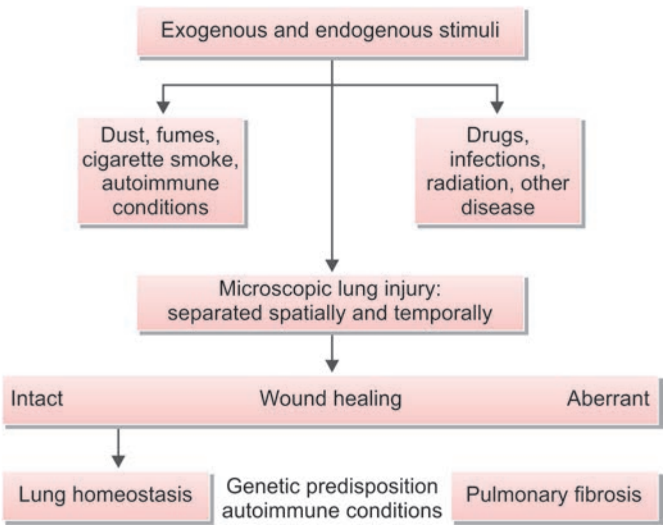


(DPLD: Diffuse parenchymal lung disease; CVD: Cardiovascular disease; LAM: Lymphangioleiomyomatosis; PLCH: Pulmonary Langerhans cell histiocytosis; COPD: Chronic obstructive pulmonary disease; IPF: Idiopathic pulmonary fibrosis; AIP: Acute interstitial pneumonia; NSIP: Nonspecific interstitial pneumonia; LIP: Lymphocytic interstitial pneumonia; DIP: Desquamative interstitial pneumonia; RBILD: Respiratory bronchiolitis interstitial lung disease; Cryptogenic organizing pneumonia)

Pathogenesis

See **Flowchart 2.18.**

FLOWCHART 2.18: Pathogenesis of pulmonary fibrosis.



Clinical Features

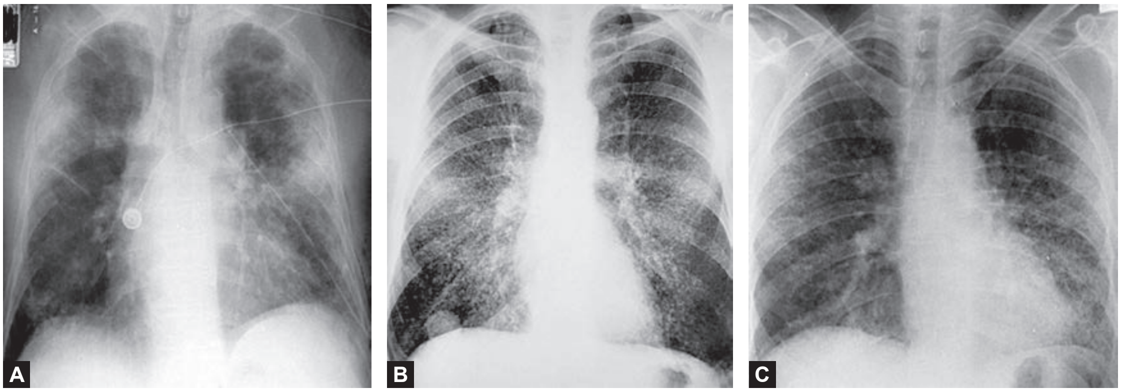
Q. Write short essay/note on clinical features of diffuse parenchymal lung disease/interstitial lung diseases.

Depending on the duration of illness and clinical presentation, ILDs are divided into acute, subacute, and chronic.

- **Acute presentation** (days to weeks):
  - This presentation is **uncommon** but may develop with drugs, acute idiopathic interstitial pneumonia and hypersensitivity pneumonitis.
  - *Chief symptoms* are cough, dyspnea, and occasionally fever.
  - *Chest X-ray* shows diffuse alveolar opacities which can be confused with "atypical" pneumonia.
- **Subacute presentation** (weeks to months):
  - *Symptoms:* Gradually increasing cough and dyspnea over weeks to months.
  - This type of presentation can occur especially with sarcoidosis, and drug-induced ILDs.
- **Chronic presentation** (months to years):
  - Most common presentation in which symptoms are present for months to years.
  - *Common symptoms:* Shortness of breath, dry cough, fatigue, weakness, loss of appetite and weight.

Physical Examination

- Clubbing
- Cyanosis
- Crackles or "velcro rales" are present on chest examination in most forms of ILD
- Scattered late inspiratory high-pitched rhonchi, so-called inspiratory squeaks
- Decreased chest expansion
- Evidence of pulmonary hypertension/cor pulmonale
- Signs of underlying cause



FIGS. 2.44A TO C: Chest X-ray showing patterns of ILD/DPLD.
(A) Alveolar filling pattern; (B) Nodular pattern; (C) Reticular pattern.

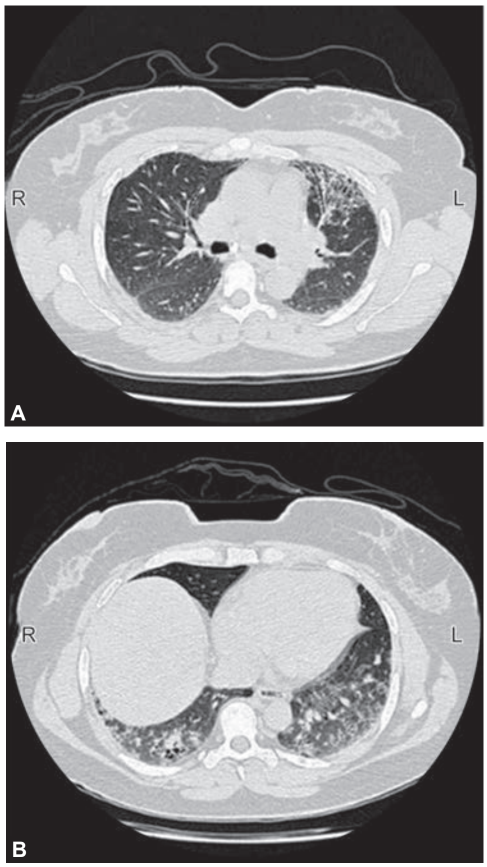
Investigations

Laboratory Investigation

- **Blood:** Total leukocyte count, ESR, renal and liver functions, antinuclear antibodies, rheumatoid factor and circulating immune complexes.
- **Serum precipitins:** Confirm exposure when hypersensitivity pneumonitis is suspected.
- **Angiotensin-converting enzyme levels:** Elevated in ILDs (e.g., sarcoidosis).
- **Antibodies:** Antineutrophil cytoplasmic antibodies and antibasement membrane antibodies are useful if vasculitis is suspected (e.g., Wegener's granulomatosis, Goodpasture syndrome).

Chest Imaging Studies

- **Chest X-ray (Figs. 2.44A to C):** Often shows a bibasilar reticular or linear pattern or ground-glass appearance. Nodular or mixed pattern of alveolar filling and reticular pattern may also be observed. Nodular opacities in the upper lung zones seen in sarcoidosis, chronic hypersensitivity pneumonitis, silicosis, rheumatoid arthritis. Honeycombing (due to small thick-walled cystic spaces and fibrosis) in long-standing conditions. It indicates poor prognosis. Basal fibrosis, pleural thickening and pleural plaques suggest asbestosis.
- **Computed tomography (Figs. 2.45A and B):** High-resolution CT (HRCT) of chest is superior to chest radiography for (i) early detection and confirmation of suspected interstitial lung disease, (ii) demonstrates the extent and distribution of disease, (iii) detection of coexisting disease (e.g., lymphadenopathy, emphysema, carcinoma), and (iv) to determine the most appropriate area for biopsy (if necessary).
- **Pulmonary function tests:** These show findings as described below.
- **Lung biopsy:** Surgical or bronchoscopic cryobiopsy can be performed in cases where CT, and other workup are non-contributory or suggestive of a non-IPF etiology.
- **Approach to evaluation for ILD:**
  - *Suspect ILD:* History, examination, lung function testing, serology (ANA, RF in all patients), high resolution CT thorax
  - *Multidisciplinary discussion:* Pulmonologist, radiologist discuss possibilities and reach a diagnosis. If diagnosis is clear treatment accordingly is initiated. If diagnosis is unclear/inconclusive biopsy is performed and a repeat discussion along with pathologist is performed and appropriate therapy initiated.



FIGS. 2.45A AND B: CT sections showing patterns of ILD/DPLD.

Q. Write short note on pulmonary function tests in interstitial lung disease.

- Restrictive ventilatory defect in the presence of reduced lung volumes and impaired gas transfer.

- Restrictive defect is characterized by reduced FVC, reduced FEV_1, normal or elevated FEV_1: FVC ratio, total lung capacity, reduced diffusion capacity.
- Obstructive pattern is seen in sarcoidosis, hypersensitivity pneumonitis, lymphangioleiomyomatosis, tuberous sclerosis, and neurofibromatosis.
  - *Exercise tests:* Help in the evaluation of severity of the disease. The 6-minute walk test records oxygen saturation before, during, and after exercise and measures the total distance walked.
- **Arterial blood gas:** It may reveal hypoxemia and respiratory alkalosis. Hypercapnia is rare and may develop in the end-stage. Increased alveolar-arterial oxygen gradient $[P(A\text{-}a)O_2]$
- **Fiberoptic bronchoscopy and bronchoalveolar lavage (BAL):** In selected cases (e.g., sarcoidosis, hypersensitivity pneumonitis, cancer) cellular analysis of BAL may be useful in narrowing the differential diagnosis. Transbronchial biopsy may aid in diagnosis.
  - *Sarcoidosis and hypersensitivity pneumonitis:* It shows T-cell lymphocytosis (increased CD4 cells in sarcoidosis and CD8 cells in hypersensitivity pneumonitis).
  - *Idiopathic pulmonary fibrosis (IPF):* It shows predominant neutrophils and eosinophils.
  - *Pulmonary alveolar proteinosis:* It is milky and microscopically contains foamy macrophages with PAS-positive material.
- **Diffuse pulmonary hemorrhage:** It shows RBCs and hemosiderin-laden macrophages.
- **Biopsy: Open or video-assisted lung biopsy is used for confirmation of diagnosis and assessment of activity.**

Treatment

Major Goals

- Treat the underlying cause, if possible.
- Permanent removal of the offending agent if known.
- General
  - Oxygen therapy: For patients with documented hypoxia, SpO_2<89%,–PaO_2<55 mm Hg. Improves exercise tolerance
  - Adequate nutrition and immunizations (pneumococcal, influenza)
  - Pulmonary rehabilitation: Treatment of pulmonary hypertension, single lung transplantation
  - Early identification and aggressive suppression of the acute and chronic inflammatory process, thereby reducing further damage to the lung.
- IPF ILD: Antifibrotics
- Non-IPF ILD: Immunomodulators → Antifibrotic with immunomodulators(If progressive)

Drugs

Immunomodulators: Used in non-IPF(NSIP, organizing pneumonia, CTD-ILD, unclassifiable) ILDs.

- **Corticosteroids** are the mainstay for the suppression of the inflammation present in ILD although not useful in the presence of fibrosis. Prednisone in the dose is 0.5–1 mg/kg once daily for 4–12 weeks. If the patient improves, it is then tapered to maintenance dose of 0.25–0.5 mg/kg level for an additional 4–12 weeks. Steroids dosing >20 mg/day is associated with scleroderma renal crisis and hence not used in SSc
- **Azathioprine:** It improved lung function in CTD-ILD, hypersensitivity pneumonitis (HP) and dermatomyositis). Use in IPF showed increased mortality
- **Cyclophosphamide:** Used in Scleroderma-ILD, Inflammatory myopathies. No data in RA-ILD, HP
- **Mycofenolate Mofetil (MMF):** Preferred over cyclophosphamide in scleroderma-ILD. Recommended 1.5–3 g/day with initial 500 mg BD and gradual uptitration over 3–4 months. Also found effective in CTD-ILD, inflammatory myopathies, HP
- **Tacrolimus:** Effective in anti-synthetase syndrome, MDA-5 associated myositis.

Antifibrotics: Most preferred in IPF.

- **Pirfenidone:** Inhibits growth (TGF-B, TNF-A) factors and procollagens I and II leading to decrease in fibroblast proliferation. It reduced risk of disease progression, 1 year mortality by 50%. Dosing is weekly increments. 1st week: 267 mg TDS (801 mg/d), 2nd week: 534 mg TDS (1602 mg/d), 3rd week onwards: 801 mg TDS
- **Nintedanib:** Inhibits tyrosine kinase thereby causing reducing lung function decline. Dosing 150 mg BD daily. Combination therapy with nintedanib with pirfenidone showed no superiority.

Idiopathic Interstitial Pneumonias

Idiopathic interstitial pneumonias (IIPs) are characterized by diffuse inflammation and fibrosis in the lung parenchyma in which the etiology is not known. They form a major subgroup of interstitial lung diseases or diffuse parenchymal lung diseases. The lung parenchyma shows varying combinations of fibrosis and inflammation.

Classification of idiopathic interstitial pneumonias (ATS/ERS joint consensus statement-2013). Based on mainly histological patterns/in order of relative frequency is presented in **Table 2.84**.

TABLE 2.84: Classification of idiopathic interstitial pneumonias based mainly on histological patterns/in order of relative frequency.

- Chronic Fibrosing: IPF, NSIP
- Smoking related: Desquamative interstitial pneumonia, respiratory-bronchiolitis ILD (RB-ILD)
- Acute/Subacute: Cryptogenic organizing pneumonia, acute interstitial pneumonia
- Rare: Lymphocytic interstitial pneumonia (LIP), idiopathic pleuro-parenchymal fibroelastosis

Idiopathic Pulmonary Fibrosis

Q. Write short essay/note on clinical features, investigations, and treatment of idiopathic pulmonary fibrosis.

Also known as usual interstitial pneumonia (UIP) and was previously called as cryptogenic fibrosing alveolitis (CFA).

Definition: It is a progressive fibrosing interstitial pneumonia of unknown cause.

Contributory factors: Cigarette smoking, chronic aspiration, antidepressants, infections (e.g., Epstein–Barr virus), exposure to occupational dusts (e.g., wood and metal dusts) and chronic gastroesophageal reflux.

Clinical features and investigations are discussed under ILD.
- **HRCT:** UIP pattern seen. Presence of UIP pattern still warrants evaluation for other causes.
- **UIP pattern:** Basal and subpleural distribution, honeycombing ± traction bronchiectasis, reticulations, absence of alternative pattern
- **Probable UIP:** No honeycombing.
- **Indeterminate:** Subtle reticulations. No specific distribution
- **Alternative diagnosis:** Mosaic attenuation (air trapping), ground glass opacities, pleural effusion/thickening, upper or midzone predominance with perilymphatic/peribronchovascular distribution.

Diagnostic Criteria
1. Exclusion of other known causes of ILD (e.g., domestic and occupational environmental exposures, CTD, drug toxicity) AND either (2) or (3).
2. The presence of the HRCT pattern of UIP as mentioned above.
3. Specific combinations of HRCT patterns and histopathology patterns in patients subjected for sampling.

Treatment
- Antifibrotics: Pirfenidone/nintedanib—reduce disease progression, mortality.
- Supportive management as mentioned in treatment of ILD

OCCUPATIONAL LUNG DISEASES

Q. Write short essay/note on common occupational lung diseases and their etiology/common causes of pneumoconiosis.

- Exposure to dusts, gases, vapors, and fumes at work can cause many types of lung disease. It is difficult to distinguish occupational and environmental lung diseases from those of nonenvironmental origin.
- **Pneumoconioses** are defined as **lung diseases produced by organic as well as inorganic particulates and chemical fumes and vapors.** However, sometimes the term occupational lung diseases and pneumoconiosis are used interchangeably.
- **Common** pneumoconiosis is due to inorganic (mineral) dusts. Occupational exposure and associated lung diseases are listed in **Table 2.85**.

TABLE 2.85: Occupational exposure and associated lung diseases.

Occupational exposure	*Disease associated with chronic exposure*
Diseases due to inorganic (mineral) dusts: Common pneumoconiosis	
➢ Coal ➢ Silica ➢ Asbestos ➢ Beryllium ➢ Iron oxide ➢ Tin dioxide	➢ Coal-worker's pneumoconiosis (CWP) ➢ CWP ➢ Silicosis ➢ Asbestos-related diseases ➢ Berylliosis ➢ Siderosis ➢ Stannosis
Diseases due to organic dusts	
➢ Cotton, flax or hemp dust ➢ Moldy hay, grain, straw ➢ Mold malting ➢ Contaminated bagasse (sugar cane)	➢ Byssinosis ➢ Farmer's lung ➢ Malt worker's lung ➢ Bagassosis
Diseases due to gases and fumes	
➢ Irritant gases, isocyanates cadmium ➢ Platinum salts	➢ Occupational asthma, bronchitis, ARDS ➢ Occupational asthma
Diseases due to biological substances	
Proteolytic enzymes, allergens from animals and insects (excreta), contaminated grain dust	Occupational asthma, bronchitis
Diseases due to chemicals and radioactive substances	
Polycyclic hydrocarbons, radon	Bronchial carcinoma

Coal-workers' Pneumoconiosis

Q. Write a short essay/note on coal-worker's pneumoconiosis.

Definition: Coal-workers' pneumoconiosis (CWP) is the **parenchymal lung disease caused by prolonged inhalation of carbon particles** (coal mine dust).

- Coal-worker's pneumoconiosis develops in coal miners with a prolonged history of inhalation of coal dust. The coal mine dust particles about 2–5 µm in diameter are retained in the small airways and alveoli of the lung. The incidence of the disease depends on the total dust exposure, and availability of ventilation and dust suppression. The risk of this disease is reduced due to improved ventilation and working conditions.
- Incidence is more in anthracite coal miners compared to bituminous miners.

Classification: Based on the size and extent of radiographic nodularity, CWP is subdivided into:
- Simple coal-worker's pneumoconiosis (SCWP)
- Progressive massive fibrosis (PMF).

Simple Coal-worker's Pneumoconiosis

- Simple coal-worker's pneumoconiosis (SCWP) refers to the deposition of coal dust in the lung and produces small radiographic (micronodular) shadows/nodules on the chest X-ray in an otherwise asymptomatic individual. It is the most common type of pneumoconiosis.
- It develops after prolonged exposure (15–20 years) to coal dust.
- SCWP does not impair lung function and does not progress if the miner leaves the industry/exposure ceases.
- Grading of simple coal-workers pneumoconiosis: Depending on the chest X-ray appearance, simple CWP is divided into three categories **(Table 2.86)**.
- Simple pneumoconiosis (categories 2 and 3) can progress to progressive massive fibrosis (PMF).

TABLE 2.86: Grading of simple coal-worker's pneumoconiosis.

Category	*Chest X-ray appearance*
1	Few small round opacities
2	Numerous small round opacities but normal lung markings still visible
3	Very numerous small round opacities and normal lung markings partly or totally obscured

Progressive Massive Fibrosis

Q. Write a short essay/note on progressive massive fibrosis.

- Progressive massive fibrosis (PMF) progresses even after the miner leaves the industry/exposure ceases.
- **Chest X-ray:** Characterized by single or multiple round fibrotic nodules/masses several centimeters in diameter (>1 cm in size to large dense masses), invariably in the upper lobes. Sometimes, these nodules show central necrotic cavities.
- **Lung function tests:** Mixed restrictive and obstructive ventilatory defect with loss of lung volume, irreversible airflow limitation and reduced gas transfer.
- **Clinical features:** Progressive breathlessness/dyspnea and cough with blackish sputum **(melanoptysis)**.
- **Complications:** Respiratory failure, right ventricular failure, and Caplan's syndrome.

Caplan's Syndrome

Q. Write a short essay/note on Caplan's syndrome (rheumatoid pneumoconiosis).

- Caplan's syndrome is the combination of pneumoconiotic nodules with seropositive rheumatoid arthritis.
- Pneumoconiotic nodules are round, fibrotic and measure 0.5–5 cm diameter and are found mainly in the periphery of the lung fields.
- Rheumatoid factor is positive and pathological features are similar to a rheumatoid nodule.
- This syndrome can also see in pneumoconiosis (e.g., silicosis) other than CWP.

Asbestos-related Lung and Pleural Diseases

Q. Write a short essay/note on the asbestos-related diseases of lungs and pleura.

- **Asbestos types:** Asbestos has the unique property of occurring naturally as a fiber. It is remarkably resistant to heat, acid, and alkali. Major types are—**chrysotile** (white asbestos), **crocidolite (blue asbestos) and amosite (brown asbestos)**. All of these types **are fibrogenic** and have the potential to cause asbestos-related diseases. It is widely used for roofing, insulation and fireproofing.
- **Occupational exposure:** It is highest with workers involved in the production of asbestos products (mining, milling, and **manufacturing), shipyard** and construction industries. Exposure can also occur in automobile and railroad workers and those involved in thermal and electrical wire insulation.

Asbestos-related Diseases

- Asbestosis (progressive pulmonary fibrosis)—highest dose is required to cause asbestosis.
- Benign localized pleural fibrous plaques—calcification of parietal pleura more pronounced on the diaphragm and mediastinum.

- Benign pleural effusions
- Carcinoma of lung
- Malignant mesotheliomas of pleura and mesothelioma of peritoneum (rare)
- Laryngeal carcinomas

Asbestosis

- **Definition:** Asbestosis is defined as interstitial fibrosis of the lung caused by exposure to **asbestos dust**. It does not include asbestos-induced pleural diseases and carcinoma of lung that are found in asbestos-exposed workers.
- **Duration of exposure:** Development of asbestosis is directly related to the intensity and duration of exposure. It develops after moderate-to-severe exposure for at least 10 years.
- **Clinical features:** Progressive exertional breathlessness, finger clubbing, and late-inspiratory crepitations/crackles over the lower zones of lung.
- **Investigations:**
  - *Radiography:*
    - Chest radiographic hallmark of asbestosis is presence of irregular or linear opacities in the lower lung fields. An indistinct heart border or a "ground-glass" appearance in the lung fields may be observed.
    - HRCT: May show subpleural curvilinear lines 5–10 mm in length which appear to be parallel to the pleural surface. Plain chest X-ray may not show pleural disease; however, CT can demonstrate pleural disease in more than 90% cases. Honeycomb lung may also be seen.
  - *Pulmonary function tests:* Restrictive pattern with a decreased both lung volumes and diffusing capacity.
  - *Lung biopsy:* May show asbestos bodies and fibrosis.
- **Complications:** Respiratory failure, right ventricular failure, and bronchial carcinoma

Lung Cancer

- Most common cancer associated with asbestos exposure. Minimum latent period ranges from 15 to 20 years between first exposure and development of cancer.
- Histological, it may be squamous cell carcinoma or adenocarcinoma.

Mesothelioma

- Mesothelioma is a primary malignant tumor of mesothelial lining of pleura and/or peritoneum and both can develop with asbestos exposure and is most commonly caused by blue asbestos.
- In contrast to lung cancer, development of mesothelioma has no association with smoking.
- Relatively short period of exposure (even 1–2 years) that occurred more than 20–40 years ago is associated with development of mesothelium.
- **Clinical presentation:** Persistent chest pain (due to involvement of chest wall), breathlessness (due to pleural effusion), and usually unilateral hemorrhagic pleural effusion. As the tumor advances, it encases the underlying lung, may invade into the lung parenchyma, the mediastinum, and the pericardium.
- Diagnosis is confirmed by video-assisted thoracoscopic biopsy of pleura.
- Mesothelioma is almost invariably fatal.

Silicosis

Q. Write a short essay/note on silicosis.

Definition: Silicosis is a **parenchymal lung disease** associated with **inhalation of crystalline silicon dioxide** (silica).

- **Occupational exposure:** High-risk occupations include mining, stone cutting, sand blasting, and quarrying (e.g., granite), pottery and ceramics industry, foundry work, boiler scaling, glass and cement manufacturing, etc.
- Silica is highly fibrogenic and causes the development of hard nodules that coalesce as the disease advances. Pulmonary fibrosis depends on dose and develops after many years of exposure.
- **Associated risk:** Silicosis is associated with increased risk of tuberculosis (silicotuberculosis), lung cancer and COPD. The increased risk is lifelong even if exposure ceases. Chemoprophylaxis using INH for 9 months is recommended if latent tuberculosis is diagnosed with a positive tuberculin test. Other diseases that can develop in silicosis include chronic renal insufficiency and autoimmune diseases (e.g., scleroderma, rheumatoid arthritis, and Wegener's granulomatosis).

  Forms of silicosis (silica-induced lung disease): Chronic, acute, or accelerated

- **Chronic or simple silicosis:**
  - Most common form of silicosis. It occurs after many decades (10 to 20 years) of exposure to relatively low levels of silica.
  - Clinically characterized by gradually progressive dyspnea and dry cough. It is often compatible with normal life.
  - **Radiological features:** Variable and similar to those of coal-worker's pneumoconiosis. It may show multiple well-circumscribed 3–5 mm-nodular opacities mainly in the mid- and upper zones. As the diseases progresses, extensive fibrosis develops and resembles PMF. The involvement of hilar lymph nodes by chronic silicosis is characteristic (but is uncommon and non-specific), with a tendency toward peripheral calcification, which produces the so-called **egg-shell calcification**.
  - **Pulmonary function tests:** Mixed pattern of obstruction and restriction with a reduced diffusion capacity.
  - No specific treatment
- **Acute silicosis:**
  - It occurs after intense exposure (very high concentration of dust) to very fine crystalline silica dust over a few months (about less than 10 months of exposure) and is usually rapidly fatal within years.
  - Characterized by pulmonary edema, interstitial inflammation, and accumulation of proteinaceous fluid rich in surfactant within the alveoli.
  - **Chest radiograph:** May show miliary infiltration or areas of consolidation. HRCT chest may show pattern known as "crazy paving".
- **Accelerated silicosis:**
  - It occurs after a few years (typically 5–10 years) of exposure to silica.
  - More aggressive course associated with rapidly progressive features of dyspnea and pulmonary fibrosis.
  - Involves mainly middle and lower zones of the lungs as compared to chronic silicosis.

	Acute silico-proteinsosis	***Chronic simple***	***Chronic complicated***	***Accelerated silicosis***
Presentation (time)	Few weeks	10–20 years	10–20 years	3–10 years
Exposure	High	Low-mod.	Low-mod.	High-mod.
Clinical features	SOB + cough	Usually asymptomatic + SOB/cough	Symptomatic (SOB/cough)	SOB/cough
Diagnosis	BAL milky (d/t surfactant) HRCT: Crazy pavement appearance	[UL] nodules (1 cm) centrilobular perilymphatic CXR-Egg shell calcifications. Conglomerated masses	[UL] nodules (>1 cm) More peripheral and grows towards hilum	Same as chronic complicated silicosis

PULMONARY FIBROSIS

A wide variety of lung disorders heals by producing fibrous tissue and results in fibrosis of the lung.

Types of Pulmonary Fibrosis

Q. **Write a short essay/note on types of pulmonary fibrosis and causes of interstitial fibrosis.**

1. **Replacement fibrosis:** In this, the damaged lung parenchyma (e.g., suppuration and infarction) replaced by fibrous tissue.
   - *Common causes:* Pulmonary tuberculosis, bronchiectasis, lung abscess, pulmonary infarcts, and necrotizing pneumonias
2. **Focal fibrosis:** In focal fibrosis, the extent of fibrosis varies from small nodular lesions to extensive areas.
   - *Common causes:* Coal-worker's pneumoconiosis (CWP), asbestosis, and silicosis
3. **Interstitial fibrosis:** In this, fibrosis is diffuse and represents the end result of interstitial lung diseases.
   - *Common causes* of interstitial lung disease **(***See* **Table 2.85)**.

Clinical Features of Replacement Fibrosis

History of cough with/without expectoration and dyspnea. Sputum may be blood-tinged.

Physical findings in the chest

- **Inspection and palpation:**
  - *Features on the affected side:* Shifting of the trachea and mediastinum toward the affected side. Drooping of the shoulder, flattening, hollowing, and crowding of the ribs, diminished movements of the chest wall and scoliosis of the spine. The hemithorax is smaller, its expansion and spinoscapular distance are diminished and spinoacromial distance is increased. Vocal fremitus is usually decreased.
  - Chest expansion is reduced.
- **Percussion: On the affected side,** there is impaired note.

- **Auscultation:**
  - When fibrosis is extensive, the intensity of breath sound is reduced and vesicular in character with prolonged expiration.
  - Vocal resonance is reduced. Crepitations are heard.
  - Sometimes if trachea or major bronchus is pulled up due to fibrosis, tubular bronchial breathing can be heard.

Fibrothorax

- **Causes of fibrothorax:** Empyema, pleural effusion, traumatic hemothorax, tuberculosis, benign asbestos pleural effusion, connective and collagen vascular disorders, uremia, paragonimiasis, and drug-induced (e.g., ergot alkaloids, bromocriptine, pergolide, methysergide, and methotrexate).
- **Clinical features** of fibrothorax: (1) marked limitation of chest movements, (2) mediastinal shift to same side, trachea may be central or shifted to affected side, (3) decrease in size of hemothorax, and (4) crowding of ribs.

Radiological findings on the affected side: Shift of the mediastinum toward the affected side. Smaller hemithorax than the other unaffected hemithorax. Crowding of the ribs, pulling of hilum upward, raised diaphragm (tenting), and lung fields show fibrous bands.

Management/treatment of replacement fibrosis
- **Symptomatic measures:** For example, breathing exercises, expectorants
- **Treatment of infections:** Antibiotics
- **Treatment of the underlying disease/cause:** For example, tuberculosis
- **Surgery:** Resection in selected cases.

LUNG CANCER (BRONCHIAL CARCINOMA)

Primary Bronchial Tumors

Q. Write a short essay/note on classification of primary bronchial tumors.

Classification of primary tumors of the lung **(Table 2.87)**.

TABLE 2.87: Classification of tumors of the lung.

Benign tumors (5%)	*Malignant tumors (95%)*
Benign epithelial tumors ➢ Papillomas ➢ Adenomas **Mesenchymal tumors** ➢ Chondroma ➢ Lipoma **Miscellaneous tumors** ➢ Pulmonary hamartoma ➢ Sclerosing hemangioma	**Nonsmall-cell carcinoma** (75%) ➢ Squamous or epidermoid carcinoma ➢ Large-cell carcinoma ➢ Adenocarcinoma – Bronchioloalveolar cell carcinoma (presently termed adenocarcinoma in situ) **Small-cell carcinoma** (oat cell carcinoma) (25%) **Adenosquamous carcinoma/Sarcomatoid carcinoma** **Salivary gland tumor:** Mucoepidermoid carcinoma, adenoid cystic carcinoma **Carcinoid tumor:** Typical carcinoid, atypical carcinoid

Q. Discuss the etiology/risk factors, clinical features, investigations, diagnosis, metastatic and nonmetastatic complications, and management of bronchial carcinoma (bronchogenic carcinoma, lung cancer).

- Carcinoma of the lung is the **most common cause of cancer death**. This is mainly due to the carcinogenic effects of **cigarette smoke.**
- In the past, the term bronchogenic carcinoma was used for primary lung cancer, to indicate the origin from the bronchi. Now it is known that about 5% of primary lung cancers do not arise from bronchus. Hence, the term lung cancer is used.
- Bronchial carcinoma is the most common primary malignant tumor of the lung and arises from the bronchial epithelium or mucous glands.

Incidence

- **Age and gender:** Mostly found between **50 and 80 years** of age. More common in males, but there is a recent increase in females due to increased smoking among females.
- More in urban than rural dwellers and more in smokers than nonsmokers.

Etiology

- **Cigarette (tobacco) smoking:**
  - There is strong evidence that tobacco smoking is the most important cause of cancer of lung. Smoking also multiplies the risk of other carcinogenic influences, such as asbestos and uranium. The **strongest association** of smoking is with **squamous-cell and small-cell carcinomas**. The risk of cancer is directly proportional to the amount of daily smoking, tendency to inhale and duration of smoking habit. Compared to nonsmokers, the smokers have 10 times and heavy smokers (more than 40 cigarettes per day for several years) are at 60 times more risk of lung cancer. Females are more susceptible to tobacco carcinogens than males. Cessation of smoking for 10 years reduces risk but not to the level in nonsmokers. However, only 11% of heavy smokers develop lung cancer, which indicates that genetic factors are involved. **Pipe and cigar smokers have lower incidence than cigarette smokers**. Bidi smoking may be associated with more risk of lung cancer than cigarette smoking. Secondhand smoke exposure (passive smoking) is also a risk factor. Adenocarcinoma is usually not related to smoking.

- More than 60% of new lung cancers occur in nonsmokers (smoked <100 cigarettes per lifetime) or former smoker (smoked 100 cigarettes per lifetime, quit 1 year), 1 in 5 women and 1 in 12 men diagnosed with lung cancer have never smoked.
- **Major carcinogens in tobacco smoke** include both initiators and promoters (such as phenol derivatives). These include **polycyclic aromatic hydrocarbons (e.g.,** Benzo[*a*]pyrene, dibenzanthracene) and **radioactive elements and other contaminants** (e.g., arsenic, nickel, cadmium, molds, and vinyl chloride).
- **Environmental tobacco smoke = Mainstream smoke + Sidestream smoke**
- Mainstream smoke refers to the smoke inhaled by smoking directly while sidestream smoke refers to the smoke emitted by the burning cigarette. Both contain carcinogens.
- **In Indians** with lung cancer, history of active tobacco smoking was found in 87% of males and 85% of females. History of passive tobacco exposure is found in 3%. Thus, 90% of all cases in India resulted from tobacco exposure.
- The relative risk of developing lung cancer is: 2.64 for beedi smokers, 2.23 for cigarette smokers and 2.45 as the overall relative risk (RR).
- **Lung cancer in never smokers:**
  - *Definition:* <100 cigarettes in lifetime. It accounts for 25% of lung cancers and is distinctly associated with—female cases, east Asian populations, positive family history, adenocarcinoma type, *EGFR* mutation, and better prognosis.

- **Industrial and occupational hazards:**
  - **Ionizing radiation:** High dose of ionizing radiation is carcinogenic (uranium mining—oat-cell carcinoma is the most common type).
  - **Uranium:** It is weakly radioactive and uranium miners (both nonsmokers and smokers) have higher incidence of lung cancer than in the general population.
  - **Asbestos:** Exposure increases the risk and risk increases when associated with smoking. It develops after a latent period of about 10 to 30 years. In insulation and shipyard workers, risk of lung cancer increases after 10 years of exposure, and with concurrent smoking, the risk increases to 90-fold.
  - **Other carcinogens:** Chloromethyl, ethers, and mustard gas (squamous and undifferentiated are the most common types), chromium, nickel, vinyl chloride, polyaromatic hydrocarbons, cadmium, formaldehyde and dioxin.
- **Atmospheric pollution:** The risk of **lung cancers is higher in urban areas** than rural areas, suggesting role of air pollution. Major air pollutants include polycyclic hydrocarbons from fossil fuels and motor vehicle exhaust, especially diesel smoke. Studies from China have shown coal burning at home is a significant risk factor for development of lung cancer in nonsmoking females. Coal smoke contains potential carcinogens SO_2, CO, TSP, B (a) P, radon, thorn.
- **Diet: Vitamin A deficiency** leads to squamous metaplasia increased **susceptibility to cancer**. Folate, carotenoid-rich fruits and vegetables, vitamin E and beta-carotene are associated with reduced risk of lung cancer.
- **Idiopathic pulmonary fibrosis:** It is associated with increased risk of lung cancer.
- **Prior lung diseases, such as** chronic bronchitis, emphysema, and **tuberculosis**
- **Genetic predisposition:** Genetic polymorphisms involving cytochrome *P-450* gene *CYP1A1* increased capacity to metabolize procarcinogens present in the cigarette smoke ®increased risk of lung cancer.
  - First-degree relatives of lung cancer probands have a two-to-three fold excess risk of lung cancer and other cancers, many of which are not smoking-related.
  - Individuals with inherited mutations in *RB* (retinoblastoma) and *p53* (Li–Fraumeni syndrome) genes may develop lung cancer.
- A rare germline mutation (T790M) involving the epidermal growth factor receptor (EGFR) may be linked to lung cancer susceptibility in nonsmokers.

Examples of the oncogenes and tumor suppressor genes involved in lung cancer are presented in **Table 2.88**.

TABLE 2.88: Examples of the oncogenes and tumor suppressor genes involved in lung cancer.

Histology	***Oncogene***	***Tumor-suppressor genes***
Adenocarcinoma	*EGFR, KRAS, ALK*	*TP53, CDKN2A/B(p16, p14), LKB1*
Squamous-cell carcinoma	*EGFR, PIK3CA, IGF-1R*	*TP53, TP63*
Small-cell carcinoma	*MYC, BCL-2*	*TP53, RB1, FHIT*
Large-cell carcinoma	(not well studied)	

Pathology

Clinical subgroups of lung cancer (Table 2.89): Lung carcinomas can be divided into two clinical subgroups on the basis of likelihood of metastases and response to therapies.

TABLE 2.89: Clinical subgroups of lung cancer.

Small-cell lung carcinomas (SCLC)—highly metastatic, high response to chemotherapy	***Nonsmall-cell lung carcinomas (NSCLC)—less metastatic, less responsive***
Small-cell (oat-cell) carcinoma	➢ Squamous (epidermoid) carcinoma ➢ Adenocarcinoma (including bronchioloalveolar carcinoma) ➢ Large-cell carcinoma

- Small-cell lung cancer (SCLC) and nonsmall-cell lung cancer (NSCLC) (includes adenocarcinoma, squamous-cell carcinoma and large-cell carcinoma) account for approximately 90% of all epithelial lung cancers.
- Among women and young adults (<60 years), adenocarcinoma tends to be the most common form of lung cancer.
- Neuroendocrine tumors of the lung arise from Kulchitsky cells of the bronchial mucosa and consist of typical carcinoid, atypical carcinoid, and small-cell lung cancer.

	Small cell cancer (oat cell cancer)	***Squamous cell carcinoma***	***Adenocarcinoma***	***Large cell cancer***
Association with smoking	Associated with smoking (almost 90%)	Associated with smoking	Most common cancer among non-smoker	Associated with smoking
Gender		Most common cancer among male	Most common cancer among females	
Location of the primary	Centrally located	Centrally located	Peripherally located	Peripherally located
Histology	Histology shows presence of scany cytoplasm with granular chromatin (**Salt and Pepper appearance**) and basophilic staining of vascular walls (**Azzopardi effect**)	Presence of keratinisation	Glandular pattern of growth seen	Cells have large nuclei with prominent nucleoli
Common paraneoplastic syndromes	Syndrome of inappropriate antidiuretic hormone secretion (SIADH) Cushing's syndrome Lambert Eaton syndrome	Hypercalcemia Grade IV clubbing Hypercoagulability	Similar to squamous cell carcinoma	Gynecomastia

Lung Cancer in India

- Nonsmall-cell lung cancer constitutes 75–80% of lung cancers. More than 70% of them are detected in stages III and IV. Thus, curative surgery cannot be done in these cases. In many Western countries, adenocarcinoma has become the most common lung cancer. However, in India, squamous-cell carcinoma is still the most common carcinoma in both males and females.
- Small-cell lung carcinoma constitutes 20% of all lung cancers. It is detected at an extensive stage in 70% of patients at the time of diagnosis.

Site: Depending on the site, the lung cancers are divided into:
- **Central (hilar):** Lung cancers arise in and around the **hilus** of the lung are the **most common**. About 75% arise from first, second, and third-order bronchi. Squamous-cell and small-cell carcinomas are generally centrally placed.
- **Peripheral:** Lung cancers may also arise in the periphery of the lung from the alveolar septal cells or terminal bronchioles. They are **usually adenocarcinomas, including bronchioloalveolar type**.

Bronchioloalveolar carcinoma (BAC) is a subtype of adenocarcinoma that grows along the alveoli without invasion. It can present with profuse, mucoid sputum (bronchorrhea).

Clinical Features

More than 50% of patients present with locally advanced or metastatic disease at the time of diagnosis. Majority patients present with signs, symptoms, or laboratory abnormalities that may be due to the (i) primary lesion, (ii) local tumor growth, (iii) invasion or obstruction of adjacent structures, (iv) metastatic tumor, or (v) as a paraneoplastic syndrome. Clinical manifestations of bronchial carcinoma can be studied under the following headings:

1. Manifestations due to the location of the primary tumor/growth **(Table 2.90).**
2. **Manifestations due to the regional spread of tumor in the thorax (Table 2.91).**

TABLE 2.90: Manifestations due to the location of the primary tumor.

Due to a central or endobronchial growth	***Due to a peripheral growth***
➢ Cough ➢ Hemoptysis ➢ Wheeze and stridor ➢ Breathlessness/dyspnea ➢ Postobstructive pneumonitis presenting with fever and productive cough	➢ Chest pain from pleural or chest-wall involvement ➢ Dyspnea on a restrictive basis ➢ Symptoms of lung abscess due to cavitation of tumor

TABLE 2.91: Manifestations of lung cancer due to regional spread of tumor in the thorax.

Pathological basis	***Local effects***
Obstruction of airway by tumor	Pneumonia, abscess, atelectasis, focal emphysema
Tracheal obstruction	Stridor (a harsh inspiratory noise) and dyspnea
Local spread into pleura	Pleuritis and malignant effusion
Local spread to pericardium	Pericarditis, effusion, tamponade

Contd...

Contd...

Pathological basis	***Local effects***
Compression of SVC by tumor	Superior vena cava syndrome
Invasion of structure	**Its effects**
Recurrent laryngeal nerve	Hoarseness of voice and "bovine" cough
Phrenic nerve	Diaphragm paralysis and dyspnea
Sympathetic ganglia	Horner syndrome
Esophagus	Dysphagia
Chest wall (direct extension)	Destruction of rib producing rib pain, pathological fractures and intercostal neuralgia
Superior sulcus tumor (destruction of the T1 and C8 roots in the lower part of the brachial plexus by an apical lung tumor)	Pancoast's syndrome (pain in the inner aspect of the arm, sometimes with small muscle wasting in the hand)
Lymphatic obstruction	Pleural effusion
Vascular obstruction	Superior vena cava (SVC) syndrome
Pericardial and cardiac extension	Pericardial effusion, cardiac tamponade, arrhythmias or cardiac failure

3. **Manifestations of extrathoracic metastasis (Table 2.92).**

Paraneoplastic Syndromes (Table 2.93)

Q. Write short essay/note on:
- **Paraneoplastic syndromes/manifestations associated with bronchogenic carcinoma.**
- **Mention the nonmetastatic manifestations/complications of bronchial carcinoma.**

- Paraneoplastic syndromes are symptom complexes in cancer patients which are not directly related to mass effects or invasion or metastasis or by the secretion of hormones indigenous to the tissue of origin.
- Lung cancer may occasionally be associated with paraneoplastic syndromes. They are summarized in **Table 2.94**.
- Paraneoplastic syndromes manifest due to the peptide hormones secreted by the tumor and may be in the first manifestations of lung cancers. They are often relieved by successful treatment of the primary lung cancer. They are common with small-cell carcinoma.

TABLE 2.92: Manifestations of extrathoracic metastasis.

Site of metastasis	***Manifestations***
Adrenal glands (~50%)	Usually asymptomatic, Addison's disease
Liver (30–50%)	Anorexia, nausea, biliary obstruction, weight loss, right upper quadrant pain, and pain
Brain (20%)	Headache, vomiting, focal neurologic deficits, epileptic seizures confusion, and raised intracranial pressure
Bone (20%)	Bony pain, pathologic fractures or raised alkaline phosphatase
Bone marrow	Cytopenias or leukoerythroblastosis
Lymph node	Lymphadenopathy—commonly to mediastinal, cervical, supraclavicular region (scalene node), and even axillary or intra-abdominal nodes
Epidural and bone metastases	Spinal cord compression (if spine is involved) syndromes

TABLE 2.93: Paraneoplastic syndromes in lung cancer.

System involved	***Manifestations***
Systemic	Anorexia, cachexia, weight loss, fever
Endocrine	➢ Syndrome of inappropriate secretion of antidiuretic hormone causing hyponatremia (small-cell carcinoma) ➢ Ectopic ACTH secretion resulting in hypokalemia, rather than full-blown Cushing's syndrome (small-cell carcinoma) ➢ Hypercalcemia due to ectopic production of parathyroid hormone or parathyroid hormone-related peptide (squamous-cell carcinoma) ➢ Carcinoid syndrome, acromegaly, hypoglycemia, hyperthyroidism ➢ Gynecomastia
Skeletal	➢ Clubbing of fingers (usually NSCLC) nonsmall-cell carcinoma ➢ Hypertrophic pulmonary osteoarthropathy (usually adenocarcinoma)—HPOA
Neuromyopathic	➢ Polyneuropathy ➢ Myelopathy encephalomyelitis, opsoclonus-myoclonus, limbic encephalitis ➢ Cerebellar degeneration ➢ Myasthenia (Lambert–Eaton syndrome) and retinal blindness (small-cell carcinoma)
Hematological	➢ Migratory venous thrombophlebitis (Trousseau's syndrome) ➢ Marantic endocarditis ➢ Disseminated intravascular coagulation (DIC) ➢ Anemia, granulocytosis, leukoerythroblastosis
Cutaneous	➢ Dermatomyositis ➢ Acanthosis nigricans
Renal	➢ Nephrotic syndrome ➢ Glomerulonephritis

Lambert–Eaton Myasthenic Syndrome (LMS)

- Autoimmune disorders of neuromuscular junction due to anti-voltage-gated calcium channel antibodies.
- Proximal muscle weakness, usually in lower extremities
- Occasionally autonomic dysfunction and rarely cranial nerve symptoms
- Frequently depressed deep tendon reflexes
- In contrast to patients with myasthenia gravis, strength improves with serial effort.

Treatment of LMS

- Chemotherapy is the initial treatment of choice.
- 3, 4–diaminopyridine: It increases duration of presynaptic action potential by blocking potassium efflux, prolonging the activation of VGCC and increasing calcium entry into nerve terminals

Differential Diagnosis of Large Bronchus Obstruction (Table 2.94)

Investigations

Main aims of investigation: (1) to confirm the diagnosis, (2) establish the histological cell type, (3) to stage the extent of the disease, and (4) to assess fitness to undergo treatment.

- **Imaging:**
  - *Chest X-ray*
  - Plain chest radiographs may show evidence of lung cancer or nonspecific appearances **(Table 2.95)**.
  - *Evidence of lung cancer:* A normal finding does not rule out the presence of an underlying tumor. Tumor is visible if greater than 1 cm diameter.
- **Computed tomography (CT) of chest and abdomen:** It is important for diagnosis and staging of carcinoma. Used—
  - To know the tumor size and extent of disease [pleural extension, to detect occult abdominal disease (e.g., liver and adrenals)].
  - To evaluate mediastinal or hilar lymph-node involvement. If nodal involvement is observed, it should be confirmed by histopathology.
  - *To plan biopsy procedures:* Used for guided biopsy of suspected lesions to know whether a tumor is accessible by bronchoscopy or percutaneous CT-guided biopsy.
  - To assess the response to treatment.
- **Cytological examination:** Specimens that may be obtained for examining for malignant cells include—sputum, bronchial brushings and washings, percutaneous (CT-guided) needle-aspiration biopsy from a peripheral tumor and fine-needle aspiration of lymph node, skin, or liver in patients with metastasis.
- **Fiberoptic bronchoscopy:**
  - It permits visualization and biopsy of an intrabronchial tumor.
  - Useful for collecting bronchial washings from the suspicious segments for cytological study.
  - It helps in assessment of operability and the proximity of central tumors to the main carina.

TABLE 2.94: Causes of large bronchus obstruction.

Common
➢ Neoplasm (bronchial carcinoma or adenoma)
➢ Enlargement of tracheobronchial lymph nodes (malignant or tuberculous)
➢ Inhaled foreign bodies (more common in the right lung)
➢ Bronchial casts or plugs composed of inspissated mucus or blood clot (especially, asthma, cystic fibrosis, hemoptysis, and debility)
➢ Collections of mucus or retained mucus in the bronchi as a result of ineffective expectoration (especially, postoperative following abdominal surgery)
Rare
➢ Aneurysm of aorta
➢ Giant left atrium
➢ Pericardial effusion
➢ Congenital bronchial atresia
➢ Bronchial stricture (e.g., following tuberculosis or bronchial surgery/lung transplant)

TABLE 2.95: Common radiological presentations of bronchial carcinoma.

Findings	*Interpretation*
Unilateral enlargement at hilum	Central tumor or enlargement of hilar lymph node or a peripheral tumor in the apical segment of the lower lobe
Peripheral lung opacity with/ without cavitation, solitary pulmonary nodule	Usually irregular but well circumscribed, and may contain irregular cavitation (squamous-cell carcinoma may cavitate)
Collapse of the whole lung, lobe or segment	Endoluminal (tumor within the bronchus) tumor or compression of the main bronchus by enlarged lymph glands may produce occlusion leading to complete collapse
Broadening of mediastinum, enlarged cardiac shadow, elevation of a hemidiaphragm	Paratracheal lymphadenopathy (widening of the upper mediastinum). Malignant pericardial effusion (enlargement of the cardiac shadow). Raised hemidiaphragm (phrenic nerve palsy). Mediastinal widening due to mediastinal invasion
Pleural effusion	Usually indicates tumor invasion of pleural cavity/space distal to a bronchial carcinoma. Usually, effusion is unilateral and large and may mask an underlying mass or pleural tumor (D/D mesothelioma)
Destruction of rib (osteolytic lesions)	Direct extension of tumor or bloodborne metastasis
Single mass/diffuse, multinodular lesion/fluffy	Bronchoalveolar carcinoma

- **Percutaneous aspiration and biopsy:** Peripheral lung tumors cannot be seen by fiberoptic bronchoscopy. Hence, samples are obtained by aspiration or biopsy through the chest wall under CT guidance.
  - Diagnosis of lung cancer rests on the morphologic or cytological features correlated with clinical and radiographic findings.
- Uses of **immunohistochemistry:**

Feature	*Adeno CA*	*Squamous cell CA*	*Small cell CA*
Smoking association	More common among non-smokers	Strong	Strongest association
Genetic alterations	EGFR and p53	p53	RB MYC 953
Location	Peripheral	Central/Hilar	Anywhere
Paraneoplastic	HPOA	Hypercalcemia	SIADH
Tumor marker	Mucin, TTF-1, Napsin A	P40, CYFRA-21 (Cytokeratin-19 soluble fragment)	Chromogranin, synaptophysin

- **Investigations for staging to guide treatment:**
  - Fine-needle aspiration: (1) Endoscopic ultrasound-guided fine-needle aspiration (FNA) of tumor or lymph node, (2) endobronchial ultrasound-guided fine-needle aspiration, and (3) endoscopic ultrasound via the esophagus.
  - Scalene node biopsy, mediastinoscopy, pleural aspiration, and biopsy and barium swallow.
  - Ultrasonography examination of liver and adrenal glands.
    If metastasis is suspected: Bone scans, bone marrow trephine biopsies, and brain CT.
    Positron emission tomography (PET) imaging /CT: It helps in detecting intrathoracic tumor, assessing the extent of mediastinal nodal involvement and detection of distant metastases. PET images combined with CT are better than CT or PET alone.
- **Other investigations:** Complete blood count for the detection of anemia, and biochemistry for liver involvement, hypercalcemia, and hyponatremia.
- **Molecular testing for guiding treatment:** Test for *EGFR* mutations and ALK fusion.

Staging

- **Staging of nonsmall-cell carcinoma (Table 2.96)**: Staged into stages I to IV depending on the size and location of primary tumor, involvement of lymph nodes, and distant metastasis (TNM International Staging System).
- **Staging of small-cell carcinoma:** It is staged into limited-stage and extensive-stage disease depending on whether the tumor can be encompassed within a tolerable radiation therapy port.

TABLE 2.96: T, N, and M: classification for lung cancer.

T: Primary tumor	
Tx	Primary tumor cannot be assessed or tumor proven by presence of malignant cells in sputum or bronchial washings but not visualized by imaging or bronchoscopy
T0	No evidence of primary tumor
Tis	Carcinoma in situ
T1	Tumor ≤3 cm in greatest dimension surrounded by lung or visceral pleura without bronchoscopic evidence of invasion more proximal than the lobar bronchus (i.e., not in the main bronchus)[1]
T1(mi)	**Minimally invasive adenocarcinoma**[2]
T1a	**Tumor ≤1 cm in greatest dimension**[1]
T1b	**Tumor >1 cm but ≤2 cm in greatest dimension**[1]
T1c	**Tumor >2 cm but ≤3 cm in greatest dimension**[1]
T2	Tumor >3 cm **but ≤5 cm** or tumor with any of the following features:[3] ➢ **Involves main bronchus regardless of distance from the carina but without involvement of the carina** ➢ Invades visceral pleura ➢ **Associated with atelectasis or obstructive pneumonitis that extends to the hilar region, involving part or all of the lung**
T2a	**Tumor >3 cm but ≤4 cm in greatest dimension**
T2b	**Tumor >4 cm but ≤5 cm in greatest dimension**
T3	**Tumor >5 cm but ≤7 cm in greatest dimension** or associated with separate tumor nodule(s) in the same lobe as the primary tumor or directly invades any of the following structures: chest wall (including the parietal pleura and superior sulcus tumors), phrenic nerve, parietal pericardium
T4	**Tumor >7 cm in greatest dimension** or associated with separate tumor nodule(s) in a different ipsilateral lobe than that of the primary tumor or invades any of the following structures: **diaphragm**, mediastinum, heart, great vessels, trachea, recurrent laryngeal nerve, esophagus, vertebral body, carina

Contd...

Contd...

N: Regional lymph node involvement	
Nx	Regional lymph nodes cannot be assessed
N0	No regional lymph node metastasis
N1	Metastasis in ipsilateral peribronchial and/or ipsilateral hilar lymph nodes and intrapulmonary nodes, including involvement by direct extension
N2	Metastasis in ipsilateral mediastinal and/or subcarinal lymph node(s)
N3	Metastasis in contralateral mediastinal, contralateral hilar, ipsilateral or contralateral scalene, or supraclavicular lymph node(s)
M: Distant Metastasis	
M0	No distant metastasis
M1	Distant metastasis present
M1a	Separate tumor nodule(s) in a contralateral lobe; tumor with pleural or pericardial nodule(s) or malignant pleural or pericardial effusion[4]
M1b	**Single extrathoracic metastasis[5]**
M1c	**Multiple extrathoracic metastases in one or several organs**

[1]The uncommon superficial spreading tumour of any size with its invasive component limited to the bronchial wall, which may extend proximal to the main bronchus, is also classified as T1a.
[2]Solitary adenocarcinoma, ≤ 3 cm in size, with a predominately lepidic pattern and ≤ 5mm invasion in any one focus.
[3]T2 tumours with these features are classified T2a if 4 cm or less in greatest dimension or if size cannot be determined, and T2b if greater than 4 cm but not larger than 5 cm in greatest dimension.
[4]Most pleural (pericardial) effusions with lung cancer are due to tumour. In a few patients, however, multiple microscopic examinations of pleural (pericardial) fluid are negative for tumour, and the fluid is non-bloody and is not an exudate. Where these elements and clinical judgement dictate that the effusion is not related to the tumour, the effusion should be excluded as a staging descriptor.
[5]This includes involvement of a single distant (non-regional) lymph node.

Lung Cancer Stage Grouping (Eighth Edition)

T/M	*Label*	*N0*	*N1*	*N2*	*N3*
T1	T1a ≤1	IA1	IIB	IIIA	IIIB
	T1b >1 to 2	IA2	IIB	IIIA	IIIB
	T1c >2 to 3	IA3	IIB	IIIA	IIIB
T2	T2a Cent, Visc Pl	IB	IIB	IIIA	IIIB
	T2a >3 to 4	IB	IIB	IIIA	IIIB
	T2b >4 to 5	IIA	IIB	IIIA	IIIB
T3	T3 >5 to 7	IIB	IIIA	IIIB	IIIC
	T3 Inv	IIB	IIIA	IIIB	IIIC
	T3 Satell	IIB	IIIA	IIIB	IIIC
T4	T4 >7	IIIA	IIIA	IIIB	IIIC
	T4 Inv	IIIA	IIIA	IIIB	IIIC
	T4 Ipsi Nod	IIIA	IIIA	IIIB	IIIC
M1	M1a Contr Nod	IVA	IVA	IVA	IVA
	M1a Pl Dissem	IVA	IVA	IVA	IVA
	M1b Single	IVA	IVA	IVA	IVA
	M1c Multi	IVB	IVB	IVB	IVB

Cent, Visc Pl: tumor involving main bronchus (not carina), atelectasis to hilum or visceral pleura; T3 Inv: tumor invading chest wall, peicardium, phrenic nerve; T3 Satell: separate tumor nodule(s) in the same lobe; T4 Inv: tumor invading mediastinum, diaphragm, heart, great vessels, recurrent laryngeal nerve, carina, trachea, esophagus, spine; T4 Ipsi Nod: tumor nodule in different ipsilateral lobe; M1a Contr Nod: separate tumor nodule(s) in contralateral lobe; M1a Pl Dissem: malignant pleural/peicardial effusion or pleural/pericardial nodules; M1b Single: single extrathroacic metastasis; M1c Multi: multiple extrathroacic metastases (1 or >1 organ).

Staging of the Small-cell and Nonsmall-cell Carcinomas (Table 2.97)

Q. Discuss the staging, diagnosis and treatment of small-cell carcinoma.

- **Assessing fitness for treatment:** Before radical treatment, an assessment of fitness for treatment should be done. These include:
  - Evaluation of performance (Karnofsky performance status: Eastern cooperative Oncology Group Performance Status).
  - Full lung function test: Specifically FEV_1 and diffusion capacity
  - Tests for any cardiovascular disease if there is evidence of disease (e.g., cardiopulmonary exercise testing, stress echo or occasionally preoperative angiography).

Treatment

1. **Nonsmall-cell carcinoma**
   a. **Surgical treatment***:* Surgical resection of the tumor is done in early stage (1A, 1B, IIA, IIB and selected IIIA) of non-small cell lung cancer.
   - **Postoperative chemotherapy** (adjuvant therapy): Where surgical staging of resected lung cancer has lymph node involvement, patients need adjuvant chemotherapy.
   - **Preoperative chemotherapy** (neoadjuvant chemotherapy: May improve survival in stage IIIB patients and can effectively"down-stage" disease. Commonly used drugs are carboplatin, paclitaxel ± bevacizumab (an antiangiogenesis drug).

   Contraindications to surgical resection
   - Stage IIIB or IV
   - **Extensive invasion into surrounding structures:** Involvement of vena cava, atrium, recurrent laryngeal or phrenic nerve and contralateral lymph nodes. Others include SVC obstruction, malignant effusion, and pericardial tamponade
   - **Medically unfit:** Poor cardiac or pulmonary status, predicted postoperative FEV1% <40%, predicted postoperative DLCO% <40% and exercise studies for marginal candidates.

   b. **Radiotherapy**

   Radiotherapy for cure: Radiotherapy is much less effective than surgical therapy. In selected cases with adequate lung function and early stage NSCLC, high-dose radiotherapy or continuous hyperfractionated accelerated regimens (CHART) is a good alternative to surgical resection. It is the treatment of choice if surgery is contraindicated due to comorbidities.

   Radiation treatment for symptoms: Radiation can be used as palliative for distressing symptoms/complications in lung cancer. Indications for palliative radiotherapy are: Bone and chest wall pain from metastases or direct invasion, recurrent hemoptysis and occluded trachea, bronchi and superior vena cava obstruction.

   Stereotactic ablative radiotherapy (SABR) is able to deliver large doses of radiation with a high precision of 1–2 mm to small lesions of <1 cm^3 using an external 3D coordinated system that is linked with movements during the respiratory cycle reserved for people with early stage cancer who have been unable/unwilling to undergo surgical resection due to medical comorbidities.

   Percutaneous radiofrequency ablation (RFA) for early stage peripherally based lung tumors or metastases in medically inoperable patients.

   c. **Chemotherapy**

   Adjuvant chemotherapy with radiotherapy improves response rate and extends median survival.
   - **Older drugs:** Drug regimen, namely cisplatin/carboplatin + 1 other (Paclitaxel/Docetaxel/Gemcitabine). Common side effects are nausea and vomiting and are best treated with 5-HT3 receptor antagonists.
   - **Newer targeted agents** against epidermal growth factor receptors and tyrosine kinases in NSCLC (in particular adenocarcinoma) in selected patients. Can also be used where intravenous chemotherapy produces severe toxicity or as second- line chemotherapy.
     - *EGFR* mutation-positive NSCLC: Erlotinib, afatanib, and gefitinib
     - Crizotinib for *ALK/ROS-1* mutation-positive NSCLC
     - Osimertinib for *EGFR T790M* mutation-positive NSCLC.
     - Immune checkpoint inhibitors: Pembrolizumab, nivolumab, and atezolizumab which act through the programmed death-ligand 1/2 (PD-L1 and PD-L2)
     - Tirapazamine, thalidomide, vandetanib
     - Antiangiogenesis (e.g., bevacizumab)
     - Nintedanib with docetaxel
     - Vinorelbine + Carboplatin or Cisplatin

   Treatment of nonsmall-cell lung cancer is summarized in **Table 2.98**.

- **Small-cell carcinoma**

  General treatment plan for small-cell carcinoma **(Table 2.99)**
  - If the patient responds to treatment plan, prophylactic radiotherapy to brain is advised. High-dose radiotherapy to the entire brain is also advised to patients with documented brain metastasis.
  - **Chemotherapeutic regimens for small-cell carcinoma**
    - Commonly used combination Cisplatin/carboplatin + etoposide, ifosfamide, irinotecan plus cisplatin, carboplatin and etoposide with/without vincristine.

  Laser therapy and tracheobronchial stents
  - In selected patients with inoperable lung cancer, palliation of symptoms due to tracheobronchial narrowing from intraluminal tumor or extrinsic compression can be achieved by bronchoscopic laser treatment.
  - It helps to clear tumor tissue and allow reaeration of collapsed lung. It is useful mainly when the tumor is the main bronchi.
  - In case of extrinsic compression by malignant nodes, endobronchial stents can be used to maintain airway patency.

TABLE 2.97: Staging of carcinomas of lung.

Stages of nonsmall-cell carcinoma based on TNM classification	
Occult carcinoma	TXN0M0
IA	T1a,bN0M0
IB	T2aN0M0
IIA	T2bN0M0; T1a,bN1M0; T2aN1M0
IIB	T2bN1M0; T3N0M0
IIIA	T1a,b, T2a,b, N2M0; T3N1, N2M0; T4N0, N1M0
IIIB	T4N2M0; any T N3M0
IV	Any T, any N, M1
Stage of small-cell carcinoma	***Description***
Limited stage	➢ 30–40% of small cell lung cancers ➢ Confined to the hemithorax, mediastinum, and ipsilateral supraclavicular lymph node ➢ Within the confines of radiation port
Extensive stage	➢ 60–70% of small cell lung cancers ➢ Any distant spread

TABLE 2.98: Treatment of nonsmall-cell lung cancer.

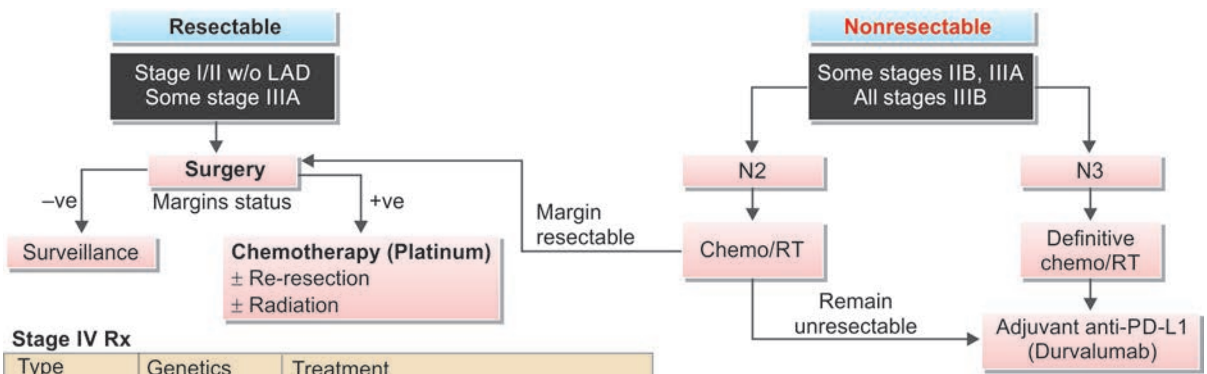
Stage IA:
- Lobectomy is treatment of choice. T1N0, lobectomy has 70% 5-year recurrence-free survival
- If inoperable:
  - 30% cure rate with XRT alone
  - Stereotactic radiosurgery (CyberKnife)
  - Radiofrequency ablation

Stage IB:
- Lobectomy
- Adjuvant chemotherapy adds a 4–12% survival benefit. Best in tumors >4 cm

Stage II:
- Lobectomy is treatment of choice
- Adjuvant chemotherapy now standard
- Consider adjuvant XRT to mediastinum

Stage III:
- Combination chemotherapy with XRT is the treatment of choice
- Surgery has yet to be established consistently as benefit in randomized trials
- Neoadjuvant therapy followed by surgical resection is option in IIIA

Stage IV: Chemotherapy



Stage IV Rx

Type	Genetics	Treatment
Adeno/others	EGFR	Osimertinib
	ALK	Alectinib
	ROS1	Crizotinib
	BRAF V600E	Dabrafenib + MEK(Trametinib)
	NTRK fusion	Larotrectinib
Squamous	PD-L1<50	Pembrolizumab + Pacli + Carboplatin
	PD-L1>50	Pembrolizumab
Brain Mets		Radiation

TABLE 2.99: General plan of treatment small-cell carcinoma.

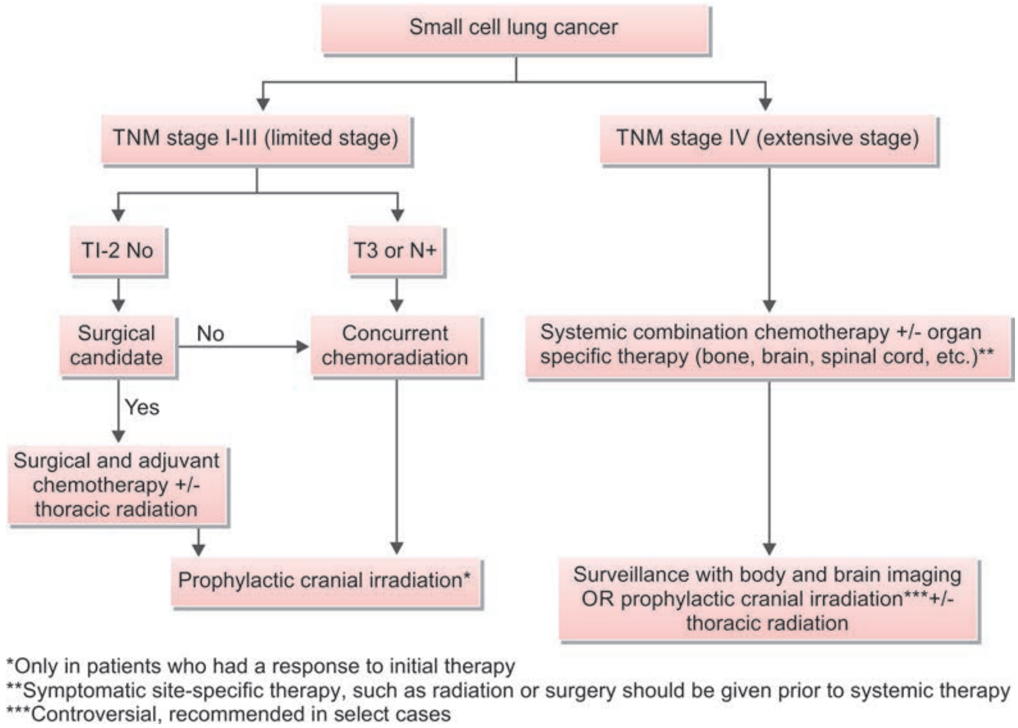
Stage	***Status of patient***	***Treatment***
Limited stage	With good performance status of patient	Chemotherapy + radiotherapy
	With poor performance status of patient	Modified chemotherapy and/or palliative radiotherapy
Extensive stage	With good performance status of patient	Chemotherapy ± local radiotherapy
	With poor performance status of patient	Modified chemotherapy and/or palliative radiotherapy

TABLE 2.100: Neurological manifestations of bronchial carcinoma.

Regional effects	***Remote effects (paraneoplastic)***
➢ Nerve paralysis – Recurrent laryngeal nerve – Phrenic nerve ➢ Intercostal neuralgia ➢ Horner's syndrome ➢ Pancoast's syndrome	➢ Polyneuropathy ➢ Myelopathy ➢ Polymyositis and dermatomyositis ➢ Eaton–Lambert syndrome ➢ Cerebellar degeneration ➢ Cortical degeneration ➢ Limbic encephalitis
Metastatic effects: Brain and spine	

Flowchart 2.19 shows treatment of small cell lung cancer.

FLOWCHART 2.19: Small cell lung cancer: treatment.



Neurological Manifestations of Bronchial Carcinoma (Table 2.100)

Q. Write short note on the neurological manifestations of bronchial carcinoma.

Hypertrophic Osteoarthropathy (Hypertrophic Pulmonary Osteoarthropathy)

Q. Write short note on pulmonary hypertrophic osteoarthropathy (hypertrophic pulmonary osteoarthropathy).

- Hypertrophic osteoarthropathy (HOA) is characterized by clubbing of digits and in more advanced stages, by periosteal new-bone formation and synovial effusions.
- Clubbing is almost always found in HOA but can occur as an isolated manifestation.
- Primary HOA is also called Pierre-Marie-Bamberger syndrome.

Clinical features

- **Sites involved:** Most common are the distal parts of the long bones of wrists (radius and ulna) and ankles (tibia and fibula).
- **Symptoms:** Pain and swelling of the wrists and ankles, to a lesser extent in knees and shin. Pain is aggravated by dependency and relieved by elevation of the affected limb.
- **Physical examination:** Digital clubbing, swollen and tender joints, and pitting edema of the anterior aspect of shin.
- **Investigations**
  - *X-ray* **(Fig. 2.46)***:* Periosteal thickening and subperiosteal new bone formation along the shaft of distal ends of long bones. The ends of distal phalanges may show osseous resorption.
  - *Radionuclide studies of bones:* It shows pericortical linear uptake along the cortical margins of long bones. These changes may be detected even before any radiographic changes.

FIG. 2.46: X-ray of the wrist showing subperiosteal new bone formation—hypertrophic pulmonary osteoarthropathy.

Treatment
- Identify and treat the associated disease.
- Vagotomy or percutaneous blocking of the vagus nerve: It may give relief in few patients.
- Analgesics: Aspirin or other nonsteroidal anti-inflammatory drugs.

Pancoast's Syndrome (Pancoast's Tumor; Superior Sulcus Tumor Syndrome)

Q. Write short essay/note on clinical features of Pancoast (superior pulmonary sulcus) tumor.

Q. Write short note on Horner's syndrome.

- Pancoast's tumor also known as Pancoast-Tobias tumor or superior sulcus tumor.
- Pancoast's **tumor is a tumor at the apex of lung which** may invade the **neural structures around the first rib, subclavian vessels**, and the **cervical sympathetic plexus. Chest X-ray appearance is shown in Figure 2.47.**
- Local extension of the tumor can involve the eighth cervical (C_8) and first thoracic (T_1) nerves.
- **Features of Pancoast's syndrome:**
  - Shoulder pain which radiates in the distribution of ulnar nerve of the arm (i.e., along the C_8, T_1 distribution).
  - Wasting of the small muscles of the hand from due to C_8, T_1 nerve involvement.
  - Pain and tenderness over the first and second ribs and radiological evidence of rib destruction due to local invasion by tumor.
  - *Horner's syndrome* (enophthalmos, ptosis, miosis, and anhidrosis) on the same side as the tumor due to involvement of the sympathetic pathway as it passes through the T_1 root.
- **Investigations:** CT scan, fine-needle aspiration, and MRI.

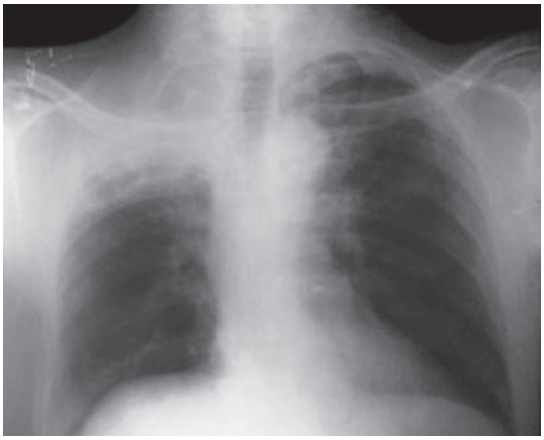


FIG. 2.47: X-ray showing mass lesion in the right upper zone with erosion of ribs—Pancoast's tumor.

Treatment: Combined chemoradiotherapy and surgery. Preoperative radiotherapy along with chemotherapy (cisplatin and etoposide) is given followed by en-bloc resection of the tumor.

Superior Vena Cava Syndrome

Q. Write short essay/note on:
- Causes, clinical features, and management of superior vena cava (SVC) syndrome.
- Superior vena cava obstruction.

It develops due to the obstruction of SVC secondary to compression and/or invasion by tumors or other lesions in the superior mediastinum **(Table 2.101)**.

Bronchial carcinoma and lymphoma are the most common causes and others are very rare causes of SVC obstruction.

TABLE 2.101: Causes of superior vena cava obstruction.

Neoplastic	*Non-neoplastic*
➢ Primary – Bronchial carcinoma (75%) – Lymphomas (20%) – Thymoma – Parathyroid tumor ➢ Metastatic tumors	➢ Tuberculosis ➢ Fibrosing mediastinitis ➢ Retrosternal goiter ➢ Central venous catheter causing thrombosis ➢ Aortic arch aneurysm

Clinical features

- Due to involvement of other structures of mediastinum (*refer* to mediastinal tumors)
- Due to involvement of SVC: Headache, visual disturbances change in the state of consciousness and orthopnea.
- **Physical signs:**
  - Distended, **nonpulsatile neck veins,** dilated veins on the upper thorax and upper limbs
  - Conjunctival edema, suffusion, and subconjunctival hemorrhage
- **Features based on location of SVC obstruction:**
  - *Preazygos or supra-azygos:* Obstruction to the return of blood above the entry of azygos vein into the SVC produces distension of veins and edema of the face, neck, and upper extremities.
  - *Postazygos or infra-azygos:* Obstruction below the entry of azygos vein into the SVC causes retrograde flow of blood through the azygos via collaterals to the inferior vena cava. This leads to the symptoms and signs of preazygos disease and dilation of the veins over the abdomen. This is usually more severe and poorly tolerated than preazygos obstruction.
  - *Grading*

- Common Terminology Criteria for Adverse Events (CTCAE) grades SVC syndrome

Adverse event	*SVC syndrome*
Grade 1	Asymptomatic; incidental finding of SVC thrombosis
Grade 2	Symptomatic, medical intervention indicated (e.g., anticoagulation, radiation, or chemotherapy)
Grade 3	Severe symptoms; multimodality intervention indicated (e.g., anticoagulation, chemotherapy, radiation, stenting)
Grade 4	Life-threatening consequences; urgent multimodality intervention indicated (e.g., lysis, thrombectomy, surgery)
Grade 5	Death

Investigation and diagnosis

- Based on **clinical features.**
- **Chest radiograph:** Widening of the superior mediastinum (mostly on the right side). Pleural effusion may be found in about 25% patients.
- **CT of the chest:** Most useful
- **Invasive procedures:** Bronchoscopy, percutaneous needle biopsy, mediastinoscopy, and thoracoscopy

For patients with severe or life-threatening symptoms

- **Catheter-based (standard) venography is preferred over computed tomography (CT) venography—**as it provides an avenue by which to provide immediate treatment to alleviate SVC obstruction with endovenous recanalization (e.g., mechanical thrombolysis, pharmacologic thrombolysis, angioplasty) and stenting, as necessary.

Techniques for venography

- **CT venography**
- **Magnetic resonance (MR) venography**
- **Superior venacavogram:** Conventional catheter-based central venography is the **gold standard for identification of SVC obstruction** and the extent of associated thrombus formation. It is superior to CT for defining the site and extent of venous obstruction and for identifying collateral pathways. However, it does not image the surrounding structures like CT venography, and thus will not identify the specific cause of SVC obstruction unless thrombosis is the sole etiology.

Management of SVC syndrome

General principles

- The goals of management for malignant SVC syndrome are to
  - Alleviate symptoms and
  - Treat the underlying disease
- The head should remain **elevated** (the higher the better, as tolerated)
  - To decrease hydrostatic pressure and head and neck edema.
- IM injections should be **avoided** to upper limbs
- IV medications can still be given (however better to avoid)
- **Avoid** the use of vesicants in this situation.
- For patients **receiving RT** on an emergency basis
- for severe airway obstruction that is not amenable to stenting, we suggest a short course of high-dose corticosteroids to minimize the risk of central airway obstruction secondary to edema
- Glucocorticoids can be effective in reversing symptomatic SVC syndrome caused by **steroid-responsive malignancies, such as lymphoma** or thymoma. However, if a suspected diagnosis of lymphoma has not yet been histologically confirmed, use of glucocorticoids is not advisable, as they are lympholytic and may obscure the diagnosis.
- The effectiveness of glucocorticoids in patients with SVC obstruction due to **other malignancies**, such as non-small cell lung cancer (NSCLC), has never been studied, and they are not indicated in these situations.

Definitive treatment

- Includes evaluation of grade of symptoms, histological diagnosis and definitive treatment of the primary disease
- Endovascular stent placement is initial treatment of choice in established SVC syndrome.

MEDIASTINUM

Q. Define mediastinum. What are its various compartments?

Mediastinum is the region of the thoracic cavity located between the pleural cavities (sacs) in the chest. It extends anteroposteriorly from the sternum to the spine (paravertebral gutter and ribs) and sagittally from the thoracic inlet (above) to the diaphragm (below) and laterally mediastinal pleura. It has numerous organs and structures which make it a veritable Pandora's box, within which congenital cysts, benign tumors, and primary and malignant neoplasms may develop.

Compartments

Mediastinum is divided into four compartments based on the lateral chest radiograph **(Table 2.102)**.

TABLE 2.102: Various compartments of mediastinum and their normal contents.

Superior mediastinum
➢ Trachea, upper esophagus, thymus gland, thoracic duct, lymph nodes ➢ Vessels: Superior vena cava (SVC), arch of aorta and its branches ➢ Nerves: Phrenic nerve, vagus nerve, left recurrent laryngeal nerve
Anterior mediastinum
➢ Thymus gland ➢ Anterior mediastinal lymph nodes, internal mammary artery and vein ➢ Adipose tissue
Posterior mediastinum
➢ Esophagus, lymph nodes ➢ Vessels: Descending aorta, azygos vein ➢ Thoracic duct, sympathetic chain
Middle mediastinum
➢ Heart, ascending aorta, arch of aorta, vena cavae, brachiocephalic arteries and veins, pulmonary arteries and veins ➢ Trachea, main bronchi, hilar lymph nodes ➢ Phrenic nerves

Common Mediastinal Tumors, Cysts, and Masses (Figs. 2.48A and B)

Q. Write short essay/note on causes of mediastinal mass.

Clinical Features

Q. Write short essay/note on clinical features, investigations, and management of mediastinal tumors.

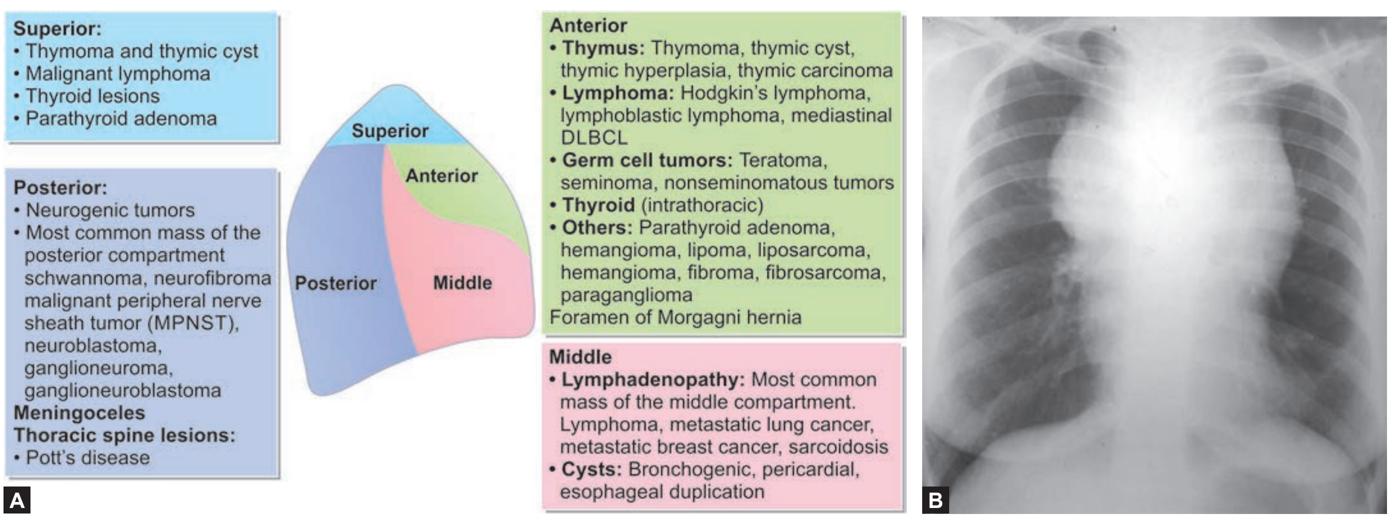
- **Age group:** Germ-cell tumors and lymphoma/leukemias are more common between 20 and 40 years of age.
- Clinical features are due to the compression and/or invasion (infiltration) of the structures of mediastinum **(Table 2.103)**. Benign tumors cause compression without invasion, whereas malignant tumors compress and invade the vital structures of mediastinum.

Investigations

- **Cytological examination of sputum** for malignant cells.
- **α-fetoprotein and β-human chorionic gonadotropin levels:** Elevated in nonseminomatous germ-cell tumors of anterior or superior mediastinum in male.
- **Radiological investigations**
- **Benign tumor** appears as a sharply circumscribed opacity in the mediastinum which may encroach on one or both lung fields.
- **Malignant tumor** appears as a lesion with ill-defined margins and often causes a generalized widening of the mediastinal shadow.

TABLE 2.103: Structures involved and clinical features of malignant mediastinal invasion.

Trachea and main bronchi: Stridor, breathlessness, cough, collapse of lung
Esophagus: Dysphagia, esophageal displacement or obstruction on barium swallow examination
Phrenic nerve: Diaphragmatic paralysis and dyspnea
Left recurrent laryngeal nerve: Paralysis of left vocal cord producing hoarseness and 'bovine' cough
Sympathetic trunk: Horner's syndrome
Superior vena cava (SVC): SVC syndrome: Nonpulsatile distension of neck veins, subconjunctival edema, and edema and cyanosis of head, neck, hands and arms. Dilated anastomotic veins on chest wall.
Pericardium: Pericarditis and/or pericardial effusion, cardiac tamponade



FIGS. 2.48A AND B : (A) Location of most common lesions of mediastinum. (B) Superior and anterior mediastinal mass—thymoma.

- **Fluoroscopic examination:** For diaphragm analysis
- **Barium swallow:** For identifying esophageal involvement
- CT scan of thorax
- **Magnetic resonance imaging (MRI):** Useful for evaluating cystic lesions
- Positron emission tomography (PET)

- **Bronchoscopy** if there is suspicion of lung cancer.
- **Mediastinoscopy:** To remove lymph node from anterior mediastinum.
- **Exploratory thoracotomy:** For removal of part or entire tumor for histopathological examination.

Management
- Benign mediastinal tumors: Surgical removal.
- SVC obstruction: Refer to SVC syndrome.
- Lung cancer with lymph nodal metastases: Radiotherapy and/or chemotherapy.
- Treatments of lymphomas and leukemias.

Solitary Pulmonary Nodule (SPN)

- **Criteria:** A single discrete pulmonary opacity surrounded by normal lung tissue. Not associated with adenopathy or atelectasis.
- **Size and nature:** Solitary pulmonary nodule must be 3 cm or less in diameter. Lesions larger than 3 cm are almost always malignant. Malignant nodules have a tumor doubling time between 30 and 300 days, whereas benign nodules, it is either less than 30 days or more than 300 days.
- **Prompt diagnosis** and resection are usually advisable. Chest X-ray appearance of solitary pulmonary nodule is shown in **Figure 2.49.**
- **Aim of evaluation** is not to miss a treatable malignant nodule early. 70% are benign and 30% of SPN are malignant.
- **Differential diagnosis** of solitary pulmonary nodules **(Table 2.104)**.

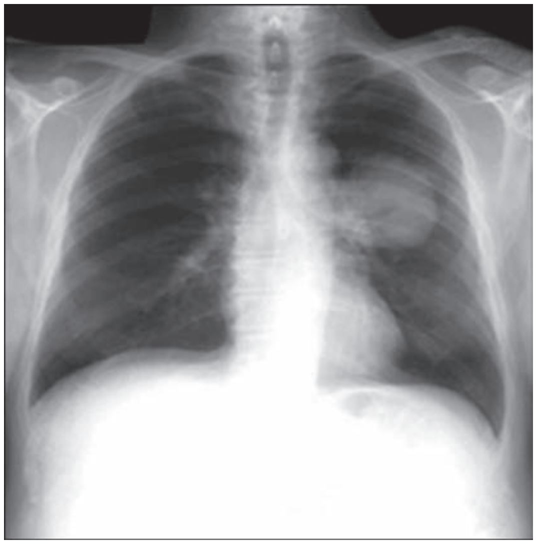


FIG. 2.49: Chest X-ray of solitary pulmonary nodule.

TABLE 2.104: Differential diagnosis of solitary pulmonary nodules.

Malignant tumors ➢ Bronchogenic carcinoma (adenocarcinoma, large cell, squamous, small cell) ➢ Carcinoid ➢ Pulmonary lymphoma	**Noninfectious granulomas** ➢ Rheumatoid arthritis ➢ Wegener's granulomatosis ➢ Sarcoidosis ➢ Paraffinoma
Benign tumors ➢ Hamartoma ➢ Granuloma ➢ Lipoma	**Others** ➢ Miscellaneous ➢ Bronchiolitis obliterans organizing pneumonia (BOOP) ➢ Abscess
Infectious granulomas ➢ Tuberculosis ➢ Histoplasmosis ➢ Coccidioidomycosis	**Pseudotumor** ➢ Spherical pneumonia ➢ Pulmonary infarction ➢ Arteriovenous malformation ➢ Bronchogenic cyst ➢ Amyloidoma

Initial evaluation

- **History:** H/o cancer, smoking, age
- **CT findings:** Size/shape, calcium levels, lymph nodes, bony destruction, pleural tags

Risk of malignancy: Can be calculated by Brock score, Mayo-clinic model which estimates probability of malignancy in SPN.

Approach to evaluation of SPN

Solid Nodule: Size >8 mm with high-risk stratification may need 3 monthly follow up or can undergo PET guided biopsy.

Subsolid Nodule:
- **Ground-glass opacity (GGO) nodule:** Size >6 mm require CT every 6–12 months follow-up
- **Part solid**: Biopsy recommended if solid component >6 mm. If solid component <6 mm CT every 3-6 months.

RESPIRATORY FAILURE

Q. Define respiratory failure. Describe in detail the causes and management of various types of respiratory failures.

Q. Discuss the role of oxygen therapy in chronic type II respiratory failure complicating chronic bronchitis.

Q. Write short essay/note on:
- Acute respiratory failure
- Clinical features of type I and type II respiratory failure.

Definition

- Respiratory failure is the term used when pulmonary gas exchange fails to maintain normal arterial oxygen and carbon dioxide levels.

- Respiratory failure is a syndrome of inadequate gas exchange due to dysfunction of one or more essential components of the respiratory system:
  - CNS
  - Peripheral nerve
  - Respiratory muscles
  - Airways
  - Alveoli
- **Criteria:** Respiratory failure is present when **PaO_2 is 60 mm Hg (8.0 kPa) and/or $PaCO_2$ is 50 mm Hg (6.5 kPa).**
- **Features of respiratory failure:** (1) breathlessness at rest, (2) central cyanosis, (3) raised respiratory rate and (4) drowsiness, confusion, or unconsciousness. In such patients, arterial blood gas analysis to be done.
- **PaO_2** cutoff of 60 mm Hg is considered as values below this correspond to steep portion of the O_2-Hb dissociation curve where small change leads to drastic fall in SpO_2 levels.

Classification

It can be classified in different ways: (1) type I, II, III and IV, (2) acute and chronic type I respiratory failure; acute and chronic type II respiratory failure, and (3) type I (hypoxemic) respiratory failure and type II (hypercapnic) respiratory failure.

Type I or Hypoxemic (PaO_2 <60 at sea level): Failure of Oxygen Exchange

Increased shunt fraction (Q_S/Q_T)
- Due to alveolar flooding
- Hypoxemia refractory to supplemental oxygen

Type II or Hypercapnic ($PaCO_2$ >45): Failure to Exchange or Remove Carbon Dioxide

- Decreased alveolar minute ventilation (V_A)
- Often accompanied by hypoxemia that corrects with supplemental oxygen.

Type III Respiratory Failure: Perioperative Respiratory Failure

- Increased atelectasis due to low functional residual capacity (FRC) in the setting of abnormal abdominal wall mechanics.
- Often results in type I or type II respiratory failure.
- Can be ameliorated by anesthetic or operative technique, posture, incentive spirometry, postoperative analgesia, and attempts to lower intra-abdominal pressure.

Type IV Respiratory Failure: Shock

Type IV describes patients who are intubated and ventilated in the process of resuscitation for shock.

Goal of ventilation is to stabilize gas exchange and to unload the respiratory muscles, lowering their oxygen consumption.

Rate of development: Respiratory failure may be
- **Acute:** In which, pH of blood may drop to below 7.2.
- **Chronic:** In which, pH is normal or slightly reduced and bicarbonate is elevated due to renal compensation. Due to associated hypoxemia, patient may develop polycythemia, pulmonary hypertension and cor pulmonale.
- Acute on chronic (e.g., acute exacerbation of advanced COPD).

It may be **transient or persistent**.

Various mechanisms of producing respiratory failure are listed in **Table 2.105.**
- **Hypoventilation:** It may be due to (1) low tidal volume, (2) increased dead space, or (3) reduced respiratory rate.

TABLE 2.105: Mechanisms of producing respiratory failure.

Decreased inspired oxygen concentration
➢ High altitude: Acute mountain sickness, chronic mountain sickness
Hypoventilation
➢ Diminished ventilatory drive, e.g., drug overdose, primary alveolar hypoventilation, stroke
➢ Impaired neuromuscular transmission, e.g., Guillain–Barre syndrome, myasthenia gravis, phrenic nerve injury, spinal cord lesion, amyotrophic lateral sclerosis
➢ Weakness of muscle, e.g., myasthenia gravis, muscular dystrophies, electrolyte disturbances
➢ Disorders of chest wall, e.g., kyphoscoliosis, ankylosing spondylitis, severe obesity
➢ Disorders of pleura, e.g., pneumothorax, pleural effusion
➢ Parenchymal lung disease
➢ Airway obstruction, e.g., chronic obstructive pulmonary disease (COPD), acute severe asthma, foreign body, vocal cord palsy
Ventilation–perfusion (V/Q) mismatch
➢ Interstitial lung disease
➢ Obstructive airway disease, e.g., chronic bronchitis, emphysema, bronchiectasis, bronchial asthma
➢ Alveolar diseases, e.g., pneumonia, acute respiratory distress syndrome (ARDS)
➢ Pulmonary vascular diseases, e.g., pulmonary embolism, pulmonary hypertension
➢ Decreased cardiac output
Shunt
➢ Intracardiac right to left shunts, e.g., tetralogy of Fallot.
➢ Intrapulmonary shunts, e.g., pulmonary arteriovenous shunts, alveolar collapse, pneumonia, pulmonary edema, pulmonary hemorrhage
Diffusion abnormality
➢ Interstitial lung disease, ARDS, interstitial pneumonias

In hypoventilation $PaCO_2$ increases, PaO_2 decreases, and alveolar arterial oxygen gradient is normal.

- **Ventilation–perfusion (V/Q) mismatch:** It is the **most common cause of hypoxemia** and administration of 100% O_2 will markedly improve hypoxemia. There is elevation of alveolar–arterial oxygen gradient.
- **Shunt:** If there is a shunt, deoxygenated blood bypasses ventilated alveoli and mixes with oxygenated blood and produces hypoxemia. In these patients, hypoxemia will persist even if patient is given 100% oxygen. Hypercapnia will not develop when shunt fraction is more than 60%. There is elevation of alveolar–arterial oxygen gradient.
- **Diffusion abnormality** is an uncommon cause of hypoxemia.

Acute and Chronic Type I Respiratory Failure

Q. Write short essay/note on the causes of acute type I respiratory failure.

Acute type I respiratory failure usually develops abruptly, often in patients with previously normal lungs. Common causes of acute and chronic type I failure are listed in **Table 2.106**.

TABLE 2.106: Causes of acute and chronic type I respiratory failure.

Acute type I respiratory failure	*Chronic type I respiratory failure*
➢ Pneumonia ➢ Pulmonary edema ➢ Acute asthma ➢ Pulmonary embolism ➢ Acute respiratory distress syndrome (ARDS) ➢ Pneumothorax	➢ Diseases with widespread pulmonary fibrosis ➢ Chronic pulmonary edema ➢ Chronic disorders of chest wall or neuromuscular diseases ➢ Chronic pulmonary thromboembolism

Acute and Chronic Type II Respiratory Failure

Occurs due to reduced alveolar hypoventilation given by the formula (v) = V (tidal volume – dead space) × RR (resp rate).

Common causes of acute and chronic type II failure are listed in **Table 2.107.**

Q. Write short essay/note on management of type I and type II respiratory failure.

Management of Respiratory Failure

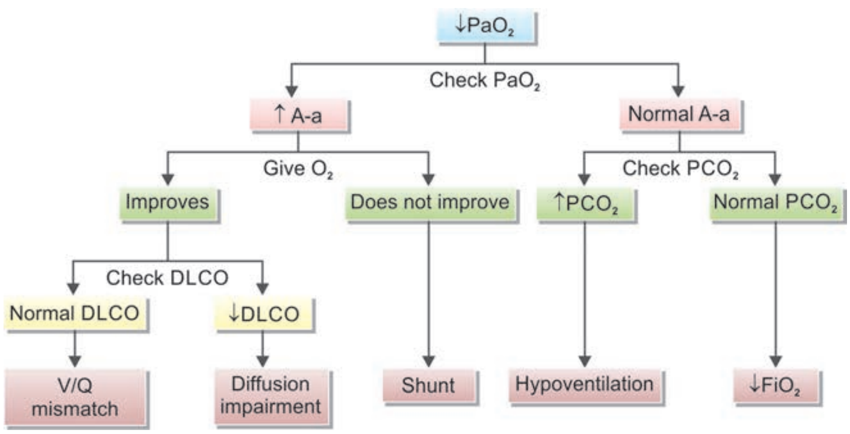
Flowchart 2.20 shows an approach to identify the mechanism of respiratory failure.

1. **Secure airway by:**
   - Endotracheal intubation or tracheostomy
   - Invasive mechanical ventilation
   - Noninvasive ventilation
2. **Supplemental oxygen as needed**
3. **Treat underlying condition:**
   - **Infection:** Antimicrobials and source control
   - **Airway obstruction:** Bronchodilators, glucocorticoids
   - **Improve cardiac function:** Positive airway pressure, diuretics, vasodilators, morphine, inotropy, and revascularization.
4. Respiratory stimulants, such as *doxapram hydrochloride, nikethamide, progesterone, modafinil, acetazolamide* can increase or maintain ventilation for a short time, at most 24 hours. However, its role is controversial.

TABLE 2.107: Causes of acute and chronic type II respiratory failure.

Acute type II respiratory failure ➢ **Respiratory depressant drugs**, e.g., diazepam, opiates, alcohol ➢ **Severe obstruction to the airflow,** e.g., severe acute asthma, laryngeal and tracheal obstruction, acute exacerbation of chronic obstructive pulmonary disease (COPD) ➢ **Disorders of respiratory muscles,** e.g., acute polymyositis ➢ **Injuries to chest,** e.g., tension pneumothorax, massive hemothorax, and flail chest ➢ **Brainstem damage,** e.g., stroke, encephalitis, trauma **Disorders of spinal cord, nerves, and neuromuscular transmission,** e.g., spinal trauma, transverse myelitis, acute GB syndrome, poliomyelitis, myasthenia gravis, and botulism
Chronic type II respiratory failure ➢ COPD (most common) ➢ Chest wall abnormalities, e.g., marked kyphoscoliosis, marked obesity ➢ Amyotrophic lateral sclerosis, muscular dystrophy ➢ Central hypoventilation

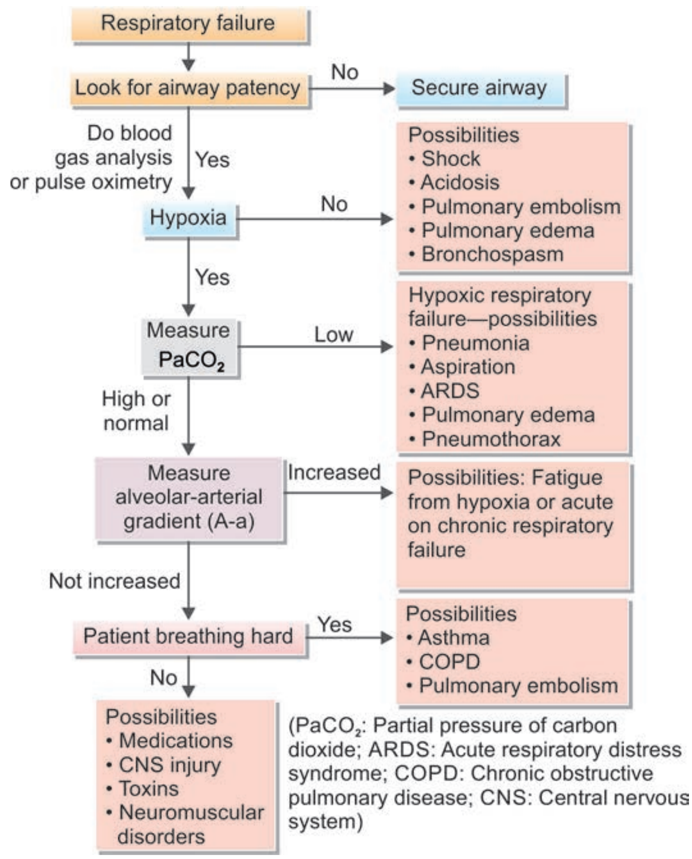
FLOWCHART 2.20: Approach to identify the mechanism of respiratory failure.



(A-a: Alveolar–arterial gradient; DLCO: Diffusing capacity of the lungs for carbon monoxide; FiO_2: Fraction of inspired oxygen; O_2: Oxygen; PaO_2: Partial pressure of oxygen; $PaCO_2$: Partial pressure of carbon dioxide; Q: Perfusion, the blood that reaches the alveoli V: Ventilation is the air that reaches the alveoli; V/Q: Ventilation/perfusion ratio)

Flowchart 2.21 shows step-by-step approach to respiratory failure.

FLOWCHART 2.21: Step-by-step approach to respiratory failure.



SLEEP APNEA/HYPOPNEA SYNDROME

Q. Write short note on sleep apnea syndromes.

- Sleep apnea is defined as an intermittent recurrent reduction or cessation of breathing (upper airway obstruction) during sleep.
- **Apnea:** Defined by the American Academy of Sleep Medicine (AASM) as the cessation of airflow for at least 10 seconds.
- **Hypopnea:** Defined as a recognizable transient reduction (but not complete cessation) of breathing for 10 seconds or longer, a decrease of greater than 50% in the amplitude of a validated measure of breathing, or a reduction in amplitude of less than 50% associated with oxygen desaturation of 4% or more.
- **Respiratory effort-related arousal** (RERA): It is an event characterized by increasing respiratory effort for 10 seconds or longer leading to an arousal from sleep. It does not fulfil the criteria for a hypopnea or apnea.

Types of sleep apnea: (1) Obstructive sleep apnea (OSA), (2) central sleep apnea, and (3) mixed sleep apnea (combination of factors).

- **Obstructive** sleep apnea is characterized by continued thoracoabdominal effort in the setting of partial or complete air flow cessation.
- **Central** sleep apnea is characterized by the lack of thoracoabdominal effort in the setting of partial or complete airflow cessation.
- **Mixed** sleep apnea has both obstructive and central features. They generally begin without thoracoabdominal effort and end with several thoracoabdominal efforts in breathing.

Obstructive Sleep Apnea

Risk Factors

See **Table 2.108**.

TABLE 2.108: Risk factors for obstructive sleep apnea.

Obesity (major)	***Mandibular retrognathia and micrognathia***
Male sex (major)	Family history
Enlarged tonsils (especially in children)	Endocrine diseases (e.g., acromegaly, hypothyroidism)
Menopause	Advancing age
Nasal obstruction (e.g., nasal deformities, rhinitis, polyps, adenoids), smoking	Respiratory depressant drugs (e.g., alcohol, sedatives, strong analgesics)

Clinical Features (Table 2.109)

Pickwickian syndrome (obesity-hypoventilation syndrome): It is diagnosed by combination of morbid obesity (BMI >30 kg/m²), chronic hypoventilation ($paCO_2$ >45 mm Hg) after excluding other causes and evidence of sleep disordered breathing. They usually have associated awake resting hypoxemia, hypersomnolence, signs of cor pulmonale (right-sided heart failure and lower extremity edema), and nocturnal hypoventilation.

TABLE 2.109: Clinical features of obstructive sleep apnea.

Nocturnal features	*Daytime features*
➢ Snoring, usually loud, habitual ➢ Witnessed apnea, which often interrupt the snoring and end with a snort ➢ Nocturnal gasping and choking sensations ➢ Nocturia ➢ Insomnia ➢ Restless sleep	➢ Nonrestorative sleep (i.e., waking up as tired as when they went to bed) ➢ Morning headache, dry or sore throat ➢ Excessive daytime sleepiness (EDS) ➢ Daytime fatigue/tiredness ➢ Cognitive deficits; memory and intellectual impairment (short-term memory, concentration) and impaired work performance ➢ Decreased vigilance ➢ Morning confusion ➢ Personality and mood changes, including depression, anxiety and irritability

Physical Examination

- Obesity: Body mass index (BMI) greater than 30 kg/m²
- Large neck circumference: Greater than 17 inches in men and 15 inches in women.

- Abnormal (increased) Mallampati score
- Narrowing of the lateral airway walls
- Other risk factors mentioned in **Table 2.108** may be found.

Diagnosis

- Screening for OSA (obstructive sleep apnea)-based on STOP-BANG score—each value 1 point
  - **S**nore loudly, **T**iredness, Anyone **O**bserved choking/gasping during sleep, Treatment for Blood **Pressure.**
  - **BMI >35, A**ge >50, **N**eck circumference >17", **G**ender (Male)
  - Score 6-8: High risk, 4-5: Intermediate risk, 1-3: Low-risk of OSA
- **Polysomnography**: Used to diagnose OSA and other sleep-related breathing disorders.
  - Sleep stages are recorded via an EEG, electro-oculogram (EOG), and chin electromyogram (EMG).
  - Heart rhythm is monitored with a single-lead ECG.
  - Leg movements are recorded via an anterior tibialis EMG.
  - Breathing is monitored, including airflow at the nose and mouth (using both a thermal sensors and a nasal pressure transducer), effort (using inductance plethysmography), and oxygen saturation.
  - The breathing pattern is analyzed for the presence of apnea and hypopneas.
  - Apnea–hypopnea index (AHI): AHI is defined as the average number of episodes of apnea and hypopnea per hour.
  - Levels of PSG (Polysomnograph):
    - *Level I:* Monitoring of EEG, EOG, EMG, airflow, respiratory effort, SpO_2, ECG by sleep technician
    - *Level II:* Monitoring of EEG, EOG, EMG, airflow, respiratory effort, SpO_2, ECG. Not attended by sleep technician
    - *Level III:* Minimum 4 channels—ECG, SpO_2, respiratory effort (1 airflow + 1 breathing). Unattenders or attended by sleep technician
    - *Level IV:* Minimum 3 channels—Airflow sensory, Oxygen saturation (SpO_2), BP. Unattended or Attended by sleep technician
    - Most preferred are Level I or II as it gives helps in the diagnosis of sleep related breathing disorders.

Diagnosis of sleep apnea is confirmed if AHI ≥5 **(Table 2.110)**.

- Excessive daytime sleepiness (EDS) assessed by **Epworth Sleepiness Scale (ESS).**
- Respiratory disturbance index (RDI): It is the average number of respiratory disturbances [(obstructive apneas, hypopneas, and respiratory event-related arousals (RERAs)] per hour.
- Evaluation for secondary causes of hypoventilation is crucial.

TABLE 2.110: Degree of obstructive sleep apnea and apnea hypopnea index.

Degree of obstructive sleep apnea	*Apnea-hypopnea index (AHI)*
Mild	5–15
Moderate	16–30
Severe	>30

Treatment

- Mild apnea: Wider variety of options
- OHS: AHI >30 – CPAP therapy. If nonresponder switch to NIV. If AHI <30 NIV considered.
- **Moderate-to-severe apnea**: Should be treated with nasal continuous positive airway pressure (CPAP).

1. Conservative nonsurgical treatment
   - Weight reduction in obese individuals
   - Avoidance of alcohol for 4–6 hours prior to bedtime
   - Avoiding bright light prior to sleep
   - Sleeping on one's side rather than on the stomach or back.
2. **Nasal CPAP therapy**
   - Most effective for OSA. Increases the caliber of the airway in the retropalatal and retroglossal regions. It increases the lateral dimensions of the upper airways and thins the lateral pharyngeal walls. Maintains upper airways patency during sleep, preventing the soft tissues from collapsing.
   - **Indications:** (1) Patients with an apnea–hypopnea index (AHI) greater than 15 regardless of symptomatology. (2) For patients with an AHI of 5–14.9, if the patient has one of the following: excessive daytime sleepiness (EDS), hypertension, or cardiovascular disease.
3. **Other modalities:** BiPAP therapy and oral appliance therapy.
4. **Treatment** of associated disease: hypothyroidism, allergic rhinitis, acromegaly
5. **Surgery for obstructive sleep apnea,** e.g., nasal surgery (septoplasty, sinus surgery, and others), tonsillectomy ± adenoidectomy, uvulopalatopharyngoplasty (UPPP), mandibular enhancement devices

6. **Pharmacotherapy:**
   - Many drugs (e.g., mirtazapine, protriptyline, theophylline, naloxone, doxapram, oxymetazoline nasal application, inhaled nasal corticosteroids, and acetazolamide) have been tried without much benefit in OSA.
   - Modafinil is used in patients who have residual daytime sleepiness despite optimal use of CPAP.
   - Selective serotonin reuptake inhibitor agents such as paroxetine and fluoxetine have been shown to increase genioglossus muscle activity and decrease REM sleep (apneas are more common in REM).
7. **Experimental:** Pharyngeal pacing, radiofrequency ablation, rapid maxillary expansion

Central Sleep Apnea

- It occurs without obstruction of the airway.
- **Causes:** Congenital central sleep apnea (Ondine's curse) and many neurological lesions.
- **Mechanisms:** Primary central alveolar hypoventilation syndrome.
- Etiology not known. Probably due to high chemoresponsiveness of the respiratory system. Apnea is terminated with an abrupt, large breath.
- Daytime somnolence is less common.
  - *Others:* High altitude, stroke, neurodegenerative diseases (e.g., Parkinson's disease), left ventricular failure with Cheyne–Stokes breathing.

Treatment
- Administration of oxygen during sleep.
- Nasal CPAP.
- Noninvasive positive-pressure ventilation (bilevel nasal positive pressure).
- Acetazolamide and theophylline may be tried though effect is modest.
- Treatment of underlying cause: β-blockers in congestive heart failure.

Oxygen Therapy

Q. Write short essay/note on different types of oxygen therapy.

Therapeutic Indications for Oxygen Therapy (Table 2.111)

Q. Write short essay/note on common therapeutic indications of oxygen therapy in clinical practice.

TABLE 2.111: Therapeutic indications for oxygen therapy.

➢ Pulmonary edema	➢ Respiratory paralysis
➢ Acute attack of bronchial asthma	➢ Anaerobic infections
➢ Chronic obstructive pulmonary disease (COPD)	➢ High altitude
➢ Acute respiratory distress syndrome (ARDS)	➢ Prevents development of severe pulmonary hypertension

Hazards of Oxygen Therapy

- CO_2 narcosis
- Pulmonary complications: Irritation of respiratory tract, pulmonary edema, ARDS, consolidation, reduction in lung compliance → fibrosis, capillary leak
- In premature infants and neonates: Retrolental fibroplasias, blindness, bronchopulmonary dysplasia
- Idiopathic epilepsy
- Reperfusion injury post MI, ↑SVR precipitate heart failure

Target Saturation: 88-92% in type 2 respiratory failure, 94-95% in type 1 respiratory failure.

Technique of Administration

- Fraction of inspired oxygen (FiO_2) while inhaling 100% oxygen depends upon rate of oxygen flow and minute ventilation of patient. Merits and demerits of various techniques of oxygen administration are presented in **Table 2.112**.
- Oxygen delivery system can be divided into:
  - Low-flow systems
  - High-flow systems

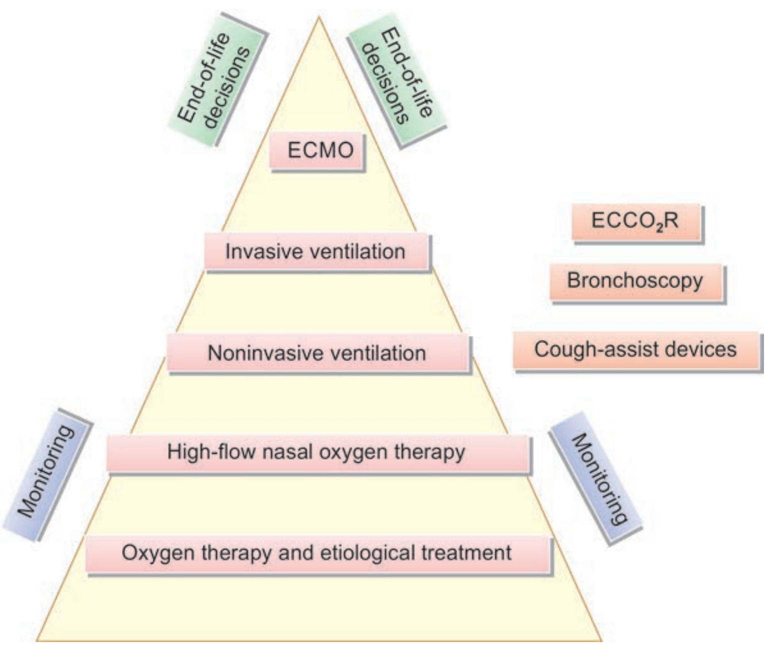


FIG. 2.50: Types of oxygen therapy.
(ECMO: Extracorporeal membrane oxygenation; ECCO$_2$R: Extracorporeal carbon dioxide removal)

TABLE 2.112: Merits and demerits of various techniques of oxygen administration.

Merits	*Demerits*
Nasal catheter, nasal cannula (simple plastic tubing + prongs)	
➢ Easy to fix ➢ Keep hands free ➢ Not much interference with further airway care ➢ Useful in both spontaneous breathing and apneic	➢ Mucosal irritation (uncomfortable) ➢ Gastric dilatation (especially with high flows) – Exact FiO_2 delivered is unknown because it is influenced by peak inspiratory flow demand
Merits	***Demerits***
Simple face mask	
➢ Simple ➢ Lightweight ➢ Delivers oxygen at FiO_2 of 35–55% using flows of 5–12 L/min	➢ Need to remove when speak, eat, drink, vomiting, expectoration of secretions ➢ Drying/irritation of eyes ➢ Uncomfortable when facial burns/trauma ➢ Application problem when *Ryle's tube* in situ
Partial rebreathing mask (polymask)	
➢ FiO_2 delivered >0.60 is delivered in moderate-to-severe hypoxia ➢ Exhaled oxygen from anatomic dead space is conserved	➢ Insufficient flow rate may lead to rebreathing of CO_2 ➢ Claustrophobia ➢ Drying and irritation of eyes
Nonrebreathing mask	
➢ Use a reservoir bag to achieve higher oxygen concentrations (up to 80%). ➢ Flow rates are generally at least 8–15 L/min	➢ Air dilution (if not fitting properly) ➢ Rebreathing (if O_2 flow is inadequate) ➢ Interfere with further airway care ➢ Uncomfortable (sweating, spitting)
Venturi masks: Allow the precise administration of oxygen via a Venturi facemask. Usual FiO_2 values delivered are 24%, 28%, 31%, 35%, 40%, and 50%.	
Heated humidified high-flow nasal cannula (HFNC): ➢ Delivers heated and humidified oxygen at high flows and concentrations such that it flushes out a significant amount of nonoxygenated air from the upper airway. ➢ The system can be titrated up to 60 L/min and 100% FiO_2 and may provide a small amount of PEEP at high-flow rates.	

Hyperbaric Oxygen Therapy

Hyperbaric oxygen (HBO) exposure (breathing oxygen at increased ambient pressure), usually 2–3 atmospheres absolute (ATA) causes an increase in PaO_2.

Indication for HBOT

- **Poisoning:** Carbon monoxide, air or gas embolism, decompression sickness
- **Infections:** Clostridial myonecrosis, refractory osteomyelitis, necrotising soft tissue infection
- **Acute ischemia:** Crush injury
- **Chronic ischemia:** Radiation necrosis
- **Ischemic ulcers:** Diabetic ulcers

Contraindications

- Untreated pneumothorax
- Patient on bleomycin (interstitial fibrosis)
- Doxorubicin toxicity (cardiotoxicity—worsen heart failure)

Noninvasive Positive Pressure Ventilation (NPPV)

- Delivers respiratory support with positive airway pressure via a sealed facemask, nasal mask, or helmet device.
- NPPV includes continuous positive airway pressure (CPAP) and bilevel positive airway pressure (BiPAP) ventilation. NPPV can be delivered by home devices or ventilators:
  - CPAP: Delivers continuous positive airway pressure throughout the respiratory cycle and prevents alveolar collapse during expiration
  - CPAP is often used in the treatment of obstructive sleep apnea and pulmonary edema
  - BiPAP: Delivers two different airway pressures during inspiration and expiration to decrease the work of breathing. BiPAP is often used for COPD exacerbations, weaning, and neuromuscular weakness
- **Benefits of NPPV:** NPPV decreases the need for mechanical ventilation in appropriately selected patients, especially in patients with neuromuscular disease, COPD, pulmonary edema, and postoperative respiratory insufficiency.

- **Disadvantages of NPPV:** NPPV is generally safe but can cause skin damage, eye irritation, claustrophobia, and aerophagia and be difficult to tolerate for some patients. Use should be limited to patients who are conscious, are cooperative, able to protect their airway, and are hemodynamically stable.

Mechanical Ventilation

Q. Write short essay on noninvasive ventilation.

Types of mechanical ventilation: (1) Invasive and (2) noninvasive. Indications for mechanical ventilation are mentioned in **Box 2.17**.

Box 2.17: Indications of mechanical ventilation.

- Apnea with respiratory arrest
- Acute lung injury
- Respiratory rate >35 breaths per minute
- Vital capacity <15 mL/kg
- PO_2 <60 at FiO_2 0.6
- Respiratory muscle fatigue
- Obtundation or coma
- Bradypnea
- PCO_2 of >50 mm Hg with pH <7.25

General Principles of Ventilation

- Endotracheal tube should be inserted to an average depth of 23 cm in men and 21 cm in women (measured at the incisor).
- Pressure in the cuff generally should not exceed 25 mm Hg.
- Tracheostomy should be done if anticipating ventilator setting for more than 3 days.
- Routine suctioning is not recommended because it may be associated with complications (e.g., desaturation, arrhythmias, bronchospasm, severe coughing, and introduction of secretions into the lower respiratory tract).

Modes of ventilatory support: Controlled mode ventilation (CMV), assist control mode ventilation (ACMV), synchronized intermittent mandatory ventilation (SIMV), and continuous positive airway pressure/positive end-expiratory pressure (CPAP/PEEP).

- **Controlled mode ventilation**
  - Used for initiation of the ventilation
  - No patient contribution
  - All variables are independent (FiO_2, TV, RR, I/E).
- **Assist control mode ventilation**
  - Inspiratory cycle is initiated either by the patient or if patient effort is not present, then by a timer signal present within the ventilator.
  - Also commonly used for initiation
  - Synchronization of ventilator cycle with patient's inspiratory effort.
  - *Drawbacks:* Respiratory alkalosis, myoclonus and seizures
- **Synchronized intermittent mandatory ventilation (SIMV)**
  - Patient is allowed to breathe spontaneously without ventilator assist in between the ventilator breaths.
  - Ventilator breath is delivered in synchrony.
  - Mandatory are the number of present breaths
  - Intermediate mode
  - Helpful in weaning
- **Continuous positive airway pressure (CPAP)**
  - Not a true mode of ventilation
  - All ventilation occurs because of patients spontaneous efforts ventilator just gives fresh gas to the breathing circuit with operator-depended positive pressure.
  - Used to access the extubation potential in the patient who requires very little ventilator support or inpatient with intact respiratory system function who require an ET tube for airway protection.

Weaning

Process of decreasing ventilator support and allowing patient greater proportion of ventilation.

- Arterial pH: 7.35–7.40
- SO_2 >90% with FiO_2 0.5
- PEEP <5 mm Hg
- Intact cough reflex (assessed during suctioning)
- Weaning index: RR/TV <105
- Patient off inotropes

Complications of Ventilation

- **Pulmonary:** Barotrauma (>50 mm Hg), interstitial emphysema, pneumomediastinum, subcutaneous emphysema, pneumothorax, **ventilator-associated pneumonia** (**VAP**), tracheal stenosis.

- **Hypotension:** Resulting from elevated intrathoracic pressure and decreased venous return and it is responds to volume repletion.
- **Gastrointestinal:** Stress ulcers and cholestasis.
  - Neuromuscular weakness
  - Pressure sores
  - Raised intracranial pressure
  - Sinusitis, oral ulcers

Extracorporeal Membrane Oxygenation

Extracorporeal membrane oxygenation (ECMO) is a type of mechanical cardiopulmonary support that is used to support patients with cardiac failure or both cardiac and respiratory failure, despite optimal conventional care, by removing blood from the venous system, oxygenating it, and returning it to either the venous (venous-venous [VV] ECMO) or arterial (veno-arterial [VA] ECMO) system.

- Veno-venous ECMO: Used to support patients with respiratory failure
- Veno-arterial ECMO: Used to support patients with cardiac failure or both cardiac and respiratory failure.

Extracorporeal Carbon Dioxide Removal

$ECCO_2R$ therapy is a useful and effective supportive treatment for adults in the ICU with both ARDS and AE-COPD.

Indications

- Acute severe heart or lung failure with high mortality risk, despite optimal conventional therapy.
- Hypoxic respiratory failure, with a partial pressure of arterial oxygen (PaO_2)/fraction of inspired oxygen (FiO_2) <150 mm Hg with FiO_2 >0.9 and Murray score of 2–3.
- Acute respiratory distress syndrome (ARDS)
- Severe cardiac failure of any cause
- Primary allograft failure following heart or lung transplantation
- As a bridge to either cardiac or lung transplantation or placement of a ventricular-assist device.

Contraindications

Most contraindications are relative:

- Conditions incompatible with normal life if the patient recovers
- Preexisting conditions which affect the quality of life (CNS status, end-stage malignancy, risk of systemic bleeding with anticoagulation).

Complications

- Bleeding due to anticoagulants used is seen in around 50% of patients and can be life-threatening.
- Systemic thromboembolism due to clots formed in the circuit
- Heparin-induced thrombocytopenia, canula-related problems, and neurological dysfunction have also been reported.

ACUTE LUNG INJURY AND THE ACUTE RESPIRATORY DISTRESS SYNDROME

Q. Describe acute respiratory distress syndrome (ARDS), noncardiogenic pulmonary edema, and acute lung injury (ALI).

Definition: Acute respiratory distress syndrome (ARDS) is a sudden and progressive form of acute respiratory failure in which the alveolar capillary membrane becomes damaged and more permeable to intravascular fluid resulting in severe dyspnea, hypoxemia, and diffuse pulmonary infiltrates.

The Berlin definition of ARDS **(Table 2.113)**: An acute, diffuse, inflammatory lung injury that leads to increased pulmonary vascular permeability, increased lung weight, and a loss of aerated tissue.

TABLE 2.113: The Berlin definition of acute respiratory distress syndrome (ARDS).

The Berlin definition of acute respiratory distress syndrome (ARDS)	
Timing	Within 1 week of a known clinical insult/new/ worsening respiratory symptoms
Chest X-ray	Bilateral opacities—not fully explained by effusions, lobar/ lung collapse or nodules
Origin of edema	➢ Respiratory failure not fully explained by cardiac failure or fluid overload ➢ Need objective assessment (e.g., echocardiography) to exclude hydrostatic edema If no risk factor is present
Oxygenation	
Mild	200 mm Hg <PaO_2/FiO_2 ≤300 mm Hg with PEEP or CPAP >5 cm H_2O
Moderate	100 mm Hg <PaO_2/FiO_2 ≤200 mm Hg with PEEP >5 cm H_2O
Severe	PaO_2/FiO_2 ≤100 mm Hg with PEEP >5 cm H_2O

In resource limited settings *Kigali Modifications* was proposed (2016).

A. Bilateral opacities could be documented by *either ultrasonography or chest radiograph*.

B. Oxygenation criteria: Pulse oximetric oxygen saturation *(SpO_2)/FiO_2 ratio ≤ 315 without the requirement for PEEP.*

ARDS and ALI are serious diseases characterized by damage to alveolar epithelium and pulmonary capillary endothelium. Alveoli become filled with edema fluid of high-protein content and inflammatory cells.

Etiology of ARDS and ALI (Table 2.114)

ARDS may develop due to diffuse lung injury from many medical and surgical disorders. The lung injury may be direct (e.g., toxic inhalation) or indirect (e.g., in septicemia).

TABLE 2.114: Disorders commonly associated with ARDS.

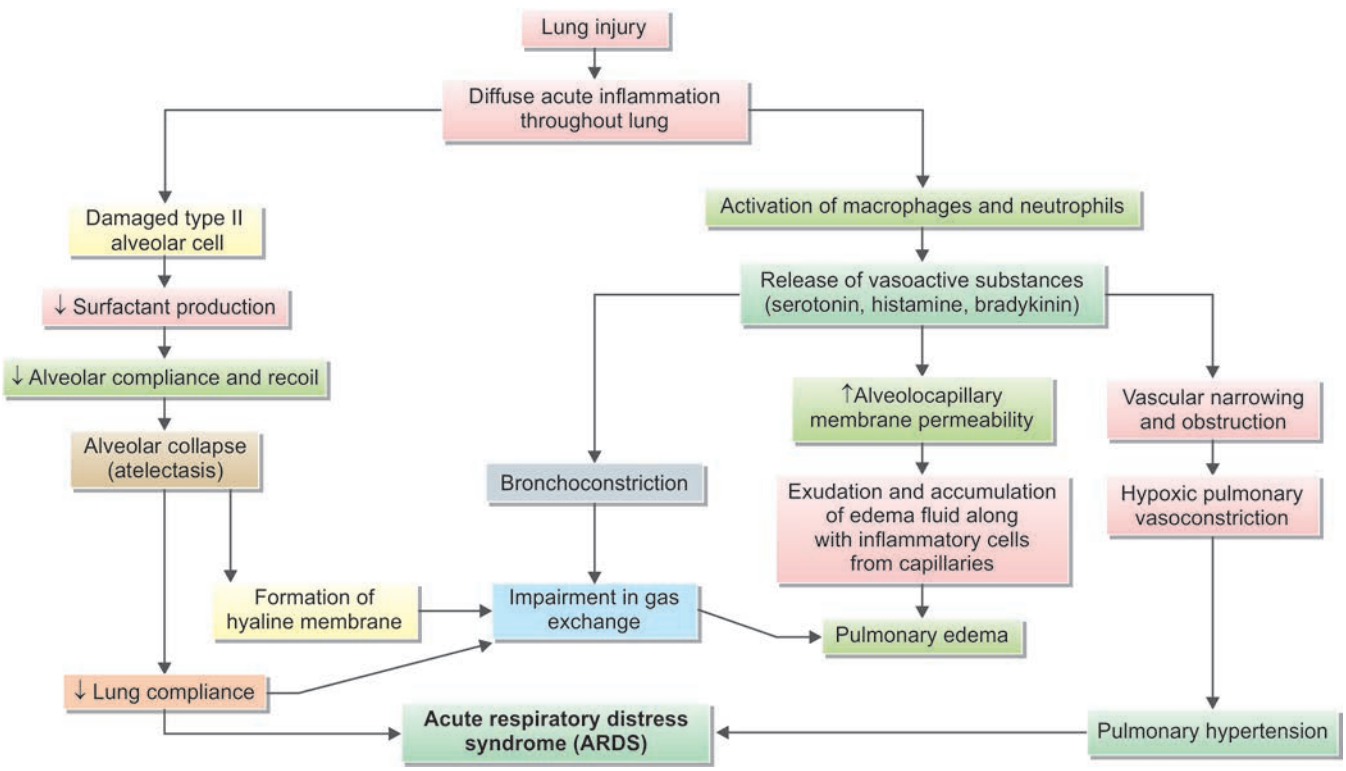
Direct lung injury	
Pulmonary infections	Pneumonia (viral, bacterial, fungal *Pneumocystis jirovecii, Mycoplasma)*
Aspiration	Aspiration of gastric contents (vomitus)
Inhalation of toxic gas	Ammonia, chlorine, nitrogen dioxide, ozone, oxygen, smoke
Blunt chest trauma	Pulmonary contusion
Near drowning	
Indirect lung injury	
Systemic disorders	Shock, septicemia, uremia, eclampsia
Severe trauma	Multiple bone fractures (fat embolism), flail chest, head trauma, burns
Blood	Multiple transfusions
Drug overdose	
➢ Narcotic overdose	Heroin, methadone, morphine, dextropropoxyphene
➢ Nonnarcotic drugs	Barbiturates, thiazides, nitrofurantoin
Others	Acute pancreatitis, cardiopulmonary bypass, trauma, Goodpasture's syndrome, SLE

Pathophysiology (Flowchart 2.22)

Natural history of ARDS can be divided into three: Exudative, proliferative, and fibrotic.

1. **Exudative phase (1–7 days):** Diffuse acute inflammation (e.g., neutrophils) occurs throughout the lungs, involving both endothelial and epithelial surfaces. Activated neutrophils are sequestered into the lungs and increase the capillary permeability as well as damages type I and II alveolar cells. The leaked proteins aggregate in the air spaces with cellular debris and dysfunctional pulmonary surfactant to form the characteristic hyaline membranes. Activated macrophages and neutrophils release cytokines and chemokines which progressively recruit more inflammatory cells and amplify the inflammatory response. There is loss of surfactant and impaired surfactant production as a secondary event. The net effect is alveolar collapse (most marked independent regions of the lung) and results in hypoxemia due to ventilation-perfusion mismatch and increased pulmonary shunt.

FLOWCHART 2.22: Pathophysiology of acute respiratory distress syndrome (ARDS).



2. **Proliferative phase (3–7 days):** Resolution often starts at this phase along with the initiation of repair process. The alveolar exudate undergoes organization and neutrophils disappear with appearance of lymphocytes. Type II pneumocytes starts proliferating. Connective tissue and other structural elements in the lungs proliferate in response to the initial injury, including development of fibroblasts. The terms "stiff lung" and "shock lung" frequently used to characterize this stage.
3. **Fibrotic phase (>10–14 days):** It occurs in some cases and is characterized by proliferation of fibroblasts resulting in interstitial fibrosis. Lung function may continue to improve for as long as 6–12 months after onset of respiratory failure, depending on the precipitating condition and severity of the initial injury.

Results

- Hypoxemia (V/Q mismatch, impaired hypoxic pulmonary vasoconstriction)
- Increase in dependent densities (surfactant dysfunction, alveolar instabilities)
- Decreased compliance (surfactant dysfunction, decreased lung volume, fibrosis)
- Collapse/consolidation (increased compression of dependent lung)
- Increased minute ventilation (increased in alveolar dead space)
- Increased work of breathing—WOB (increased elastase, increased minute volume requirement)
- Pulmonary hypertension (vasoconstriction, microvascular thrombi, fibrosis, PEEP (positive end expiratory pressure).

Clinical Features

- Development of **acute dyspnea and hypoxemia within hours to days** of an inciting event.
- **Tachypnea, tachycardia,** cyanosis and the need for a high-fraction of inspired oxygen (FiO_2) to maintain oxygen saturation.
- Febrile or hypothermic
- Sepsis—hypotension and peripheral vasoconstriction with cold extremities.
- Bilateral rales/crepitations
- Manifestations of the underlying cause.
- *Because cardiogenic pulmonary edema must be distinguished from ARDS, carefully look for signs of congestive heart failure or intravascular volume overload, including jugular venous distension, cardiac murmurs and gallops, hepatomegaly, and edema.*

Investigations

- **Chest radiograph (Fig. 2.51):** Diffuse, bilateral alveolar infiltrates consistent with pulmonary edema
  - *Early stage:* Infiltrates associated with ARDS **may be variable:** Mild or dense, interstitial or alveolar, patchy or confluent.
  - Initially, the **infiltrates** may have a **patchy peripheral distribution**, but soon they progress to **diffuse bilateral involvement with ground-glass changes** or **frank alveolar infiltrates.**
  - *Cardiogenic edema:* Increased heart size, increased width of the vascular pedicle, vascular redistribution toward upper lobes, the presence of septal lines, or a perihilar ("bat's wing") distribution of the edema. Lack of these findings, in conjunction with patchy peripheral infiltrates that extend to the lateral lung margins, suggests ARDS.
- **Arterial blood gas analysis:**
  - PaO_2/FiO_2 ratio and severity **(Table 2.115)**
  - In addition to hypoxemia, arterial blood gases often initially show a respiratory alkalosis.
  - However, in ARDS occurring in the context of sepsis, a metabolic acidosis with or without respiratory compensation may be present.
  - As the condition progresses and the work of breathing increases, the partial pressure of carbon dioxide (PCO_2) begins to rise and respiratory alkalosis gives way to respiratory acidosis.
- **To exclude cardiogenic pulmonary edema:**
  - Echocardiogram: Left ventricular ejection fraction, wall motion, and valvular abnormalities
  - Plasma B-type natriuretic peptide (BNP) value.

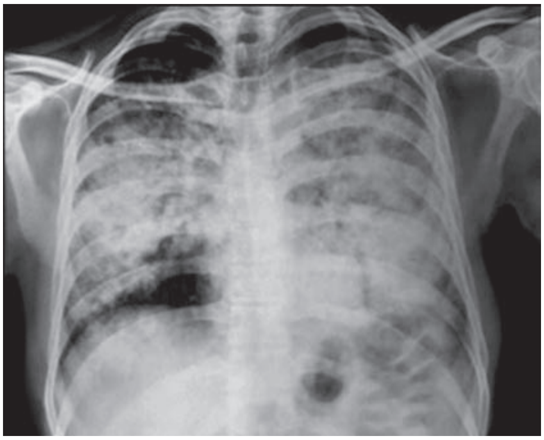


FIG. 2.51: Bilateral alveolar shadows in ARDS on chest X-ray.

TABLE 2.115: RDS severity and PaO_2/FiO_2.

ARDS severity	*PaO_2/FiO_2*
Mild	200–300
Moderate	100–200
Severe	<100

- **CT scan:** Diffuse consolidation with air bronchograms, bullae, pleural effusions, pneumomediastinum and pneumothorax. Later may show lung cysts.
- **Clinico-Radiological phenotypes in COVID-19 ARDS**
  - **Low-phenotype:** Low elastance, V/Q ratio, lung weight, lung recruitability
    - PEEP nonresponsive
  - **High-phenotype:** High elastance, V/Q ratio, lung weight, lung recruitability
    - PEEP responsive.

Differential Diagnosis

- Acute eosinophilic pneumonia
- Acute interstitial pneumonia (AIP), also known as Hamman-Rich syndrome
- Organizing pneumonia: Drugs, radiation, infection
- Diffuse alveolar hemorrhage (DAH): Vasculitis, pulmonary renal syndrome

Management: Summarized in Figure 2.52

Goals of management of patients with ARDS

- Treatment of respiratory system abnormalities
- Diagnose and treat the precipitating/initiating cause of ARDS. Support or treat other organ system dysfunction or failure.
- Maintain oxygenation.
- General critical care, adequate early nutritional support, prophylaxis against deep vein thrombosis (DVT) and gastrointestinal (GI) bleeding.

Maintaining adequate oxygenation

- Positive end-expiratory pressure (PEEP) is employed.
- When utilized in sufficient amounts, PEEP allows FiO_2 to be lowered from high potentially toxic concentrations.
- Mild ARDS: High-flow nasal oxygen (HFNO) trial can be attempted with close monitoring of ratio of oxygen saturation (ROX) index
- **Lung-protective mechanical ventilation**
  - Mechanical ventilation using limited tidal volumes
  - The goals of lung-protective ventilation are to avoid injury due to overexpansion of alveoli during inspiration ("volutrauma") and injury due to repetitive opening and closing of alveoli during inspiration and expiration ("atelectotrauma").
- **Low tidal volume ventilation (LTVV) (Mortality Benefit)**
  - Set initial tidal volume to 8 mL/kg IBW.
  - Reduce tidal volume to 7 mL/kg IBW then 6 mL/kg IBW over the next 1–3 hours.
  - Set respiratory rate to <35 bpm to match baseline minute ventilation.
  - Target plateau pressure (pressure in respiratory zone) <30
  - Driving pressure (Pplat – PEEP) <15 (better target)
- **Recruitment maneuvers:** Indicated in refractory ARDS
  - High PEEP administered stepwise for short time
  - Opens up collapsed alveoli and improved oxygenation
- **Permissive hypercapnia** is defined as clinician-allowed hypercapnia during assisted ventilation, despite an ability to achieve a level of minute ventilation sufficient to maintain homeostasis.

Fluid management

- Primary ARDS due to aspiration, pneumonia, or inhalational injury, usually can be treated with fluid restriction.
- Secondary ARDS due to remote infection or inflammation requires initial fluid and potential vasoactive drug therapy.
- If BNP >300 – consider IV diuretics (target urine output 4–8 mL/kg/hr x 3 hours)

Prone positioning ventilation

- About two-thirds of patients with ARDS improve their oxygenation after being placed in a prone position.
- 16 hours/day of proning proven to ***reduce mortality***.

Neuromuscular blockade

- Neuromuscular blockade upto 48 hours preferred drug being cisatracurium (tested in trials)
- Improves patient-ventilator dyssynchrony

Adjuncts to lung protective mechanical ventilation: Inhaled nitric oxide, inhaled prostacyclin, tracheal gas insufflation, extracorporeal membrane oxygenation (ECMO) or extracorporeal CO_2 removal ($ECCO_2R$).

Corticosteroids: Corticosteroids have little or no role in acute phase of ALI or ARDS. Proven benefit has been seen in COVID-19 H1N1 ARDS.

Experimental therapies

- Surfactant replacement therapy: Improves oxygenation but no improvement in mortality.
- Ketoconazole: Antifungal that inhibits thromboxane synthase and 5-lipooxygenase

Future nonventilatory therapeutic options

- Gene therapy for ALI/ARDS
- Mesenchymal stem cells (MSC)

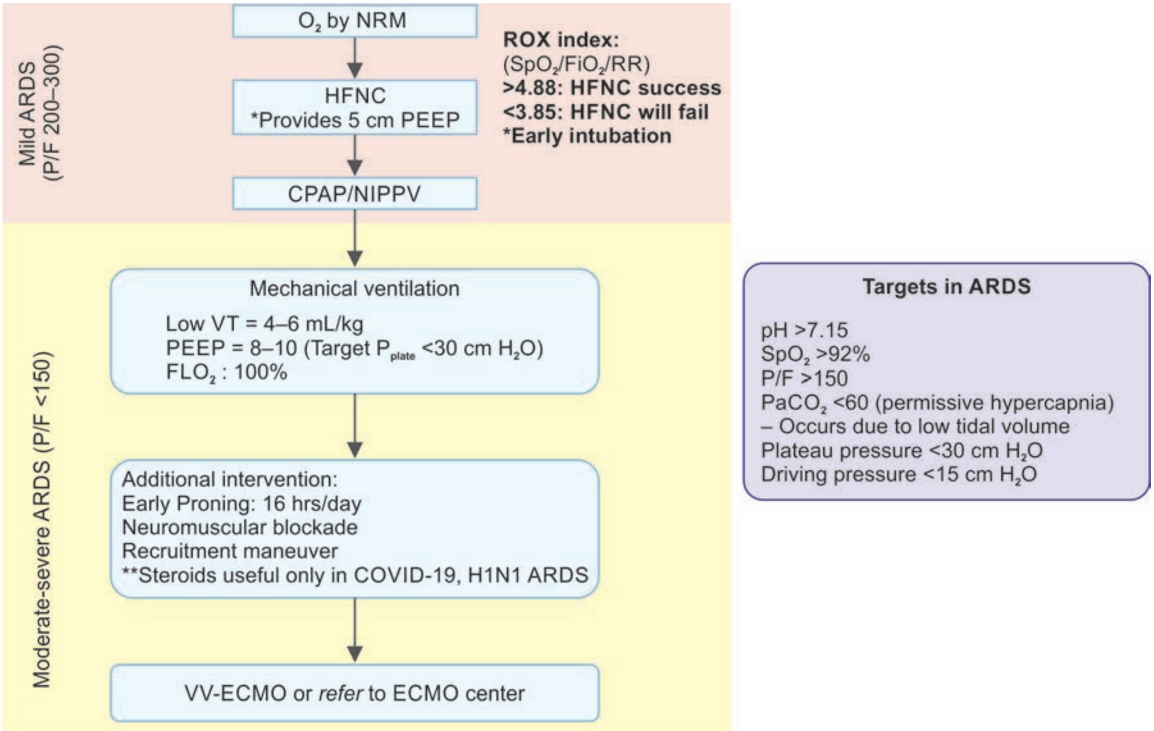


FIG. 2.52: Diagnosis and management of acute respiratory distress syndrome.

(ARDS: Acute respiratory distress syndrome; ROX: Ratio of oxygen saturation; NRM: Non-rebreather mask; HFNC: High-flow nasal cannula; CPAP: Continuous positive airway pressure; NIPPV: Noninvasive positive pressure ventilation; PEEP: Positive end-expiratory pressure; VV-ECMO: Venovenous extracorporeal membrane oxygenation)

Ventilator Associated Lung Injury

Application of mechanical ventilation can cause injury to the lung which is termed as VALI (ventilator associated lung injury). VILI (ventilator induced lung injury) is synonymously used with VALI.

Types of Injury

- **Volutrauma:** Occurs due to over-distension of normal alveolar units to transpulmonary pressure >30 cm H_2O resulting in stretch of baby lung (functionally normal lung unit in ARDS). This results in stretch and damage of basement membrane.
- **Barotrauma:** Occurs due to excess transpulmonary pressures. High pressures/PEEP leads to damage of basement membrane resulting in hypoxemia/pneumothorax.
  Both volutrauma and barotrauma can be minimized by using **low VT (4–6 mL/kg IBW) and driving pressure <15 cm H_2O (Peak – plateau pressure).**
  Evidence from ARDSnet ARMA Trial
- **Atelectotrauma:** Also known as recruitment/decruitment injury. Repeated alveolar collapse and expansion results in inflammation leading to lung injury
  - High PEEP, recruitment maneuvers, **prone position** minimizes atelectotrauma.

Evidence from PROSEVA Trial

- **Oxygen toxicity:** Higher than required target generated increased free radical worsening damage.
  - **Maintain SpO_2** 88–94%

Sarcoidosis

Q. Discuss the clinical features, investigations, and management of sarcoidosis.

Definition: Sarcoidosis is a multisystem disorder of unknown etiology characterized by noncaseating granuloma which mainly affects lung but can also affect any other organs.

Etiology

- Etiology of sarcoidosis **remains unknown**, but several lines of evidence suggest that it is a disease of disordered immune regulation in genetically predisposed individual.
- Factors that play a role in its pathogenesis are: (1) **Environmental agents** (infectious and noninfectious), (2) **genetic** (e.g., HLA-A1 and HLA-B8), and (3) **immunological** (cell-mediated immune response).
- The granuloma is the pathologic hallmark of sarcoidosis. The sarcoid granulomas develop as an exaggerated cell-mediated immune response to an unidentified environmental antigen. Infectious or noninfectious agent in a genetically susceptible individual. Probable infectious inciting agent includes—mycobacteria (both *Mycobacterium tuberculosis* and nontuberculous mycobacteria), *Propionibacterium acnes*, *Borrelia burgdorferi*, viruses, fungi, spirochetes, and *Rickettsia*.

Pathology

- **Noncaseating granulomas** are composed of an **aggregate of epithelioid cells and lymphocytes** (mainly CD4+ T-cell).
- Epithelioid cells are modified macrophages and characteristically have abundant eosinophilic cytoplasm and vesicular nuclei. Schumann bodies and stellate inclusion as asteroid bodies enclosed within giant cells are formed in approximately 60% of granulomas.
- **Organs involved:** Sarcoidosis can affect every organ of the body. **It most commonly** affects the **lung** and the **lymph nodes** in the **mediastinum and hilar regions**. Other organs commonly affected are the liver, spleen, skin, parotid glands, and eye.

Clinical Features

- **Age and gender:** It occurs mainly in 3rd or 4th decade of life. More predominant in women.
- **More prevalent** in Swedes, Danes, and US blacks.
- **Systems involved:** Pulmonary (90%), lymph nodes (70%), hepatic (50–80%), cardiac (30%), cutaneous (25%), and ocular (20%)

Manifestations of Sarcoidosis

- **Asymptomatic form** (30–45%): It is usually detected incidentally on a routine chest X-ray (~30%) or abnormal liver function tests.
- Acute or subacute form (10–15%)
- Chronic form (40–60%).

Pulmonary manifestations

- First site of involvement. Begins with alveolitis involving small bronchi and small blood vessels. Alveolitis either clear-up spontaneously or lead to granuloma or fibrosis.
- **Other features:** Interstitial lung diseases, atelectasis, cavitation, unilateral pleural effusion, pulmonary nodules, miliary mottling, lymphadenopathy.
- Most patients have pulmonary manifestations, and most presenting with incidental findings on CXR as an interstitial disease.
- Symptoms include dry cough, dyspnea, and chest discomfort.
- Unpredictable course.

Extrapulmonary Involvement

- **Lymph nodes:** Lymph nodes are involved in almost all cases (mainly hilar and mediastinal nodes), but any nodes may be involved. Involved nodes are characteristically enlarged, discrete, and sometimes calcified.
  - *Typical:*
    - Bilateral hilar and right paratracheal lymph nodes
    - Middle mediastinal lymph node (**50%** of cases).
    - Left paratracheal, aortopulmonary, and subcarinal lymph nodes **1-2-3** sign present in **95%** of cases. This is called **Garland triad**.

- *Atypical:*
  - Unilateral hilar lymph node
  - Anterior or posterior mediastinal lymph nodes
  - Lymph node calcification (amorphous, punctate, popcorn or egg-shell calcification).
- Tonsil may be affected in about quarter to one-third of the cases.

- **Skin (33%):** Erythema nodosum, plaques, maculopapular eruptions, subcutaneous nodules, lupus pernio, keloid, infiltration of previous scars by granuloma.
- **Eyes (25%):** Anterior uveitis, iridocyclitis, retinitis, phlyctenular conjunctivitis, lacrimal gland involvement—keratoconjunctivitis sicca (dry eyes).
- **Salivary glands:** Parotid gland enlargement.

Q. Write short note on Heerfordt–Waldenström syndrome.

- **Heerfordt's syndrome**/uveoparotid fever/Heerfordt–Mylius syndrome/Waldenström's uveoparotitis is a rare manifestation of sarcoidosis. It is characterized by uveitis, swelling of the parotid gland, chronic fever, and facial nerve palsy (in some cases).

- **Heart:** Cardiac arrhythmias, heart block, sudden death, and CHF
- **Liver and spleen:** Granulomatous liver disease, hepatosplenomegaly, Budd–Chiari syndrome.
- **Nervous system:** Cranial nerve palsy, pachymeningitis, space-occupying lesion, diabetes insipidus due to hypothalamic involvement, mononeuritis multiplex, peripheral neuropathy
- **Kidneys:** Nephrocalcinosis, hypercalciuria, and renal stones
- **Musculoskeletal:**
  - Arthropathies, osteoporosis, phalangeal bone cysts, polymyositis, chronic myopathy
  - Endocrine
  - Diabetes insipidus, anterior pituitary dysfunction, Addison's syndrome.
- **Lupus pernio:**
  - It is a specific complex involvement of skin around nose, eye and cheeks.
  - These chilblains-like lesions often cause is figuration by eroding the cartilage and bone, especially around the nose.
  - It is diagnostic for chronic form of sarcoidosis and is associated with more sever pulmonary disease.
  - Associated with poor prognosis of sarcoidosis.

Löfgren's syndrome: It is an *acute* triad of erythema nodosum, joint pains, and radiographic evidence of bilateral hilar adenopathy. It is often seen in young women and resolves over 6 months to 2 years with NSAIDs.

Investigations

Q. Write short note on radiological/X-ray findings in sarcoidosis.

- **Radiological investigations**
  - **Classic radiographic patterns of pulmonary sarcoidosis (Table 2.116 and Fig. 2.53):** The chest X-ray is the most commonly used tool to assess the lung involvement in sarcoidosis. According to the pattern of involvement on X-ray findings, sarcoidosis is classified into four stages.
  - **Computed tomography of the chest (contrast-enhanced):** Helps in better delineation of the hilar and mediastinal lymph nodes.

FIG. 2.53: X-ray showing bilateral hilar adenopathy.

TABLE 2.116: Scadding staging of pulmonary sarcoidosis according to radiographic patterns and corresponding clinical features.

Stage	*X-ray finding*	*Clinical features*
I	Bilateral hilar lymphadenopathy (BHL) without any parenchymal abnormalities (50% cases)	Often asymptomatic but may be associated with erythema nodosum and arthralgia. Majority resolve spontaneously within 1 year
II	BHL with diffuse pulmonary infiltrates (30% cases)	May present with breathlessness or cough. Majority resolve spontaneously
III	Diffuse pulmonary infiltrates without BHL (10% cases)	Disease is less likely to resolve spontaneously
IV	Evidence of diffuse pulmonary fibrosis	Can progress to ventilatory failure, pulmonary hypertension and cor pulmonale

- **HRCT:** Characteristic appearance is presence of reticulonodular opacities that follow a perilymphatic distribution, centered on *bronchovascular bundles* and the subpleural areas. Confluence of many interstitial granulomas can produce large, irregular, mass-like opacities (alveolar sarcoid). Small satellite nodules are usually found at the periphery of these large nodules termed the "galaxy sign" (a collection of stars). Honeycombing, pleural effusion, necrosis is not common.
- **Gallium 67 scan:** Shows diffuse uptake.
- **PET scan:** Using radiolabeled fluorodeoxyglucose is more useful than gallium 67 scan to detect areas of granulomatous disease in the chest and other sites and select potential area for biopsy.
- **MRI:** May be useful in the assessment of extrapulmonary sites (e.g., brain, heart, and bone).

- **Complete blood count:** There may be a mild normochromic, normocytic anemia, lymphocytopenia (characteristic), eosinophilia, raised ESR, hyperglobulinemia, elevated serum alkaline phosphatase.
- **Serum biochemistry:** Serum calcium is often raised (hypercalcemia due to increased production of 1,25-dihydroxy-vitamin D by the granuloma) and there is hypergammaglobulinemia.
- **Serum angiotensin-converting enzyme (ACE) level** is elevated in more than 75% of patients with untreated sarcoidosis. Elevated levels are not of diagnostic value because it may also be elevated in lymphoma, pulmonary tuberculosis, asbestosis, and silicosis. However, it is useful in assessing disease activity and response to treatment.
- **Mantoux test:** Skin sensitivity to tuberculin is reduced or absent. So, Mantoux test is a useful screening test and a strongly positive reaction virtually rules out sarcoidosis.
- **Lung function tests:** They show atypical restrictive lung defect in patients with pulmonary infiltration or fibrosis. There is a decrease in TLC, FEV, and FVC, and gas transfer. Lung function is usually normal in patients with extrapulmonary disease or those with only hilar lymphadenopathy on chest X-ray.
- **Bronchoscopy:** May show a "cobblestone" appearance of the mucosa.
- **Bronchial and transbronchial biopsy:** Most useful investigation. It usually shows noncaseating granulomas composed of epithelioid cells and multinucleate giant cells.
- **Bronchoalveolar lavage (BAL):** Shows increased proportion of lymphocytes, mostly activated Th1 (subset of CD4+ cells) cells. BAL fluid also shows an increased CD4:CD8 T-cell ratio.
- **Kveim (or Kveim-Siltzbach) test:** It involves intradermal injection of 0.1 mL of an antigen obtained from sarcoidosis spleen extract. Development of small nodule at the injection site indicates the test as positive. The nodule is biopsied at 4–6 weeks and shows typical noncaseating epithelioid granuloma. However, this test is obsolete now.
- Mediastinoscopic biopsy of mediastinal or hilar lymph nodes.
- **Exercise tests:** May show oxygen desaturation.
- Bone marrow examination may show granulomas in about 30% cases.

Differential diagnosis of bilateral hilar lymphadenopathy **(Table 2.117).**

Various causes of granuloma are listed in **Table 2.118**.

TABLE 2.117: Differential diagnosis of bilateral hilar lymphadenopathy.

Lymphoma	Involvement of only the hilar lymph nodes is rare
Pulmonary tuberculosis	Symmetrical enlargement of hilar lymph nodes is rare
Carcinoma of the bronchus	Malignant spread to the contralateral hilar lymph nodes is rarely symmetrical

TABLE 2.118: Causes of granulomas.

Infectious diseases	***Noninfectious diseases***
➢ Mycobacterial infections ➢ *Mycobacterium tuberculosis* ➢ Nontuberculous mycobacteria (NTM) ➢ Fungal infections: *Histoplasma, Cryptococcus, Coccidioides, Blastomyces, Pneumocystis jirovecii, Aspergillus* ➢ Other infections: *Brucella, Chlamydia*, cat scratch disease	Sarcoidosis, lymphoma, chronic beryllium disease, hypersensitivity pneumonitis, Wegener's granulomatosis, Churg–Strauss syndrome, talc granulomatosis, rheumatoid nodule, foreign body

Treatment

Treatment should be based on symptoms or presence of organ or life-threatening disease.

Minimal or no symptoms

- No treatment is required when there is only hilar lymph adenopathy.
- Persisting infiltration found on the chest X-ray with normal lung function tests requires careful monitoring.

Symptomatic single organ disease

Symptoms limited to only one organ: Topical steroid therapy is preferable.

Symptomatic multiple organ disease

Multiorgan or disease too extensive for topical therapy requires systemic therapy and are usually immunosuppressive including glucocorticoids, cytotoxics, or biologics.

- **Glucocorticoids:** Remain the drugs of choice and given immediately in the presence of hypercalcemia, pulmonary impairment, renal impairment and uveitis. Prednisolone is given 20–40 mg/day for 6 weeks, followed by a maintenance dose of 7.5–10 mg daily for 6–12 months.
- **Steroid-sparing agents:** In patients with severe disease, methotrexate (10–20 mg/week), azathioprine (50–150 mg/day) and the use of specific tumor necrosis factor (TNF)-α inhibitors is useful in patients developing toxicity with glucocorticoids. Chloroquine, hydroxychloroquine, and low-dose thalidomide may be useful for skin, bone, and joint involvement.
- Selected patients may be considered for single lung transplantation.

Bronchopulmonary Aspergillosis

Q. Write short essay/note on bronchopulmonary aspergillosis and classify bronchopulmonary aspergillosis.

Bronchopulmonary aspergillosis is the term used for the bronchopulmonary diseases caused by fungus *Aspergillus* species, the most common being *Aspergillus fumigates*. Others fungus in *Aspergillus* species include *A. clavatus*, *A. niger*, *A. flavus* and *A. terreus.*

Classification of Bronchopulmonary Aspergillosis (Table 2.119)

Spectrum of *Aspergillosis* depends on the *host immune* status

- **Immune hypersensitivity:** ABPA, severe asthma with fungal sensitization (SAFS)
- **Immunocompetent:** Pre-existing cavity/structural lung disease (TB, NTM, COPD, bronchiectasis)
  - Aspergilloma (fungal ball)
  - Chronic necrotising aspergillosis
  - Chronic cavitatory pulmonary aspergillosis
  - Subacute invasive aspergillosis
- **Immunocompromised** (malignancy, HIV, etc.): Invasive pulmonary aspergillosis (IPA).

TABLE 2.119: Classification of bronchopulmonary aspergillosis.

➢ Allergic bronchopulmonary aspergillosis (ABPA)
➢ Extrinsic allergic alveolitis *(Aspergillus clavatus)* (hypersensitivity pneumonitis)
➢ Intracavitary aspergilloma
➢ Invasive pulmonary aspergillosis
➢ Chronic and subacute pulmonary aspergillosis

Q. Write short note on classification of bronchopulmonary aspergillosis.

Allergic Bronchopulmonary Aspergillosis (ABPA)

Q. Discuss the clinical features, diagnostic criteria and management of allergic bronchopulmonary aspergillosis (ABPA-asthmatic pulmonary eosinophilia).

Allergic bronchopulmonary aspergillosis (ABPA) occurs due to hypersensitivity reaction against germinating fungal spores in the wall of the airway. Usually hypersensitivity reactions are seen to *A. fumigatus* and rare cases are due to other Aspergilli and other fungi.

- It can complicate bronchial asthma or cystic fibrosis.
- It is one of the causes of pulmonary eosinophilia.

Pathogenesis

Exact pathogenesis of ABPA is not known but probably occurs as a result of a hypersensitivity reaction to germinating fungal spores in the airway wall. Thus, it may be due to *Aspergillus*-specific IgE-mediated type I hypersensitivity reactions or specific IgG-mediated type III hypersensitivity reactions and abnormal T-lymphocyte cellular immune response.

Clinical features

- Fever, breathlessness, coughing up of thick sputum casts and worsening of asthmatic symptoms.
- Aspergillus may grow in the walls of proximal bronchi and may produce **proximal bronchiectasis.** It may cause repeated episodes of eosinophilic pneumonia which manifest as wheeze, cough, fever, and expectoration with sputum-containing fungal mycelia.

Investigations

The diagnosis of ABPA is primary serological. Patients may have no radiological findings in the early stages.

- **Radiological signs on chest X-ray:**
  - Radiographically show infiltrates which may be mistaken for pneumonia. Segmental or lobar collapse on chest X-rays in patients where asthma symptoms are stable, suggestive of ABPA.
  - Repeated pneumonic episodes may show fleeting shadows of infiltrates on chest radiographs. Other findings include transient area of opacification (due to mucoid impaction of the airways), band-like opacities with rounded distal margin (gloved-finger appearance), and “ring sign” and “tramlines” (due to thickened and inflamed bronchi). Eventually, it may result in central bronchiectasis and progressive pulmonary fibrosis.

- **HRCT:** Reveal proximal bronchiectasis, mucus plugging, high attenuation mucus (HAM), mosaic attenuation due to air trapping.
  - **HAM**: Mucus plugging with density greater than paraspinal muscle is **pathognomonic of ABPA.**

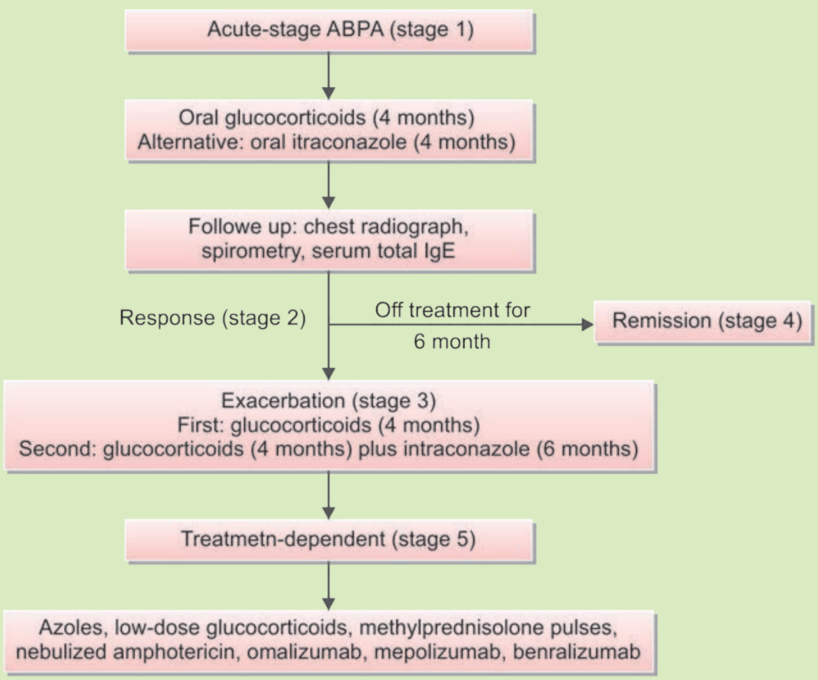
Diagnostic Criteria (Isham criteria)

All patients with bronchial asthma must be screened for ABPA.

- **Predisposing condition:** Asthma or cystic fibrosis.
- **Obligatory (both** should be present**)**
  - **Evidence of Type 1 hypersensitivity to *Aspergillus fumigatus*** (any 1 method)
    - Skin prick test positive to *Aspergillus fumigatus* (Sens 88–94%). Lack of antigen standardization limits its use.
    - Aspergillus specific IgE: **>0.35kUA/L :** Most sensitive investigation (Sensitivity 100%, Specificity 70%). No role in follow up
  - **Elevated IgE titers:** Usually >1000 IU/mL: Marker of disease activity.
- **Other criteria (2 out of 3):**
  - Increased Aspergillus specific IgG/Precipitins
  - Radiology consistent with ABPA
  - Total eosinophil >500 cell/mm^3

Management

- **Clinical response** = clinical, spirometric and radiological improvement and ↓total IgE by 25% every 8 weeks initially.
- **Corticosteroids:** Oral prednisolone 30 mg daily for 7–10 days causes rapid clearing of the pulmonary infiltrate. Prednisolone should be gradually tapered to a maintenance dose of 5–10 mg/day for long term. Asthma responds to inhaled corticosteroids.
- **Antifungal agents:** Oral itraconazole dose: 200 mg bid for 16 weeks, then once a day for 16 weeks, or voriconazole should be used in patients on high doses of steroids. It reduces exacerbations and requirement of steroids.
- **Bronchoscopic extraction of the casts:** If there is persistent lobar collapse, bronchoscopic (usually under general anesthesia) removal of impacted mucus and casts may result in reinflation of the collapsed lobe.
- **Non-responders** treated with combination of biologicals, steroids.



Aspergilloma

Q. Discuss the etiology, clinical features, diagnosis, and management of intracavitary aspergilloma (fungal ball).

Aspergilloma is the growth of *Aspergillus* fungus within previously damaged lung tissue and forms a ball of fungus (mycelium) within lung cavities. Most commonly the fungal ball is produced by *Aspergillus fumigatus* and rarely by other fungi (e.g., Zygomycetes and Fusarium).

Etiology

- An intracavitary aspergilloma may develop in any area of damaged lung with abnormal cavity. Inhaled *Aspergillus* may reach and germinate in this cavity forming "aspergilloma."
- **Causes of cavitation associated with Aspergillum:** Colonization and proliferation of fungus in a preexisting lung cavity.
  - **Common cause:** Tuberculosis cavity occurs most often in the upper lobes.
  - **Less common causes:** Lung abscess, sarcoidosis, histoplasmosis, blastomycosis, AIDS pneumonia, bronchiectasis, rheumatoid nodules, pulmonary infarction, and lung cancer
- When there are multiple aspergilloma cavities in a diseased area of lung, it is termed as a "complex aspergilloma."
- Aspergilloma (fungus ball) consists of fungal hyphae, inflammatory cells, fibrin, mucous, and tissue debris.

Clinical Features

- Simple aspergilloma is usually asymptomatic.
- Occasionally it may cause recurrent scanty-to-**massive hemoptysis.** The source of hemoptysis is usually bronchial blood vessels. It may be due to local invasion of blood vessels lining the cavity, endotoxins released from the fungus or mechanical irritation of the vessels inside the cavity during rolling of the fungus ball.
- Nonspecific systemic features such as lethargy and weight loss may be present.

Diagnosis

- **Chest X-ray:** Shows a round cavity with a tumor-like opacity inside. It is distinguished from a carcinoma by the presence of a crescentic air shadow (halo) between the fungal ball and the upper wall of the cavity. If the radiographs are repeated in a different position, the fungal ball occupies the dependent portion of the cavity. **Crescent sign of Monad (Fig. 2.54)**.
- **HRCT** is more sensitive for demonstration of the fungal ball.
- Serum precipitins to *Aspergillus fumigatus* can be detected.
- **Sputum:**
  - Microscopic examination shows fungal hyphal fragments.
  - Culture usually grows the fungus.
- Positive skin tests (hypersensitivity) to extracts of *Aspergillus fumigatus* suggestive of activity.

Management

- No treatment required for asymptomatic patients.
- Antifungal drugs have not been useful and corticosteroids may predispose to invasion.
- Bronchial artery embolization in massive hemoptysis.
- **Surgical removal:**
  - Aspergillomas complicated by hemoptysis should be excised surgically in suitable patients.
  - For patient unfit for surgery, palliative procedures such as ultrasound or CT-guided local injection of amphotericin B is given into the cavity.

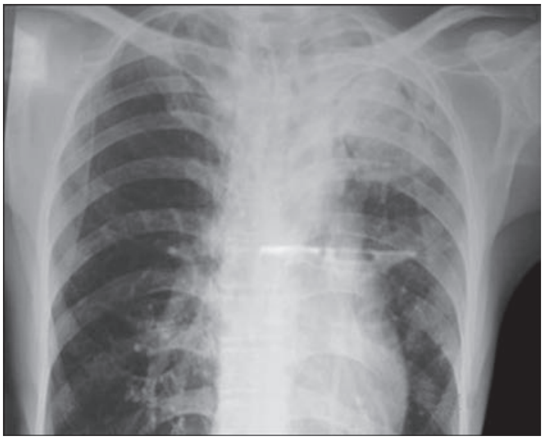


FIG. 2.54: Chest X-ray shows air crescent sign of aspergilloma in the left upper zone.

Invasive Pulmonary Aspergillosis

Q. Discuss the etiology, clinical features, diagnosis, and management of invasive pulmonary aspergillosis.

Invasive pulmonary aspergillosis (IPA) is invasion of previously healthy lung tissue by *Aspergillus fumigates* and usually develops as a complication in patients who are immunocompromised (with profound neutropenia) either by drugs (especially immunosuppressants) and/or disease.

Risk factors for invasive aspergillosis

See **Table 2.120**.

TABLE 2.120: Risk factors for invasive aspergillosis.

➢ Prolonged neutropenia (<500 cells/mm^3 for >10 days) most common and risk is related to duration and degree
➢ Solid-organ (particularly lung transplantation) or allogeneic stem cell transplantation
➢ Prolonged (>3 weeks), high dose corticosteroid therapy
➢ Leukemia and other hematological malignancies
➢ Cytotoxic chemotherapy
➢ Advance AIDS
➢ Critically ill patients on intensive care units
➢ Chronic granulomatous disease
➢ COVID-19 infection (termed as COVID associated pulmonary aspergillosis CAPA)
➢ Uncontrolled diabetes, cirrhosis, malnutrition, severe COPD

Clinical features

Acute IPA (≤1 month): Causes severe necrotizing pneumonia.

Recent studies show increase IPA post influenza, COVID-19 infections and immunocompetent individuals. They present with refractory fever (may persist despite anti-fungals), cough (sometimes productive), chest discomfort, mild-to-massive hemoptysis, and shortness of breath and metastatic spread (multiple abscesses, skin lesions).

Nonneutropenic patients present with dyspnea, *copious sputum.* Fever is less common.

Subacute (1–3 months):

- Local invasion of pulmonary vessels causes thrombosis and infarction.
- **Epithelial spread:** Involvement of tracheobronchial mucosa produces fungal plaques and ulceration.
- Systemic spread through blood may occur to the brain (causes seizures, ring-enhancing lesions, cerebral infarctions, intracranial hemorrhage, meningitis, and epidural abscess), heart, kidneys, and other organs.

Diagnosis

- **Sputum:**
  - Microscopic examination shows fungal hyphal fragments.
  - Culture usually grows the fungus.
- Serum precipitins to *Aspergillus fumigatus* can be detected.
- **Chest X-ray:** Nonspecific and includes round densities, pleural-based infiltrates (suggestive of pulmonary infarctions) and cavitations.
- **HRCT:** Very useful for diagnosis particularly in neutropenic patients. Characteristically it shows macronodules (usually ≥1 cm) surrounded by a *"halo"* (low attenuation due to hemorrhage surrounding the pulmonary nodule) during first 5 days.
  - HRCT in immunocompetent: Ground glass opacities, crazy paving appearance, halo, reverse halo sign.
- Detection of cell wall components **(galactomannan and β-1, 3-glucan)** in blood or BAL fluid is useful for early confirmation of the diagnosis, and also its serial estimation may be useful in assessing the evolution of infection during treatment.
  - *Serum galactomannan* (Sensitivity 50–75%, Specificity 100%): >0.5—may cross react with other fungals (*Histoplasma*, *Blastomyces*, Penicillium). Piperacillin-tazobactam previously was considered to produce false-positive results, however newer preparations do not interfere with results.
  - *BAL galactomannan* (Sn 80 Sp 85): BAL superior to serum in nonneutropenic patients.
  - β-D *glucan*: Component of cell wall of all fungi (except crypto, *Mucor*). Positive in *Pneumocystis carinii* pneumonia (PCP), *Candida*, *Histoplasma*. Rarely in bacteria (*Pseudomonas*, *Mycobacterium*)
- PCR to detect *Aspergillus* DNA in BAL fluid and serum.
- Bronchoscopy and BAL transbronchial biopsy is of little value and may produce complications. However, biopsy may be needed for confirmation.
- Tissue biopsy is **gold standard**: However not always feasible to obtain in sick patients.

Management

Treatment should not be delayed because IPA has a high mortality rate. It requires aggressive antifungal therapy and immunosuppression should be reduced if possible.

Voriconazole and liposomal amphotericin B allow a safer and more effective treatment of invasive aspergillosis when compared with amphotericin B-deoxycholate.

First-line agents: The treatment of choice is intravenous voriconazole (6 mg/kg every 12 hours for two doses and then 4 mg/kg twice a day). It is better tolerated and more effective than amphotericin B. Antifungal therapy with amphotericin (1.0–1.5 mg/kg/day) with or without flucytosine is equally effective.

Second-line agents: Intravenous echinocandin derivatives such as caspofungin, micafungin, and anidulafungin are used in refractory cases or if the patient cannot tolerate first-line agents.

In less immunosuppressed patients, oral itraconazole (200 mg twice a day) may be given.

Duration of therapy: 6–12 weeks. Rarely may extend for months in severely immunocompromised.

Box 2.18: Fungal infections of the lung.

Endemic fungal pneumonia pathogens in healthy and in immunocompromised individuals

- *Histoplasma capsulatum*
- *Coccidioides immitis*
- *Blastomyces dermatitidis*
- *Paracoccidioides brasiliensis*
- *Sporothrix schenckii*
- *Cryptococcus neoformans*

Opportunistic fungal infections organisms in patients with congenital or acquired defects in the host immune defenses

- *Candida spp.*
- *Aspergillus spp.*
- *Mucor spp.*
- *Cryptococcus neoformans*

Fungal Infections of Lung

See **Box 2.18**.

HEMOPTYSIS

Q. Write essay on hemoptysis (its definition, causes, clinical features, investigations, and management).

Definition

- Hemoptysis is defined as coughing of blood originating from below the vocal cords.

- Hemoptysis can range from blood-streaking of sputum to the presence of gross blood in the absence of any accompanying sputum.
- Life threatening (or) massive hemoptysis is defined as coughing of blood >150 mL/time (or) >600 mL/24 hours.
- Only 5% of hemoptysis is massive but mortality is 80%.
- Clinical definition of massive hemoptysis is any bleeding that result in a threat to life because of airway or hemodynamic compromise due to airway obstruction and bleeding.

Causes of Hemoptysis (Tables 2.121 and 2.122)

Q. Write short essay/note on common causes of hemoptysis.

TABLE 2.121: Causes of hemoptysis.

Structure involved	*Common causes*	*Uncommon causes*
Bronchial disease	Bronchial carcinoma, bronchiectasis, acute and chronic bronchitis	Bronchial adenoma, foreign body
Parenchymal disease of lung	Pulmonary tuberculosis (Rasmussen's aneurysm—dilation of a pulmonary artery in a tuberculous cavity), lung abscess, pneumonia (particularly *Klebsiella*), fungal infections (aspergilloma and invasive aspergillosis, pulmonary contusion/laceration(traumatic)	Parasites (e.g., hydatid disease, flukes), trauma, actinomycosis, mycetoma
Vascular diseases of lung	Pulmonary infarction	Goodpasture's syndrome, polyarteritis nodosa, idiopathic, pulmonary hemosiderosis, primary pulmonary hypertension
Cardiovascular disease	Acute left ventricular failure	Mitral stenosis, aortic aneurysm, pulmonary thromboembolism
Hematological disorders		Leukemia, hemophilia, anticoagulants, hemorrhagic diathesis

TABLE 2.122: Causes of massive hemoptysis.

➢ Pulmonary tuberculosis	➢ Cystic fibrosis	➢ Mitral stenosis
➢ Pulmonary infarction	➢ Lung abscess	➢ Pulmonary arteriovenous malformation
➢ Bronchiectasis	➢ Necrotizing pneumonia	
➢ Bronchogenic carcinoma		

Clinical Clues for the Diagnosis of Cause of Hemoptysis (Table 2.123)

TABLE 2.123: Clinical clues for the diagnosis of cause of hemoptysis.

Clinical clues	*Suggested diagnosis*
Anticoagulant use	Medication effect, coagulation disorder
Tobacco use	Acute bronchitis, chronic bronchitis, pneumonia, lung cancer
Dyspnea on exertion, fatigue, orthopnea, paroxysmal nocturnal dyspnea, frothy pink sputum	Congestive heart failure, left ventricular failure, mitral stenosis
Fever, productive cough	Upper respiratory tract infection, acute bronchitis, pneumonia, lung abscess
History of cancer (e.g., breast, colon, or kidney)	Endobronchial metastasis from carcinoma
History of chronic lung disease, recurrent lower respiratory tract infection, cough with copious purulent sputum	Bronchiectasis, lung abscess
Pleuritic chest pain, calf tenderness	Pulmonary embolism or infarction
Toxic symptoms	Tuberculosis
Weight loss	Emphysema, lung cancer, tuberculosis, bronchiectasis, lung abscess
Melena, alcoholism, chronic use of NSAIDs	Gastritis, gastric or peptic ulcer, esophageal varices
Association with menses	Catamenial hemoptysis
Cachexia, clubbing, hoarseness	Lung cancer, small-cell carcinoma
Clubbing	Lung cancer, bronchiectasis, lung abscess
Dullness to percussion, fever, crepitations	Pneumonia

Differences between True and False Hemoptysis (Table 2.124)

TABLE 2.124: Differences between true and false hemoptysis.

True hemoptysis	*False hemoptysis (Spurious)*
Below vocal cords	Above vocal cords
Persists as blood-tinged sputum	Does not persist
May be mixed with sputum	Not mixed with sputum
History of cardiopulmonary disease	Obvious by ENT examination
Chest X-ray may be abnormal	Normal chest X-ray

Differences between Hemoptysis and Hematemesis (Table 2.125)

Q. Write short essay/note on differences between hematemesis and hemoptysis.

Investigations

When a patient comes with massive hemoptysis, initial diagnostic tests **(Table 2.126)** must begin in concert with efforts to stabilize the patient and control the bleeding.

- Other test includes antineutrophil cytoplasmic antibody, antiglomerular basement membrane antibody, and antinuclear antibody.
- **Chest X-ray:** Posteroanterior and lateral views may provide important diagnostic clues. It may provide evidence of a localized lesion, including tumor (malignant or benign), pneumonia, mycetoma or tuberculosis.

Chest X-ray may be useful in pulmonary thromboembolism, mitral stenosis, primary pulmonary hypertension, pulmonary hemosiderosis, and bronchial adenoma.

- **Computed tomography of the chest:** Useful in delineating lesions that are not seen on a plain chest X-ray and it defines **the lesions better than seen on X**-ray.
  - It is valuable in selected cases to show the presence of lung cavities, solid masses, and mediastinal and hilar lymphadenopathy.
  - Along with fiberoptic bronchoscopy, it gives a greater positive yield of pathology and is useful for excluding malignancy in high-risk patients.
  - Allows application of special imaging techniques, for example, HRCT (1–3 mm thickness section) → **bronchiectasis** and spiral CT with pulmonary angiography → **pulmonary embolism**.
- **Electrocardiogram:** It may be useful in unsuspected mitral stenosis, pulmonary thromboembolism and pulmonary hypertension.
- **Bronchoscopy:** Most important diagnostic procedure.
- Angiography
- **Isotope lung scans:** Useful when pulmonary embolism is suspected in a patient with a normal chest radiograph.

TABLE 2.125: Differences between hemoptysis and hematemesis.

Hemoptysis	*Hematemesis*
Coughing of blood. Cough precedes hemoptysis	Vomiting of blood. Nausea and vomiting precedes hematemesis
History of cardiopulmonary disease	History of gastrointestinal disease
Bright red in color	Dark brown in color
Sputum remains blood stained after the attack for few days	Usually followed by melena
Mixed with sputum	Mixed with gastric contents
Blood is frothy due to admixture of air	Airless and not frothy
Alkaline	Acidic
Sputum contains hemosiderin-laden macrophages	No
Melena absent	Melena present

TABLE 2.126: Common laboratory tests and their interpretation.

Test	*Diagnostic findings*
Total leukocyte count with differential count	Raised leukocyte count and shift to the left in upper and lower respiratory tract infections
Hemoglobin and hematocrit	Decreased in anemia
Platelet count	Reduced in thrombocytopenia
Prothrombin and partial thromboplastin time	Increased in anticoagulant use, disorders of coagulation
Arterial blood gas	Hypoxia, hypercarbia
D-dimer	Raised in pulmonary embolism
Sputum	
➢ Gram stain, and culture and sensitivity	*Klebsiella pneumoniae,* identify organism in acute exacerbations of chronic bronchitis, bronchiectasis and lung abscess
➢ AFB (Ziehl–Neelsen stain) and culture	Pulmonary tuberculosis
➢ Cytological examination	Neoplasm (bronchial carcinoma)
Tuberculin test	Positive in TB
Erythrocyte sedimentation rate (ESR)	↑ in infection, autoimmune disorders (e.g., Wegener's syndrome, SLE, Goodpasture's syndrome) and malignancy
Urine microscopy	RBCs and red cell casts in hemorrhagic diathesis and Goodpasture's syndrome

Q. Write short essay/note on:
- **Outline the management of a case of hemoptysis.**
- **Management of massive (potentially lethal) hemoptysis.**

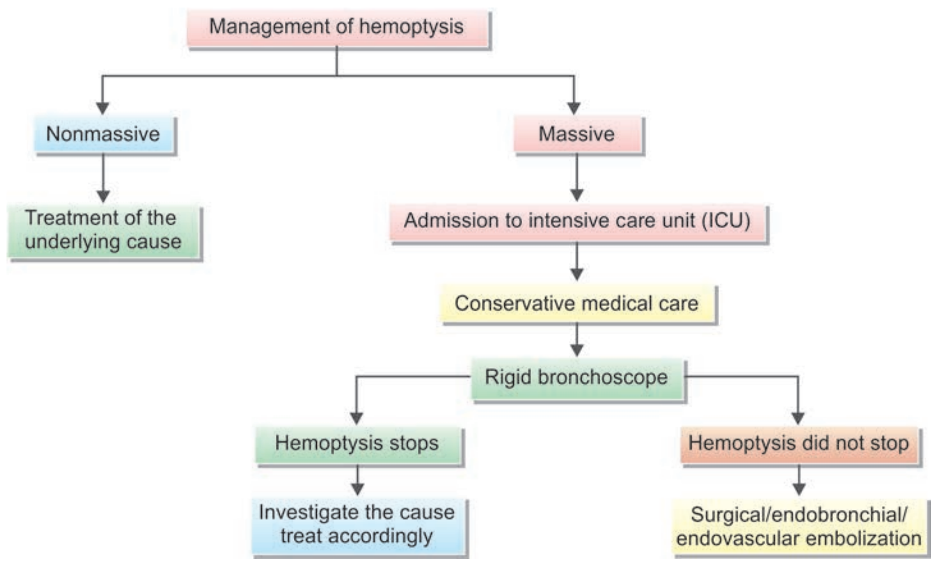
Key principles In management:
- Massive hemoptysis kills by **large airway obstruction** and not by hemorrhagic shock
- Bronchoscopic localization is **extraordinarily difficult** during active bleeding.
- An awake patient with strong cough and respiratory reserve will prevent large airway occlusion better than Intubation, bronchoscopy.
- Definitive therapy in massive hemoptysis is by **embolization**.
- All patients with underlying bronchiectasis and hemoptysis *must* be treated with antibiotics despite absence of fever, raised counts. To cover for GNRs (gram negative rods), GPC (gram positive cocci).
- Optimal position to place a nonintubated patient is usually upright and side of the bed (best position to cough and clear blood, clots).
- **Medical**
  - Endotracheal tube [single wide bore (or) double lumen]
  - Large-bore IV line for fluids, blood transfusion

- Avoid cough suppressants (if necessary benzodiazepine)
- Position of the patient in sitting (or) bleeding side down
- Supplemental oxygen/mechanical ventilation
- Vasopressin 0.2–0.4 units/min IV
- IV antibiotics in *all patients with underlying bronchiectasis* to cover GNR, GPC

- **Surgical**
  - Emergency resection for bronchogenic mass
  - Surgical resection for aspergilloma if recurrent hemoptysis
  - Resection of bronchogenic mass after patient stabilization
- **Endobronchial**
  - Identify: **S**ource, **R**ate and to **S**low (or) **A**rrest bleeding.
  - Preferred if patient unstable for CT scan to maintain airway patency and localize the bleeding.
  - The rigid bronchoscope is preferred as it enables blood to be aspirated more easily.
  - The fiberoptic bronchoscopy may be used for cold saline lavage, which may sometimes arrest bleeding. The iced saline is instilled in 50–100 mL aliquots followed by suctioning and repeated until there is noticeable improvement.
  - Other techniques used to control bleeding include: Topical thrombin or fibrinogen, topical coagulation with laser photocoagulation (Nd:YAG), argon plasma coagulator, endobronchial brachytherapy in high doses (10–12 Gy/h for a total of 500–4,000 Gy), endobronchial cryotherapy.
  - A balloon catheter passed through the bronchoscope can be inflated proximally in the bleeding bronchus. This will isolate the source of bleeding from the rest of the lung and the contralateral lung, preventing asphyxiation by blood flooding.
- **Endovascular**
  - In most patients, the bleeding originates from bronchial arteries rather than pulmonary arteries.
  - Transcatheter embolization is effective in immediate control of massive hemoptysis (73–98%).
  - Causes of recurrence: Incomplete embolization of artery, recanalization of previously embolized artery, revascularization through collateral circulation, progression of basic lung disease.

Management of hemoptysis is summarized in **Flowchart 2.23.**

FLOWCHART 2.23: Management of hemoptysis.



DYSPNEA

Q. Discuss briefly the differential diagnosis of acute-onset dyspnea.

Q. Write short note on respiratory causes of acute breathlessness (dyspnea).

Dyspnea or breathlessness is defined as **subjective experience of breathing discomfort** (i.e., uncomfortable need to breathe). It is a sense of awareness of increased respiratory effort that is unpleasant and recognized by the patient as being inappropriate. Patient often complains of tightness in the chest.

Causes of Dyspnea (Table 2.127)

Acute dyspnea: Dyspnea arising over the course of a few minutes to 24–28 hours is termed acute dyspnea.

TABLE 2.127: Causes of acute and chronic dyspnea.

Acute dyspnea	***Chronic dyspnea***
Cardiovascular system	
Cardiogenic acute pulmonary edema	Chronic heart failure, myocardial ischemia
Respiratory system	
Acute severe bronchial asthma	COPD
Acute exacerbation of COPD	Chronic bronchial asthma
Spontaneous pneumothorax	Bronchial carcinoma
Pneumonia	Interstitial lung disease (e.g., sarcoidosis, fibrosing alveolitis, extrinsic allergic alveolitis, pneumoconiosis)
Acute pulmonary embolism	
ARDS	
Inhaled foreign body (especially in children)	
Lobar collapse	Chronic pulmonary thromboembolism
Laryngeal edema (e.g., anaphylaxis) or obstruction	Lymphatic carcinomatosis
	Large pleural effusion(s)
Metabolic acidosis (e.g., diabetic ketoacidosis, lactic acidosis, uremia, overdose of salicylates, ethylene glycol poisoning)	Severe anemia
	Obesity
Psychogenic hyperventilation (anxiety or panic-related)	Deconditioning

Bronchial Obstruction

Q. Write short note on the causes, clinical presentation, and management of bronchial obstruction.

Causes

See **Table 2.128**.

TABLE 2.128: Causes of b ronchial obstruction.

Causes	***Disease/lesion***
Tumors	Bronchial adenoma or carcinoma
Enlarged tracheobronchial lymph nodes	Malignancy, Hodgkin lymphoma, non-Hodgkin lymphoma, tuberculosis
Inhaled foreign bodies	Common foreign bodies nuts, peas, beans, small pieces of toys, articles of food
Bronchial casts or plugs	Inspissated mucus (in asthma or pulmonary eosinophilia) or blood clots (severe hemoptysis)
Ineffective expectoration	Results in collection of mucus or mucopus in the bronchus
Rarely	Congenital bronchial atresia, post-tuberculous bronchial stricture, aortic aneurysm

Clinical Features

Depends on the following:
- Cause of obstruction
- *Degree of obstruction:* Complete or partial obstruction.
- Presence or absence of secondary infection
- Effect on function of lung.

Degree of Obstruction

- **Complete obstruction:**
  - *Consequences of complete obstruction:* Absorption of air distal to the obstruction, closure of alveolar spaces, collapse and solidification of the affected lung distal to obstruction.
  - *Physical signs on involved side:* Mediastinal shift to the side of collapse, dull percussion note, and absent/diminished breath sounds.
  - *Radiological features on involved side:* Mediastinal shift to the side of collapse, elevation of the diaphragm and a dense pulmonary opacity. When only small portion of the lung is collapsed, there may not be mediastinal displacement and abnormal physical signs, but a characteristic opacity will be seen on the radiograph.
- **Partial obstruction:**
  - *Consequences of partial obstruction:* Less resistance to the airflow during inspiration than during expiration and results is trapping of air distal to the obstruction. This leads to overdistension of the part of the lung distal to the obstruction (obstructive emphysema).
  - *Physical signs on involved side:* Percussion note is hyperresonant and breath sounds are diminished.
  - *Radiological features on involved side:* Hypertranslucency.

Features of Secondary Infection

- Secondary bacterial infection usually develops distal to the obstruction and is usually caused by microorganism of low virulence. Sometimes suppuration can produce lung abscess.
- It may manifest as a recurrent pneumonia in the same segment/lobe of the involved lung.

Effect on Function of Lung

- Symptoms usually follow if there is obstruction of a main or lobar bronchus.
- Sudden occlusion may produce severe breathlessness and hypoxemia.

Management
- Identify the cause of obstruction: By chest X-ray, bronchoscopy and biopsy.
- Treat the underlying cause.
- Remove foreign bodies, bronchial casts, plugs and secretions by bronchoscopy.

PULMONARY EOSINOPHILIC SYNDROMES

Q. Write short essay/note on:
- **Causes of eosinophilia**
- **Diagnosis, differential diagnosis and management/treatment of tropical pulmonary eosinophilia.**
- **Infective causes of pulmonary eosinophilia.**

Heterogenous group of pulmonary disorders characterized by pulmonary parenchymal or peripheral blood eosinophilia. They may occur due to primary pulmonary eosinophilic lung disease or secondary to specific causes.

- **Primary pulmonary eosinophilic disorders:** Acute eosinophilic pneumonia, chronic eosinophilic pneumonia, hypereosinophilic syndrome, eosinophilic granulomatosis with polyangiitis (Churg-Strauss)
- **Pulmonary disorders of known causes associated with eosinophilia:** ABPA, drug/toxin reaction, infection-parasitic/nonparasitic
- **Lung diseases associated with eosinophilia:** Cryptogenic organizing pneumonia, hypersensitivity pneumonitis, idiopathic pulmonary fibrosis, pulmonary Langerhans cell granulomatosis
- **Malignant neoplasms associated with eosinophilia:** Leukemia, lymphoma, lung cancer, adenocarcinoma/squamous cell carcinoma of various organs
- **Systemic disease associated with eosinophilia:** Postradiation pneumonitis, rheumatoid arthritis, sarcoidosis, Sjogrens syndrome

- **Infective causes of pulmonary eosinophilia.**

- **Loeffler syndrome:** Ascaris, hookworm, *Dirofilaria*, *Schistosoma*
- **Heavy parasite burden;** *Strongyloides*
- **Pulmonary penetration:** Paragonimus (endemic hemoptysis), visceral larva migrans
- **Immunologic response to organisms in lung:** *Filaria*, *Dirofilaria*
- **Cystic disease:** Echinococcus, Cysticercosis
- **Nonparasitic:** Coccidia, Paracoccidioides, tuberculosis

1. ***Loeffler's syndrome (simple pulmonary eosinophilia)***
   - It is a clinical syndrome characterized by:
     - Mild respiratory symptoms
     - Peripheral blood eosinophilia, and
     - Transient, migratory pulmonary infiltrates.
   - Affects all ages
   - **Immune hypersensitivity to *Ascaris lumbricoides*** is the likely cause. Other parasites such as *Necator*, *Ancylostoma*, and *Dirofilaria*, can be associated
   - **Chest X-ray:** Transient, migratory, nonsegmental interstitial and alveolar infiltrates (often peripheral or pleural based).
   - **Pulmonary function tests:** Typically reveals mild-to-moderate restrictive ventilatory defect with a reduced diffusing capacity of the lungs for carbon monoxide (DLCO).
2. ***Drug and toxin-induced pulmonary eosinophilic syndromes***
   - Onset: Acute or subacute.
   - *Respiratory symptoms:* Vary widely in severity.
     - Mild Loeffler's-like illness with dyspnea, cough, and fever
     - Severe fulminant respiratory failure
   - Drugs implicated are acetyl salicylate, nitrofurantoin, bleomycin, methotrexate, minocycline, sulfa drugs, gold salts, INH, etc.
3. ***Idiopathic acute eosinophilic pneumonia***
   - More common in younger men (mean age about 30 years).
   - Occurs commonly in previously healthy persons. Also seen in persons with history of chronic myeloid leukemia (CML), HIV infection, recent commencement of **smoking**, etc.
   - Commonly presents like ARDS
   - *Diagnostic criteria:*
     - Acute onset of febrile respiratory manifestations (≤1 month duration before consultation).
     - Bilateral diffuse infiltrates on chest X-ray.
     - Hypoxemia, with PaO_2 on room air <60 mm Hg, and/or PaO_2/FiO_2 ≤300 mm Hg, and/or oxygen saturation on room air <90%.

- Lung eosinophilia, with >25% eosinophils in BAL (or eosinophilic pneumonia at lung biopsy).
- Absence of infection, or of other known causes of eosinophilic lung disease (especially, exposure to drug known to induce pulmonary eosinophilia).

Treatment
- Initial doses of methyl-prednisolone used in the range from 60 to 125 mg every 6 hours.
- After resolution of respiratory failure, oral prednisolone (in doses of 40–60 mg per day) may be continued for 2–4 weeks with a subsequent slow taper over the next several weeks.
- Idiopathic acute eosinophilic pneumonia (AEP) carries an excellent prognosis.

4. ***Tropical pulmonary eosinophilia***

Q. Write short note on tropical pulmonary eosinophilia, its diagnosis and management.

Tropical pulmonary eosinophilia (TPE) was first described in the early 1940s by Weingarten, in India. It is seen mainly in South and South-east Asia and Africa and previously termed *"pseudotuberculosis with eosinophilia".*

Definition: Tropical pulmonary eosinophilia is an occasional atypical host response (hypersensitivity reaction) of an individual to a mosquito-borne filarial infection by tissue-dwelling human nematode (microfilariae) *Wuchereria bancrofti* and *Brugia malayi.*

Pathogenesis: When filarial parasites are destroyed, the antigens [**Bm22-25, gamma-glutamyl transferase (GGT)** of *B. malayi*], released initiate an immediate IgE-mediated reaction (**type 1 hypersensitivity reaction)**. Subsequent eosinophilic inflammation and degranulation of eosinophilic enzymes result in lung damage and clearance of microfilariae ultimately resulting in granuloma formation and fibrosis. In children, marked enlargement of lymph nodes and spleen **(Meyers-Kouwenaar syndrome)** may be evident. In adults, symptoms are predominantly due to lung involvement. Microfilariae are trapped in the pulmonary capillaries and produce symptoms of lung involvement.

Clinical features
- Paroxysmal dry cough, fever, malaise, anorexia, weight loss, dyspnea or wheeze/nocturnal bronchospasm (asthma-like symptoms) and miliary pulmonary infiltrates **(Weingarten syndrome or tropical pulmonary eosinophilia).**
- Tropical pulmonary eosinophilia is a complication seen mainly in India. If untreated, it may progress to chronic interstitial lung disease. Spontaneous resolution over several weeks.
- *Other features:*
  - Presentation can be similar to status asthmaticus
  - Chest pain, muscle tenderness, pericardial and CNS involvement
  - Rarely, patients remain asymptomatic.
- *Chest examination:* Coarse crackles and rhonchi.
- Generalized lymphadenopathy and hepatosplenomegaly may be present.

Evaluation
- **CBC:** Marked peripheral blood eosinophilia >3000 cell/mm^3
- **Stool exam:** To rule out other parasitic causes of eosinophilia
- **IgE levels:** Usually >1000 u/L
- **Indirect ELISA:** Rising titers helps confirm diagnosis
- **CXR:** Miliary mottling in middle, lower lobes, reticulonodular opacities. 20–30% have normal CXR
- **HRCT:** Bronchiectasis, lymphadenopathy, pleural effusion. Late stages fibrosis and interstitial lung disease (ILD)
- **PFT: Show *mixed pattern* with predominant restriction and mild-moderate obstruction.**

Diagnosis
- History of travel to filarial **endemic area**
- Paroxysmal and **nocturnal cough** with dyspnea
- Leucocytosis with marked eosinophilia >3000
- Serology: ↑IgE, filarial antibodies
- Pulmonary infiltrates on CXR
- **Response to diethylcarbamazine (DEC) (6 mg/kg/day)** within 7–10 days of therapy

Treatment: Diethylcarbamazine in a dose of 6 mg/kg orally three times a day for 14–21 days or for as long as 4 weeks (better outcomes). It is directly filaricidal to both adultworms and microfilariae.
Oral steroids may be tried in nonresponders (20–40%) after ruling out *Strongyloides* infection

5. ***Chronic eosinophilic pneumonia***
   - Typically subacute presentation.
   - Symptoms present for several months before diagnosis.
   - Common presenting complaints include: Low-grade fevers, drenching night sweats, moderate (10–50 pound) weight loss, cough.
   - History of atopy, allergic rhinitis, or nasal polyps may be found.
   - About 2/3 develop adult-onset asthma: Preceding or concurrent with the occurrence of CEP.
   - No major extrapulmonary manifestations
   - Chest X-ray: Bilateral opacities in upper and mid zone. Photographic negative of pulmonary edema.
   - Corticosteroids are the mainstay of therapy for CEP. Dramatic clinical, radiographic, and physiological improvements.
6. ***Allergic bronchopulmonary aspergillosis*** *(discussed earlier)*
7. ***Churg–Strauss syndrome:*** It is a form of necrotizing vasculitis in several organs, associated with eosinophilic tissue inflammation and extravascular granulomas. It occurs in asthmatics and presents with fever and peripheral hypereosinophilia.
8. ***Idiopathic hypereosinophilic syndrome***
   - **Several names:** Eosinophilic leukemia, Loeffler's fibroblastic endocarditis, disseminated eosinophilic cardiovascular disease.
   - **Clinical features:** Often nonspecific and include:
     - Weakness, fatigue, low-grade fevers, myalgias, cough, angioedema, rash, retinal lesions, and dyspnea.
     - Can affect every organ system.
     - Cough is nocturnal, either nonproductive or productive of small quantities of nonpurulent sputum.
     - Wheezing and dyspnea without evidence of airflow obstruction on spirometry.
     - Pulmonary hypertension, ARDS, and pleural effusions.
     - Progressive chronic heart failure due to eosinophilic myocarditis and endocarditis, intracardiac thrombi and endocardial fibrosis.
     - Encephalopathy with neuropsychiatric dysfunction, thromboembolic events such as hemiparesis. Peripheral neuropathy is extremely common in IHS.
     - Hepatosplenomegaly, lymphadenopathy
   - **Investigations:**
     - Anemia, thrombocytopenia
     - Elevated vitamin B_{12} levels
     - Bone marrow: Universally affected with a striking eosinophilia (up to 25–75% of the differential count).
     - Mutation testing: PDGFR-A, B, FIP1L1 testing.

Treatment
- Glucocorticoids are the first-line therapy in all patients without FIP1L1/PDGFRA mutation.
- Patients with FIP1L1/PDGFRA mutation: Imatinib is the drug of choice with a very good response rate.
- Other drugs include: Hydroxyurea, vincristine, interferons, anti–IL-5 monoclonal antibody (e.g., mepolizumab) and an anti-CD52 antibody (alemtuzumab).

Causes of Calcification in Lung Parenchyma (Box 2.19)

Q. Write short note on causes of calcification in lung parenchyma.

Box 2.19: Calcification in lung parenchyma.

Micronodules
Pulmonary alveolar microlithiasis, occupational lung diseases (e.g., silicosis, coal-worker's pneumoconiosis)
Large nodules or masses
- Granulomatous disease (tuberculosis, histoplasmosis)
- Primary lung cancer (5–10%)
- Metastasis to lung (e.g., osteosarcoma, chondrosarcoma)
- Nonmalignant metastatic calcification in lung (e.g., renal failure, hyperparathyroidism)
- Multiple pulmonary chondroma (e.g., with Carney triad)
- *Others:* Chickenpox pneumonia, hemosiderosis, pulmonary hemorrhage

Breath Sounds (Fig. 2.55)

Q. Write short note on:
- **Causes of bronchial breathing, tubular and amphoric breathing. Three components of tubular breathing.**
- **Cavernous breathing.**

Normal breath sounds are produced by vibration of vocal cords due to turbulent flow in the larynx. This sound is harsher anteriorly over the upper lobes (particularly on the right).

FIG. 2.55: Types of breath sounds.

- **Vesicular breath sounds:** Healthy lungs filter out most of the high-frequency component, and the resulting low-pitched, rustling sounds are called vesicular. Inspiration is longer than expiration and there is no gap in between and rustling in character.
  - Conditions with diminished vesicular breath sounds are heard in bronchial asthma, tumors, pleural effusion (small), pleural thickening, collapsed lung with occluded bronchus, and emphysema.
- **Bronchial breath sounds:** They are produced by the passage of air through the trachea and large bronchi. It is loud and high pitched. Inspiration and expiration are of equal duration, there is a gap between inspiration and expiration, and the quality is guttural or aspirate.
  - *Types of bronchial breathing*
    - **Tubular:** High pitched and heard in pneumonia with consolidation, partially collapsed lung or lobe, above the level of pleural effusion (relaxation atelectasis) and Ewart's sign (pericardial effusion compressing left main bronchus).
    - **Cavernous:** Low pitched and heard in thick-walled cavity with communicating bronchus.
    - **Amphoric:** Low pitched with high tone and a metallic quality. It is heard in bronchopleural fistula, rarely over a superficial cavity and tension pneumothorax.
- **Bronchovesicular breath sound:** This is classically seen in obstructive lung disease where expiration is prolonged; however, there is no gap in between.

Indications of Bronchoscopy (Table 2.129)

Q. List the indications for bronchoscopy.

TABLE 2.129: Indications of bronchoscopy.

Diagnostic	*Therapeutic*
➢ Diagnosis of lung cancer ➢ Chest radiographic abnormality (lung tumor or changes suggestive of bronchial obstruction, such as appearance of early volume loss or undoubted collapse, unresolved pneumonia or hemidiaphragmatic paralysis) ➢ Evaluation of hemoptysis ➢ Evaluation of persistent or recurrent cough ➢ Evaluation of paralyzed vocal cord ➢ To obtain positive sputum cytology ➢ Staging of lung cancer ➢ Diagnosis of diffuse lung disease ➢ Identification of infecting agents	➢ Insertion of an endotracheal tube for general anesthesia in patients in whom extension of neck may be dangerous (atlantoaxial subluxation) ➢ Tamponade of endobronchial bleeding, either with end of bronchoscope itself or by using a Fogarty ➢ Removal of foreign bodies ➢ Aspiration of secretions in acute inflammatory lobar atelectasis where physiotherapy has proved unsuccessful in achieving this ➢ Relief of tracheobronchial narrowing by laser treatment ➢ Treatment of lung cancer by placement of stents or delivery of endobronchial radiotherapy (brachytherapy)

Bronchoalveolar Lavage

Q. Write a short note on bronchoalveolar lavage.

Bronchoalveolar lavage (BAL) is a diagnostic procedure used to recover cellular and noncellular components of the epithelial lining fluid from the alveolar and bronchial air spaces.

Common Findings on BAL

Increased neutrophils (>5%) (Normal <3%)	Idiopathic pulmonary fibrosis (IPF), acute respiratory distress syndrome (ARDS), infections, connective tissue disorders, pneumoconiosis
Eosinophilia (Normal <1–2%)	>25%: Suggesting eosinophilic lung diseases such as acute eosinophilic pneumonia, chronic eosinophilic pneumonia, and Churg–Strauss syndrome

Contd...

Contd...

Lymphocytosis (Normal <15%)	>50%: Hypersensitivity pneumonitis When greater than or equal to 15% lymphocytes are present, CD4/CD8 ratios can be assessed. ➢ Elevated CD4/CD8: Hypersensitivity pneumonitis (chronic or smoker), sarcoidosis, berylliosis, asbestosis, Crohn disease, connective tissue disorders ➢ Normal CD4/CD8: Tuberculosis, malignancies ➢ Low CD4/CD8: Acute hypersensitivity pneumonitis, silicosis, drug-induced lung disease, HIV infection, bronchiolitis obliterans with organizing pneumonia (BOOP) [cryptogenic organizing pneumonia (COP)]
Alveolar macrophages (normal >80%)	Decreased in sarcoidosis (to less than 50%)
RBCs	Alveolar hemorrhage
Diagnosis of infection	*Mycobacterium tuberculosis, Aspergillus, Pneumocystis carinii, Toxoplasma gondii, Strongyloides stercoralis, Legionella pneumophila, Cryptococcus neoformans, Histoplasma capsulatum, Mycoplasma pneumoniae,* influenza A and B viruses, respiratory syncytial virus
Diagnosis of malignancy	
Progressively darkening aliquots of bloody BAL fluid is highly suggestive of diffuse alveolar hemorrhage	
Pulmonary alveolar microlithiasis—calcospherites can be demonstrated in BAL fluid	